SUITABILITY OF THE RSP2PRO SOFTWARE-DEFINED RADIO FOR PRECOMPLIANCE RADIATED EMISSIONS TESTING

By

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ABSTRACT

SUITABILITY OF THE RSP2PRO SOFTWARE-DEFINED RADIO FOR PRECOMPLIANCE RADIATED EMISSIONS TESTING

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This thesis demonstrates the ability to use an existing software-defined radio for accurate RF measurements related to the precompliance testing of radiated emissions. The software-defined radio chosen for this work is the SDRplay Radio Spectrum Processor 2 Pro (RSP2pro). The RSP2pro was chosen because of its ability to provide absolute received power measurement capabilities at a price much lower than a traditional spectrum analyzer.

The absolute power level and frequency performance of the RSP2pro has been characterized by direct comparison with an electromagnetic compatibility analyzer and the results are shown. A method for automating measurements taken with the RSP2pro companion software is described. The RSP2pro is also used to perform precompliance radiated emissions tests with an electromagnetic compatibility antenna and the results are discussed. Throughout this work a sample of three RSP2pro units is compared and variations in measurements across the three units is examined.

A common very-high frequency "rabbit ear" television antenna and an ultra-high frequency "bowtie" television antenna are used with the RSP2pro to create a complete radiated emissions precompliance test setup. A procedure for characterizing these antennas and determining their suitability for radiated emissions precompliance testing is also described. The RSP2pro test setup for measuring radiated emissions is compared to a traditional precompliance test setup consisting of an electromagnetic compatibility spectrum analyzer and an electromagnetic compatibility antenna. This comparison includes a discussion of the relative cost, measurement accuracy, and overall convenience of using each test setup.

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For Uncle Mike

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KEY TO SYMBOLS AND ABBREVIATIONS

RSP2pro	Radio Spectrum Processor 2 Pro
SDR	software-defined radio
EMC	electromagnetic compatibility
EUT	equipment under test
EMI	electromagnetic interference
LED	light-emitting diode
dBFS	dB full-scale
FCC	Federal Communications Commission
EU	European Union
CISPR	Comité International Spécial des Perturbations Radio
QP	quasi-peak
OATS	open area test site
IF	intermediate frequency
RBW	resolution bandwidth
A/D	analog-to-digital
IQ	in-phase and quadrature
GPP	general purpose processor
DSP	digital signal processor
FPGA	field-programmable gate array
ASIC	application-specific integrated circuit
TCXO	temperature-compensated oscillator
SNR	signal-to-noise ratio
CW	continuous wave
GPIB	general-purpose interface bus
ADC	analog-to-digital converter

CHAPTER 1

INTRODUCTION

This thesis investigates using the SDRplay Radio Spectrum Processor 2 Pro (RSP2pro) software-defined radio (SDR) to perform accurate RF measurements related to the precompliance testing of radiated emissions.

Chapter 2 gives an overview of electromagnetic compatibility (EMC) with a focus on radiated emissions measurements, regulations, and precompliance techniques. The basics of SDR implementation and typical SDR implementations are also described. The methodology for selecting the RSP2pro for this research is explained. A description of the RSP2pro and its companion software, SDRuno, is provided.

Chapter 3 assesses the suitability of the RSP2pro for precompliance radiated emissions testing by directly comparing its measurement capabilities to an EMC analyzer. The process for characterizing both the absolute power and frequency performance of the RSP2pro is explained and the results are examined. A method for automating measurements with SDRuno is also described.

The RSP2pro is used for precompliance radiated emissions tests in Chapter 4. A detailed description of the custom circuit serving as the equipment under test (EUT) is given. Radiated emissions measurements using the RSP2pro and an EMC antenna are compared with measurements using an EMC analyzer and the same EMC antenna.

Chapter 5 describes the use of two common television antennas for radiated emissions measurements in the RF range of 30 MHz to 1000 MHz. A detailed description of each antenna is given and the results obtained using these antennas are compared to results obtained with an EMC antenna. The utility of the RSP2pro test setup for measuring radiated emissions is then compared to a traditional precompliance test setup consisting of an EMC analyzer and an EMC antenna.

Chapter 6 is the conclusion of this thesis and includes both a summary of the results and a proposal for future work related to the use of the RSP2pro for more accurate radiated emissions measurements.

CHAPTER 2

BACKGROUND

An electronic product is any electronic system that is intended for use by an end-user in a domestic or industrial environment. Section 2.1 summarizes the importance of EMC and one of its main areas, electromagnetic interference (EMI) in the form of the radiated emissions created by an electronic product. Governmental regulations related to radiated emissions and the differences between official EMC testing procedures and precompliance testing procedures are explained in Section 2.2. This includes a discussion about existing "low-cost" precompliance techniques. An introduction to SDR technology is given in Section 2.3. Finally, a description of the RSP2pro and its companion software, SDRuno, is given Section 2.4.

2.1 Electromagnetic Compatibility and Radiated Emissions

An electronic product meets electromagnetic compatibility (EMC) requirements when it does not interfere with its surrounding RF environment while also being able to accept electronic interference without losing proper functionality [1]. The former describes minimizing the EMI created by the product itself while the latter describes ensuring that the product is immune to EMI. According to Paul [1], all of the EMC problems related to an electronic product can be summarized into the four generalized cases shown in Figure 2.1. Conducted emissions and susceptibility describe EMI that is generated (emissions) or received (susceptibility/immunity) by the product through its power cord, while radiated emissions and susceptibility describe EMI that is generated or received by the product in the form of RF radiation. Designing for immunity is generally built into the development of an electronic product because it is apparent to the designers (and the end-user) when a product is not operating correctly [2]. This same "self-regulation" does not apply to the EMI created by the product, as the product may not interfere with itself and will operate normally. Emissions from a product are therefore more likely to go unnoticed until an EMC issue is discovered much later in the design process. The insidious nature of radiated and conducted

emissions, combined with the need for an environment as free from EMI as possible, is why many agencies around the world set compliance standards for the acceptable amount of EMI created by an electronic product. These compliance standards must be met before a product can be placed on the market. While conducted emissions are an important part of the compliance process, the techniques and procedures in this thesis will be focused solely on radiated emissions. This is because radiated emissions are both poorly understood by most electronic product designers and are the most difficult part of passing an official compliance test [3].

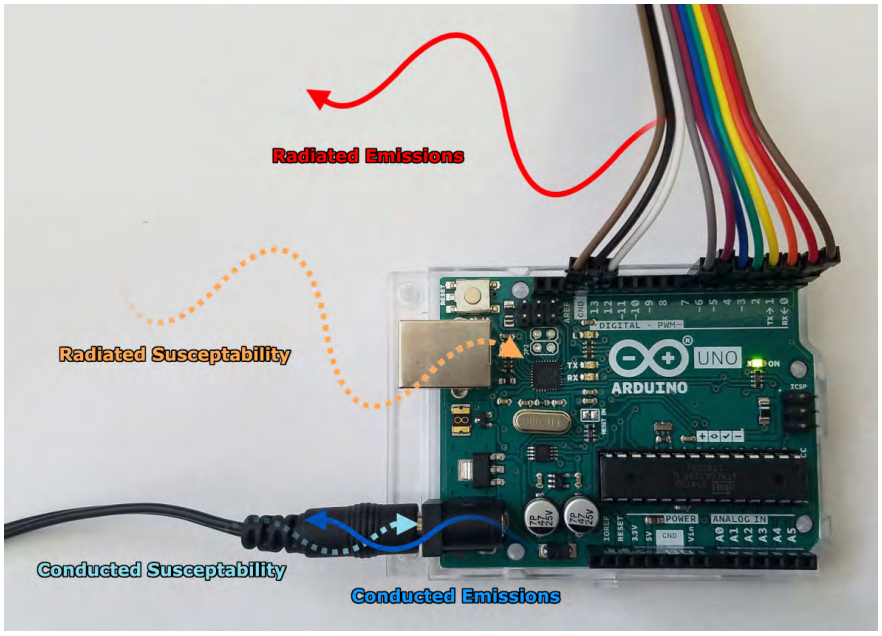


Figure 2.1: The four main EMC issues that can effect an electronic product.

Radiated emissions have a very specific definition within EMC and are not to be confused with intentional RF signals such as those from a radio station or wireless internet router. While intentional radiators can cause EMI issues, radiated emissions in an EMC context most commonly refers to the unintentional RF signals that originate from the digital circuits within an electronic product [1]. When measuring radiated emissions for comparison to regulatory limits, the strength of the electric field at a given distance from the electronic product is measured. The units of this measurement are in dBμV/m (decibels relative to one microvolt, per meter). Before reviewing the

compliance standards and measurement practices related to radiated emissions, it is important to first understand where these types of emissions come from.

Any metallic component that is capable of carrying electric current has the potential to become a radiating structure that will create EMI [1]. Common examples of such components include PCB traces, ribbon cables, and the product enclosure. When these components become large enough relative to the wavelength of the current they are carrying, the components can produce radiated emissions [4]. Most of the current in a conductor travels in a closed loop like the one shown in Figure 2.2. Here the total current that originates from the integrated circuit pin circled in magenta

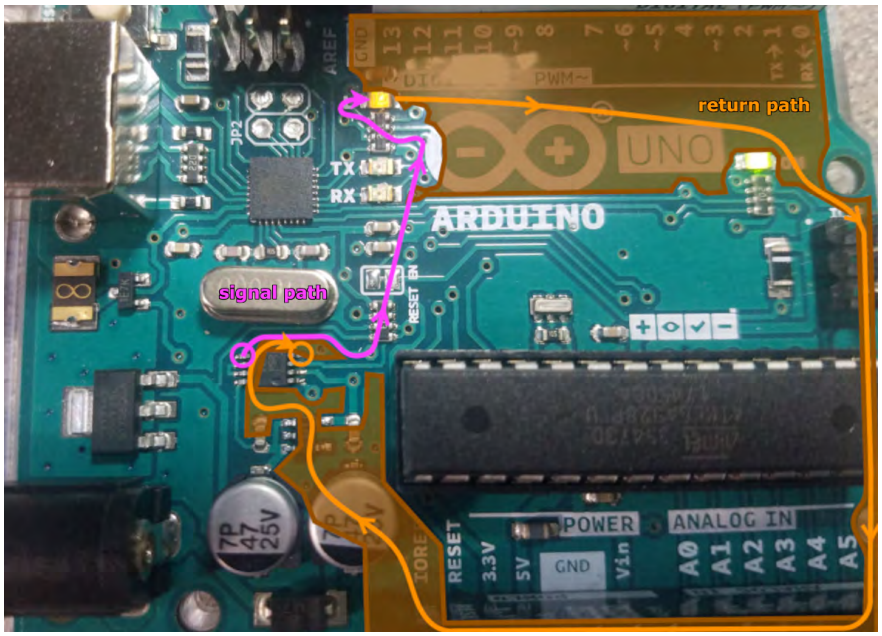


Figure 2.2: An example of a current loop created by a signal on a PCB.

creates a loop consisting of the pink path that goes to the yellow LED and the orange return path through the ground copper fill region shown in light orange. This current loop exists because a predominant portion of the current continues along the path of least impedance within the ground region to the point from which it originated [5]. This type of current is called a differential-mode current because it is returning to its source, creating a closed loop. The loop shown in Figure 2.2 is rather large because the electronic product shown uses a two-layer PCB. This loop could be made

considerably smaller if there was a separate ground layer, but this adds manufacturing cost and increases the complexity of the design. Note that the loop here is only being used as an example of how the current travels in the circuit; the actual switching current on the path shown is well below 50 kHz and is not a concern for creating radiated emissions [5]. This is why two signal layers are sufficient for this PCB.

A PCB trace above a ground plane and the individual wires in a ribbon cable are both examples of parallel conductors that are commonly found in electronic products. There are two types of electric current that exist along parallel conductors that contribute to radiated emissions: differential-mode current and common-mode current [6]. Differential-mode currents are easily described with circuit theory and exist in equal but opposite directions along parallel conductors (recall the current loop shown in Figure 2.2). These currents are the ideal signals being carried by the conductors in a circuit [7].

Common-mode currents are more difficult to predict because they exist in the same direction along parallel conductors and are created by the parasitic impedances within the conductors in the circuit [2]. When a cable is connected to a digital circuit, a common-mode current can exist along the cable and will cause it to behave like an antenna that effectively radiates electric fields. Common-mode currents do not only exist in the cables connected to an electronic product. For example, common-mode currents are a major source of noise in microcomputer circuits that do not use external memory [5]. Special attention is given to the cabling of electronic products because they can easily cause the common-mode currents from within the product to escape into the RF environment. This is especially true if the cable connectors are either poorly bonded to the product's metallic enclosure or if the product uses a nonmetallic enclosure [8].

The total current in a conductor can be represented as the sum of the differential and common-mode parts. The relationship between the currents I_1 and I_2 carried by two parallel conductors and the common-mode and differential-mode current, I_C and I_D, respectively, is

$$I_1 = I_C + I_D, \tag{2.1}$$

$$I_2 = I_C - I_D. \tag{2.2}$$

By subtracting (2.2) from (2.1), an equation for the differential-mode current I_D along two parallel conductors can be expressed as

$$I_D = \Big(I_1 - I_2\Big)/2. \tag{2.3}$$

The total currents I_1 and I_2 subtract in (2.3) and thus most of the radiated emissions created by the differential-mode current cancel out when I_1 and I_2 are similar in both phase and magnitude. The radiated emissions are not exactly zero even when I_1 is identical to I_2 because the parallel conductors are separated by some distance. This is illustrated in Figure 2.3, where the differential current is shown in magenta and the parallel conductors are separated by a small distance $d << \lambda$ where λ is the wavelength of the signal creating the current.

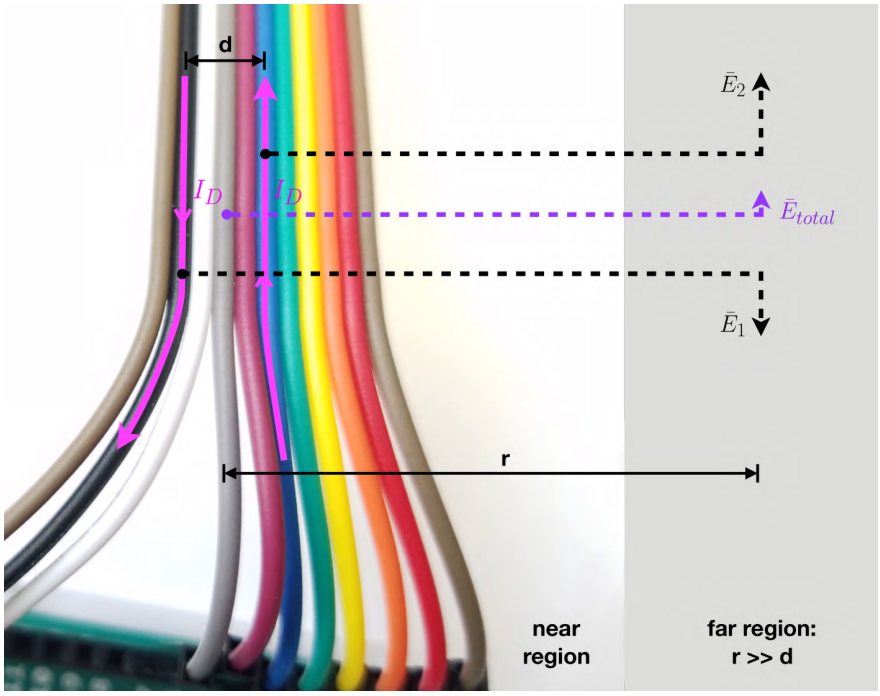


Figure 2.3: The electric fields created by the differential-mode current along parallel conductors.

For simplicity, only the electric field created parallel to the ribbon cable has been considered. The second conductor is closer to the observation point r by a distance of d, so the magnitude of the electric field created by the second conductor, $|\bar{E}_2|$, is slightly larger than the magnitude of the electric field created at this point by the first conductor, $|\bar{E}_1|$. The value of the net field

$\bar{E}_{total} = \bar{E}_2 - \bar{E}_1$ created by the differential-current at r is small, but nonzero. Differential-mode currents also create an electric potential when they travel through the parasitic impedances within the conductors of a circuit [2]. This "common-mode potential" then creates the troublesome common-mode currents that radiate electric fields much more effectively.

To show that common-mode currents create much larger radiated emissions, we add and subtract (2.2) and (2.1) to obtain an equation for the common-mode current I_C along two parallel conductors. This equation can be expressed as

$$I_C = \Big(I_1 + I_2\Big)/2. \tag{2.4}$$

Unlike before with the differential-mode current, the fields produced by the currents I_1 and I_2 now add together. The effect of this addition is shown in Figure 2.4. Here, the common-mode current

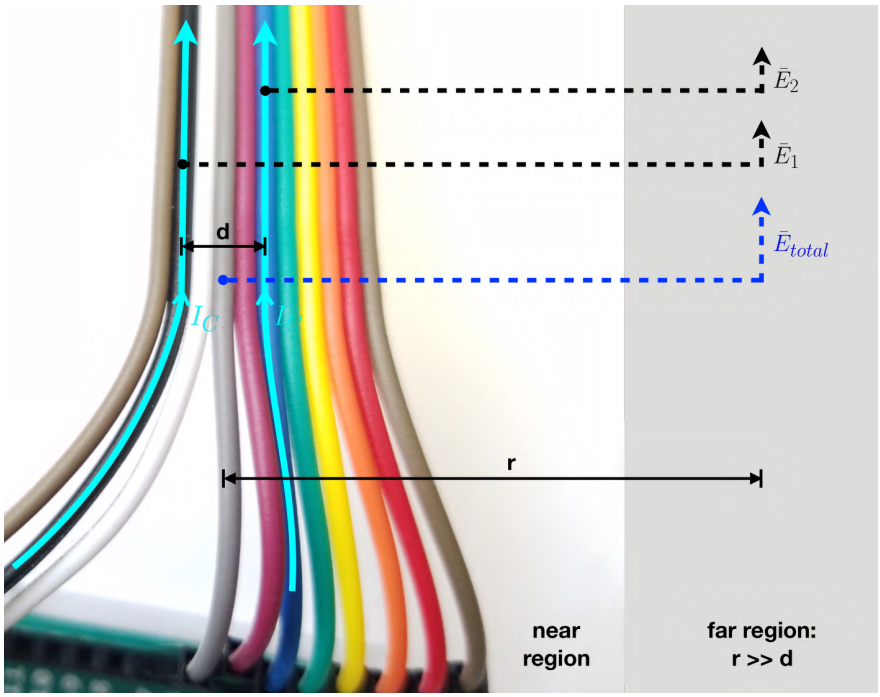


Figure 2.4: The electric fields created by the common-mode current along parallel conductors.

is shown in cyan and the parallel conductors are again separated by a small distance $d << \lambda$ where λ is the wavelength of the signal creating the current. Just like before, only the electric field created parallel to the ribbon cable is considered. The second conductor is closer to the observation point

r by a distance of d, so the magnitude of the electric field created by the second conductor, $|\bar{E}_2|$, is slightly larger than the magnitude of the electric field created at this point by the first conductor, $|\bar{E}_1|$. However, now the value of the net field is $\bar{E}_{total} = \bar{E}_2 + \bar{E}_1$, which is larger than the fields created by the individual portions of the common-mode current (as long as $d << \lambda$). This is significant because it shows that even small common-mode currents can create electric fields. The electric fields shown in Figure 2.3 and Figure 2.4 are provided only for illustrative purposes and represent a very simplistic way of modeling the electric fields created by differential-mode and common-mode currents. More detailed methods for predicting the electric fields created by these currents can be found in [1], [2], and [7]. In Section 4.1 of this thesis, one such method is used to predict the radiated emissions created by the common-mode currents in the cabling of the custom equipment under test (EUT) that was designed to exceed regulatory limits.

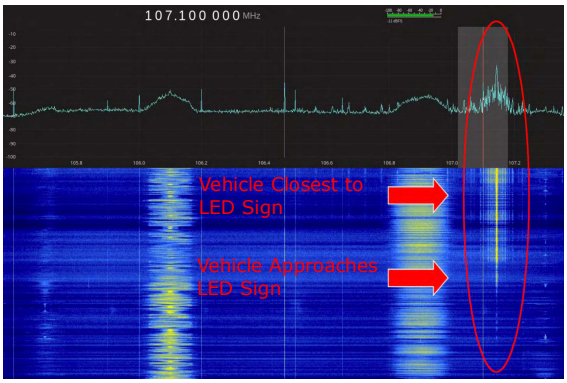
Radiated emissions can cause a variety of issues in the local RF environment when left unchecked. Light-emitting diode (LED) devices like the sign shown in Figure 2.5 have come under scrutiny in recent years because they can create sporadic, broadband radiated emissions that interfere with other electronic products [9], [10], [11], [12].

The LED sign shown in Figure 2.5 generates consistent broadband emissions centered at a frequency of approximately 107.1 MHz [13]. These emissions are powerful enough to interfere

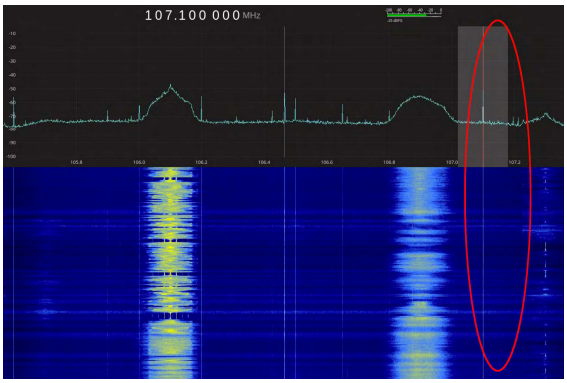


Figure 2.5: An example of a LED sign that creates EMI.

with FM radio broadcasts and short range transmissions from vehicle key fobs [13]. A sample of these radiated emissions are shown in Figure 2.6. In this figure, the x-axis is the frequency of the signal in MHz and the y-axis for the black spectrum display is the RF signal magnitude in relative units, or "dB full-scale" (dBFS). The y-axis for the blue waterfall plot has units of seconds and has units of seconds and shows how the measured signal varies with time, with the most recent measurement being shown at the top.



(a)



(b)

Figure 2.6: (a) Radiated emissions created by the LED sign from Figure 2.5 and (b) a control sample measured 1 km away from the LED sign.

Figure 2.6a shows a measurement of the RF spectrum near the LED sign. The radiated emissions from the sign are circled in red. The other two peaks towards the left of the spectrum display are the signals from nearby FM radio stations. Notice that the radiated emissions from the LED sign are stronger than the FM radio signals. This measurement was taken from a moving vehicle that passed by the sign at a distance of approximately 15 meters. The approach of the vehicle towards

the sign is shown by the blue waterfall plot. The intensity of the radiated emissions appeared and then greatly increased about halfway through the measurement which indicates that the vehicle was approaching the source of the radiated emissions (the LED sign).

Figure 2.6b shows a control measurement of the same part of the RF spectrum approximately one kilometer away from the LED sign. The radiated emissions are completely absent in the control sample while the FM radio signals are still present. This is because the radiated emissions are being created by the LED sign. Since the radiated emissions are strong enough to cause noticeable interference in the area near the LED sign, the regulatory limits are most likely being violated and the sign should be reported to the appropriate authorities.

Large LED signs are not the only unusual source of this type of interference. There are cases where vehicle key fobs and cell phones have been affected by non-compliant switch-mode power supplies found in other types of electronic signs [14]. There are also cases where EMI from radiated emissions can have dangerous consequences. In August 2019, an open-heart surgery blood pump was the subject of a major product recall after forty-five reports of injuries related to the system malfunctioning when in the presence of EMI [15]. Cases like this illustrate how important it is for electronic product designers to fully understand the effects of radiated emissions and why it is equally important for regulatory agencies to enforce strict emission limits.

2.2 EMC Regulations and Precompliance Testing

Different organizations from individual countries develop and enforce standards for what constitutes an acceptable amount of radiated emissions from an electronic product [6]. In the United States, the regulatory organization is the Federal Communications Commission (FCC). A summary of the regulatory organizations in the Americas, Europe, China, Taiwan, and Japan can be found in [16]. This reference also includes a summary of the regulatory procedures used in each of these regions.

While each country may have its own regulatory procedures related to radiated emissions, many countries are slowly moving towards the standards that were created by the Comité International Spécial des Perturbations Radio (CISPR) in 1985 [2]. These standards were formally adopted by the European Union (EU) as EN 55022, but are more commonly referred to as CISPR-22 [17]. These types of standards are often updated to improve harmonization. For example, in 2017 EN 55022 was combined with EU EMC directive EN 55032 [18]. While this change did improve standard harmonization, it also reclassified many electronic products that were originally exempt from testing. This reclassification required many product designers to retroactively fix the designs of existing products before they could continue to be marketed in the EU. Situations like this will continue to happen as more countries move towards unified EMC standards, so it is important for designers to ensure that their products meet the most ubiquitous standards before the end of the design process. Focusing on CISPR-22 and similar EU EMC standards is good practice because they are considered the "most likely vehicle for accomplishing [international] harmonization" [2].

The CISPR-22 standard applies to radiated emissions in the frequency range from 30 MHz to 1000 MHz and designates two broad categories for electronic devices: Class A and Class B. The Class A standards specify measurements 30 m from the EUT and are applied to "industrial" devices, while the Class B standards specify measurements 10 m from the EUT and are applied to "residential, commercial, and light-industrial" devices [16]. The values of the limits are identical for both categories, but the difference in measurement distance makes the Class B limits more stringent. The FCC uses identically named categories that are much broader in scope: FCC Class A devices

are designated as "non-residential" and FCC Class B devices are designated as "residential" [16]. Just like the CISPR-22 classifications, the FCC Class B limits are more stringent. Figure 2.7 shows the radiated emission limits for Class B devices in both the EU and the United States.

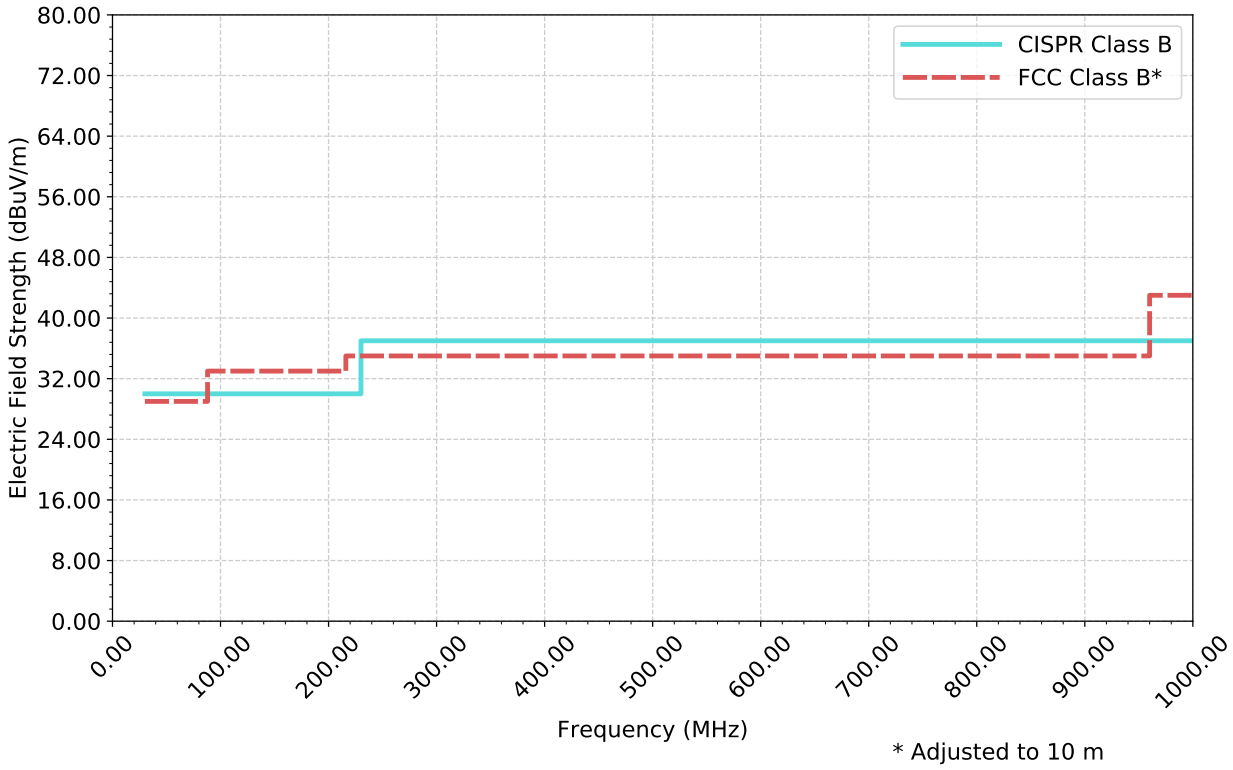


Figure 2.7: Radiated emission limits measured at a distance of 10 meters for the European Union (solid) and the United States (dashed).

The FCC specifies that the radiated emissions from Class B devices are to be measured at a distance of 3 m, so the values in Figure 2.7 have been adjusted to 10 m for a better comparison to the CISPR-22 Class B limits. From this comparison one can see that the limits between the two standards are very close.

The limits have units of dBμV/m because the standards apply to the strength of the electric field, not the voltage measured by the receiver. To convert between the measured voltage and the electric field strength of the received signal, EMC antennas are provided with a special parameter known as the antenna (or transducer) factor. This parameter is defined over the entire usable frequency

range of the antenna. The antenna factor AF has units of 1/m and is defined as the ratio between the electric field incident on the antenna E and the voltage at its terminals V [19]:

$$AF = E \big/ V \tag{2.5}$$

A direct result of this definition is that a smaller antenna factor indicates that an antenna is more sensitive (and antenna sensitivity is especially important for measurements close to 1 m) [2]. The manufacturer will often provide the antenna factor in decibel units. This is convenient because the received voltage V can then be measured in units of dBμV and added to the decibel antenna factor AF to calculate the received electric field strength E in units of dBμV/m:

$$E\left(dB\mu V/m\right) = V\left(dB\mu V\right) + AF\left(dB\ 1/m\right) \tag{2.6}$$

The relationship between the antenna factor and the measured radiated emission strength is summarized in Figure 2.8.

Before discussing CISPR-22 further, it is important to mention that receivers must meet strict hardware requirements in order to be used for full compliance measurements. These hardware

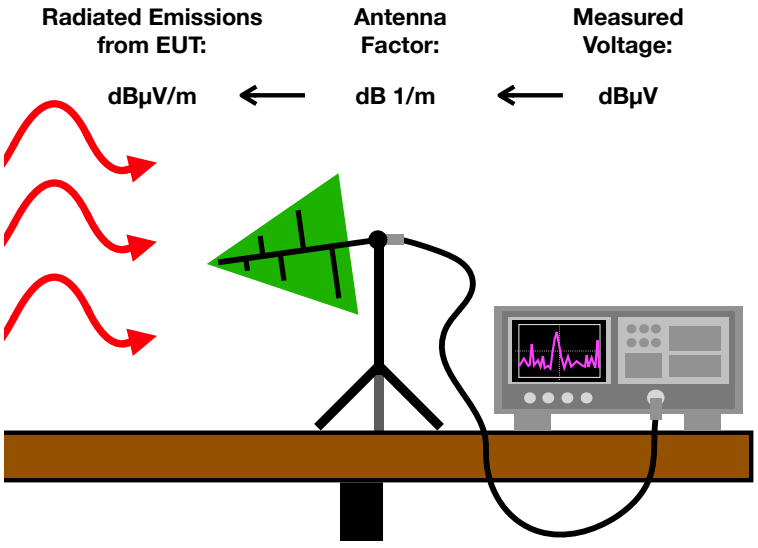


Figure 2.8: The antenna factor can be used to easily convert a measured voltage back into the electric field strength, which can then be directly compared to the regulatory limits.

requirements are specified by CISPR-16-1 [17]. For the purposes of this thesis, only the CISPR-16-1 requirements on receiver detectors will be discussed. In order to use the radiated emission limits shown in Figure 2.7, the emissions have to be measured with a quasi-peak (QP) detector. The QP detector accounts for the repetition rate, or annoyance factor, of a received signal [20]. This means that intermittent signals are weighted less than continuous signals and that the displayed voltage level of an intermittent signal can be significantly lower than the peak signal level. The QP detector is implemented with a short charging time constant followed by a slow discharging time constant [20]. These time constants are specified in CISPR-16-1 [17]. The charging and discharging behavior of the QP detector is illustrated by the dotted blue line in Figure 2.9.

A normal receiver uses peak detection and will display intermittent signals at their peak values. The effect of each of these detectors on the trace displayed by the receiver is illustrated in Figure 2.9. A range is shown for the peak detector because powerful intermittent signals can over-saturate the receiver input and lead to an inaccurate measurement [17]. Notice that because of the low repetition rate of the signal shown in Figure 2.9, the QP trace is significantly lower than the peak values of the signal. Another important aspect of the QP detector is that it responds to signals in a linear fashion; large amplitude, low-repetition signals will produce the same output as low amplitude, high-repetition signals [21]. For signals with a constant amplitude, the QP detector output is the same as the peak detector's output. QP detection requires very long sweep times, so it is often easier to first identify radiated emissions with peak detection and to then verify individual emissions for compliance with QP detection.

It is acceptable for designers in the US to focus on the CISPR-22 Class B limits (despite the slight difference between the FCC and CISPR-22 limits shown in Figure 2.7) because the United States and the EU have a multilateral agreement that covers EMC and several other market sectors [16]. This means that in most cases the FCC and CISPR-22 standards are interchangeable for many of the electronic products that are marketed in both regions. Where major differences occur are in the legal enforcement of the standards. Unlike the United States, which exempts certain market sectors from compliance testing, the EU has strictly enforced its EMC standards for nearly all electronic

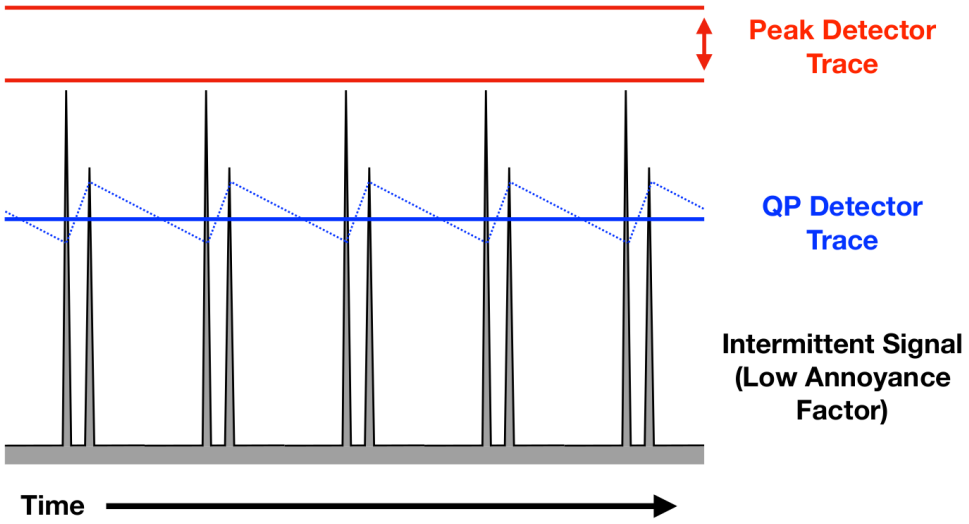


Figure 2.9: The difference between the displayed output trace of regular peak detection and quasi-peak (QP) detection when measuring a spurious signal with a low annoyance factor [17].

devices since the mid-1990s [22]. This means that there is usually an application-specific radiated emissions limit for an electronic product that is enforced by the EU. Nevertheless, CISPR-22 is broad enough to cover a large majority of consumer electronic products. This broad coverage (combined with the EU's strict enforcement policy and the tendency for global harmonization towards the EU EMC standards) is why the CISPR-22 Class B limits are used in Chapters 3, 4, and 5 of this thesis.

CISPR-22 includes detailed specifications for radiated emissions measurement techniques [17]. The most import specification it mentions is the acceptable level of test site attenuation where the radiated emissions measurement is performed. The preferred RF test environment described by CISPR-22 is the open area test site (OATS) that consists of a vertically scanning, tuned dipole antenna, an elliptical area surrounding the test site free of metallic objects, and a metallic ground plane on which the antenna and EUT are placed [23]. As the name implies, the test site is also required to be located outdoors. Large semianechoic chambers can be calibrated to meet the normalized site attenuation specified by CISPR-22 and are an acceptable alternative to an OATS for full compliance testing [17].

Replicating the exact radiated emissions measurement techniques specified in CISPR-22 is only necessary when performing a full compliance test. This kind of testing is not typically performed by the product designers, but by an independent third party that is licensed to do so. Failure to meet EMC standards at this point in the design process is extremely costly for three reasons: (1) paying for a third party company to test a product is expensive and often requires scheduling weeks in advance, (2) if a product fails the test it must be redesigned, and (3) if a product fails the test it must be retested until it meets the necessary EMC standards. This third point is especially important because solely relying on expensive third party testing can greatly increase the cost it takes to bring a product to market. It is best to periodically test a product for EMC compliance throughout the design process so that a successful full compliance test can be guaranteed [3]. This form of iterative testing is referred to as precompliance testing.

Product designers at large companies will often use a precompliance test setup like the one depicted in Figure 2.10 [6]. The EUT is placed upon a non-metallic support 10 m away from the EMC antenna. A specialized spectrum analyzer with QP detection (referred to as an EMC analyzer) is then used to measure the radiated emissions. The semianechoic chamber is lined with RF absorbers that create a controlled RF environment free from external interference. The best chambers are large enough to allow for vertically scanning the EMC antenna, but smaller anechoic chambers where this is not possible can also be used. Measurements may also be performed with a regular spectrum analyzer that is only capable of peak detection. Compromises such as these are common when performing precompliance measurements.

It is important to understand that the goal of precompliance testing is not to recreate full compliance testing. Rather, the goal of precompliance testing is to aide product designers in identifying EMC issues before a product is sent out for a full compliance test [24]. With this goal in mind, the concept of ignoring the CISPR-22 measurement standards can be extended to create low-cost precompliance techniques that are still useful.

Many product designers do not have on-site access to the equipment shown in Figure 2.10. In these situations, performing radiated emission measurements in an uncontrolled RF environment

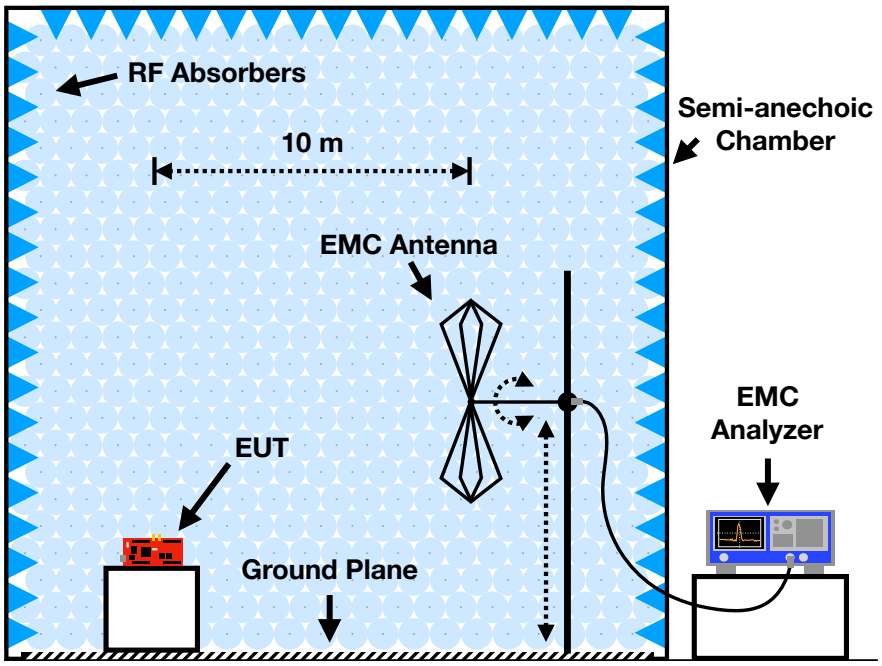


Figure 2.10: A traditional precompliance radiated emissions measurement test setup using an EMC analyzer and a shielded RF chamber.

can still help to identify problem signals when the measurement is compared to a background measurement of the same RF environment. One can use a spectrum analyzer and common VHF and UHF television antennas to measure signals in the 30 MHz to 1000 MHz range [25]. Figure 2.11 shows an example of a simple on-site test setup for measuring radiated emissions from an EUT. This setup is easy to construct and only requires a single operator, which means that many measurements can be performed throughout the design process.

A major issue with the setup shown in Figure 2.11 is that the distance between the antenna and the EUT is much shorter than 10 m. This means that the measurement is being taken in the near field of the antenna where coupling effects will drastically effect the measured emission strength [2]. While this problem can be solved by increasing the distance between the antenna and the EUT, the uncontrolled RF environment and the lack of a metallic ground plane can introduce other errors as the distance between the antenna and the EUT is increased. For example, a distance of 10 m allows for frequency-dependent errors such as signal reflections with objects in the uncontrolled

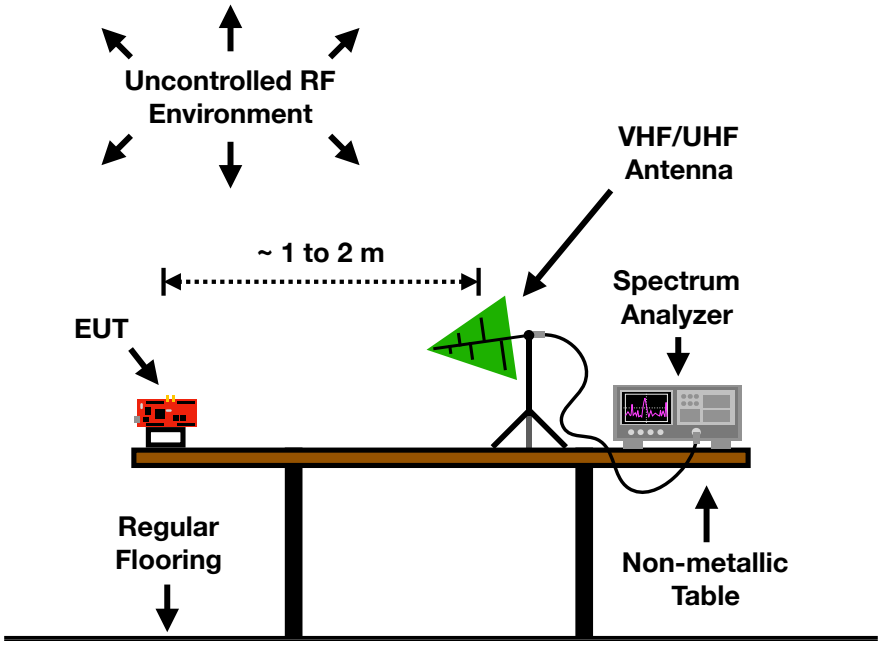


Figure 2.11: A low-cost precompliance test setup for measuring radiated emissions in an uncontrolled RF environment such as a hallway or office space [25].

environment. These errors can be very large, sometimes ranging from +6 dB to -25 dB for each reflection [23]. These large inaccuracies are difficult to predict because they vary greatly depending upon the type of uncontrolled RF environment that is used for testing. However, performing many radiated emission measurements in the same RF environment allows for a relative comparison between the results of each test. When combined with iterative changes to the electronic product, reducing a particular emission by at least 2 or 3 dB can be considered a significant improvement [25]. This is an example of how the primary goal for low-cost precompliance testing should be measurement repetition with gradual design improvement.

Many of the compromises made by the test setup shown in Figure 2.11 significantly reduce the cost of precompliance measurements. However, this test setup still requires a receiver. In most cases this is a spectrum analyzer with measurement capabilities in the frequency range of 30 MHz to 1000 MHz. A spectrum analyzer will cost less then an EMI receiver or EMC analyzer, but it will not have a QP detector. Another issue is that spectrum analyzers are still expensive relative to the

rest of the test equipment shown in Figure 2.11. Used spectrum analyzers can range in price from $1000.00 to $5000.00, while new spectrum analyzers can cost well over $10,000.00 [25]. More affordable USB-based spectrum analyzers exist, but they still generally cost well over $300.00 [25]. It is this author's opinion thatsuch spectrum analyzers typically have minimal functionality and a non-intuitive user interface that can greatly increase the time it takes to perform even basic measurements. If one could replace the spectrum analyzer shown in Figure 2.11 with a comparable receiver that costs significantly less, a complete low-cost precompliance test setup could be created.

According to Keysight Technologies [26], a spectrum analyzer is a device that performs a Fourier transform and then displays the spectral components of the measured RF signal as a voltage. Figure 2.12 shows the block diagram for a generic superheterodyne spectrum analyzer as described in [26]. A sweep generator is used to create the horizontal movement of the displayed voltage. This generator is also used to tune the local oscillator. The orange shaded area shows all of the signal conditioning that is necessary before the measured RF signal can be displayed.

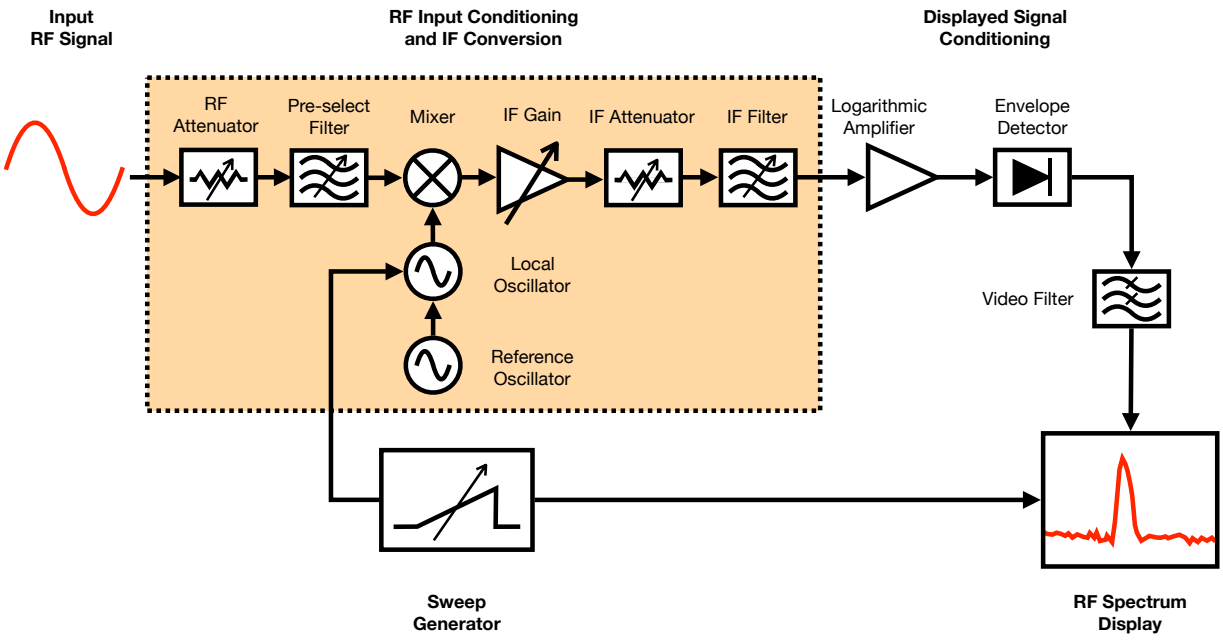


Figure 2.12: Block diagram of a typical superheterodyne spectrum analyzer as described in [26].

The input RF signal is first sent through an attenuation stage to prevent the mixer input from being overloaded. This attenuated version of the RF signal is then filtered depending on the frequency range of interest so that out-of-band signals are not displayed by the spectrum analyzer. The most important part of the process is the intermediate frequency (IF) conversion stage which begins at the mixer. The downconversion to an IF allows for the processing of the RF input signal at a much lower frequency. The IF conversion stage is also what determines the tunable frequency range of the spectrum analyzer.

The processing performed on the IF signal consists of an IF gain stage, an optional attenuation stage, and a filter stage. These are all shown just after the mixer in Figure 2.12. The IF filter is where the resolution bandwidth (RBW) of the spectrum analyzer can be adjusted. The RBW defines the smallest displayed frequency separation between two adjacent signals [27]. A narrow resolution bandwidth allows for the accurate measurement of RF signals that are very close together in frequency at the cost of greatly increasing the trace sweep time.

The last stage of the spectrum analyzer modifies the output of the IF stage so that the proper signal level is displayed. This involves an envelope detector that tracks the envelope of the signal while suppressing any instantaneous variations [26]. The output of the envelope detector is a video signal that is passed through a video filter which smooths out the final displayed signal trace. The output of the envelope detector is also where an EMC analyzer would implement a QP detector. Closely replicating this measurement process with a low-cost, off-the-shelf software-defined radio (SDR) receiver can greatly reduce the cost of a low-cost precompliance test setup.

2.3 Software-defined Radio (SDR)

A software-defined radio (SDR) is defined by the Wireless Innovation Forum [28] as a "radio in which some or all of the physical layer functions are software-defined." According to Grayver [29], the ideal SDR receiver would be able to receive "any waveform at any carrier frequency and any bandwidth" while consisting of only an antenna, an analog-to-digital (A/D) converter, and a digital processor. Such a receiver is physically impossible to construct with current technology, so a practical SDR consists of an analog signal processing stage combined with a digital processor as shown in Figure 2.13 [30]. SDR receivers (and transmitters) are predominantly used for communications systems; in 2011 they were utilized by 93% of the mobile infrastructure market [31]. The output of the SDR receiver shown in Figure 2.13 has been shown as a frequency spectrum display for comparison to a traditional spectrum analyzer.

The analog stage of the SDR receiver is similar to the RF input conditioning stage of the spectrum analyzer shown in Figure 2.12. The key difference is that the SDR's analog stage must digitize the signal with an A/D converter before further signal processing can be performed. This is most commonly done by first separating the RF signal into in-phase and quadrature (IQ) components with a quadrature mixer and then sampling each of these components with a dedicated A/D converter [32]. After the A/D conversion, the IQ samples are passed through the digital downconversion stage shown in the green shaded area of Figure 2.13. This type of design is referred to as a digital low-IF architecture. Implementing this last conversion stage digitally has several advantages, including immunity from the operating conditions that can corrupt more sensitive analog circuitry [32]. A digital downconversion stage also reduces system cost and increases system integration when compared with a traditional analog downconversion stage.

The IQ samples from the A/D conversion stage in Figure 2.13 can also be directly converted to DC to increase system simplicity by avoiding additional digital mixing and filter circuitry [32]. This is called a direct conversion (or zero-IF) architecture and is commonly used by SDR receivers (as discussed at the end of this section). The zero-IF architecture has several significant disadvantages when compared to the digital low-IF architecture shown in Figure 2.13. For example, leakage from

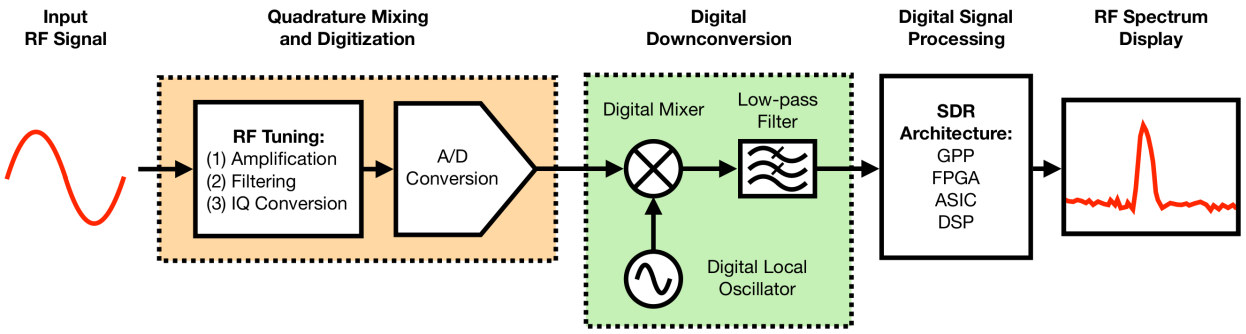


Figure 2.13: Generalized block diagram for a digital low-IF SDR receiver as described in [32].

the local oscillator used in the analog stage can create a DC offset and intermodulation products that will appear as signals on the spectrum display [32]. These false signals can make identifying the signals of interest much more difficult for the operator. An SDR receiver that uses either a traditional analog low-IF or the digital low-IF architecture shown in Figure 2.13 is able to avoid these issues.

The digital signal processing stage shown in Figure 2.13 is where the SDR receiver transforms the IQ samples into a meaningful result. When using an SDR receiver to measure the strength of RF signals, this stage is comparable to the envelope detector and video filtering used by a traditional spectrum analyzer. There are several different device architectures that can be used to perform the digital signal processing in an SDR receiver. Deciding on an architecture is largely dependent upon the intended application. The four main types described by Grayver [29] are the general purpose processor (GPP), the digital signal processor (DSP), the field-programmable gate array (FPGA), and the application-specific integrated circuit (ASIC). Table 2.1 shows the primary advantages of each of these architectures.

GPPs are the most common type of digital signal processor used today, but for SDR applications they are unable to process high-throughput signals on their own [29]. For SDR applications that require high data throughput, the ASIC and the FPGA are the best choice [29]. However, both require a significant amount of expertise to develop when compared to a GPP. The major design

trade-offs between these four architectures are shown in Figure 2.14. In this figure, the development effort shown along the x-axis does not account for the savings from mass-production but instead is representative of the design expertise required to create and prototype a new design with a particular architecture.

SDR Architecture	Primary Advantage(s)	Example
General purpose processor (GPP)	maximum processing flexibility easy development	Intel® Core™ i7 CPU family
Field-programmable gate array (FPGA)	highest data throughput high bandwidth waveforms	Xilinx® Spartan®-6 FPGA family
Digital signal processor (DSP)	low power consumption	Freescale™ MSC8144 multi-core DSP
Application-specific integrated circuit (ASIC)	high data throughput moderate power consumption	Mirics MSi2500 FlexiTV™ USB Interface IC

Table 2.1: The advantages of four SDR digital signal processing architectures with examples.

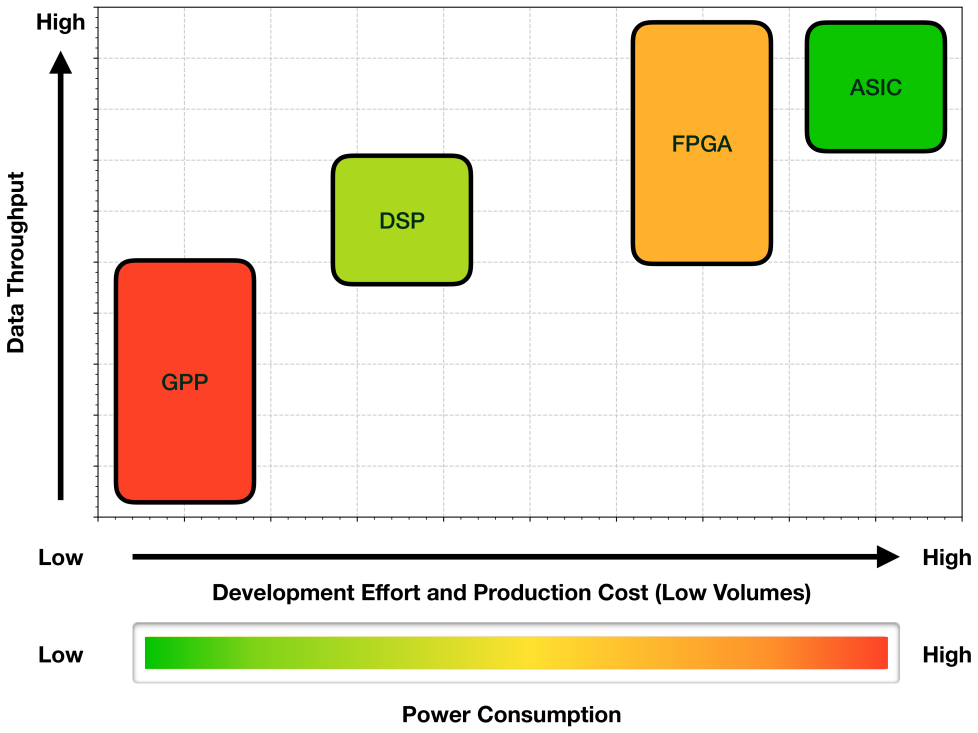


Figure 2.14: The relationships between development effort, cost, power consumption, and data throughput for the most common SDR digital signal processing architectures (adapted from [29]).

For the purpose of creating an SDR receiver capable of making radiated emission measurements in the frequency range of 30 MHz to 1000 MHz, a GPP architecture would be ideal. This is because an SDR receiver designed for this purpose would not have any strict power requirements and would not need to process high-throughput signals. In fact, modern GPPs are fast enough to implement all of the necessary digital signal processing for practical radios [29]. If a GPP is used for all of the digital signal processing, the SDR receiver will only require additional circuitry for the orange and green shaded areas shown in Figure 2.13. This circuitry will become the SDR frontend [32] and the combination of the SDR frontend with a GPP creates a "software-centric" SDR receiver [29].

One such software-centric SDR is shown in Figure 2.15. This SDR utilizes several ICs: the Analog Devices 9364 ASIC performs the analog RF signal handling while a Xilinx® Spartan-6® FPGA performs both the receiver digital down-conversion and the transmitter digital up-conversion [33]. The final IQ samples from the B200mini receiver are sent over USB to a GPP (usually the CPU of a host PC). This unique IC arrangement allows the GPP to focus solely on digital signal processing, which is a great advantage when implementing complex wireless communication protocols [29]. However, the B200mini has many features that are unnecessary for simply performing RF measurements. These features make it too expensive for replacing the spectrum analyzer in the low-cost test setup shown in Figure 2.11. Fortunately, many other types of software-centric SDRs are available thanks to their popularity in the hobbyist radio market [34]. A selection of software-centric SDR frontends are shown in Figure 2.16.

Note that while it is technically the combination of the SDR frontend and a GPP that makes a complete software-centric SDR, the term SDR is commonly used in reference to just the frontend by itself. Throughout the rest of this thesis, the term SDR will refer to the SDR frontend.

The SDRs shown in Figure 2.16 were all considered as potential replacements for a spectrum analyzer. They each are capable of receiving RF signals in the frequency range from 30 MHz to 1000 MHz and (at the time of writing) cost significantly less than a spectrum analyzer.

The NESDR SMArt SDR was the most compact model tested, but its low sampling rate and simplistic hardware design make it too inaccurate for making consistent RF signal measurements.



Figure 2.15: The B200mini from Ettus Research™ is a compact FPGA-based SDR frontend with transmit and receive capabilities over a wide frequency range.

Another issue related to this SDR's low sample rate is the user experience; searching for radiated emissions over a range of 970 MHz in roughly 1.8 MHz increments is a very slow and tedious process. This 1.8 MHz span further makes identifying real signals difficult because of the artifacts that are created by the zero-IF tuner IC used by the NESDR SMArt SDR.

The HackRF One is the only SDR shown in Figure 2.16 with transmission capabilities. It also covers a much larger frequency range, much like the B200mini. However, both of these properties make the HackRF One an unsuitable spectrum analyzer replacement for the same reasons the B200mini was not considered. The HackRF One also has a considerably worse signal-to-noise ratio (SNR) than the other SDRs shown in Figure 2.16 [34]. It is this author's opinion that the HackRF One should only be considered for low-cost applications focusing on RF signal transmission.

The Airspy R2 is an interesting SDR because it has the same zero-IF RF tuner IC used by the NESDR SMArt SDR combined with a more advanced 12-bit A/D converter [36]. This allows the Airspy R2 SDR to have a much more usable span of almost 10 MHz. While this frequency span is still much smaller than what is possible with the tracking generator of a spectrum analyzer, it makes separating real RF signals from the zero-IF artifacts very easy for the operator. The Airspy R2 is also very compact and is smaller than the RSP2pro, HackRF One, and even the B200mini.

All of the SDRs discussed thus far are only capable of measuring RF signal strength in relative

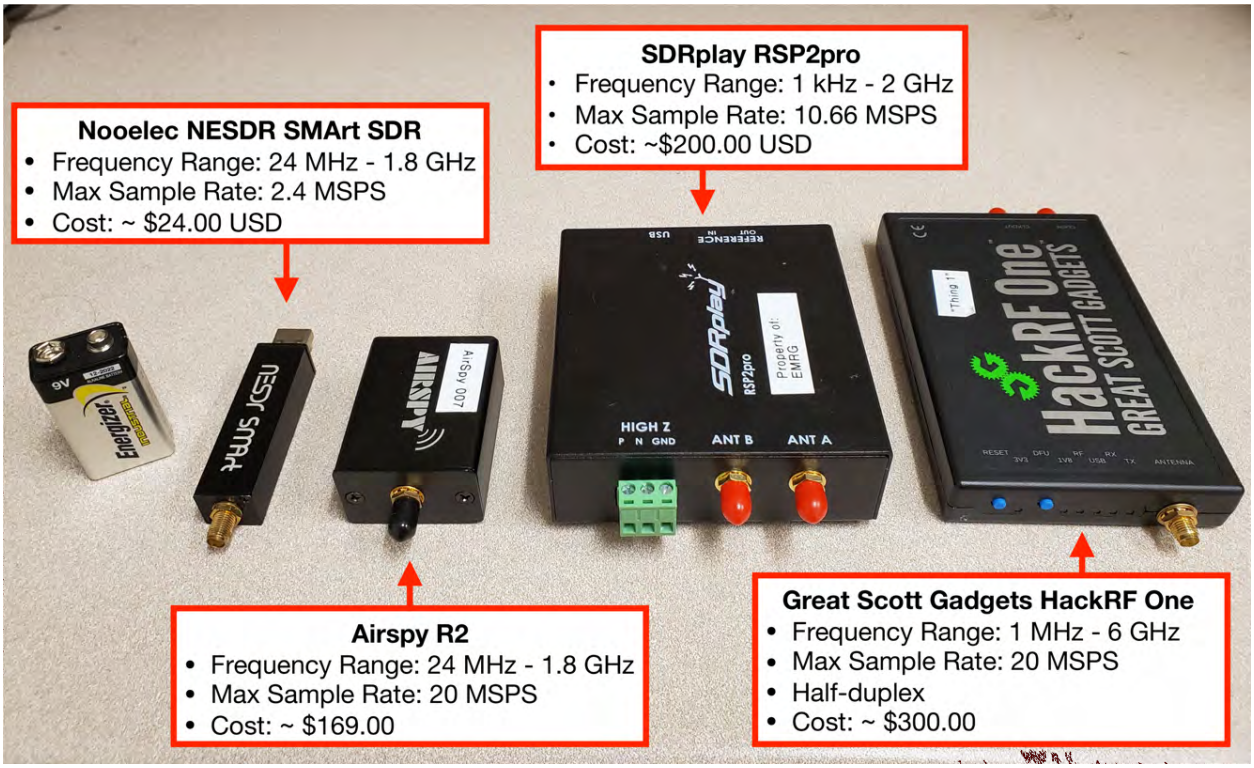


Figure 2.16: Several software-centric SDRs that were considered for this thesis. The hardware characteristics shown were retrieved from the manufacturer websites ([35] [36] [37] [38]).

units of dBFS. This means that it is impossible to know the actual power of the received signal, which is unacceptable for precompliance radiated emission measurements. Any of these SDRs could be used to measure absolute signal strength by calibrating the dBFS reading with a known input power, but an off-the-shelf SDR capable of such measurements would be much more convenient.

2.4 The Radio Spectrum Processor 2 Pro (RSP2pro)

The RSP2pro SDR shown in Figure 2.17 is a software-centric SDR capable of performing absolute signal strength measurements thanks to its proprietary companion software, SDRuno. Figure 2.18 shows a simplified block diagram of the RSP2pro and Figure 2.19 shows how these components are arranged.



Figure 2.17: The RSP2pro from SDRplay is a software-centric SDR receiver with many unique RF signal conditioning capabilities.

The Port A and B signal chain contains many of the RF signal conditioning features that make the RSP2pro unique when compared to other software-centric SDRs. The selectable RF filter network reduces out-of-band signals, just like a spectrum analyzer. There is also an optional AM/FM notch filter for suppressing powerful broadcast signals. This feature is ideal for the uncontrolled RF environment used during a low-cost precompliance test, where such broadcast signals may mask emissions from the EUT. The RSP2pro has a dedicated RF gain control stage (shown after the AM/FM notch filter in Figure 2.18) and an LNA. Both can be controlled by the operator via software, although in practice it is best to use automatic gain control for the LNA [39].

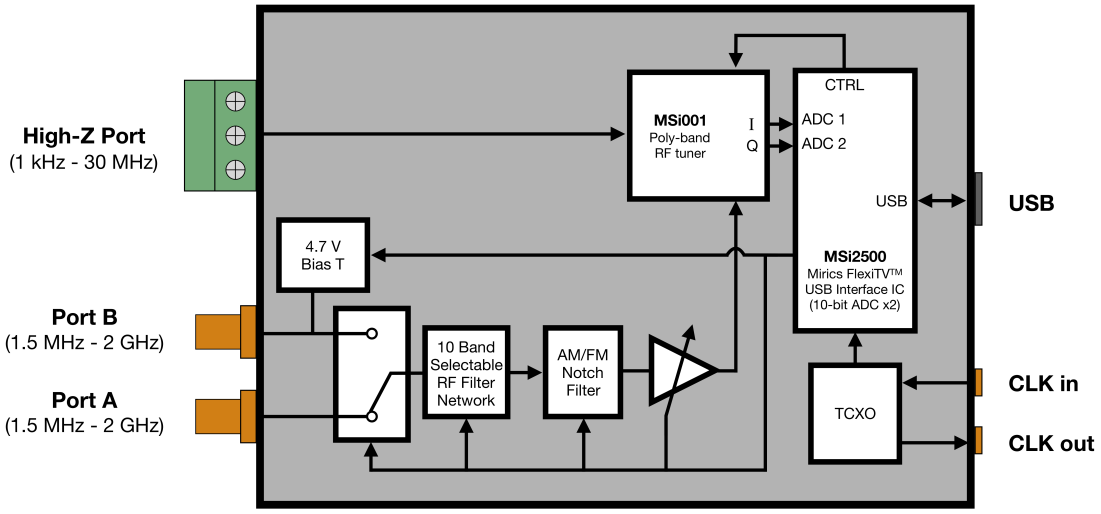


Figure 2.18: A simplified block diagram showing the main components of the RSP2pro SDR (adapted from [39] and [40]).

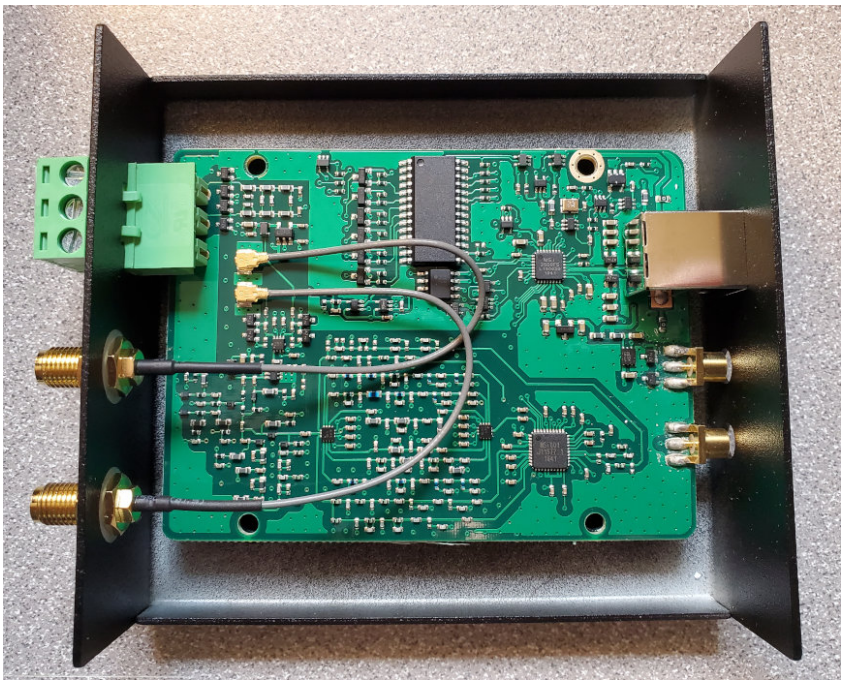


Figure 2.19: Internal view of the RSP2pro; the 10 band selectable RF filter network can be seen on the lower-center of the PCB, while the RF tuner IC can be seen on the lower-right.

All of the receiver functionality of the RSP2pro is handled by two ASICs: the Mirics Semiconductor, Inc., MSi001 RF tuner and MSi2500 USB interface IC. The MSi001 features both zero-IF and low-IF operation, as well as a dedicated LNA for each of its RF inputs. It is important to note that the low-IF implementation used by the RSP2pro is not the same as the digital low-IF implementation shown by the green shaded area in Figure 2.13. This is because the selection between zero-IF and low-IF mode on the RSP2pro takes place in the RF tuner chip (i.e. the orange shaded area in Figure 2.13). This difference does not significantly impact the operation of the RSP2pro because both architectures condition an RF signal in the same way.

In low-IF mode, the MSi001 first amplifies the conditioned RF signal and then uses a quadrature mixer to separate the signal into I and Q components at an IF. The IF IQ components are then separately digitized by the MSi2500's integrated A/D converters and the final IQ datastream is sent over USB to a GPP for further digital signal processing.

The flagship feature of the RSP2pro is its ability to measure the absolute power of RF signals. Modern SDRs measure signal level at the input of the A/D converter [39] where the incoming voltage is sampled at twice the rate of the IQ bandwidth. Modern A/D converters have very accurate reference voltages, so the voltage reading at the A/D input is very precise. However, this voltage is not the same as the received signal strength but is instead the resulting voltage from all of the preceding RF stages. This is why most SDRs cannot measure the absolute power of the RF input. The designers of the RSP2pro were able to solve this issue by creating a complex data set of the gain effects for every usable configuration of the RF stages shown in Figure 2.18 [39]. The appropriate adjustment is automatically made to the spectrum display based on how the operator has configured the RSP2pro so power measurements can be made seamlessly. Unfortunately these data sets are not available directly, so the RSP2pro can only make absolute power measurements when it is used with its companion software (see the end of this section for more information).

The RSP2pro can also make very accurate frequency measurements. It does this by using a precision 0.5 ppm temperature-compensated oscillator (TCXO) as a reference for the IQ conversion and A/D sampling. This feature, the absolute power measurement capability, and the all metal

shielded enclosure make the RSP2pro more comparable to a piece of laboratory equipment than a hobbyist SDR receiver [41]. Together, they make the RSP2pro the best candidate for successfully replacing a spectrum analyzer for low-cost precompliance measurements. Table 2.2 summarizes all of the key hardware specifications of the RSP2pro.

Parameter	Conditions	Min.	Typ.	Max.	Unit
Frequency Range	Port A and Port B	1.5		2000	MHz
ADC Sample Frequency (f_s)			8	10.66	MSPS
ADC ENOB	f_s = 8 MSPS		10.4		bits
ADC SNR			60		dB
ADC SFDR			67		dB
Receiver Sensitivity (MDS)*	f_c = 50 MHz		-147		dBm
	f_c = 144 MHz				
	f_c = 222 MHz				
	f_c = 440 MHz				
	f_c = 902 MHz				
	f_c = 1296 MHz		-141		dBm
Maximum RF Input Power	Typ. Continuous and Max. Short Duration		0	+10	dBm
Noise Figure (NF)†	f_c = 20 MHz		1.6		dB
	f_{in} = 40 MHz		1.5		dB
	f_{in} = 100 MHz		1.5		dB
	f_{in} = 200 MHz		1.9		dB
	f_{in} = 360 MHz		5.0		dB
	f_{in} = 600 MHz		2.5		dB
	f_{in} = 1300 MHz		3.5		dB
IIP3	Minimum LNA Gain		+15		dBm
IF Bandwidths	Zero-IF and Low-IF		200		kHz
			300		kHz
			600		kHz
			1.536		MHz
	Zero-IF Only (4 levels in 1 MHz increments)	5.0		8.0	MHz
Power Consumption			170		mA

* Measured Independently by ARRL [44]
† Measured with Maximum RF Gain Setting in SDRuno

Table 2.2: The RSP2pro hardware specifications that are relevant to this thesis. All information taken from [42] and [41].

Software-centric SDRs rely heavily on a GPP for signal processing, and the RSP2pro is no exception. As mentioned previously, one must use the RSP2pro's companion software in order to make absolute power measurements. This software is called SDRuno and is available on the SDRplay website [37]. At the time of writing, SDRuno only supports Windows operating systems.

Figure 2.20 shows the SDRuno user interface. It is composed of multiple blocks that mimic the appearance of a traditional amateur radio receiver. The spectrum display has two components: a traditional spectrum display and a waterfall plot. The x-axis is the frequency of the received

signal in MHz and the y-axis for the spectrum display is the magnitude of the signal in units of dBm (decibels relative to one milliwatt). The y-axis for the waterfall plot has units of seconds and has units of seconds and shows how the measured signal varies with time, with the most recent measurement being shown at the top.

The waterfall plot is convenient when comparing background signals to the radiated emissions from the EUT. By powering the EUT on halfway through a complete waterfall capture, the operator can quickly see if new peaks have appeared or if the noise floor has been affected by the EUT. This is especially true for weak or intermittent signals, which can be hard to notice with just the spectrum display.

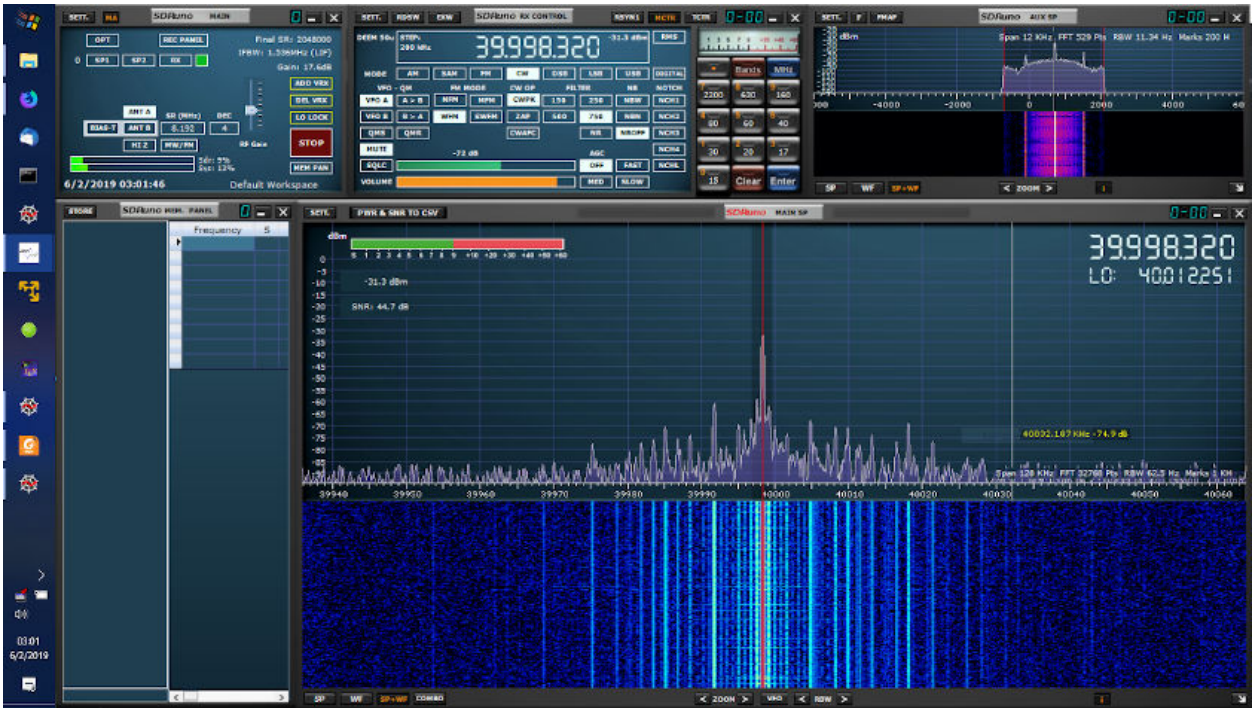


Figure 2.20: The SDRuno user interface.

The power meter in SDRuno works by measuring the peak signal strength within the bandwidth of the secondary spectrum display in the top right corner of Figure 2.20. The bandwidth of this secondary spectral display is also shown by the gray region in the main spectrum display. The power meter measures the peak value within the selected bandwidth. For radiated emission measurements,

using the continuous-wave (CW) mode works best because it allows for the narrowest bandwidth. CW mode also includes a button that will move the receiver frequency to the measured peak value. The power level (in dBm) and the frequency of the measured peak can be continuously saved to a csv file where the time interval between individual measurements can be set by the operator.

Before the RSP2pro can be used as a spectrum analyzer, the accuracy of its measurements must be verified. The tests performed in [39] found that the absolute signal accuracy of the RSP2pro between 1 to 2 dB. The next chapter will focus on performing a large number of measurements under conditions that are closer to the intended use case (i.e. measurements that would be expected during a radiated emissions test).

CHAPTER 3

PERFORMANCE OF THE RSP2PRO

This chapter focuses on describing the test equipment and procedures that were used to characterize the RSP2pro's measurement accuracy. A description of the test equipment that was used is given in Section 3.1. Section 3.2 then outlines the semi-automated characterization procedures that were used to obtain a large number of readings from both the EMC analyzer and three RSP2pro units. Sections 3.3 and 3.4 show the characterization results obtained from these procedures. These results are then used to show the margin of error for measurements taken with the RSP2pro.

3.1 Equipment Information

The test equipment described in this section was used to characterize the RSP2pro. The important information for each piece of test equipment has been provided here for convenience.

Measurements taken with the Agilent E7410A EMC analyzer were used to assess the measurement performance of the RSP2pro throughout this thesis. The EMC analyzer was also used in this section to calibrate and characterize the signal generator.

The antenna factor (AF) values of several EMC antennas were used to determine the output frequency and amplitude combinations used by the signal generator in Section 3.2. The BicoLOG 30100X EMC antenna was also used for performing narrow-band and broad-band radiated emission tests with both the EMC analyzer and the RSP2pro in Chapters 4 and 5.

The HP 8657A signal generator was used to generate continuous wave (CW) signals with specific frequencies and amplitudes for the characterization procedure described in Section 3.2. The signal generator was also combined with a standard whip antenna to create narrow-band radiated emissions that were measured by both the EMC analyzer and the RSP2pro in Section 4.2.

3.1.1 Agilent E7401A EMC Analyzer

The Agilent E7401A EMC analyzer shown in Figure 3.1 is a superheterodyne spectrum analyzer with several specialized EMC features, including a CISPR-16-1 compliant quasi-peak (QP) detector [43]. It has a 50 Ω RF input that is capable of measurements in the frequency range from 9.5 kHz to 1.5 GHz. According to the data sheet [43], the E7401A has an absolute amplitude measurement accuracy of ± 0.30 dB.

The E7401A EMC analyzer is programmable via the IEEE general-purpose interface bus (GPIB). This allowed for the automation of nearly all of the EMC analyzer measurements presented in this thesis. To automate measurements, an EMC analyzer state file was created for each measurement type. Each state file can be loaded by the EMC analyzer to quickly preset the analyzer settings before a measurement is taken. These state files are described in Table 3.1. The leftmost column shows the name of each specific state file while the remaining columns show the analyzer settings for that state file. This table will be referred to throughout the rest of this thesis as necessary.

The "Default Sweep" state file was used whenever a QP measurement was necessary. The variable f_{in} represents the desired center frequency for a particular measurement. This state file



Figure 3.1: The Agilent E7104A EMC analyzer that was used for all measurement comparisons with the RSP2pro.

has the smallest frequency span and the finest resolution bandwidth (RBW) because the QP detector mode of the E7401A has an extremely long sweep time, making larger spanning QP measurements unrealistic given the amount of measurements that was necessary for this thesis. The combination of the small span and fine RBW make the "Default Sweep" state ideal for capturing the peak value of the narrow band signals created by the signal generator source that was used for the measurements in this chapter. The "Spectrum Sweep Part 1" through "Spectrum Sweep Part 4" state files were used together whenever a full sweep of the CISPR-22 frequency range of 30 MHz to 1000 MHz was necessary. Separating this frequency range into four state files allows the signal trace resolution of a full sweep to be 32,768 points rather than 8192 points.

The last state file, "RE Sweep," was used to get an initial measurement before the QP measurements taken with the "Default Sweep" state file settings. The "RE Sweep" state has a much larger span and RBW so that a measured signal's peak value can be acquired quickly. Once a peak has been acquired, the frequency of the peak is used as the center frequency of the "Default Sweep." This allows the QP measurement to always be centered at the appropriate frequency for a given input signal.

State File	Frequency Range (MHz)	Span (kHz)	Resolution Bandwidth (Hz)	Reference Level (dBμV)	Attenuation (dB)	Detector Mode	Number of Trace Points
Default Sweep	(centered at f_{in})	3.0	300.0	(automatic)	0.0	Quasi-peak	8192
Spectrum Sweep Part 1	25.0 to 270.0	245000.0	3000.0	(automatic)	10	Peak	8192
Spectrum Sweep Part 2	270.0 to 515.0	245000.0	3000.0	(automatic)	10	Peak	8192
Spectrum Sweep Part 3	515.0 to 760.0	245000.0	3000.0	(automatic)	10	Peak	8192
Spectrum Sweep Part 4	760.0 to 1005.0	245000.0	3000.0	(automatic)	10	Peak	8192
RE Sweep	(centered at f_{in})	100.0	9000.0	120.0	25	Peak	8192

Table 3.1: Parameters of the EMC analyzer state files that were used to control the settings of the EMC analyzer.

3.1.2 EMC Antennas

As mentioned in Chapter 2, radiated emission measurements are usually performed with specialized antennas called EMC antennas. EMC antennas are provided with a parameter called the antenna factor, which is defined by Equation 2.5 and can be used to convert the measured voltage to the strength of the electric field incident upon the EMC antenna (this is shown in Figure 2.8). Manufacturer provided antenna factors are usually given in a table or chart and will be listed at a number of uniformly-spaced frequencies; for example, at every 10 MHz for frequencies from 30 MHz to 1000 MHz. Figures 3.2c and 3.4a show the antenna factors of several different EMC antennas that were used for this thesis.

The EMC antennas shown in Figures 3.2a and 3.2b are able to cover the entire CISPR-22 frequency range when used together. The Agilent 11955A EMC antenna shown in Figure 3.2a is a biconical antenna designed for receiving signals from 20 MHz to 200 MHz. The antenna factor data provided by Agilent [44] for this antenna is in 10 MHz increments and this data was used to plot the 11955A antenna factor shown in Figure 3.2c. This antenna is most sensitive to RF signals measured in the frequency range from 50 MHz to 100 MHz as indicated by the very low antenna factor between these frequencies [2]. At 300 MHz, the sensitivity of the 11955A EMC antenna is more than 10 dB below its peak sensitivity. This sharp decrease in sensitivity happens because of the null that develops directly in front of the antenna's boresight as this frequency is approached [45]. The 11955A EMC antenna is best suited for measurements between 80 MHz and 200 MHz because it is well matched in this range, as indicated by the low variation of the antenna factor between these frequencies [45].

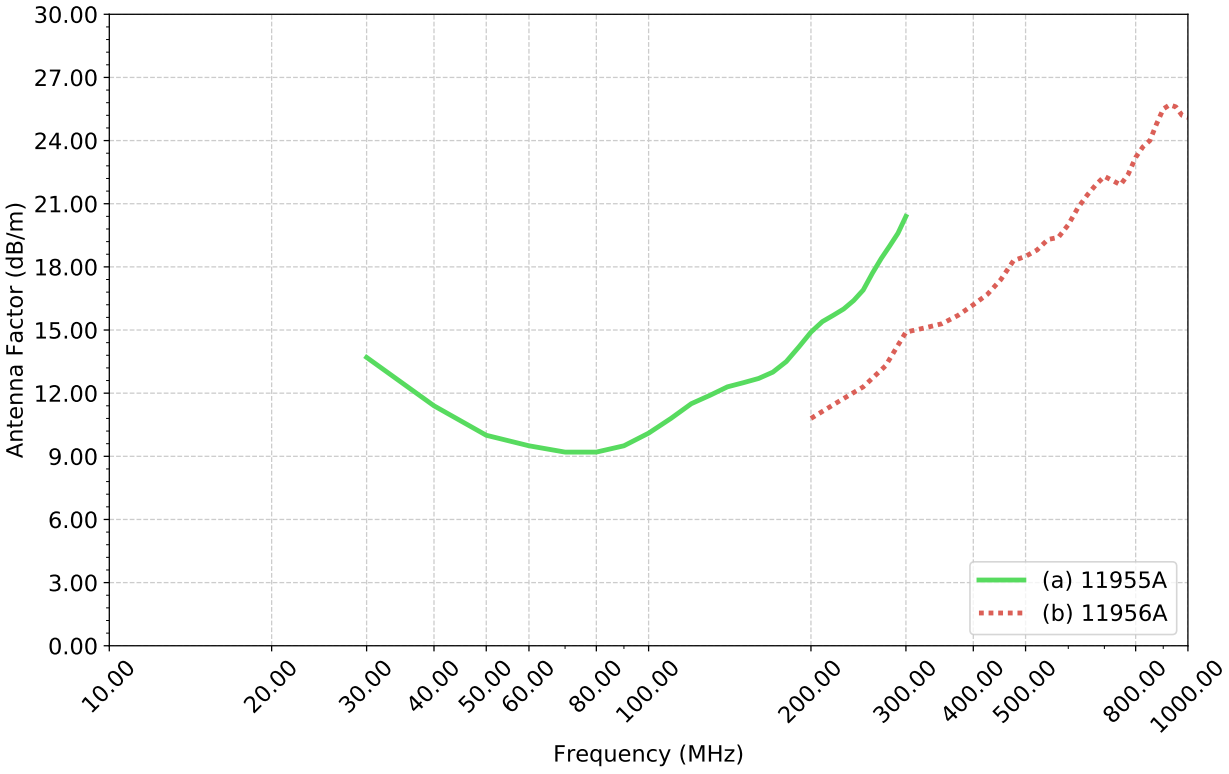
The Agilent 11956A EMC antenna shown in Figure 3.2b is a log periodic antenna with an antenna factor that increases nearly linearly from 200 MHz to 900 MHz. The sharp decline observed at 900 MHz in Figure 3.2c is most likely caused by the losses in the antenna's feeder coaxial transmission line [45]. The 11956A antenna factor in Figure 3.2c was plotted using the data provided by Agilent [44] and is in 25 MHz increments. It should be noted that this 11956A antenna is provided with an antenna factor up to 2000 MHz for use with other types of EMC measurements.



(a)



(b)



(c)

Figure 3.2: (a) The Agilent 11955A and (b) 11956A EMC antennas and (c) their antenna factors [44].

The antenna factor shown in Figure 3.2c has been truncated at 1000 MHz because that is the highest signal frequency relevant to this thesis.

While both of these antennas are ideal for full compliance tests, their large element size can lead to issues when used in a low-cost precompliance setup like the one shown in Figure 2.11. This is because the measurement distance used by this setup is comparable to the element length of these antennas and thus the beam width of the radiated emissions from the equipment under test (EUT) may not fully illuminate these antennas completely [2]. For measurement distances close to 1 m, Ott [2] recommends an antenna that meets the four following criteria:

- Have a wide bandwidth (e.g. the 30 MHz to 1000 MHz required for CISPR-22)
- Be small in physical size (ideally less than 12 inches long)
- Be very sensitive (i.e. an antenna factor less than 6 dB)
- Have a (reasonably) flat frequency response

Ott [2] acknowledges that these criterion can be contradictory (e.g. a sensitive antenna at 30 MHz cannot also be physically small). However, by carefully making compromises with these criteria, one can have an EMC antenna that is better suited for low-cost precompliance testing at short distances than the traditional EMC antennas like the ones shown in Figures 3.2a and 3.2b. A physically small antenna has a much better chance of avoiding the aforementioned illumination issue that is inherent to the larger EMC antennas at distances near 1 m. So an example of a good compromise would be forgoing a flat frequency response or high sensitivity to ensure a small physical antenna size.

The BicoLOG 30100X EMC antenna from Aaronia AG shown in Figure 3.3 is a physically small, active antenna that is designed to receive signals from 30 MHz to 1000 MHz. This antenna is able to meet two of Ott's [2] four criteria by compromising on the overall sensitivity and response flatness. The BicoLOG 30100X's antenna factor is shown in Figure 3.4a. This data was provided by Aaronia AG [46] and is in 10 MHz increments. Unlike the antenna factors shown in Figure 3.2c,

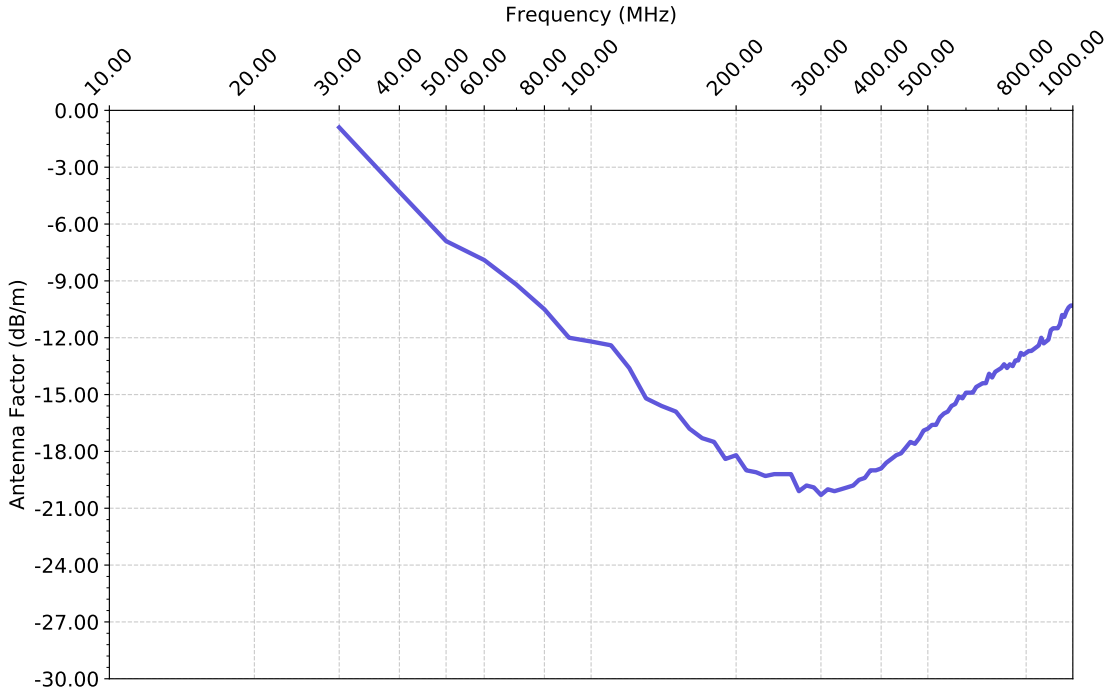
the BicoLOG 30100X antenna factors are all negative. This is because the BicoLOG 30100X is an active antenna that uses a built-in preamplifier to achieve its wide bandwidth [47]. The preamplifier gain for the unit that was used in this thesis is shown in Figure 3.4b. This data was taken from the preamplifier using the method described in [48] and represents the values measured by Aaronia AG for this specific preamplifier.

The BicoLOG 30100X EMC antenna is most sensitive to measurements in the frequency range from 150 MHz to 500 MHz. This is indicated by the low antenna factor shown between these frequencies [2] in Figure 3.4a. One can also see that the BicoLOG 30100X is much like the Agilent 11956A in the frequency range from 300 MHz to 1000 MHz in that the antenna factors of both increase linearly with the frequency in this range.

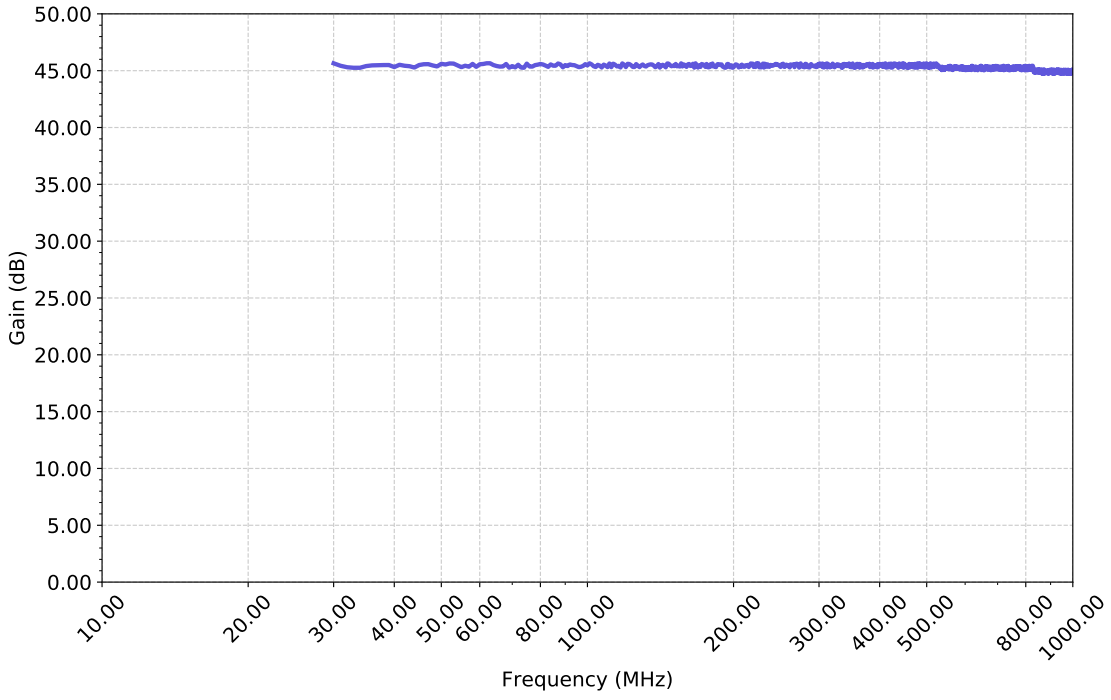


Figure 3.3: The BicoLOG 30100X EMC antenna.

Figure 3.4b shows that the gain of the preamplifier is very flat across the entire frequency range, with the exception of an approximate reduction of 0.5 dB in the frequency range from 800 MHz to 1000 MHz. This flat response means that the large variation in the BicoLOG 30100X antenna factor shown in Figure 3.4a is almost entirely caused by the antenna construction itself. The use of a preamplifier does complicate using this antenna in that there is a usable battery life of only three



(a)



(b)

Figure 3.4: (a) The BicoLOG 30100X antenna factor and (b) the gain of its preamplifier. The antenna factor shown in (a) is negative because of the large gain of the preamplifier shown in (b) [47].

to four hours [48]. Using the antenna while attached to the charger is not recommended by this author, as the charger introduces significant background radiated emissions that can interfere with measurements.

The small physical size of the BicoLOG 30100X combined with its ability to perform measurements across the entire CISPR-22 frequency band make it ideal for performing low-cost precompliance measurements at distances between 1 to 2 m. This antenna was used for all of the radiated measurements in Chapter 4 of this thesis. This antenna was also used for the low-cost antenna characterizations presented in Sections 5.2 and 5.3. Lastly, this antenna was used as part of the more traditional precompliance test setup in Section 5.4.

3.1.3 HP 8657A Signal Generator

The HP 8657A signal generator shown in Figure 3.5 is a synthesized signal generator with a carrier frequency range from 100 kHz to 1040 MHz and an amplitude range of -127 dBm to +13 dBm into 50 Ω [49]. The HP 8657A has a 50 Ω RF output and can be controlled from a computer via its GPIB interface. This feature was used to automate nearly all of the measurements in this thesis involving the signal generator. All measurements in this thesis involving the signal generator required signal levels less than 0 dBm, so the possible amplitude error caused by the signal generator is assumed to be less than 1.0 dB [49]. This error is not critical to the results of this thesis because this thesis is only interested in the measurement accuracy of the RSP2pro relative to the EMC analyzer.



Figure 3.5: The HP 8657A signal generator.

An important limitation on the accuracy of the frequency measurements in Sections 3.4 and 4.2 are caused by the specific HP 8657A signal generator that was used for this thesis. According to the manufacturer [49], the signal generator's output center frequency should never stray more than 500 Hz away from the center frequency setting. However, the unit used for this thesis was found to have a frequency error much greater than this. Since another signal generator was not available, a rudimentary calibration process was implemented by first measuring the actual frequency for each desired frequency that was going to be used and then calculating how far off the measured frequency was for each desired frequency.

The center frequency setting f_c of the signal generator required for the desired center frequency f_{in} can be represented as

$$f_c = f_{in} \pm f_o \tag{3.1}$$

where f_o is the unknown frequency offset from f_{in} that must be accounted for. There were 114 separate values for f_{in} required for the measurements involving the signal generator presented in this thesis. The procedure for finding the value of f_c corresponding to each value of f_{in} was automated via the Python script outlined in Figure 3.6. The EMC analyzer and the signal generator were automated via their GPIB interfaces using commands from the PyVISA library [50].

Before the procedure begins, the EMC analyzer and the signal generator are allowed to warm

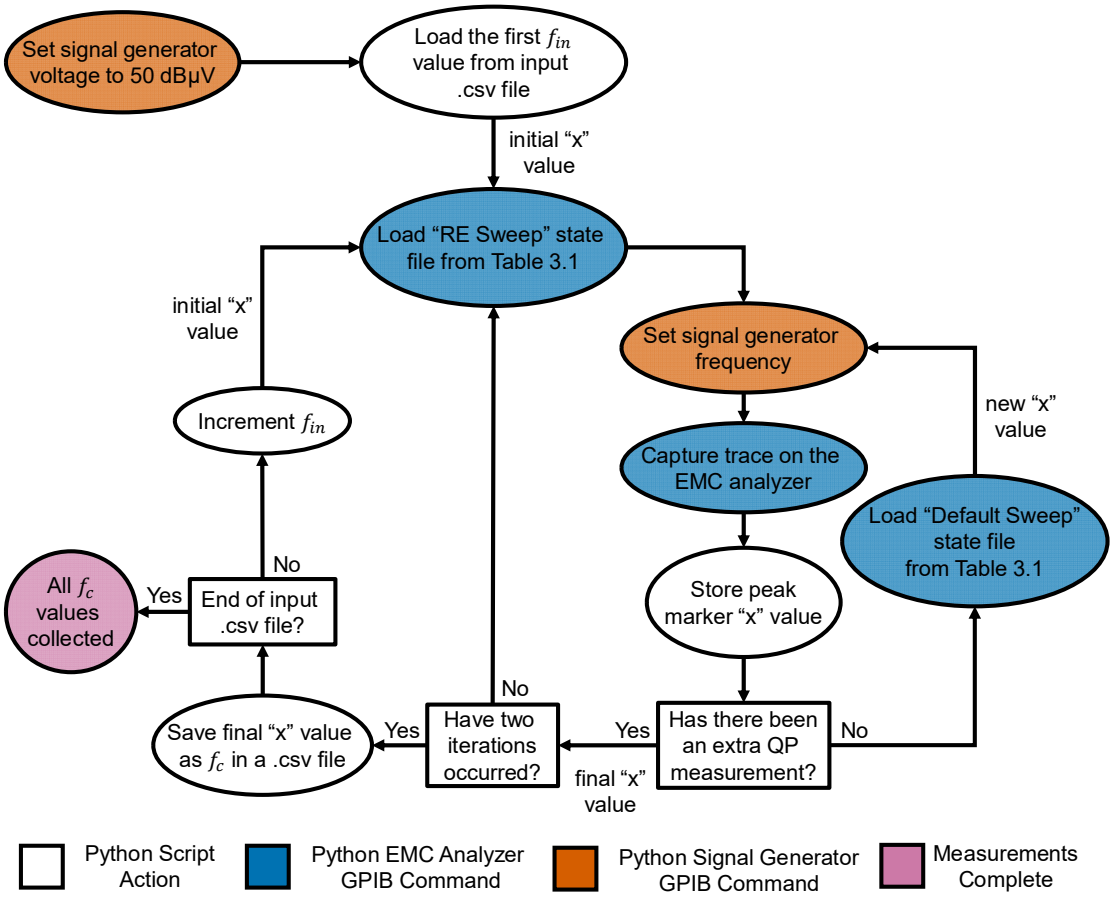


Figure 3.6: Flowchart for the Python script that was used to find the frequency offsets of the uncalibrated HP 8657A signal generator.

up for one hour. The signal generator is then set to an output level of 50 dBμV and the first of the 114 f_{in} values is loaded from an input csv file. The EMC analyzer loads the "RE Sweep" state file from Table 3.1 and uses the loaded value of f_{in} as the center frequency of the spectrum sweep. The EMC analyzer's built-in peak search function then marks the peak value within the 100 kHz measurement span. The frequency of this peak is stored as a temporary variable "x" and used to set the center frequency of the signal generator for a second measurement. This second measurement is performed with the "Default Sweep" state file settings. This means that the frequency span of the second measurement is much narrower (3 kHz), resulting in a finer peak frequency measurement. The frequency of this peak measurement is stored as the new value of "x." These measurements are then repeated using the new value of "x" as the initial input frequency.

The final value of "x" after this repetition is the result of four measurements. The final value of "x" is stored in a new csv file as the f_c corresponding to the desired frequency f_{in} and the entire process is repeated for the next value of f_{in} from the input csv file. This procedure resulted in a set of frequencies that was used for all of the measurements in this thesis involving the signal generator. This means that the actual signal generator center frequency setting for these measurements is not the desired frequency f_{in}, but the frequency f_c from Equation 3.1. Figure 3.7 shows how the value of f_o from Equation 3.1 increased linearly as the desired frequency f_{in} was increased.

This whole procedure had to be performed twice in different sessions, as the values of f_{in} that were only multiples of 25 MHz were not included in the first session. One can see from Figure 3.7 that both datasets increase linearly with f_{in}, but that the conditions particular to each session affected the value of f_o. This kind of inaccuracy is unavoidable and indicates the importance of measurement conditions when using the signal generator.

One such measurement condition is the operating temperature of the signal generator, which increases over the time that the signal generator is powered on. As this temperature increases, it can cause the center frequency of the signal generator output to vary over time. The characterization measurement results in Sections 3.3 and 3.4 all involved the signal generator and repeating the same measurements to acquire a large number of samples for each of the 114 values of f_{in}. Sometimes

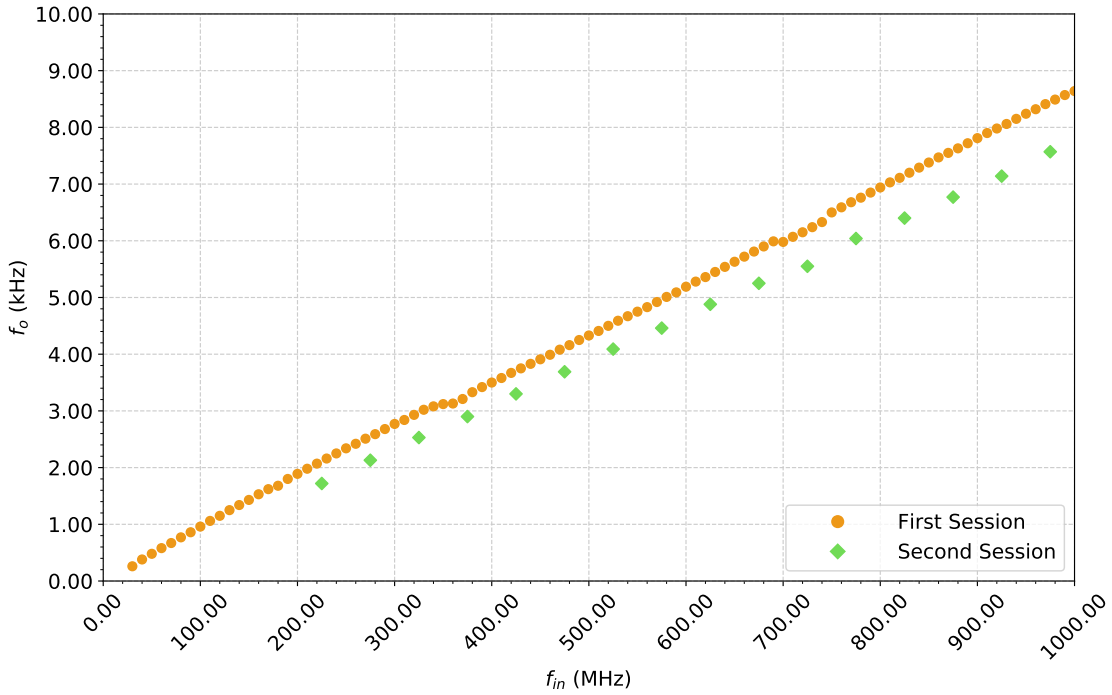


Figure 3.7: Plot of the f_o associated with each value of f_{in} from Equation 3.1.

these measurements took as long as 14 hours to complete, so measurements at the same frequency would occur hours apart. Automating these measurements made this kind of data collection possible, but the long measurement times meant that the equipment was subject to periods of continuous operation and thus potential frequency drift from changes in the operating temperature. The EMC analyzer has a built-in calibration routine that was used to prevent potential frequency drift, but such a routine was not an available for the signal generator. To better understand the impact of continuous operation, a procedure was used to characterize the frequency drift of the HP 8657A signal generator over a 24-hour period. The Python script used to automate this procedure is outlined in Figure 3.8. The EMC analyzer and the signal generator were automated via their GPIB interfaces using commands from the PyVISA library [50].

Before the procedure starts, the EMC analyzer is allowed to warm up for one hour. The first frequency f_c from the frequency table created by the Python script from Figure 3.6 is used as the center frequency of the EMC analyzer. The signal generator is then set to a constant output level

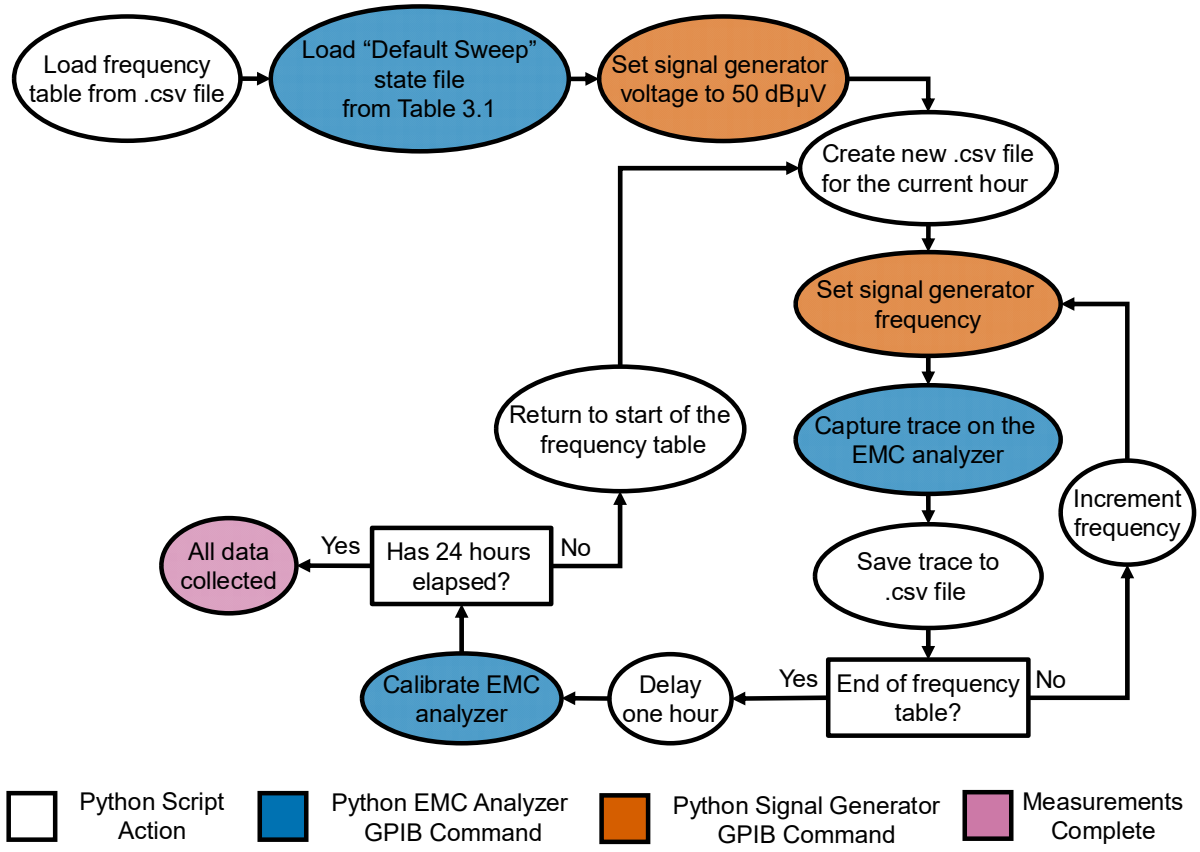


Figure 3.8: Flowchart for the Python script that was used to characterize the frequency drift of the signal generator over a 24-hour period.

of 50 dBμV and a new csv file is created for the current hour. The signal generator frequency is set to f_c and the EMC trace is stored to the new csv file. This process is repeated until the end of the frequency table is reached, after which there is a delay of exactly one hour. During this hour the EMC analyzer and the signal generator are left on.

After an hour has passed, the EMC analyzer is calibrated and the entire process is repeated until there are data sets for each of the 114 f_c values over a full 24-hour period. The recorded traces for each hour can be compared to see how the signal generator center frequency at a particular frequency setting drifts from hour-to-hour.

Figure 3.9 shows the frequency drift of the signal generator at all frequency settings over the first twelve hours of a 24-hour period. One can see that all frequency settings have significant

drift over the first hour, but then the frequency drift from hour-to-hour greatly decrease. Figure 3.9 also shows that higher frequencies have a much larger frequency drift, whereas lower frequencies become very stable after only two hours. Note that the constant value observed for the $t = 0$ hour data from approximately 620 MHz onward occurs because they frequency settings above this frequency drifted by more than the measurement bandwidth of the EMC analyzer.

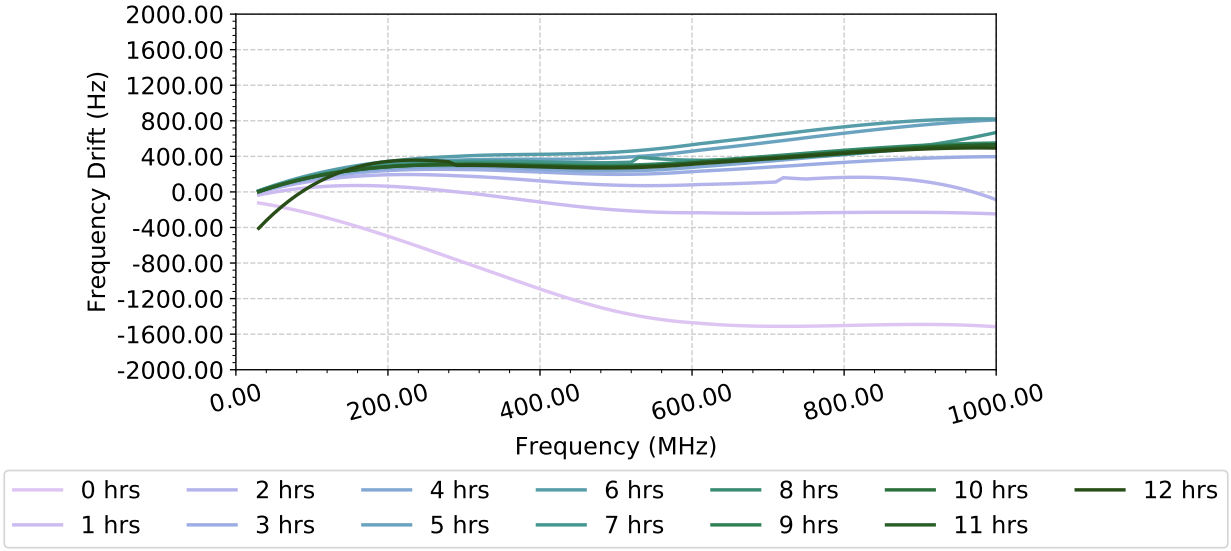


Figure 3.9: Measured signal generator frequency drift for $t = 0$ to $t = 12$ hours.

Figure 3.10 shows the frequency drift for all frequency settings over the second twelve hours of a 24-hour period. One can see that the frequency drift at the lower frequency settings is almost non-existent while the frequency drift at the higher frequency settings is much more significant. However, the frequency drift at all frequency settings is still well below 2 kHz.

There are two key conclusions from the results of the frequency drift characterization process. The first is that the frequency drift of the HP 8657A is well below 1000 Hz if the signal generator is allowed to warm up for at least one hour. While this amount of drift is not within the specification [49], it is close enough to have confidence when comparing the relative frequency measurements between the EMC analyzer and the RSP2pro. For this reason, a warm up time of one hour was used for all of the measurements involving the signal generator. The second conclusion is that

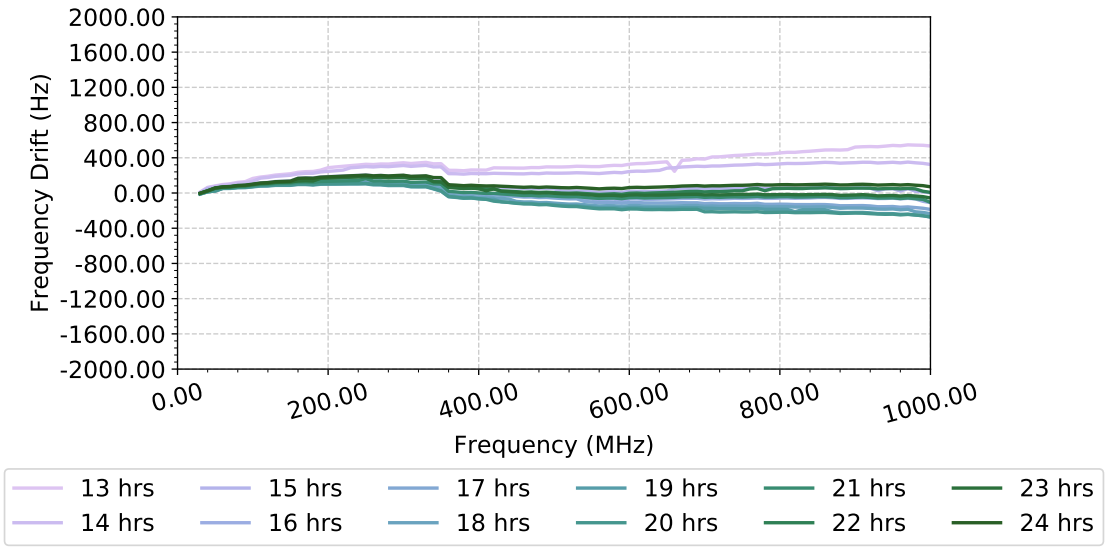


Figure 3.10: Measured signal generator frequency drift for $t = 13$ to $t = 24$ hours.

extended periods of operational time increase the stability of the signal generator center frequency. However, the amount of stability varies significantly with the signal generator frequency setting. This variance is well below 2 kHz at all frequencies, so long measurement times do not significantly affect the results presented in Section 3.4.

Figure 3.11 shows the full CISPR-22 Class B spectrum measured by the EMC analyzer directly connected to the signal generator with the signal generator set to an output level of 50 dBμV at a frequency of 500 MHz. From this figure one can see that the signal generator output has no unexpected RF signals. Figure 3.12 shows a detailed view of the 500 MHz signal shown in Figure 3.11. This figure shows that the signal generaotr output is extremely narrowband, with a 3-dB bandwidth of only 300 Hz.

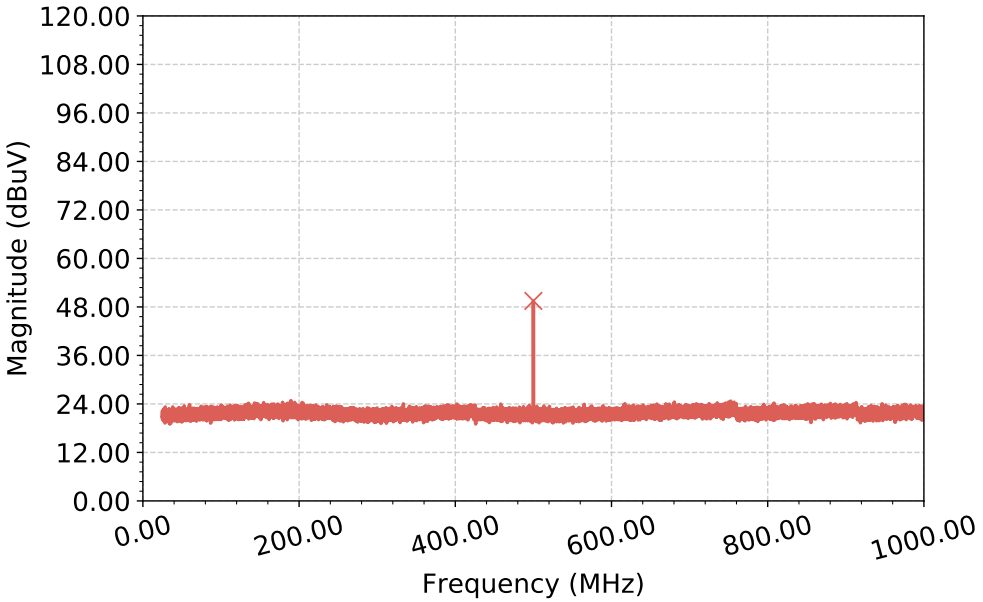


Figure 3.11: The output of the signal generator with a frequency setting of 500 MHz and a signal level of 50 dBμV.

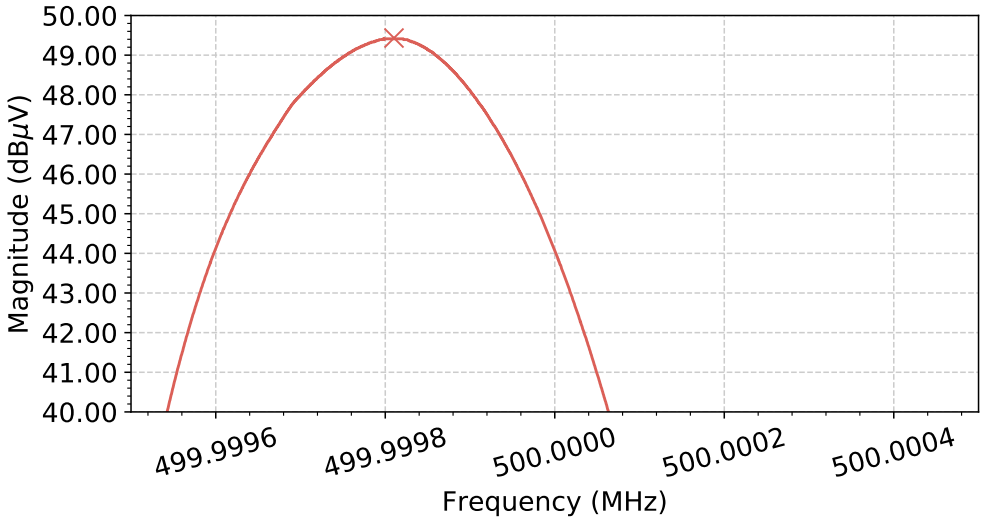


Figure 3.12: Detailed view of the signal in Figure 3.11 showing how narrowband the signal generator output is.

3.2 Characterization Procedure and Measurement Automation

All three of the antenna factors shown in Figures 3.2 and 3.4a were used to generate data sets for controlling the signal generator amplitude in the procedure discussed in Section 3.2. These specific antennas were chosen because they represent three common types of EMC antennas. Rearranging Equation 2.5 and solving for the receiver voltage V yields:

$$V\left(dB\mu V\right) = E\left(dB\mu V/m\right) - AF\left(dB\ 1/m\right). \tag{3.2}$$

Radiated emissions near the CISPR-22 Class B limit in Figure 2.7 is where the RSP2pro needs to be the most accurate when compared to the EMC analyzer because this is the level expected from an EUT that is breaking the regulatory limit. Testing the accuracy in this region is achieved by using the CISPR-22 Class B radiated emissions limit to set the signal generator voltage. By replacing V in Equation 3.2 with the signal generator voltage V_g and replacing E with the CISPR-22 Class B radiated emission limit at a particular frequency E_{limit}, the equation for the signal generator voltage at a particular frequency can be represented as

$$V_g\left(dB\mu V\right) = E_{limit}\left(dB\mu V/m\right) - AF\left(dB\ 1/m\right) + C\left(dB\right) \tag{3.3}$$

where the antenna factor AF can be the antenna factor value at a particular frequency for one of the three EMC antennas described in Section 3.1.2 and C is an adjustment factor that is initially set to 0 dB. To understand the performance at other potential radiated emission levels, several extra data sets were created by adjusting the value of C by ± 10 dB to create a total of nine data sets. All nine data sets have been summarized in Table 3.2. In this table, V_g and C are from Equation 3.3 and their entries represent the values of these variables for a particular antenna factor (or AF as shown in Equation 3.3). The csv files for these data sets are available in the Digital Appendix that is included with the digital version of this thesis (refer to Appendix B for more information).

The fifth column of Table 3.2 describes the source used for the value of AF in Equation 3.3. For example, Data Set 1 was created with the antenna factor for the Agilent 11955A EMC antenna (shown in Figure 3.2) and a value of 10 dB for C. The frequency range and step size for each

Data Set	Frequency Range (MHz)	Frequency Step Size (MHz)	Number of Data Points	V_g Range (dBμV)	Source for AF	C (dB)	File Name
1	30 to 300	10	28	24.0 to 30.8	Agilent 11955A AF	10	agilent_11955A_plus10.csv
2	30 to 300	10	28	14.0 to 20.8	Agilent 11955A AF	0	agilent_11955A.csv
3	30 to 300	10	28	4.0 to 10.8	Agilent 11955A AF	-10	agilent_11955A_minus10.csv
4	200 to 1000	25	33	21.3 to 34.7	Agilent 11956A AF	10	agilent_11956A_plus10.csv
5	200 to 1000	25	33	11.3 to 24.7	Agilent 11956A AF	0	agilent_11956A.csv
6	200 to 1000	25	33	1.3 to 14.7	Agilent 11956A AF	-10	agilent_11956A_minus10.csv
7	30 to 1000	10	98	40.9 to 67.3	BicoLOG 30100X AF	10	bicolog30100X_plus10.csv
8	30 to 1000	10	98	30.9 to 57.3	BicoLOG 30100X AF	0	bicolog30100X.csv
9	30 to 1000	10	98	20.9 to 47.3	BicoLOG 30100X AF	-10	bicolog30100X_minus10.csv

Table 3.2: Description of the nine data sets that were used to characterize the EMC analyzer and the RSP2pro.

data set was determined by the frequency component of the antenna factor particular to that data set. From the fourth column of Table 3.2, one can see that the signal generator amplitude level V_g varied over a wide range across the nine data sets. This highlights how a large range of input voltage levels were tested by using just nine input data sets. Each data set in Table 3.2 has been assigned a number that is used when they are referred to in Sections 3.3 and 3.4.

The setup used to characterize the RF signal measurement accuracy of the RSP2pro is shown in Figure 3.13. In this figure, the setup has been configured for RSP2pro measurements but the same setup was also used for the EMC analyzer measurements. The laptop pictured was connected to the EMC analyzer and the signal generator via a GPIB-to-USB adapter. The RSP2pro was connected to the laptop via USB. The test setup's two RF signal chains are shown in Figure 3.14. The same SMA cable was used to for both the EMC analyzer and RSP2pro measurements. The only difference between the measurement chains shown in Figure 3.14 was the additional RF adapter that was necessary for connecting the EMC analyzer to the SMA cable. Three separate RSP2pro units were characterized to investigate how the measurement performance varied across several units.

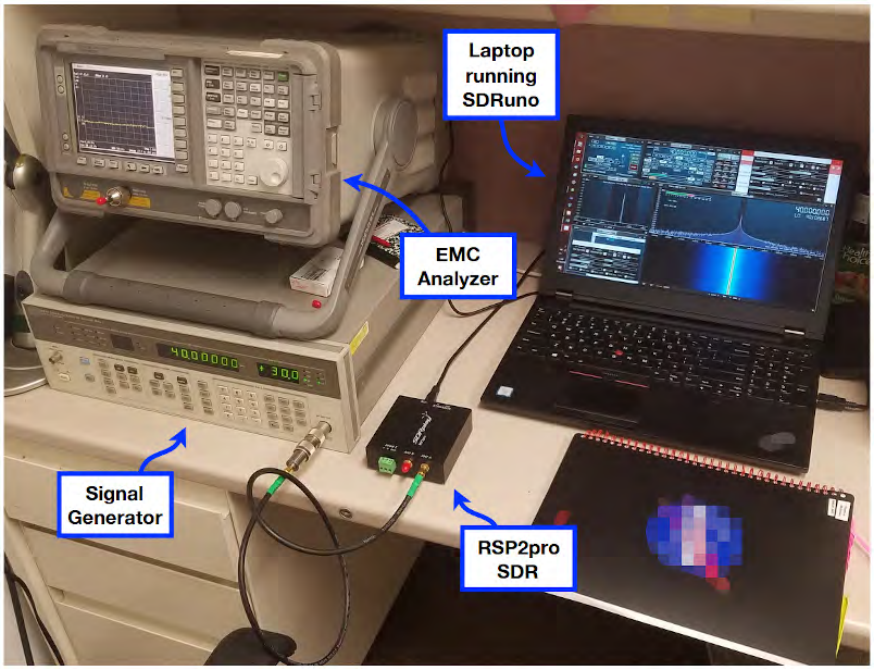


Figure 3.13: The characterization test setup consisting of the EMC analyzer, signal generator, RSP2pro, and laptop. Here the signal generator is shown connected to the RSP2pro.

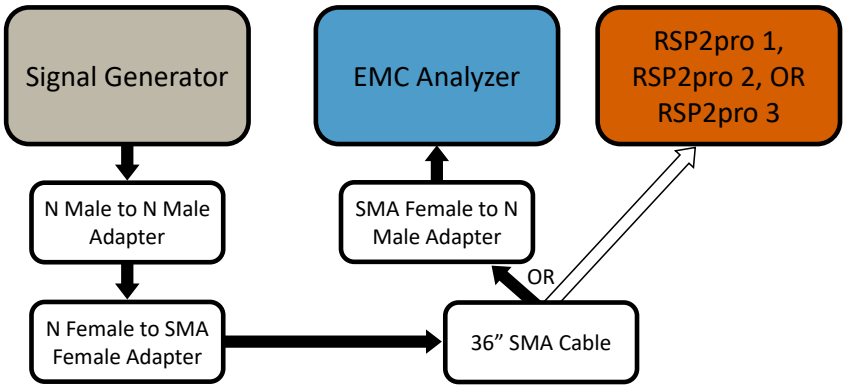


Figure 3.14: Diagram showing the RF signal chains used for the characterization test setup.

For the four devices that were characterized (the EMC analyzer and three RSP2pro units), the measurement results from the nine input data sets shown in Table 3.2 were collected five separate times. Each of these times is referred to as a "set" (for a total of five measurement "sets" per input data set) and each of these "sets" were collected at a different time, usually days apart from one another. For each of these five sets, the individual measurements within a set were repeated ten times. Each of these measurements is referred to as a "trial." This means that a total of 50 measurements were performed for each data point of the nine input sets. Thus for each device that was characterized, 23,850 individual measurements were performed for a grand total of 95,400 individual measurements. To collect such a large amount of data, the characterization process was automated as much as possible. Differences in the user interfaces of the EMC analyzer and the RSP2pro units made the characterization process for each slightly different. For this reason, two separate, semi-automated processes were developed for performing characterization measurements.

The semi-automated procedure for performing the EMC analyzer characterization measurements is outlined in Figure 3.15. The EMC analyzer and the signal generator were automated via their GPIB interfaces using commands from the PyVISA library [50]. The process begins with the user manually selecting the measurement set number, which starts at one. The user then starts the automated Python script.

The Python script first commands the EMC analyzer to load the "Default Sweep" setup file. The first of the nine input data sets from Table 3.2 is then loaded and the frequency and voltage settings from this data set are used to set the output of the signal generator. The EMC analyzer captures the trace data and records the measured quasi-peak (QP) value. This process is repeated until data has been recorded for all of the data points from the input data set. After all of the data points are collected, they are saved to a new csv file and the first trial is complete. After five trials, the EMC analyzer is calibrated before the next five trials are recorded. Once ten trials are collected, the EMC analyzer is calibrated again and the whole process starts over with the next input data set. After collecting ten trials for each of the nine input data sets, the automated Python script ends and the user must manually enter the measurement set number and start the Python script again. After five

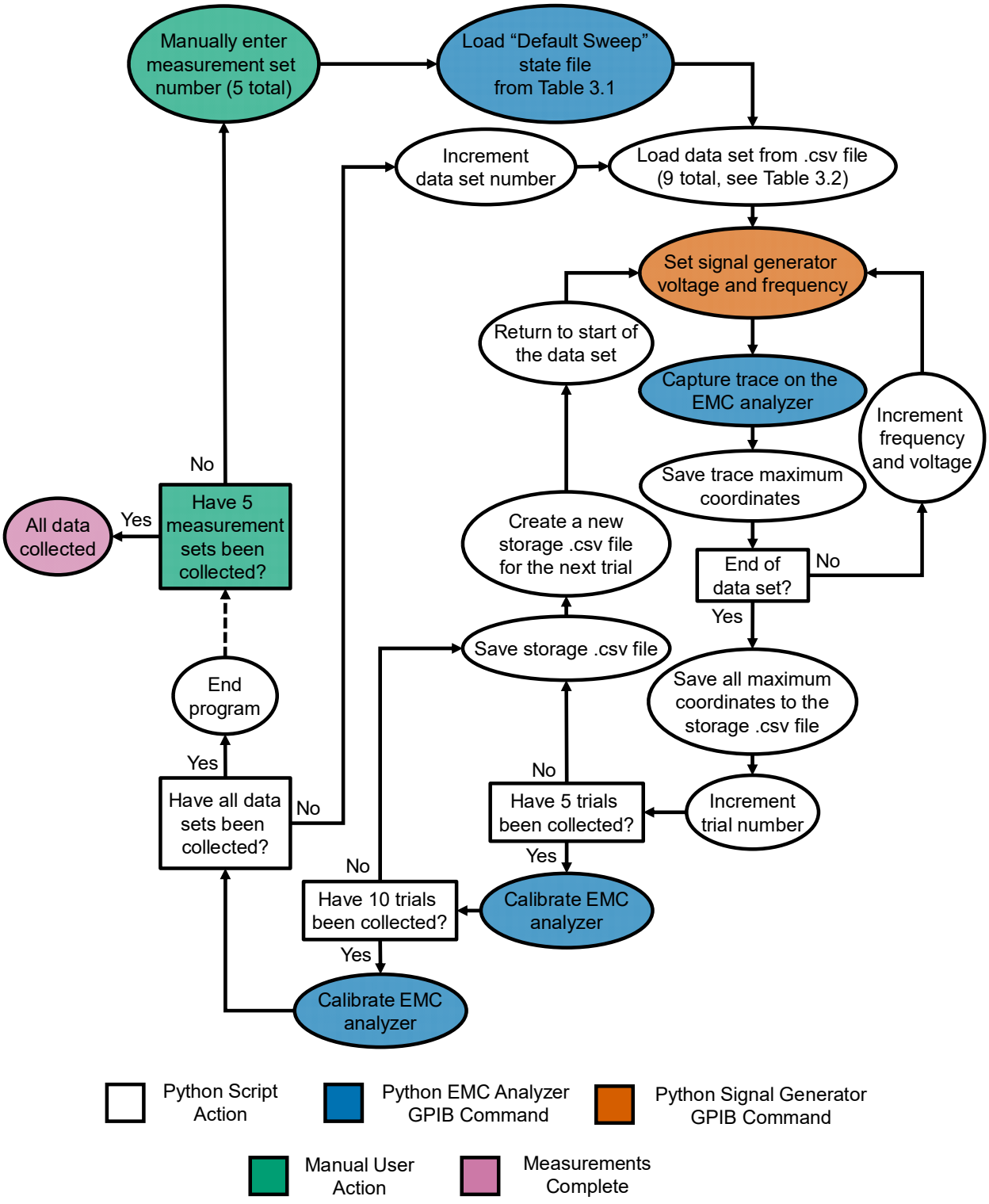


Figure 3.15: Flowchart for the semi-automated procedure that was used to collect the EMC analyzer characterization data.

measurement sets have been collected, the EMC analyzer characterization procedure is complete.

Semi-automating this process was relatively straight-forward thanks to the EMC analyzer and signal generator GPIB interfaces. This is because the GPIB commands in the Python script are just mimicking the user inputs on the EMC analyzer and the signal generator front panels. To slightly improve the speed of each measurement, the EMC analyzer display was disabled. This prevents the EMC analyzer from having to waste time creating the trace on the display. The few seconds that this saved for each individual measurement resulted in a much faster overall collection time given the large number of individual measurements that were performed.

The semi-automated procedure for performing the RSP2pro characterization measurements is outlined in Figure 3.16. The signal generator was automated via its GPIB interface using commands from the PyVISA library [50]. The RSP2pro units required use of the SDRuno companion software described in Section 2.4. Table 3.3 summarizes the SDRuno settings that were used for all of the RSP2pro measurements presented in this thesis. These settings were chosen to have a measurement span comparable to the "Default Sweep" used by the EMC analyzer. The RSP2pro was also used in the low-IF mode, which has fewer options for the RSP2pro sampling rate compared to the options available with zero-IF mode operation. As discussed in Section 2.3, the low-IF operation reduces the amount of intermodulation products at the output of the SDR making it more suitable than the zero-IF mode in spectrum analysis applications. While SDRuno supports several configurations for the low-IF mode, it was important to use the same settings for all measurements so that accurate comparisons could be made between the results presented throughout this thesis.

Since the SDRuno user interface is designed for manual user input via a mouse and keyboard, a specialized control software called AutoIt was used to automate the RSP2pro measurements. According to the the AutoIt website [51], AutoIt is a "scripting language designed for automating the Windows GUI" that "uses a combination of simulated keystrokes, mouse movement and window/control manipulation in order to automate tasks in a way not possible or reliable with other languages." By combining a custom AutoIt script with SDRuno's built-in ability to save measurement data to a csv file, it was possible to completely automate SDRuno measurements. The AutoIt

SDRuno Parameter	Setting Used	Note(s)
Sample Rate	8.192 MHz	IF Mode: Low-IF
Decimation Level	4	
Final Sample Rate	2.048 MHz	
csv Save Interval	1 s	
FFT Size	65536 pts	Rectangular Window, No Averaging
Span	128 kHz	
Peak Search Bandwidth	5 kHz	Tuner Mode: CW
RBW	31.25 Hz	
IF Gain	Automatic	
RF Gain	(varied)	Refer to Figure 3.17

Table 3.3: Summary of the SDRuno settings used for all of the RSP2pro measurements presented in this thesis.

script is outlined by the yellow elements shown in Figure 3.16.

The procedure in Figure 3.16 begins with the user manually connecting the first RSP2pro unit's Port A SMA connector to the signal generator as shown in Figure 3.14. The user then connects the RSP2pro to the host laptop and manually selects the measurement set number, which starts at one. Finally, the user starts the AutoIt script.

The AutoIt script first starts a Python script that controls the signal generator. This Python script then loads the first input data set from the nine shown in Table 3.2 and the frequency and voltage settings from this data set are used to set the output of the signal generator. The Python script then waits for the Autoit script to measure a data point. While the Python script is waiting, the AutoIt script opens SDRuno and applies the settings shown in Table 3.3. It then creates a new storage csv file for the current trial. Next, the AutoIt script sets the RSP2pro tuner frequency and optimizes the RF gain (the RF gain optimization process is shown separately in Figure 3.17).

Using SDRuno's built-in peak measurement button, AutoIt records the measured frequency and amplitude of the strongest signal in the 5 kHz measurement span. This peak signal is the output from the signal generator for the current data point. When the data point is recorded, the Python script detects the file change in the storage csv file and updates the signal generator settings with

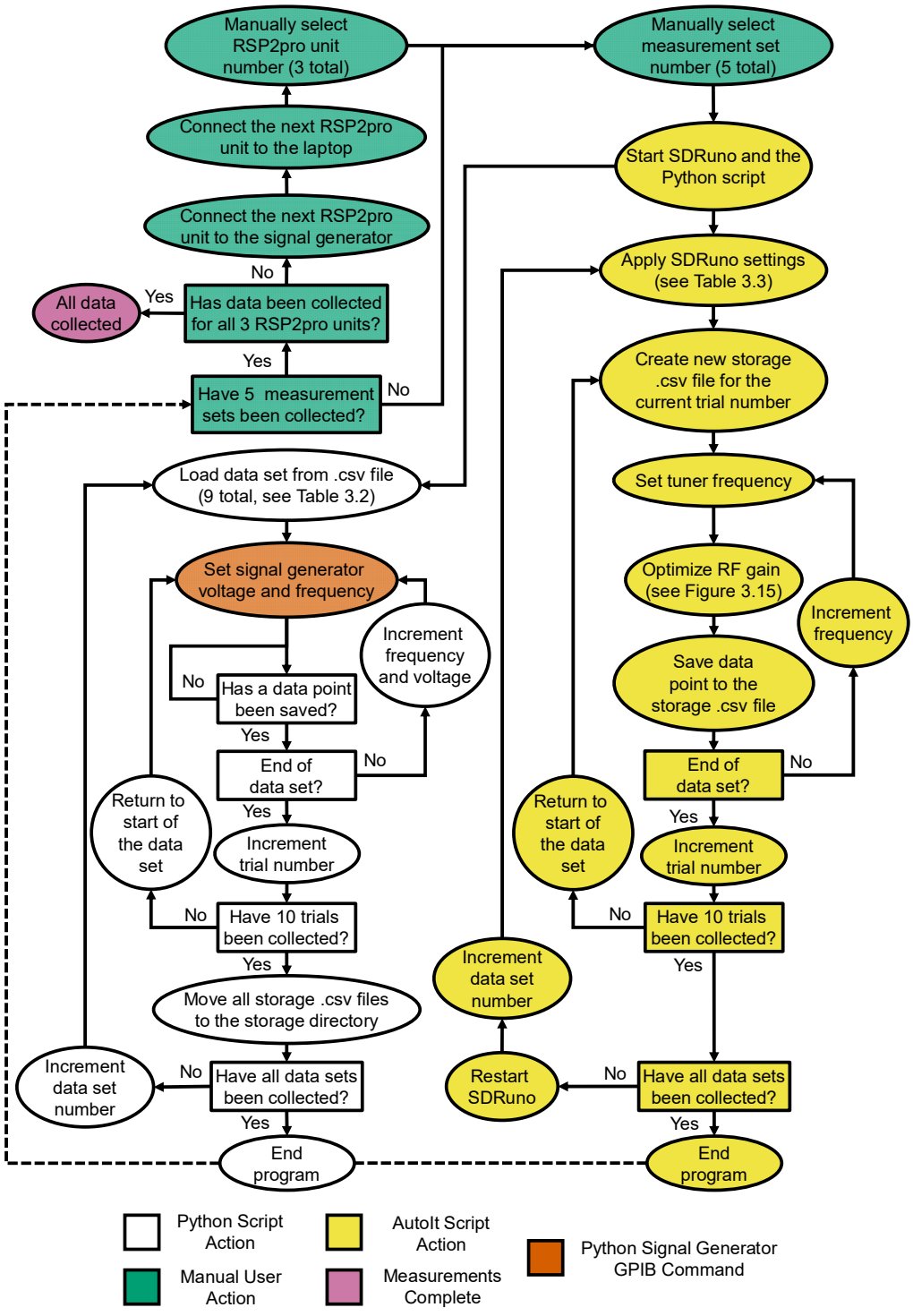


Figure 3.16: Flowchart for the semi-automated procedure that was used to characterize the RSP2pro units.

the next values from the input data set. This process is repeated until ten trials have been collected.

After ten trials have been collected, the AutoIt script restarts SDRuno and the entire process is repeated until ten trials have been completed for each of the nine input data sets. After all of the trials have been collected, the AutoIt script terminates the Python script and closes. The user manually updates the measurement number and then starts the AutoIt script again. After five measurement sets have been collected, the user restarts the entire procedure with the next RSP2pro unit. After five measurement sets have been collected for all three of the RSP2pro units, the RSP2pro characterization procedure is complete.

Figure 3.17 shows how the RF gain was optimized for each RSP2pro measurement. According to [40], the best way to optimize the RF gain setting is to increase the RF gain until an analog-to-digital converter (ADC) overload occurs. From here, the gain can be reduced by one or two stages until the ADC overload has cleared. The yellow ADC overload indicator can be seen in the SDRuno window shown in Figure 3.17. Thanks to AutoIt's ability to use screen pixel colors as function inputs, the AutoIt script can detect when an ADC overload occurs by monitoring the pixels near the ADC overload indicator.

The RF gain optimizer in Figure 3.17 starts with the AutoIt script setting the RF gain to zero. AutoIt then increases the slider (shown in the center of the SDRuno window shown in Figure 3.17) by one stage. Next, the script detects the screen pixel color near the ADC overload indicator to see if an overload has occurred. If the checked pixel is not yellow, the gain stage is increased until an ADC overload occurs and the pixel is yellow. The gain setting is then reduced by one stage and the RF gain has been optimized. When the ADC overload occurs at the lowest gain setting, the RF gain cannot be optimized. This happens when the input RF signal is too strong. The number of RF gain stages available vary depending on the tuner frequency [52], so the amount of time it took to optimize the RF gain at a particular frequency varied throughout the characterization process.

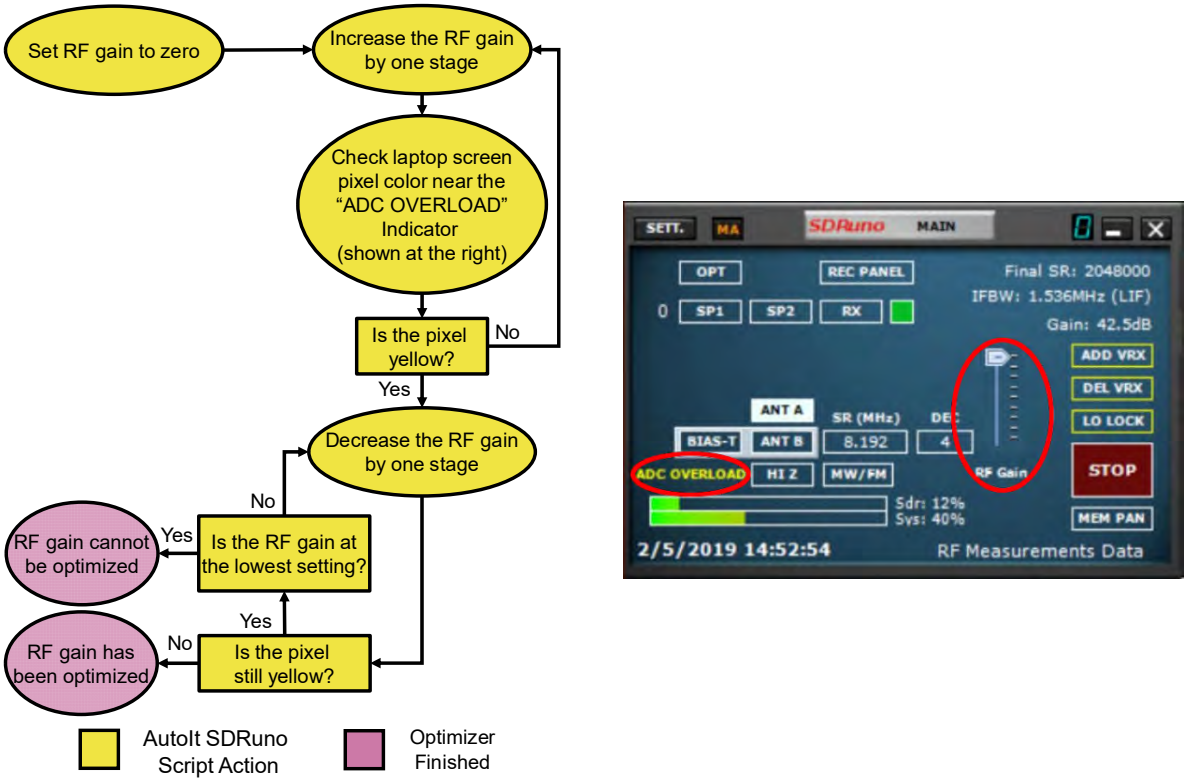


Figure 3.17: The RF gain optimizer that is part of the AutoIt script outlined in Figure 3.16. An outline of the AutoIt code is shown on the left and the SDRuno window with the RF gain control and ADC overload indicator is shown on the right.

3.3 Signal Level Measurement Performance

In this section the accuracy of the RSP2pro's signal level measurement capability is examined by direct comparison to the EMC analyzer. One key difference between the two devices is the ability to display measured RF signals in different units. The EMC analyzer is able to display the amplitude of measured signals in a variety of units such as dBm (decibels relative to one milliwatt) and dBμV. However, SDRuno is only capable of displaying the amplitude of measured signals in dBm.

According to the RSP2pro data sheet [42], the RSP2pro's Port A RF input has an impedance of 50 Ω. With this knowledge, SDRuno measurements recorded in dBm can be converted to dBμV so that they can be compared to the measurements made with the EMC analyzer. Note that the EMC analyzer measurements were made in units of dBμV because the goal of this thesis is to use the RSP2pro to measure radiated emissions, which have units of dBμV/m. The conversion from dBm to dBμV can be accomplished by deriving an equation that relates dBm and dBμV for a system that is assumed to have a purely resistive impedance, R.

According to Paul [1], the power of a signal expressed in dBm, P_{dBm}, is defined as

$$P_{dBm} = 10\log_{10}\left(P/10^{-3}\right)$$

where P represents the power of the signal in Watts. Rewriting this equation in terms of P_{dBm} yields

$$P = (10^{-3})10^{P_{dBm}/10}. \tag{3.4}$$

Also according to Paul [1], the voltage of a signal expressed in dBμV, $V_{dB\mu V}$, is defined as

$$V_{dB\mu V} = 20\log_{10}\left(V/10^{-6}\right) \tag{3.5}$$

where V represents the voltage in Volts. Rewriting this equation in terms of $V_{dB\mu V}$ yields

$$V = (10^{-6})10^{V_{dB\mu V}/10}. \tag{3.6}$$

For a system that is assumed to be purely resistive, the time average power P is related to the RMS voltage of the signal V by Ohm's law and can be expressed as

$$P = V^2/R. \tag{3.7}$$

The derivation begins by substituting Equations 3.4 and 3.6 into Equation 3.7 and rearranging terms so that the P_{dBm} and $V_{dB\mu V}$ terms are on the same side of the equation:

$$P = V^2 / R$$

$$(10^{-3})10^{P_{dBm}/10} = \left[(10^{-12})10^{V_{dB\mu V}/10}\right] / R$$

$$\left[(10^{-3})10^{P_{dBm}/10}\right]10^{-V_{dB\mu V}/10} = 10^{-12} / R$$

$$10^{(P_{dBm}-V_{dB\mu V})/10} = 10^{-9} / R.$$

The next steps in the derivation are to take the logarithm of each side of the equation and to then use the logarithm product and power rules to simplify each term:

$$\log_{10}\left[10^{(P_{dBm}-V_{dB\mu V})/10}\right] = \log_{10}\left(10^{-9} / R\right)$$

$$(P_{dBm} - V_{dB\mu V}) / 10 = \log_{10}\left(1/R\right) + \log_{10}\left(10^{-9}\right)$$

$$(P_{dBm} - V_{dB\mu V}) / 10 = \log_{10}\left(1/R\right) - 9.$$

The final step is to simplify the remaining terms further and to solve for $V_{dB\mu V}$:

$$P_{dBm} - V_{dB\mu V} = 10\log_{10}\left(1/R\right) - 90$$

$$V_{dB\mu V} = P_{dBm} - 10\log_{10}\left(1/R\right) + 90. \tag{3.8}$$

Equation 3.8 shows that for a 50 Ω system

$$V_{dB\mu V} = P_{dBm} - 10\log_{10}\left(1/R\right) + 90\Big|_{R=50\Omega}$$

$$= P_{dBm} - 10\log_{10}\left(1/50\right) + 90$$

$$= P_{dBm} + 16.9897004... + 90$$

$$= P_{dBm} + 106.9897004...$$

$$\approx P_{dBm} + 107.$$

Thus the conversion between dBm and dBμV for a 50 Ω system can be performed with the equation

$$V_{dB\mu V} = P_{dBm} + 107. \tag{3.9}$$

This equation was used to convert all of the RSP2pro measurements presented in this thesis into dBμV so that they could be directly compared with the measurements taken with the EMC analyzer.

Equation 3.9 is very commonly used when performing radiated emission measurements because most spectrum analyzers and receivers have a 50 Ω RF input. This equation is also how the HP 8657A signal generator converts its output signal units from dBm to dBμV [49]. Note that Equation 3.9 only holds true if the system has a 50 Ω impedance. If the system has some other impedance (for example, 75 Ω to match with some types of co-axial cable), Equation 3.8 can be used instead.

Equation 3.9 also assumes that the impedance of the system is matched to 50 Ω at all frequencies. While the RSP2pro does have a 50 Ω RF input, its input will not be perfectly matched at all frequencies. This is true for all instruments with a 50 Ω input or output [26]. According to Agilent [26], the measurement error ME in dB due to the impedance mismatch between a spectrum analyzer and the RF source it is measuring can be calculated as

$$ME = -20\log_{10}\left(1 \pm \left|\left(\rho_a\right)\left(\rho_s\right)\right|\right) \tag{3.10}$$

where ρ_a and ρ_s are the magnitude of the reflection coefficients for the spectrum analyzer input and the RF source, respectively. The reflection coefficient magnitude ρ is related to the voltage

standing wave ration (VSWR) with the following equation:

$$\rho = (\text{VSWR} - 1) \big/ (\text{VSWR} + 1). \tag{3.11}$$

The E7401A EMC analyzer has a characteristic RF input VSWR of 1.35 for signals measured from 1 to 1500 MHz [43] and the HP 8657A signal generator's RF output has a VSWR of (at most) 1.5 for power output level settings less than +5 dBm [49]. Using Equation 3.11 to calculate the reflection coefficient for both the EMC analyzer (ρ_a) and the signal generator (ρ_s) yields

$$\begin{aligned} \rho_a = (1.35 - 1) \big/ (1.35 + 1) &= 0.1489362... \\ &\approx 0.15, \\ \rho_s = (1.5 - 1) \big/ (1.5 + 1) \quad &= 0.20. \end{aligned}$$

These values can then be used with Equation 3.10 to calculate both the negative error due to mismatch

$$\begin{aligned} ME_- &= -20 \log_{10} \left(1 + 0.030\right) \\ &= -20(0.128372...) \\ &= -0.256744... \\ &\approx -0.26 \text{ dB} \end{aligned}$$

and the positive error due to mismatch

$$\begin{aligned} ME_+ &= -20 \log_{10} \left(1 - 0.030\right) \\ &= -20(-0.132282...) \\ &= 0.264565... \\ &\approx 0.26 \text{ dB} \end{aligned}$$

Thus all of the EMC analyzer measurements where the signal generator was used as a source have an error due to mismatch, ME_E, of at least

$$ME_E = \pm 0.26 \text{ dB}. \tag{3.12}$$

This loss is comparable to the overall amplitude measurement accuracy of the EMC analyzer, which according to [43] is approximately ±0.30 dB.

There is additional signal loss caused by the SMA cable that was used with both the RSP2pro units and the EMC analyzer (refer to Figure 3.14). Using a network analyzer, the S-parameters of the SMA cable were measured and used to compute the VSWR at both ends of the SMA cable from 30 MHz to 1000 MHz. These are shown in Figure 3.18. The measured S-parameters were also used to plot the insertion loss of the cable in Figure 3.18b. Note that the y-axis shown in Figure 3.18a starts at 0.90 so that a detailed view of the measured VSWR can be shown.

One can see that at both ends of the SMA cable there is almost a perfect impedance match for frequencies up to 200 MHz. This is indicated by the VSWR being very close to one over this frequency range. At frequencies above 200 MHz, the impedance mismatch begins to increase in an oscillatory manner that varies between a VSWR of 1.00 and 1.04. The conclusion here is that any impedance mismatch errors in the measurements presented in this thesis are predominantly caused by the devices connected to the SMA cable, not the SMA cable itself.

Figure 3.18b shows that the insertion loss of the SMA cable does become significant relative to the ±0.30 dB measurement accuracy of the EMC analyzer when the signal frequency is greater than or equal to approximately 600 MHz. The insertion loss of the cable is greatest around 1000 MHz, where it has a value of almost 0.40 dB. When performing compliance radiated emission measurements, the insertion loss of any cables used for said measurements must be accounted for before using the antenna factor to convert the measured voltage into the radiated emission field strength. However, the measurements in this thesis are focused on comparing the RSP2pro to the EMC analyzer and all of the comparison measurements were performed with the same SMA cable. This means that the insertion loss shown in Figure 3.18b can be ignored when directly comparing the measurement accuracy of the RSP2pro to the measurement accuracy of the EMC analyzer.

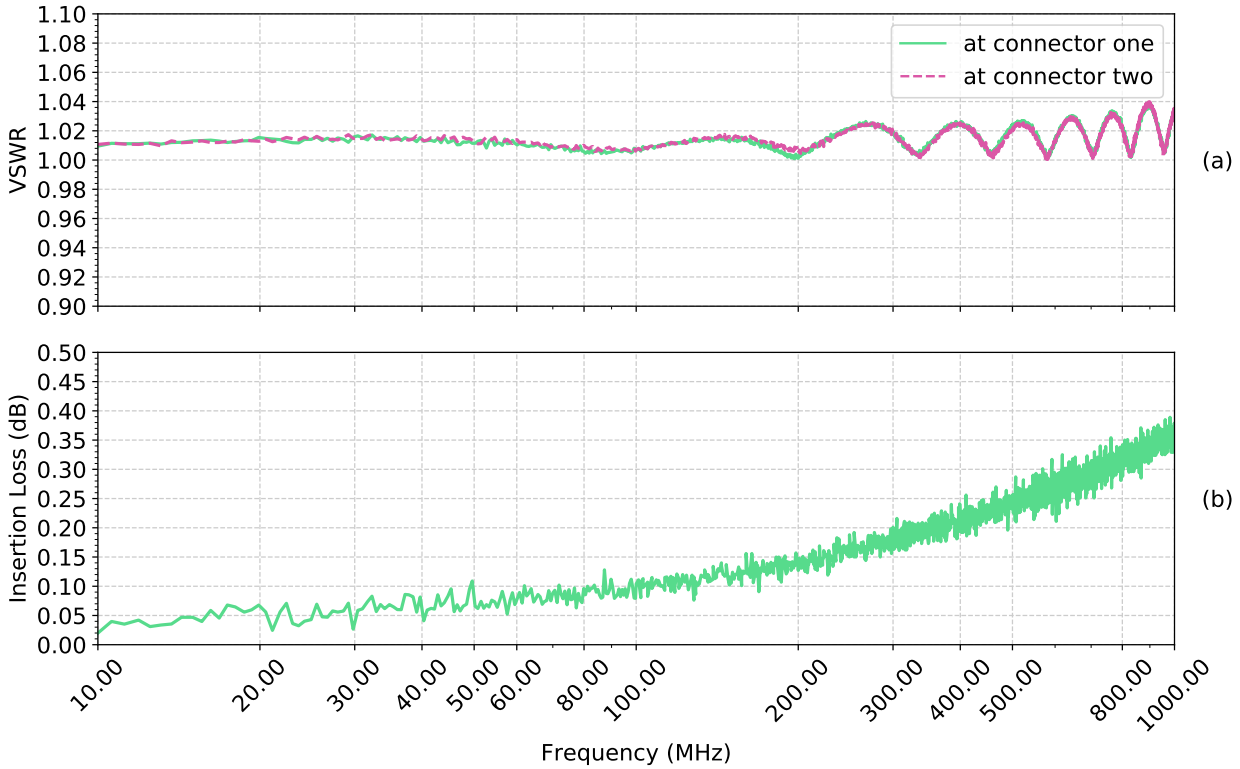


Figure 3.18: (a) The VSWR and (b) the insertion loss of the SMA cable that was used for all of the measurements presented in this thesis.

To characterize the RF signal measurement accuracy of the RSP2pro, the process described in Section 3.2 was used to collect measurements with the EMC analyzer and three RSP2pro units. The measurements used a variety of CW signals that each had a specific frequency and signal level (refer to Table 3.2). Figure 3.19 shows the average voltage from 50 separate measurements made using the EMC analyzer for each of the nine input data sets.

The signals shown in Figure 3.19a and 3.19b represent the data sets corresponding to the Agilent 11955A and 11956A antenna factors (respectively), while Figure 3.19c shows the signals corresponding to the BicoLOG 30100X data sets. On each of these plots, the middle trace shows the voltage level for a received signal that would be exactly at the CISPR-22 Class B limit when measured with that particular EMC antenna. The other traces on each plot correspond to the data sets that were created by using C values of ± 10 dB with Equation 3.3. One can see that the signals

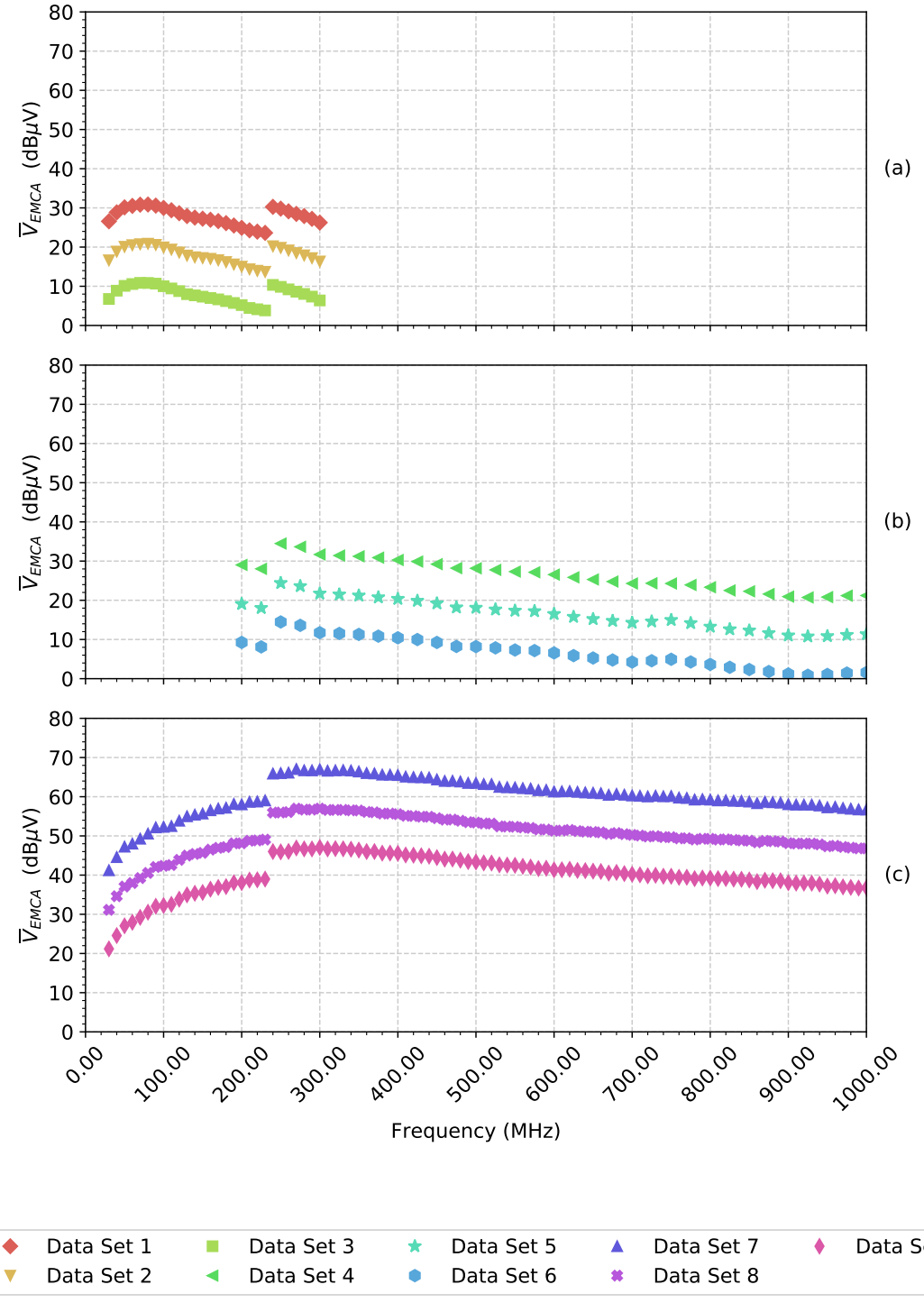


Figure 3.19: Average EMC analyzer voltage measurement for: (a) Data Sets 1, 2, and 3, (b) Data Sets 4, 5, and 6, and (c) Data Sets 7, 8 and 9. Each data point represents the average of 50 measurements.

in Data Sets 1 through 6 carry significantly less power than those in Data Sets 7, 8, and 9. The abrupt shift in signal power observed at 230 MHz on all of these plots is from the CISPR-22 Class B radiated emission limit that was shown in Figure 2.7; at 230 MHz, the limits increase from 30 dBμV/m to 37 dBμV/m.

The measurements shown in Figure 3.19 were also performed using three RSP2pro units. Note that the frequency of each data point presented in this section has been adjusted to the desired signal generator frequency f_{in} so that the differences in the measured signal level of each data point can be more easily compared.

Figures 3.20, 3.21, and 3.22 show the standard deviation of every data point from the RSP2pro and EMC analyzer measurements that was collected using the input data sets from Table 3.2. Each of these figures corresponds with one of the three EMC antennas discussed in Section 3.1.2 (i.e. each figure shows the results from three of the data sets in Table 3.2).

Figure 3.20 shows the standard deviation results for Data Sets 1, 2, and 3. These correspond with the Agilent 11955A antenna factor and C values of +10 dB, 0, and -10dB (respectively). One can see from Figure 3.20a, 3.20b, and 3.20c that the standard deviation of the RSP2pro units is very consistent across the entire frequency range, with an average value of approximately 0.10 dBμV at all frequencies across the CISPR-22 Class B band. The EMC analyzer has a lower standard deviation than the RSP2pro units in Figure 3.20a and a similar standard deviation to the RSP2pro units in Figure 3.20b. Interestingly, Figure 3.20c shows that the EMC analyzer has a much larger standard deviation than the RSP2pro units. This is most likely because Data Set 3 has the second lowest overall signal power level of the nine input data sets and the RSP2pro has a better receiver sensitivity (-147 dBm) than the EMC analyzer, which has a displayed average noise level of approximately -119 dBm for the frequency range shown in Figure 3.20 [43].

This same trend can be seen in Figure 3.21, which shows the standard deviation results for Data Sets 4, 5, and 6. These input data sets correspond with the Agilent 11956A antenna factor and C values of +10 dB, 0, and -10dB (respectively). All three plots in this figure show that the RSP2pro units each have a standard deviation of approximately 0.10 dBμV while the EMC analyzer has a

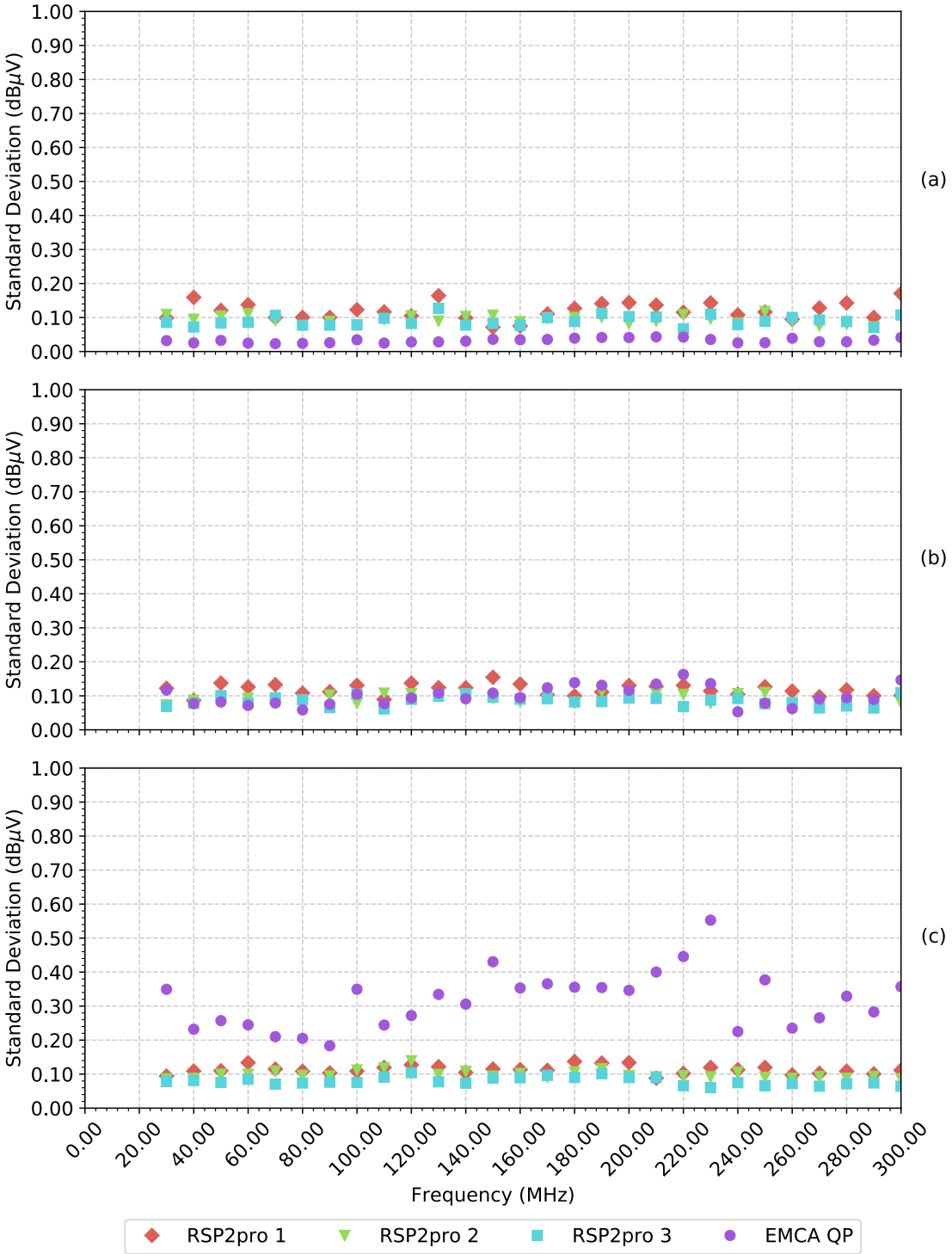


Figure 3.20: RSP2pro and EMC analyzer voltage measurement standard deviation for: (a) Data Set 1, (b) Data Set 2, and (c) Data Set 3. Each data point represents the standard deviation for 50 measurements.

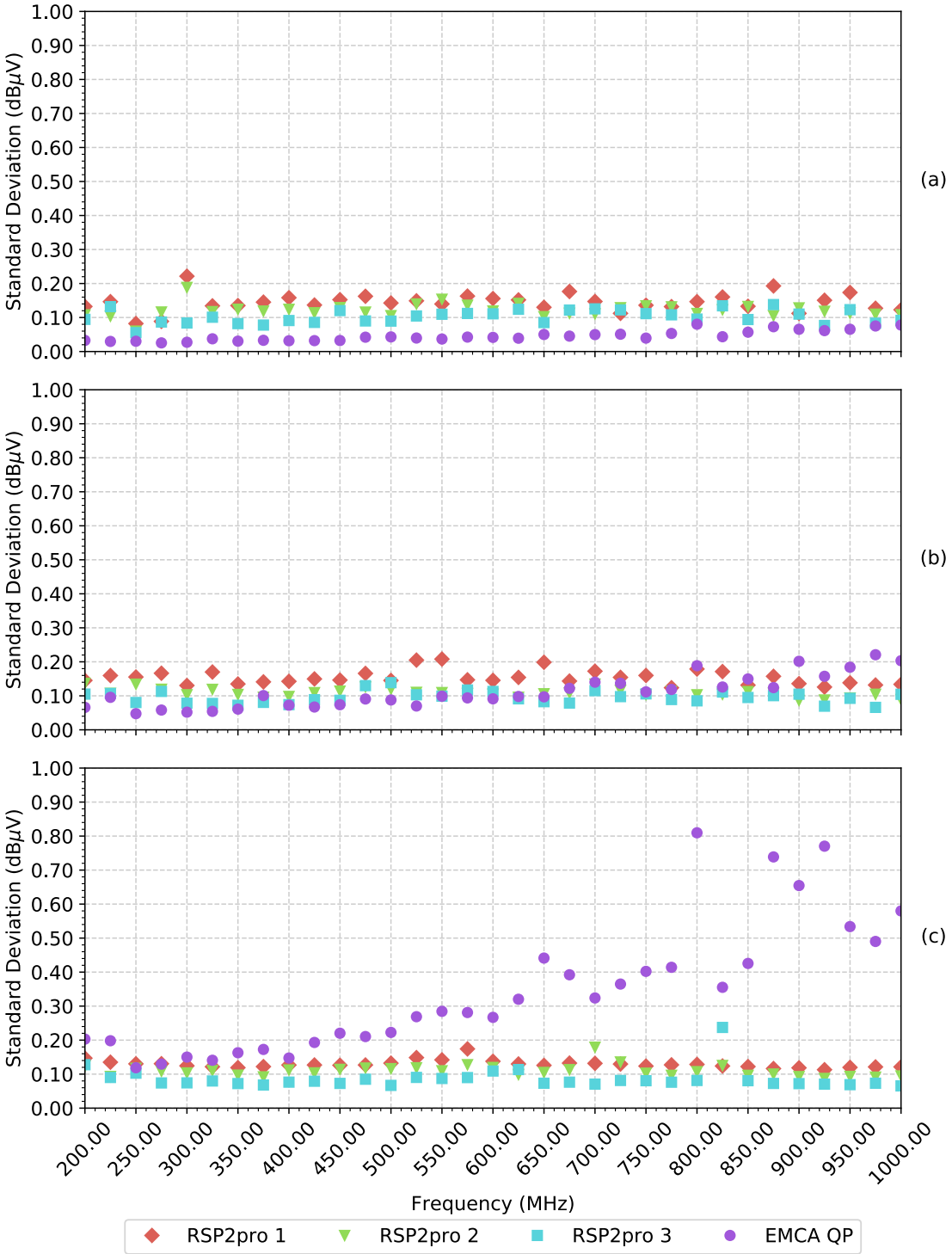


Figure 3.21: RSP2pro and EMC analyzer voltage measurement standard deviation for: (a) Data Set 4, (b) Data Set 5, and (c) Data Set 6. Each data point represents the standard deviation for 50 measurements.

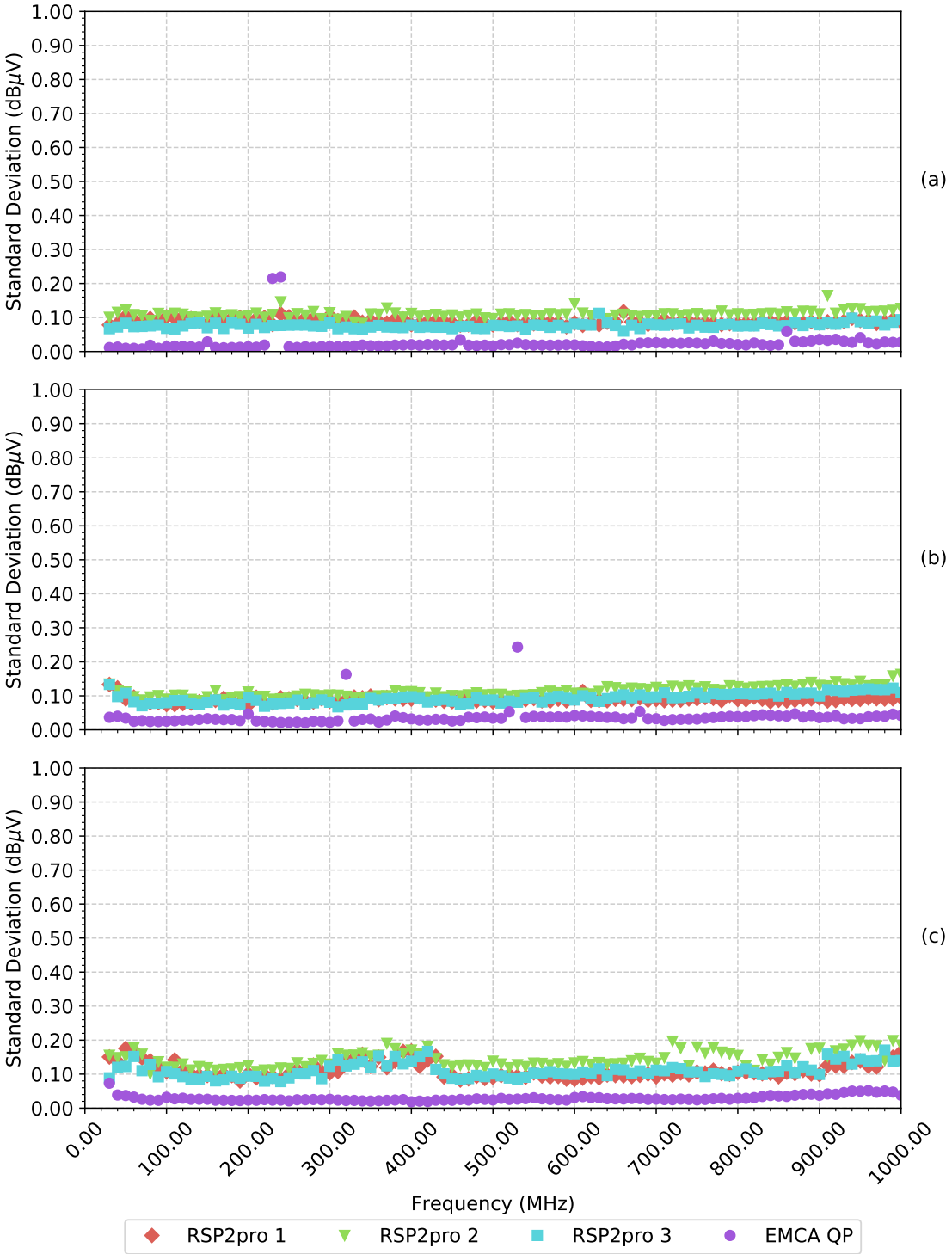


Figure 3.22: RSP2pro and EMC analyzer voltage measurement standard deviation for: (a) Data Set 7, (b) Data Set 8, and (c) Data Set 9. Each data point represents the standard deviation for 50 measurements.

significantly higher standard deviation, as shown in Figure 3.21c. Data Set 6 has the lowest overall signal power level of the nine input data sets, and Figure 3.21c is where the standard deviation of the EMC measurements varies the most.

Figure 3.22 shows the standard deviation results for Data Sets 7, 8, and 9. These input data sets correspond with the BicoLOG 30100X antenna factor and C values of +10 dB, 0, and -10dB (respectively). Figure 3.19 shows that Data Set 7 is the input data set with the highest overall signal level, and Figure 3.22a shows that the EMC analyzer has the least amount of standard deviation with this data set. Data Sets 8 and 9 are the second and third most powerful input data sets, and Figure 3.22b and 3.22c show that for these data sets the EMC analyzer measurements have a better standard deviation than the RSP2pro units at all frequencies. This means that the input power level has a large impact on the measurement consistency of the EMC analyzer. Surprisingly, the RSP2pro units each appear to have the same standard deviation value of 0.10 dBμV at all frequencies despite the higher overall power of these input data sets. Figure 3.22b and 3.22c also show that the standard deviation of the EMC analyzer and the RSP2pro slightly increase with frequency. However, this increase is negligible compared to the average standard deviation observed at all frequencies across the CISPR-22 Class B band.

The main conclusion from Figures 3.20, 3.21, and 3.22 is that the standard deviation for individual RF signal measurements is less than 0.25 dBμV for the three RSP2pro units that were tested. This performance is only slightly worse than the EMC analyzer, and in the case of low power RF signal measurements, the standard deviation performance of the RSP2pro units is actually better than that of the EMC analyzer. This is most likely due to the RSP2pro's excellent receiver sensitivity at the frequency ranges that were tested.

Figures 3.23, 3.24, and 3.25 show the average measured amplitude difference between the EMC analyzer and each of the three RSP2pro units for every data point collected (using the input data sets from Table 3.2). Each of these figures correspond with one of the three EMC antennas discussed in Section 3.1.2 (i.e. each figure shows the results from three of the data sets in Table 3.2). For

these figures, the average offset value $\overline{V}_d$ is defined as

$$\overline{V}_d = \overline{V}_R - \overline{V}_E \tag{3.13}$$

where $\overline{V}_R$ represents the average voltage measured by the RSP2pro at a particular input data point and $\overline{V}_E$ represents the average voltage measured by the EMC analyzer at that same input data point. The value of $\overline{V}_d$ has been plotted for all nine of the input data sets from Table 3.2 so that the amplitude measurement accuracy of all three RSP2pro units can be compared with the amplitude measurement accuracy of the EMC analyzer. Recall that the EMC analyzer amplitude has an absolute measurement accuracy of ±0.30 dB [43] for all of the input data points tested. By using the EMC analyzer for comparison, a new margin of error can be developed for measurements performed with the RSP2pro.

Figure 3.23 shows the average offset value of Equation 3.13 for Data Sets 1, 2, and 3. The first observation is that measurements from all three RSP2pro units remain very consistent across all three plots. This consistency is indicated by the near identical trend of each of the RSP2pro data sets shown in all three plots of Figure 3.23. All three RSP2pro units also have the same relative measurement accuracy. The RSP2pro 1 measurements are slightly higher than the EMC analyzer measurement, while the RSP2pro 2 measurements are consistently smaller than the EMC analyzer measurements. For all three plots, one can see that the measurements for all of the RSP2pro units are within 1.00 dB of the EMC analyzer measurement across the entire frequency range. Each of the RSP2pro units also performed nearly identically at frequencies between 260 and 300 MHz.

Figure 3.24 shows the average offset value from Equation 3.13 for Data Sets 4, 5, and 6. While the plots shown in Figure 3.24a and 3.24b show that the measurements from all three RSP2pro units are consistent, there is less consistency observed in Figure 3.24c. This plot corresponds to the input data set with the lowest overall signal level. All three plots show that RSP2pro units 1 and 3 have about the same measurement accuracy relative to the EMC analyzer, while RSP2pro 2 measurements are consistently lower than the EMC analyzer measurements. Unlike Figure 3.23, one can see from Figure 3.24 that the measurements for all three units are within 2.00 dB of the EMC

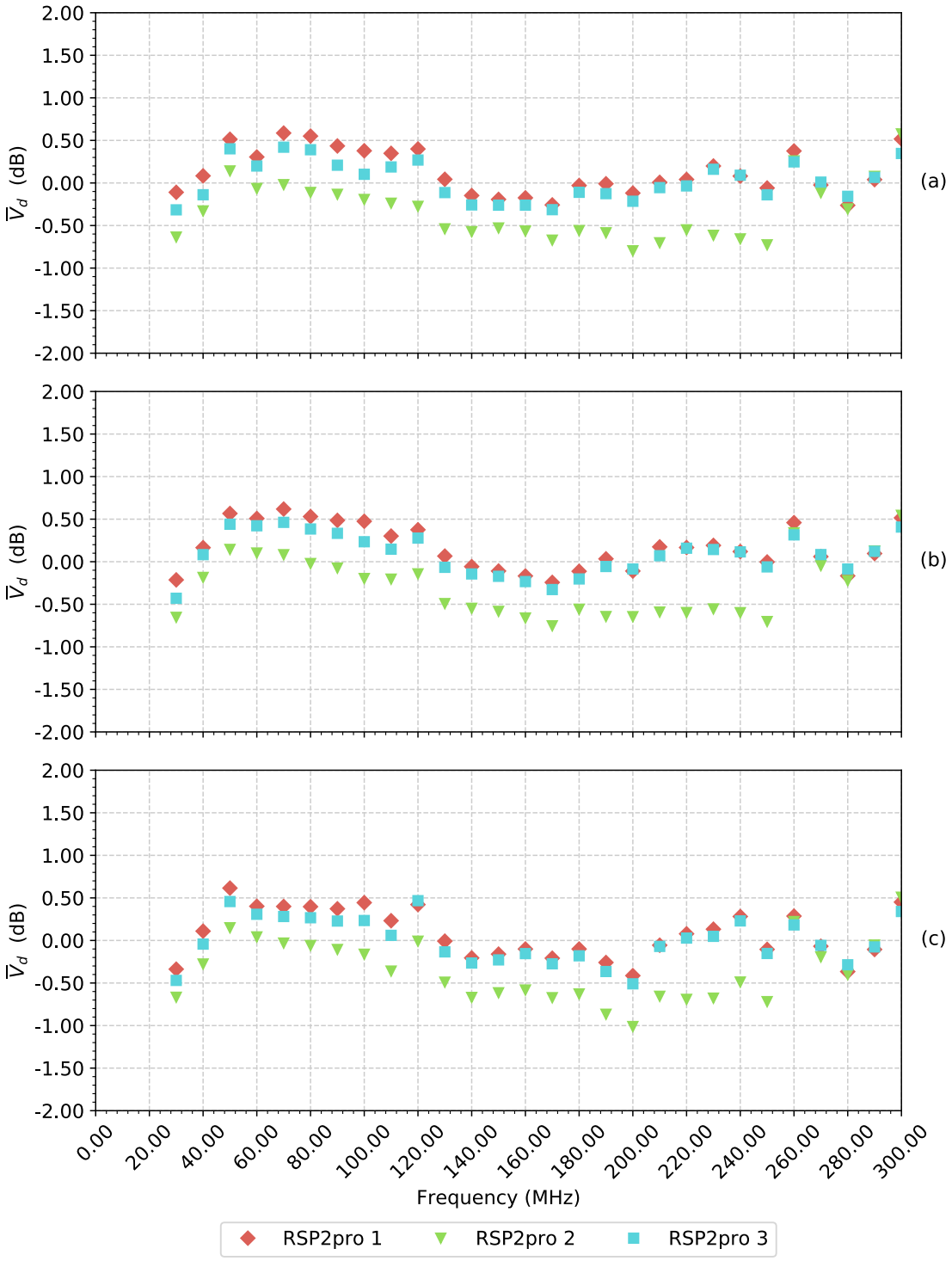


Figure 3.23: Average RSP2pro voltage measurement offset for: (a) Data Set 1, (b) Data Set 2, and (c) Data Set 3. Each data point represents the average of 50 measurements.

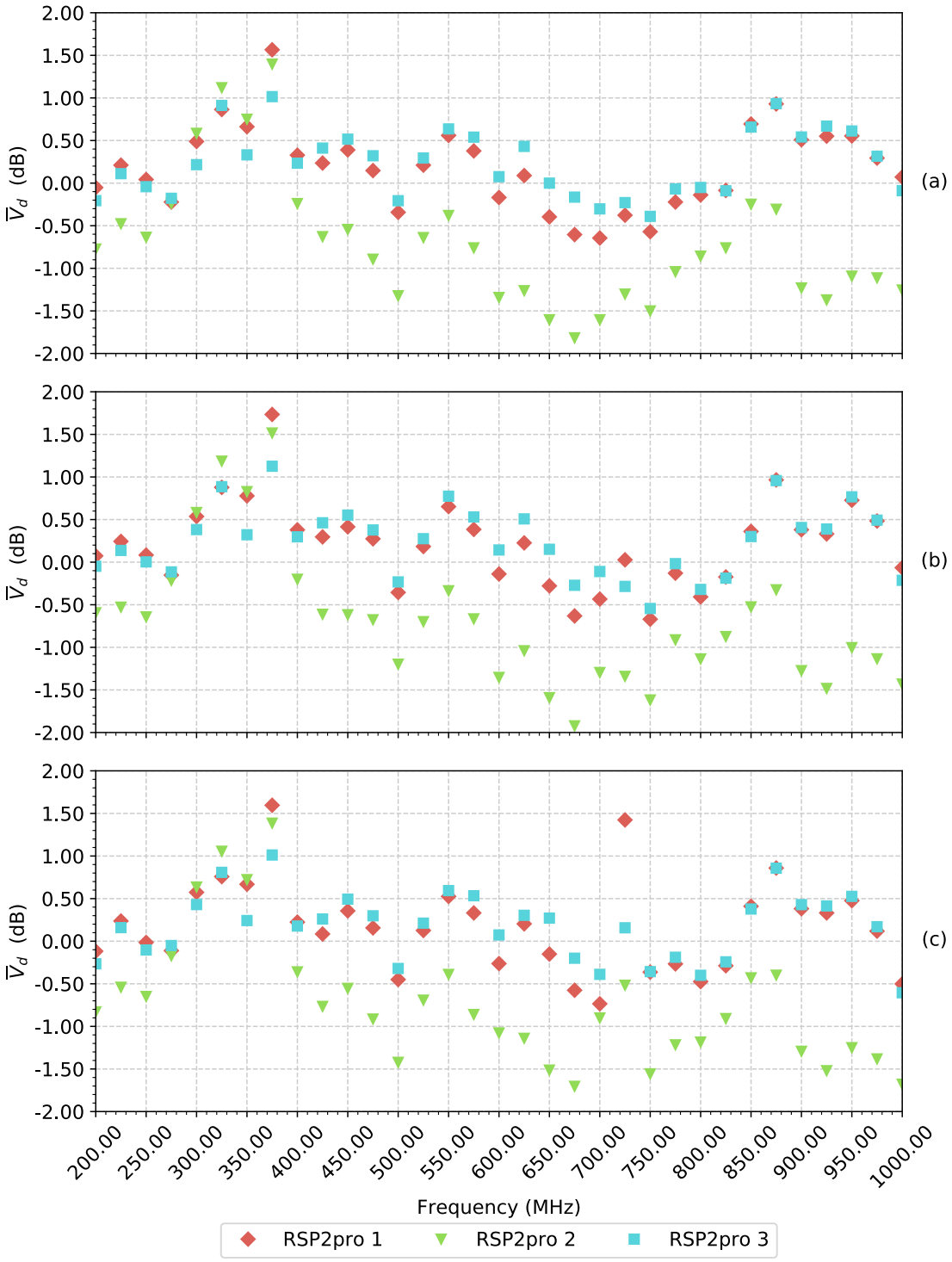


Figure 3.24: Average RSP2pro voltage measurement offset for: (a) Data Set 4, (b) Data Set 5, and (c) Data Set 6. Each data point represents the average of 50 measurements.

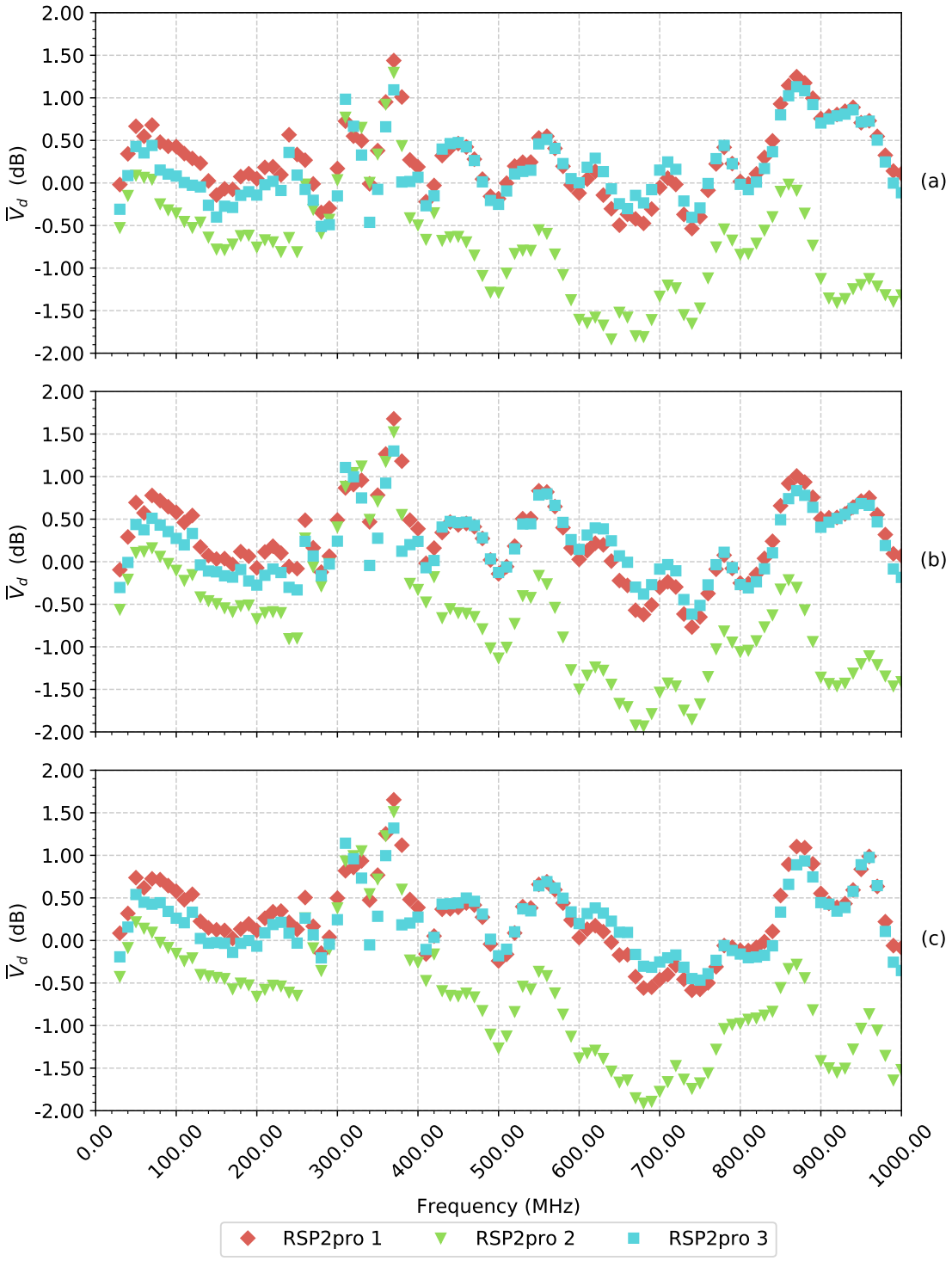


Figure 3.25: Average RSP2pro voltage measurement offset for: (a) Data Set 7, (b) Data Set 8, and (c) Data Set 9. Each data point represents the average of 50 measurements.

analyzer measurement across the entire frequency range. This larger overall average measurement offset can be attributed to the low power level of Data Sets 4, 5, and 6 (refer to Figure 3.19b).

Figure 3.25 is the final offset voltage comparison and shows the average offset value from Equation 3.13 for Data Sets 7, 8, and 9. Figure 3.25a and 3.25b show nearly identical trends for the RSP2pro 1 and 3 measurements. In these plots, the RSP2pro 1 and 3 measurements also have nearly identical offsets from the EMC analyzer measurement from 400 MHz to 600 MHz and 750 MHz to 1000 MHz. Figure 3.25c shows a similar trend, but the average offset for both units is actually better in the range from 600 MHz to 800 MHz. All three of the plots in Figure 3.25 highlight that the measurements from RSP2pro 2 were significantly different from the other two units. While this trend can also be observed in Figures 3.23 and 3.24, the larger frequency range in this figure more clearly illustrates the extent this difference. Despite RSP2pro 2's poorer performance, the overall average offset for all three RSP2pro units is still within 2.00 dB of the EMC analyzer measurements.

The results shown in Figure 3.23, 3.24, and 3.25 indicate that for all of the signal levels tested, the three RSP2pro units were able to have a measurement accuracy within ±2.00 dB of the EMC analyzer. Measurements from two of the units were able to achieve an accuracy of just over ±1.50 dB relative to the EMC analyzer. This level of accuracy is acceptable for low-cost precompliance testing, where the radiated emission measurement errors caused by the uncontrolled RF environment can range from +6 dB to -25 dB [23].

While the procedure in Section 3.2 has been used here to compare several RSP2pro units to an EMC analyzer, it can also be used to calibrate the RSP2pro. Within SDRuno, the user can manually account for additional RF gain external to the RSP2pro unit by adjusting the measured power level by a constant value in dB [53]. This feature is intended to account for additional gains and losses in the RF signal chain connected to the RSP2pro, but can also be used to manually calibrate the measured signal level if said signal level is compared to the level measured by another calibrated device (e.g. a spectrum or EMC analyzer). However, this adjustment is constant at all frequencies, so it would need to be manually adjusted for each measurement to account for any frequency-dependent differences. For example, in Figure 3.25 the average offset for the RSP2pro

2 measurements is only -0.50 dB in the frequency range from 30 MHz to 200 MHz but is nearly -2.00 dB in the frequency range from 600 MHz to 800 MHz.

The accuracy of ±2.00 dB observed here is sufficient for replacing a spectrum analyzer for low-cost precompliance radiated emission measurements. The characterization test setup allowed for a very high SNR because the RSP2pro units and the EMC analyzer were directly connected to the signal generator via an SMA cable. This direct connection minimized the noise floor for all of the measurements, providing a high SNR for every measurement.

According to Horowitz [54], the SNR of a measured signal is a ratio in dB that is defined as

$$\text{SNR} = 10 \log_{10} \left(v_s^2 / v_n^2 \right) \tag{3.14}$$

where v_s and v_n represent the rms voltages of the desired signal and the signal noise that is also present (respectively). Horowitz [54] explains that "the squared amplitudes [of Equation 3.14 suggest] a ratio in terms of power, which is the origin of the decibel ratio definition" used for SNR. The SNR of a measured signal is directly related to the bandwidth of the signal and the measurement bandwidth used by the receiver (i.e. the RSP2pro). For example, a signal with a narrowband frequency response will continue to have the same power level as the measurement bandwidth is increased. This means that the SNR of the measurement can decrease because the receiver amplifier will continue to add noise power as the measurement bandwidth becomes larger than the signal bandwidth [54]. In this context, the measurement bandwidth is equivalent to the span setting of a spectrum analyzer or receiver. Recall that Figure 3.11 from Section 3.1.3 shows that the signal generator used for the characterization measurements had a narrowband output signal with a 3-dB bandwidth of only 300 Hz. This bandwidth is much smaller than the span used by both the EMC analyzer (3 kHz) and the RSP2pro (128 kHz).

SDRUno also records the SNR when saving the measured signal peak and frequency. The SNR data from all the measurement results shown in Figures 3.23, 3.24, and 3.25 can be used to understand how the SNR relates to the RSP2pro signal measurement accuracy.

Figures 3.26, 3.27, and 3.28 show the standard deviation of the SNR for each of the data points that was measured from the input data sets shown in Table 3.2. Each of these figures correspond

with one of the three EMC antennas discussed in Section 3.1.2 (i.e. each figure shows the results from three of the data sets in Table 3.2).

Figures 3.26 and 3.27 show the standard deviation of the SNR for Data Sets 1 through 6. With the exception of one frequency (825 MHz), these plots show that the SNR for every data point tested had a standard of deviation less than 0.5 dB. Figure 3.19 shows that these data sets all have signal output levels below 40 dBμV. This implies that for narrowband signal levels less than 40 dBμV, the RSP2pro experiences a very consistent SNR. This same level of consistency did not occur with the much higher input signal levels from Data Set 7, 8, and 9. Figure 3.28 shows the standard deviation for the SNR when measuring these data sets.

All three plots in Figure 3.28 show a significant amount of deviation in the SNR measurements from all three RSP2pro units. The largest deviation occurs in the frequency range from 230 MHz to 500 MHz in Figure 3.28a. This frequency range corresponds with the largest input signal levels from Data Set 7, which is the input data set with the highest overall amplitude levels (refer to Figure 3.19). This shows that the SNR observed by the RSP2pro varies by more than 1 dB when the measured signal is a narrowband signal with a magnitude above 60 dBμV. The frequency range from 30 MHz to 200 MHz is where Data Set 9 has a signal level below 40 dBμV, and here the standard deviation of the measured SNR is comparable to what can be observed in Figures 3.26 and 3.27. While this may seem like further evidence that narrowband signal levels below 40 dBμV provide a much more consistent SNR, this discrepancy actually occurs because the analog-to-digital (A/D) converters of the RSP2pro are being overloaded when measuring the signals higher than 200 MHz.

An A/D converter overload can be shown by plotting the difference between the measured SNR of an input signal and the SNR of the same input signal reduced by 10 dB. The difference between the SNR measurements should be approximately 10 dB but will be different when the A/D converters are overloaded. This method works well with the test setup used for the characterization process because the noise floor is minimized by the direct connection to the signal source. Using the input data sets from Table 3.2, the difference D_{xy} between data sets that are 10 dB apart can be

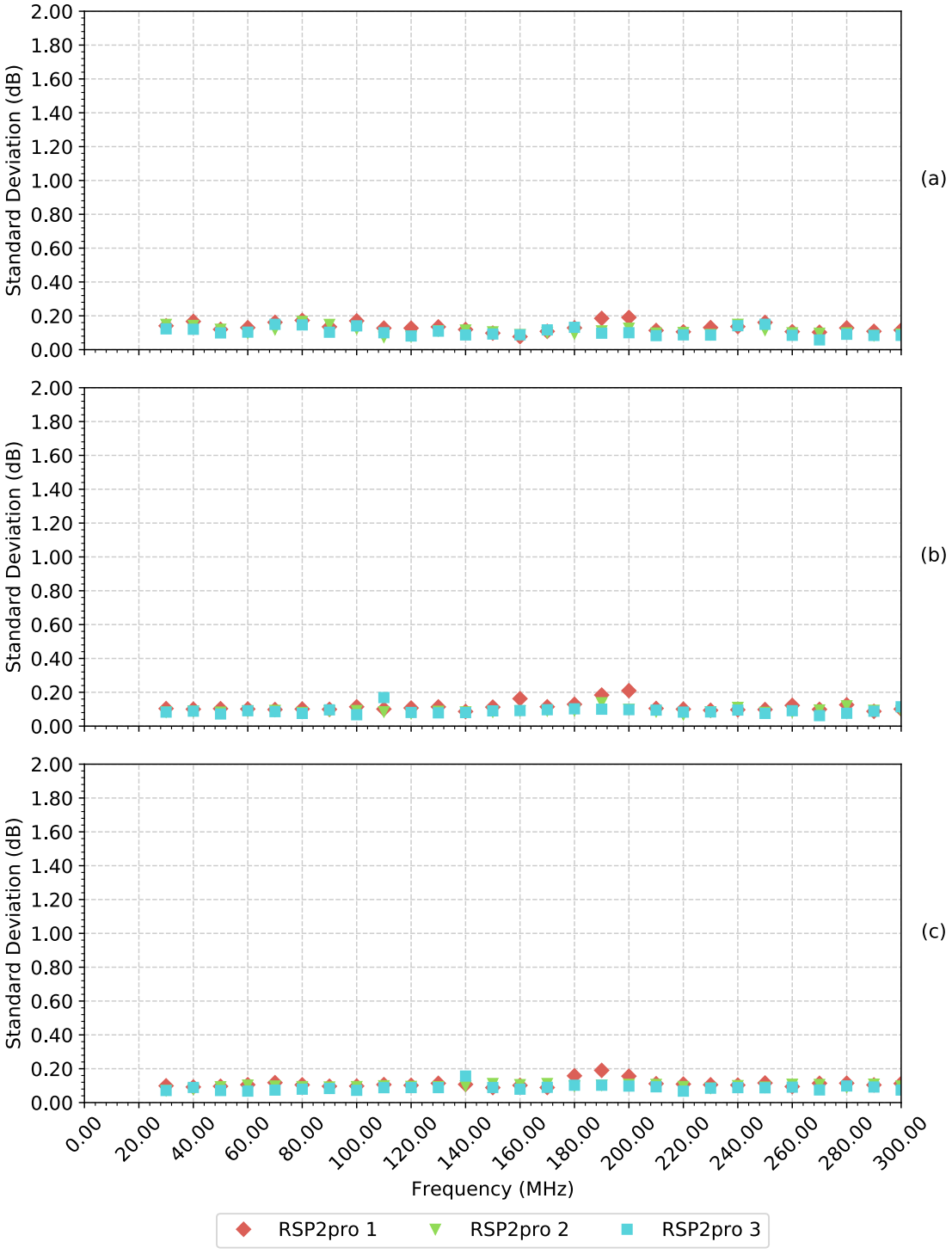


Figure 3.26: RSP2pro SNR measurement standard deviation for: (a) Data Set 1, (b) Data Set 2, and (c) Data Set 3. Each data point represents the standard deviation for 50 measurements.

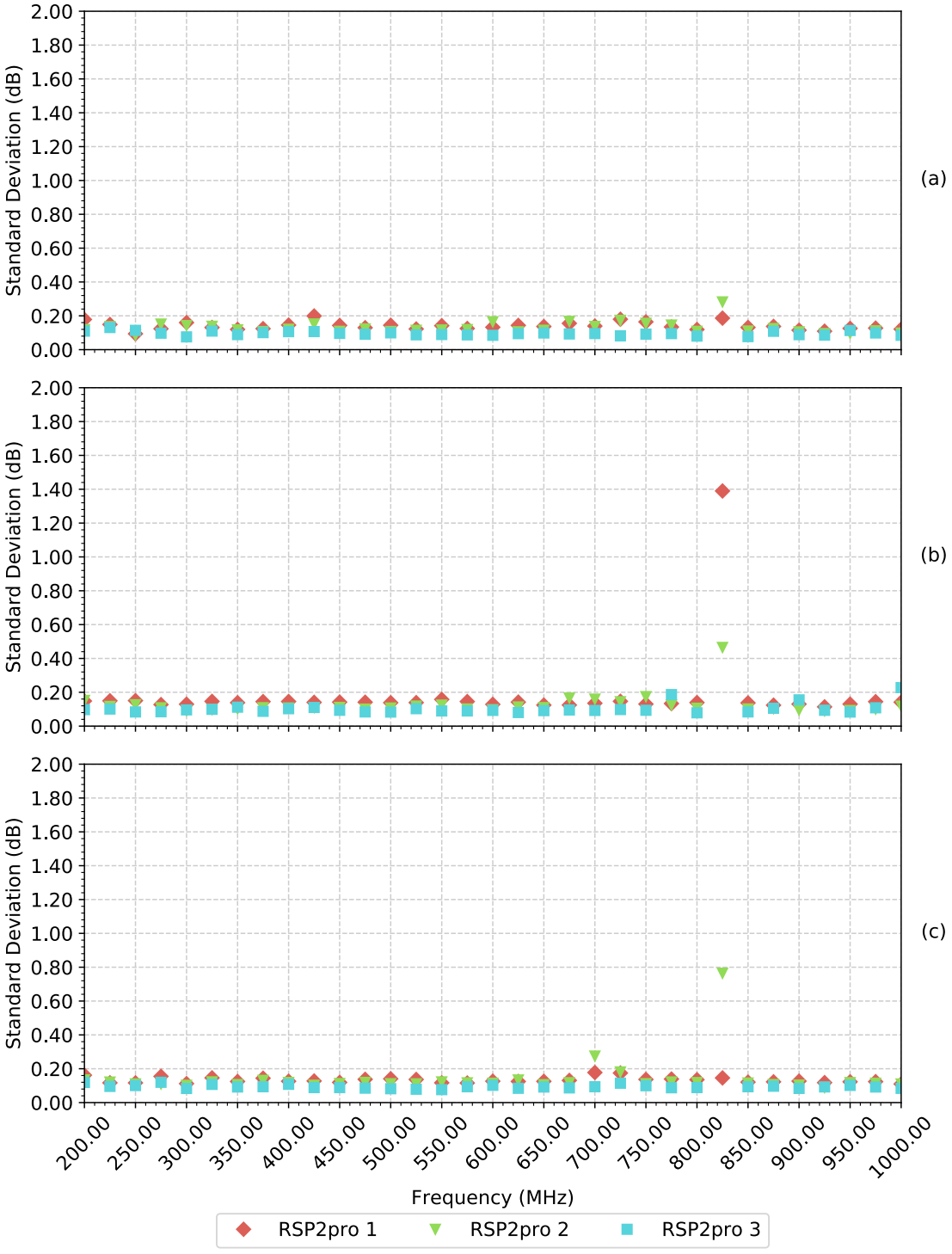


Figure 3.27: RSP2pro SNR measurement standard deviation for: (a) Data Set 4, (b) Data Set 5, and (c) Data Set 6. Each data point represents the standard deviation for 50 measurements.

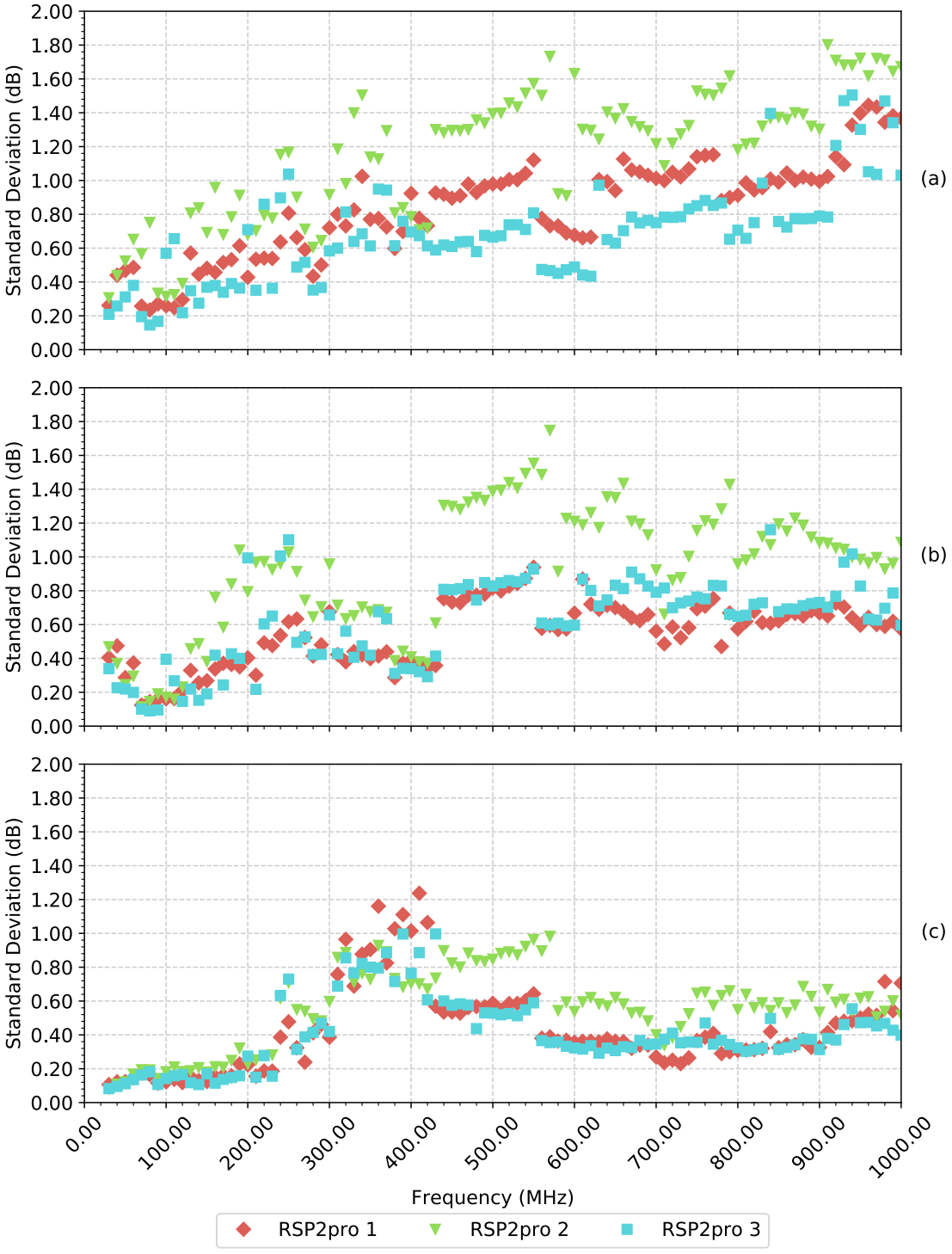


Figure 3.28: RSP2pro SNR measurement standard deviation for: (a) Data Set 7, (b) Data Set 8, and (c) Data Set 9. Each data point represents the standard deviation for 50 measurements.

used to create such plots. This difference is defined as

$$D_{xy} = \text{SNR}_x - \text{SNR}_y \tag{3.15}$$

where x and y correspond to the data set numbers from Table 3.2. This equation was used with the data sets from Table 3.2 that used the same EMC antenna and were separated by 10 dB to create Figures 3.29 and 3.30.

Figure 3.29 shows that for Data Sets 7 and 8, the value of Equation 3.15 is much less than 10 dB at nearly every frequency. This indicates that the A/D converters of the RSP2pro units were likely overloaded when performing measurements with these input data sets. Figures 3.29a and 3.30 show that the other data sets did not overload the A/D converters since the difference shown in these figures is close to the expected value of 10 dB. Figures 3.30a and 3.30b each show a similar trend for the value of D_{23} and D_{56}. Figure 3.30c shows that the value of Equation 3.15 for Data Sets 8 and 9 is only close to 10 for signals less than 200 MHz. This explains why the standard deviation of the SNR in Figure 3.28c was more comparable to Data Sets 1 through 6; the A/D converters are not being overloaded for input data points from Data Set 9 that are at a frequency below 200 MHz.

Figures 3.31, 3.32, and 3.33 show the average SNR for each of the data points that was measured from the input data sets shown in Table 3.2. Each of these figures correspond with one of the three EMC antennas discussed in Section 3.1.2 (i.e. each figure shows the results from three of the data sets in Table 3.2).

Figure 3.31 shows the SNR for measurements where Data Sets 1, 2, and 3 were measured. The plots in this figure show that the measured SNR at every frequency was almost identical for each RSP2pro unit. All of the SNR values in Figure 3.31 are between 24 and 56 dB. Figure 3.32 shows that the SNR for the Data Set 4, 5, and 6 measurements are also between 16 and 56 dB. For frequencies up to 600 MHz, all three RSP2pro units observed almost exactly the same SNR for every data point. Figure 3.32 also shows that the SNR observed by each RSP2pro varied significantly at 700 MHz.

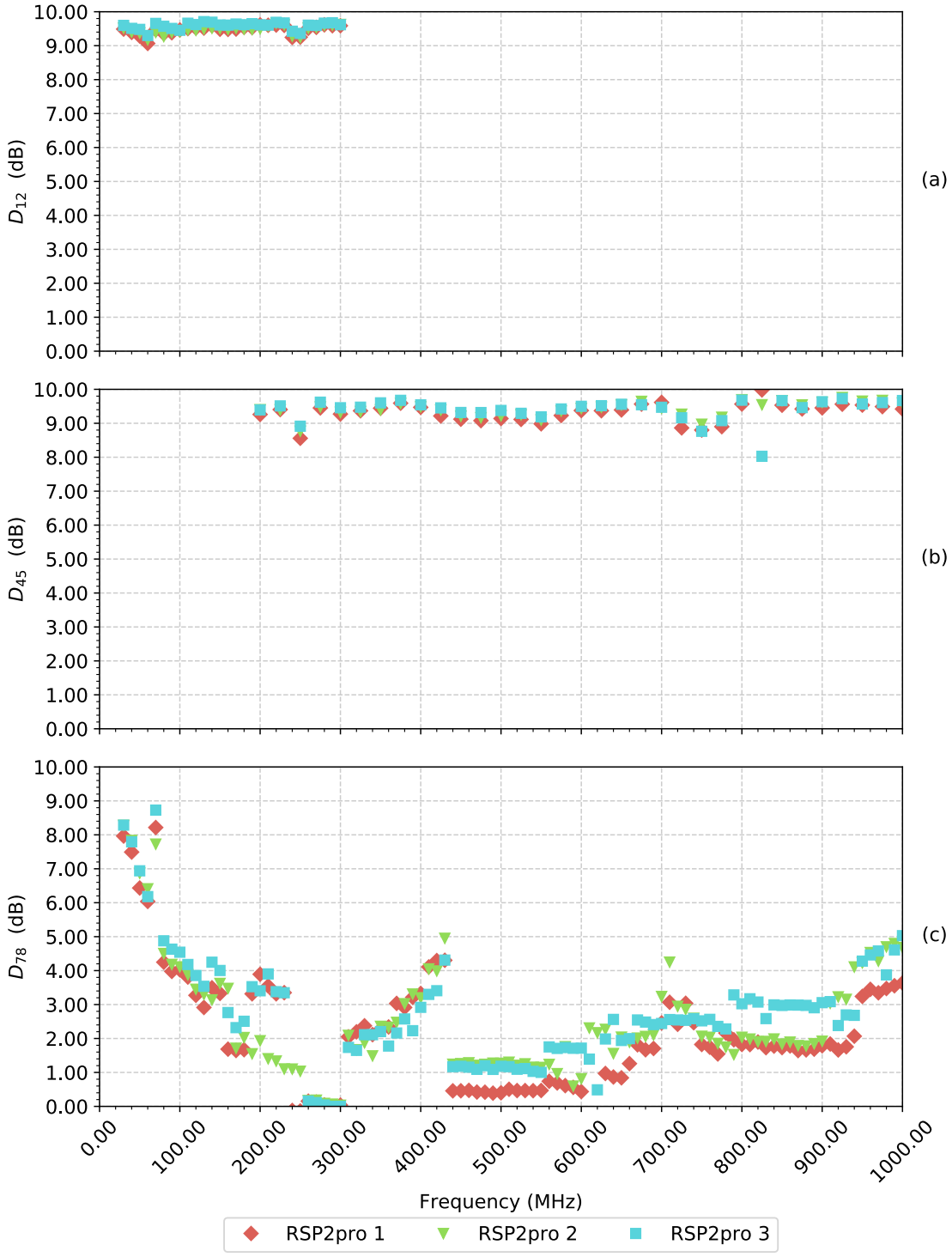


Figure 3.29: The value of Equation 3.15 using (a) Data Set 1 and 2, (b) Data Set 4 and 5, and (c) Data Set 7 and 8.

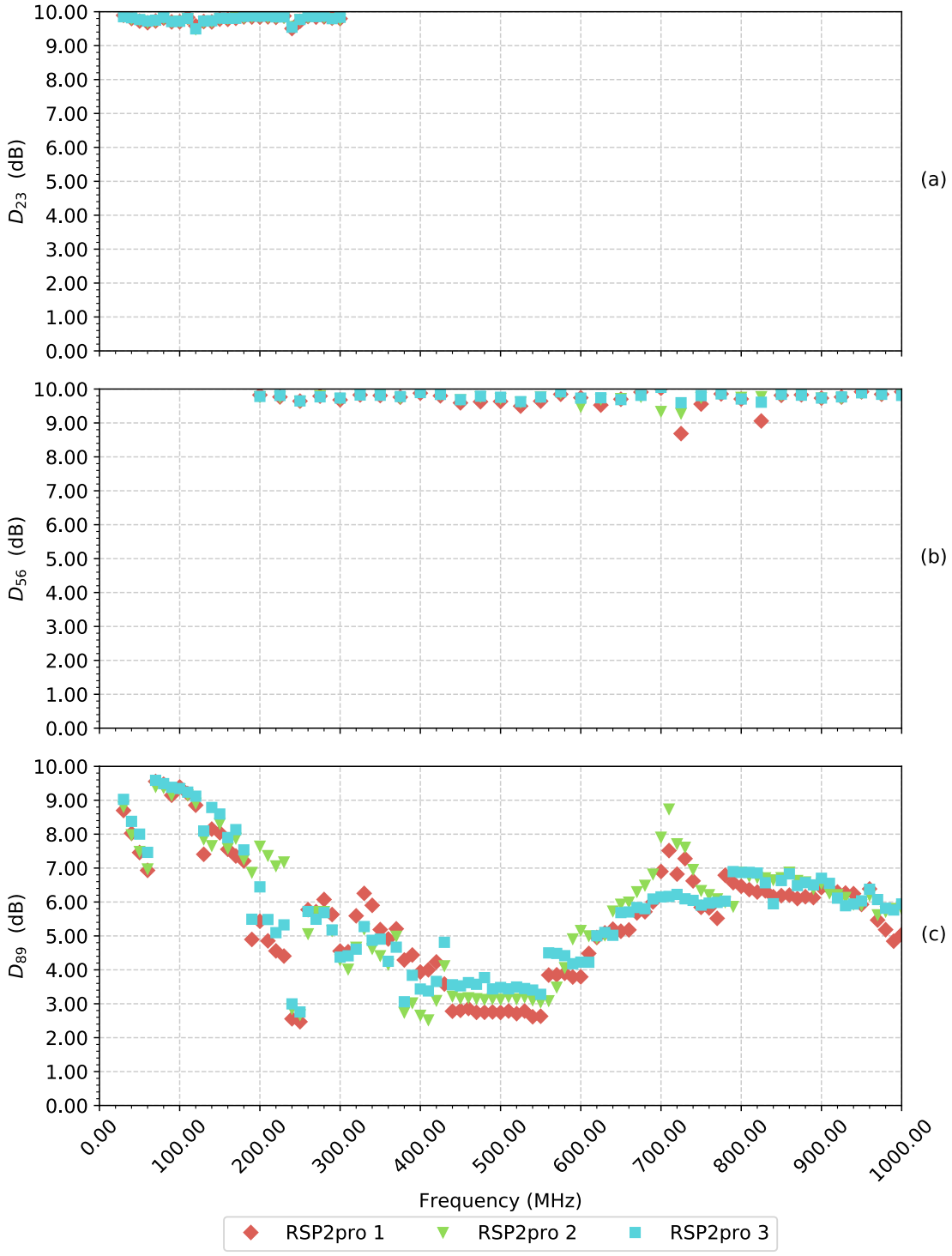


Figure 3.30: The value of Equation 3.15 using (a) Data Set 2 and 3, (b) Data Set 5 and 6, and (c) Data Set 8 and 9.

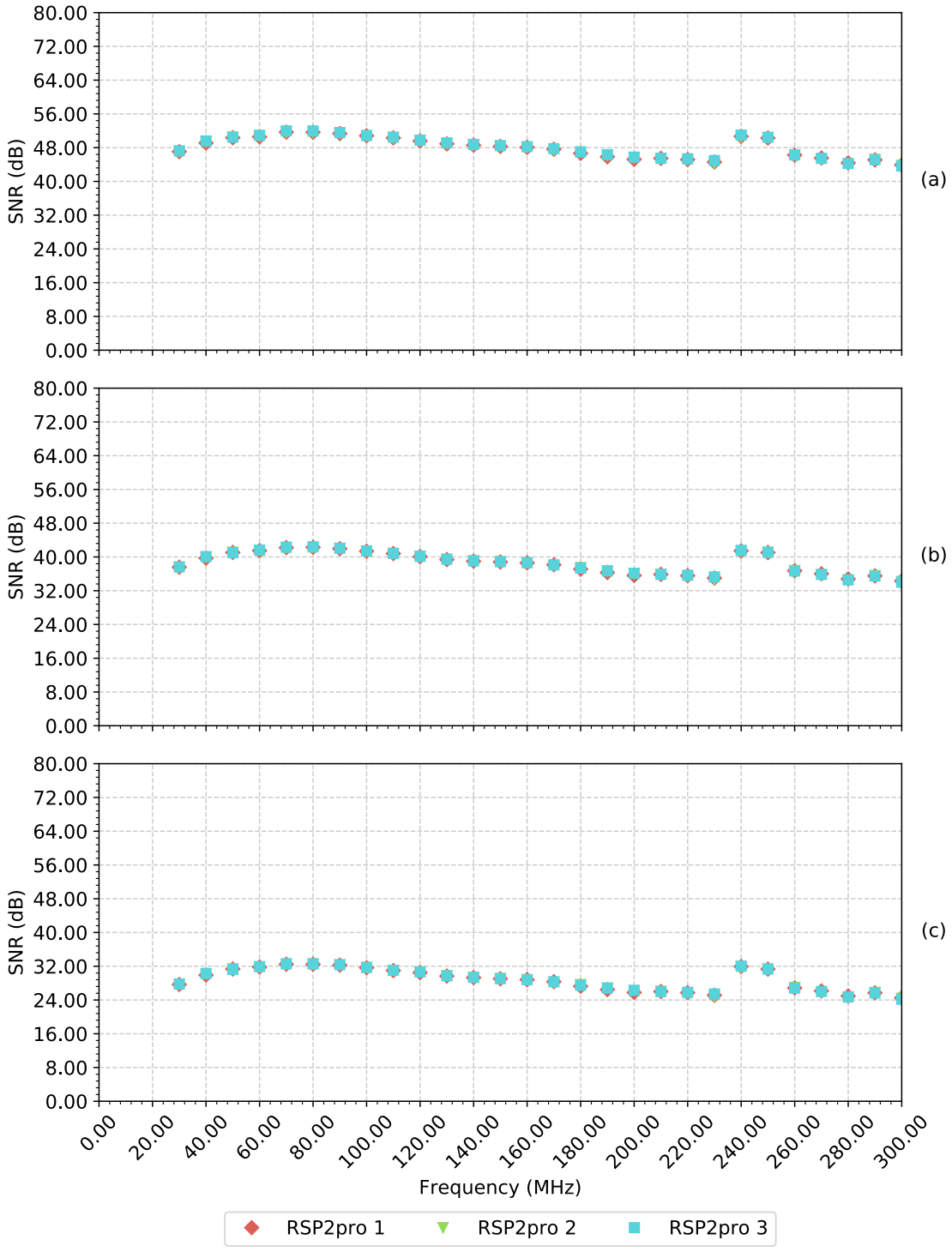


Figure 3.31: RSP2pro SNR measurement for: (a) Data Set 1, (b) Data Set 2, and (c) Data Set 3. Each data point represents the average of 50 measurements.

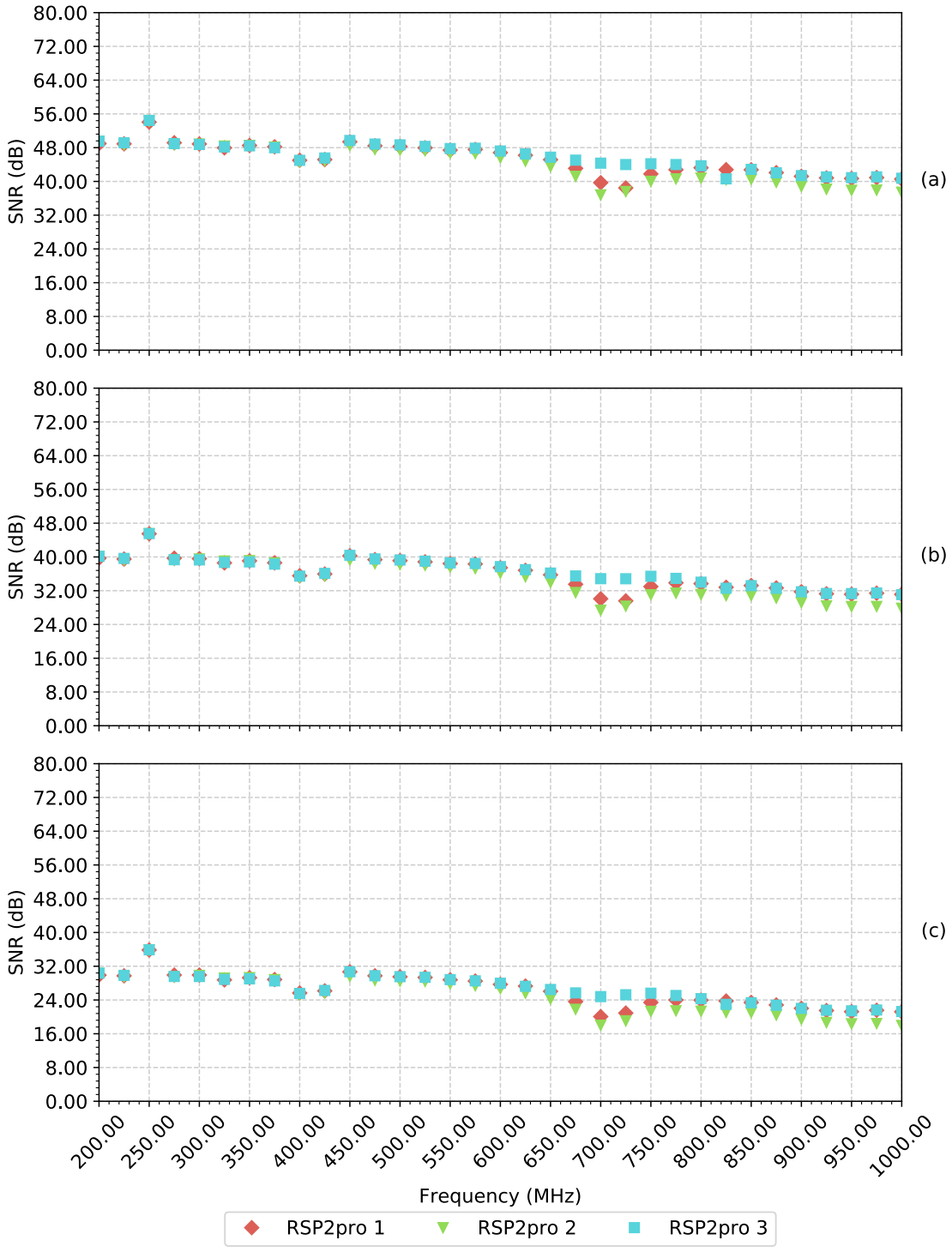


Figure 3.32: RSP2pro SNR measurement for: (a) Data Set 4, (b) Data Set 5, and (c) Data Set 6. Each data point represents the average of 50 measurements.

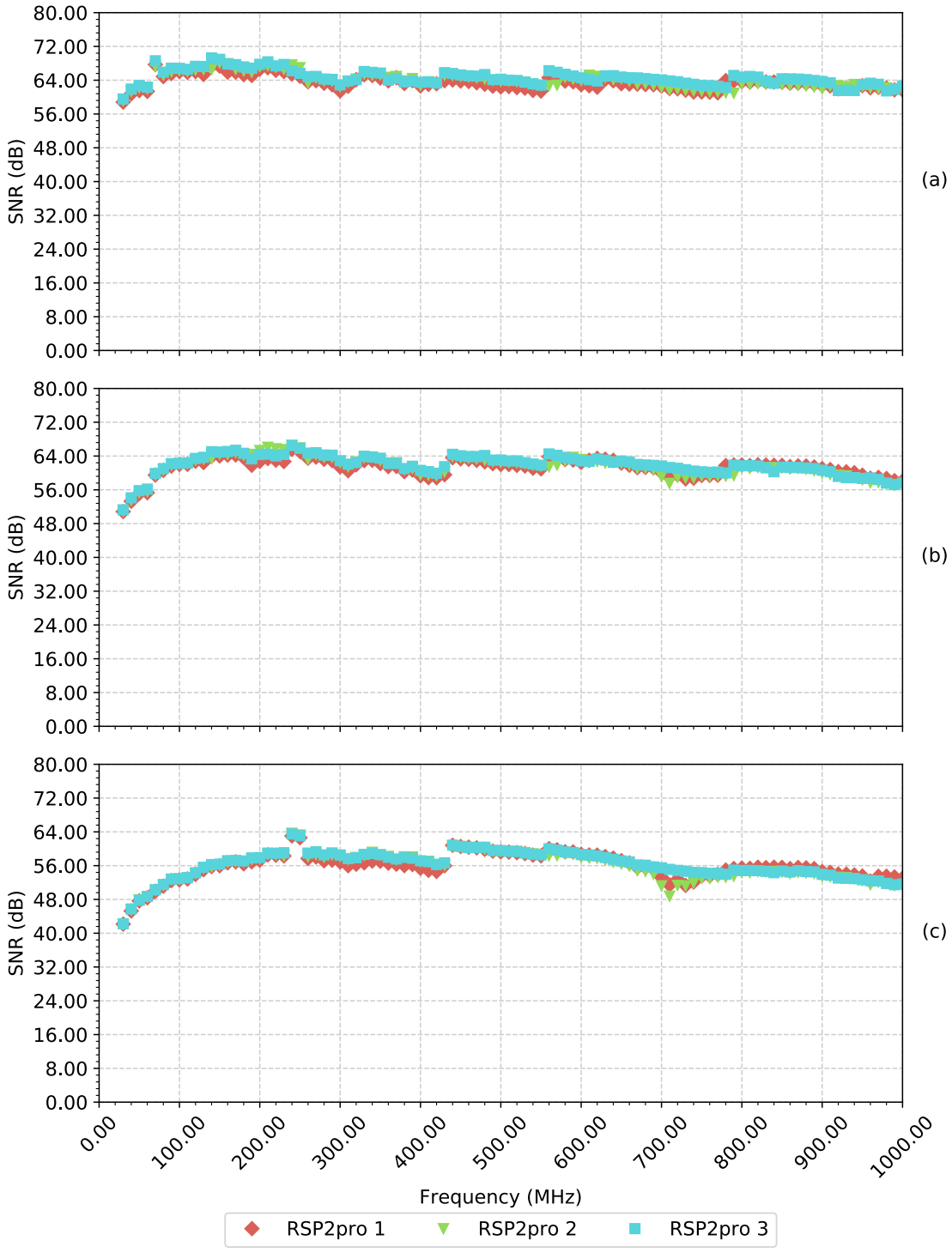


Figure 3.33: RSP2pro SNR measurement for: (a) Data Set 7, (b) Data Set 8, and (c) Data Set 9. Each data point represents the average of 50 measurements.

Figure 3.33 shows the SNR for measurements where Data Sets 7, 8, and 9 were measured. In this figure, the overall SNR shown on all three plots is much higher than the SNR shown in Figures 3.31 and 3.32, with a range spanning from 40 to 72 dB. All three RSP2pro units appear to have the same SNR relative to one another, only differing by a few dB at nearly every data point. The two exceptions to this are in Figure 3.33c, where the SNR recorded by RSP2pro 1 and 3 is approximately 6 dB lower than the SNR recorded by RSP2pro 2 at 700 MHz. As mentioned previously, Data Sets 7, 8, and 9 were all able to overload the A/D converters of the RSP2pro units that were tested.

The main conclusion from this section is that the RSP2pro can measure RF signals with an amplitude accuracy of ±2.00 dB so long as the A/D converters of the RSP2pro are not overloaded by the measured signal. This amount of accuracy can be maintained with an SNR as low as 16 dB.

3.4 Frequency Measurement Performance

The frequency measurement performance of the RSP2pro was characterized by directly comparing to the frequency measurement accuracy of the EMC analyzer. As explained in Section 3.1.3, the EMC analyzer was also used to calibrate the output frequencies of the signal generator. This means that the results presented in this section are completely relative to the EMC analyzer, as an independent calibrated frequency source was not used. Regardless, the standard deviation and difference in measured frequency between the RSP2pro and the EMC analyzer is still useful for understanding how suitable the RSP2pro is for determining the frequency of radiated emissions during a low-cost precompliance test. The frequency of each data point presented in this section has been adjusted to the desired signal generator frequency f_{in} using Equation 3.1 so that the differences in the actual frequency measured at each data point can be compared.

Figures 3.34, 3.35, and 3.36 show the standard deviation of every data point from the RSP2pro and EMC analyzer measurements that were collected using the input data sets from Table 3.2. Each of these figures correspond with one of the three EMC antennas discussed in Section 3.1.2 (i.e. each figure shows the results from three of the data sets in Table 3.2).

Figure 3.34 shows the standard deviation results corresponding to Data Sets 1, 2, and 3. All three plots in this figure show that the frequency measured for each data point was extremely consistent for all three of the RSP2pro units and the EMC analyzer. The standard deviation for the RSP2pro units is below 20 Hz, and the EMC analyzer only has one outlier with a standard deviation of 120 Hz. The results from using Data Sets 4, 5, and 6 indicate a similar level of consistency and are shown in Figure 3.35. Figure 3.35c represents the results from using the input data set with the lowest overall signal level. Here, the EMC analyzer becomes less consistent than the RSP2pro units as the measured frequency increases. However, the EMC analyzer still has a standard deviation that is less than 60 Hz which is more than sufficient when identifying the frequency of radiated emissions.

Figure 3.36 shows the standard deviation results corresponding to Data Sets 7, 8, and 9. The plots in this figure show that all three of the RSP2pro units remained very consistent in measuring

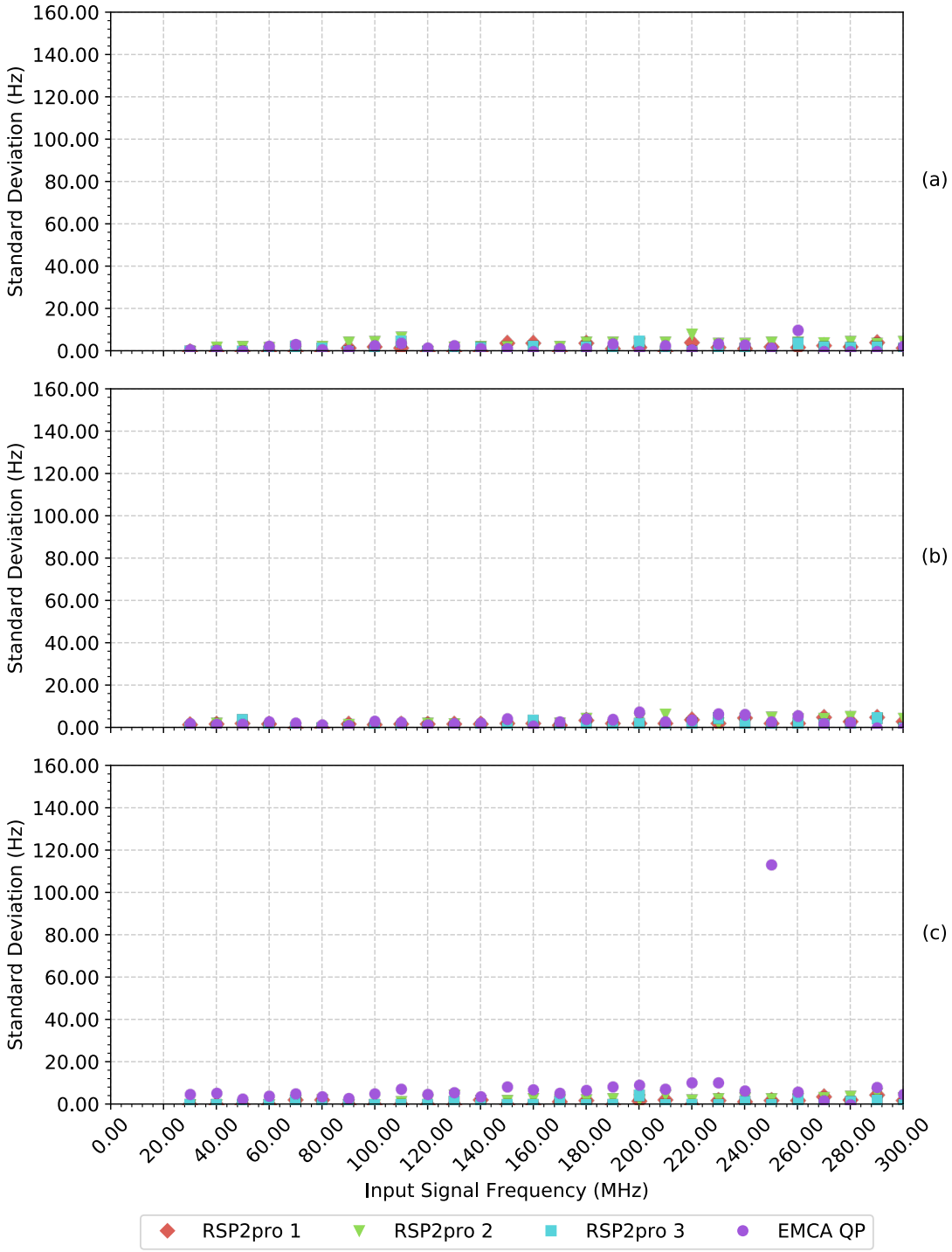


Figure 3.34: RSP2pro and EMC analyzer frequency measurement standard deviation for: (a) Data Set 1, (b) Data Set 2, and (c) Data Set 3. Each data point represents the standard deviation for 50 measurements.

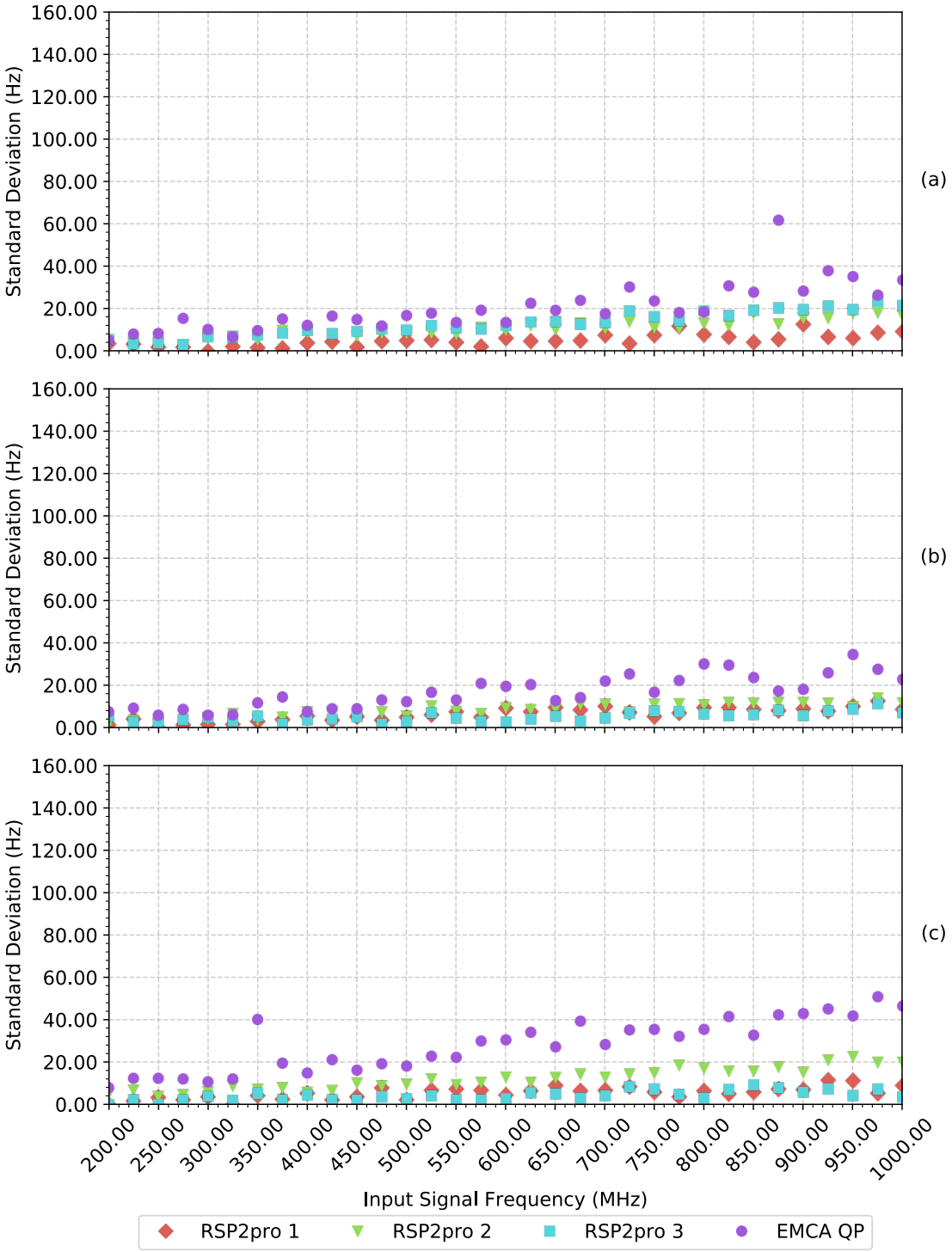


Figure 3.35: RSP2pro and EMC analyzer frequency measurement standard deviation for: (a) Data Set 4, (b) Data Set 5, and (c) Data Set 6. Each data point represents the standard deviation for 50 measurements.

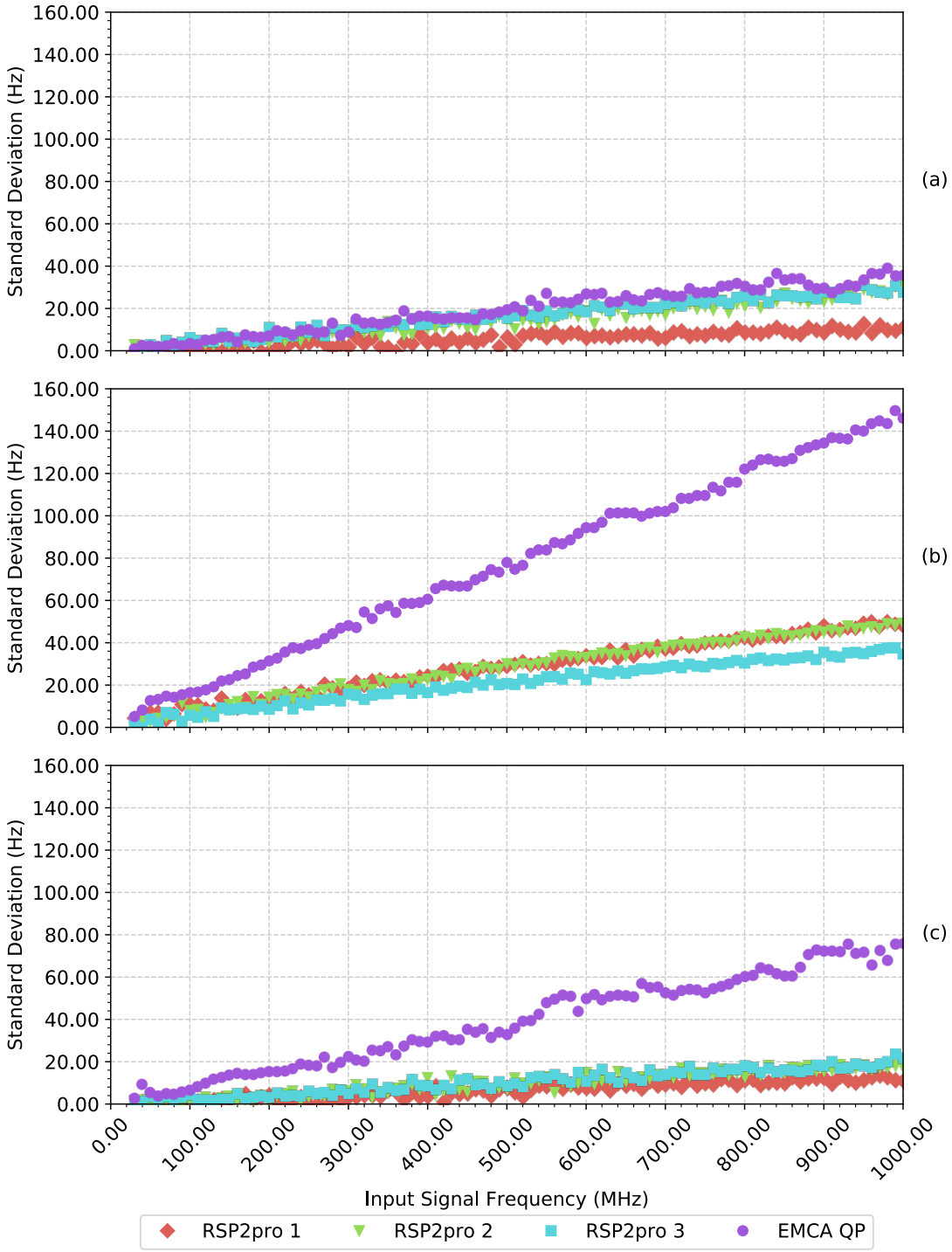


Figure 3.36: RSP2pro and EMC analyzer frequency measurement standard deviation for: (a) Data Set 7, (b) Data Set 8, and (c) Data Set 9. Each data point represents the standard deviation for 50 measurements.

the frequency of the input signal at all frequencies. This is indicated by the small separation between the RSP2pro data sets in all three plots, as the largest separation between the RSP2pro data sets in these plots is less than 40 Hz. Figure 3.36b has the largest increase of standard deviation for the EMC analyzer frequency measurement, which reaches a maximum value of 150 Hz. This is amount of inconsistency is still inconsequential for radiated emission measurements, which have a center frequency in the range from 30 MHz to 1000 MHz.

Figures 3.37, 3.38, and 3.39 show the average difference between the EMC analyzer measured frequency and the measured frequency of the three RSP2pro units for every data point collected from using the input data sets from Table 3.2. Each of these figures correspond with one of the three EMC antennas discussed in Section 3.1.2 (i.e. each figure shows the results from three of the data sets in Table 3.2). For these figures, the average frequency offset $\overline{f}_d$ is defined as

$$\overline{f}_d = \overline{f}_R - \overline{f}_E \tag{3.16}$$

where $\overline{f}_R$ represents the average frequency measured by the RSP2pro for a particular input data point and $\overline{f}_E$ represents the average frequency measured by the EMC analyzer for that same input data point. The value of $\overline{f}_d$ directly compares the frequency measurement accuracy of all three RSP2pro units with the frequency measurement accuracy of the EMC analyzer for all of the input data points tested.

The results shown in Figures 3.37, 3.38, and 3.39 show that the frequency accuracy of the RSP2pro (relative to the EMC analyzer) was very consistent for all of the input data sets. Every plot in these figures show that the frequency measurement of RSP2pro 2 was significantly more inaccurate than the frequency measurements taken with the other two units. However, this inaccuracy was still less than 1.6 kHz for nearly every data point that was tested. This frequency is very small compared to the frequency of the signals encountered during CISPR-22 Class B radiated emissions testing, which have center frequencies in the range of 30 MHz to 1000 MHz. Figures 3.38 and 3.39 show that the value of $\overline{f}_d$ for all three RSP2pro units increased linearly with the input signal frequency for signals from 200 MHz to 1000 MHz. One can also observe this linear behavior occurring for input frequencies of 30 MHz to 200 MHz in Figure 3.37. The value of $\overline{f}_d$ in

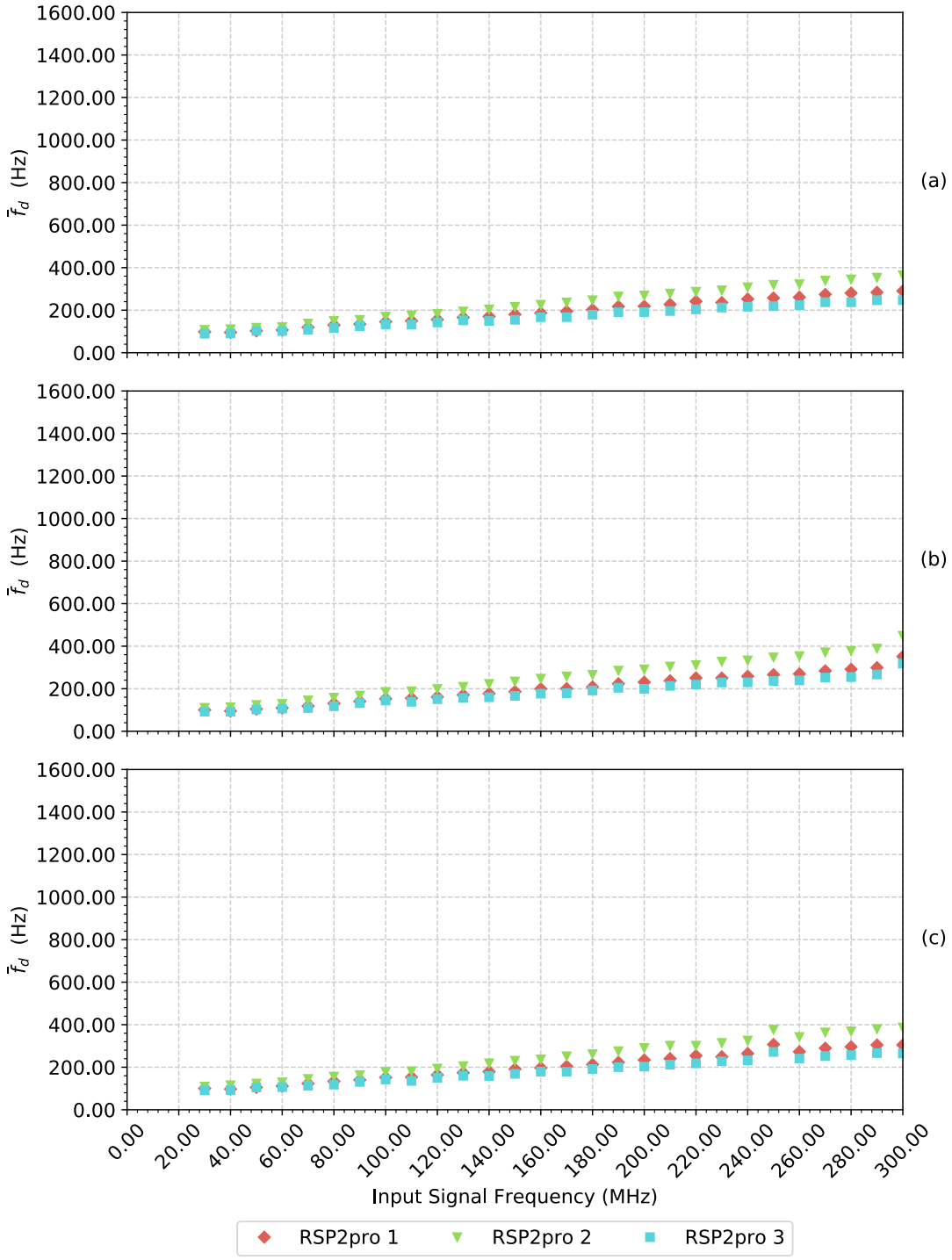


Figure 3.37: Average RSP2pro frequency measurement offset for: (a) Data Set 1, (b) Data Set 2, and (c) Data Set 3. Each data point represents the average of 50 measurements.

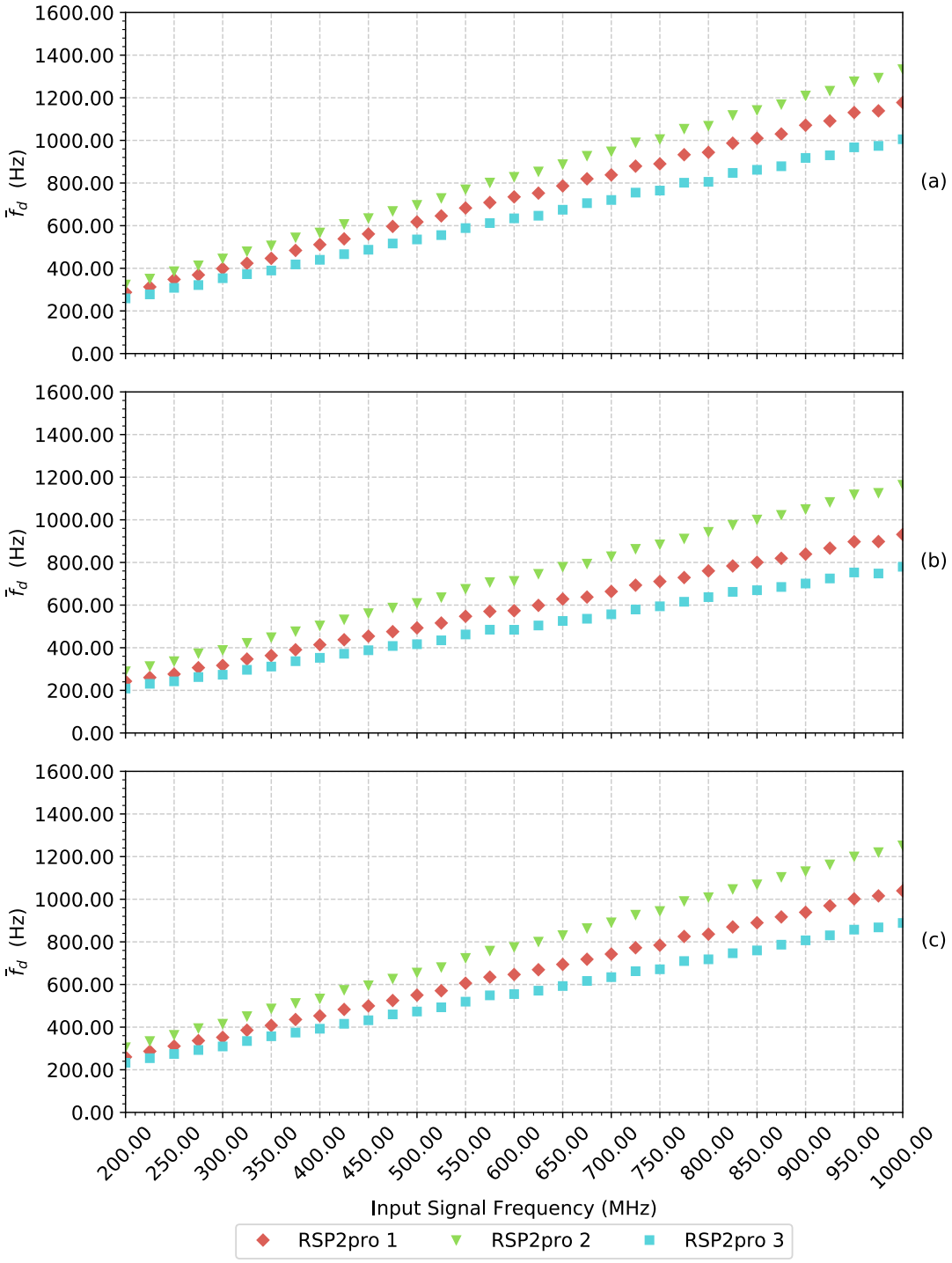


Figure 3.38: Average RSP2pro frequency measurement offset for: (a) Data Set 4, (b) Data Set 5, and (c) Data Set 6. Each data point represents the average of 50 measurements.

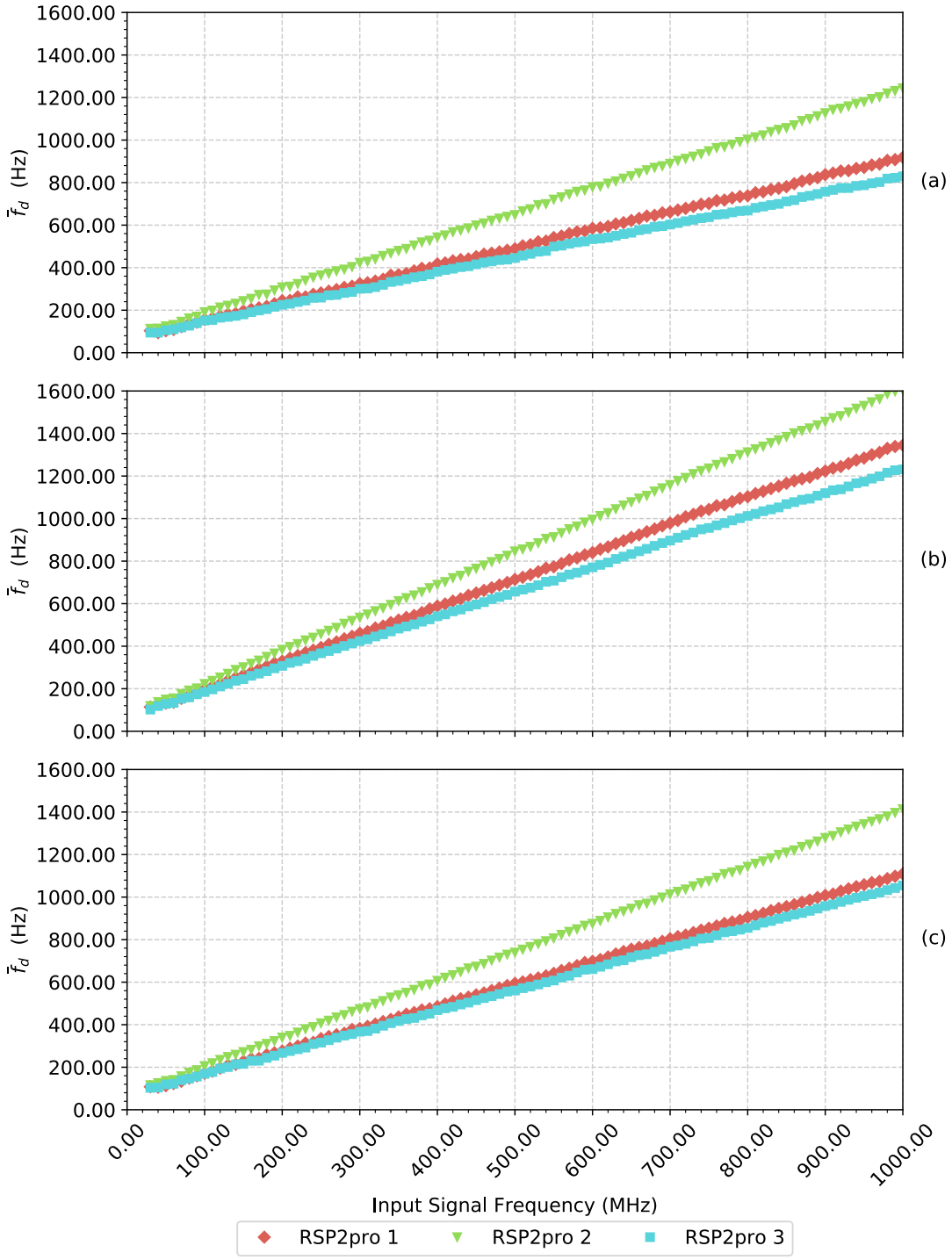


Figure 3.39: Average RSP2pro frequency measurement offset for: (a) Data Set 7, (b) Data Set 8, and (c) Data Set 9. Each data point represents the average of 50 measurements.

this figure is less than 400 Hz for input frequencies up to 200 MHz. This is identical to the results shown in Figures 3.38 and 3.39.

There are two main conclusions from the results presented in this section. First, that the frequency measured by the RSP2pro is very consistent for signals from 30 MHz to 1000 MHz, varying less than 60 Hz for all of the input data points that were tested. This is most likely due to the accuracy of the RSP2pro's temperature-compensated oscillator (TCXO). The second conclusion is that the value of $\overline{f}_d$ increased linearly with an increase in the measured frequency. This indicates that there was likely an inaccuracy with the frequency measured by the EMC analyzer, not the RSP2pro units, because the EMC analyzer was the only common factor between all of the $\overline{f}_d$ values shown.

CHAPTER 4

USING THE RSP2PRO FOR RADIATED EMISSIONS TESTING

Having established a margin of error for the RSP2pro's absolute signal level measurements in the previous chapter, the RSP2pro is now used to measure radiated emissions from two different sources and the results are directly compared to the EMC analyzer. Section 4.1 outlines the radiated emissions measurement procedure used with both the EMC analyzer and the RSP2pro. This section includes a description of the RF environment that was selected for testing and provides a detailed description of the two signal sources that were used to create radiated emissions. Section 4.2 presents the results of the radiated emission tests, and the received RF power margin of ±2.00 dB developed for the RSP2pro in the previous chapter is compared to these results.

4.1 Radiated Emissions Measurement Procedure

Before the procedure for measuring radiated emissions was developed, several locations were examined to find an RF environment suitable for radiated emissions tests. Since the primary goal of this thesis is to show that the RSP2pro can replace the spectrum analyzer used in a low-cost precompliance setup like the one shown in Figure 2.11, only uncontrolled RF environments were considered. An uncontrolled RF environment in this context is defined as an area where the RF signals external to the EUT being evaluated cannot be controlled. These external RF emissions are typically from sources like nearby FM radio stations or other telecommunications equipment that operate in the frequency range from 30 MHz to 1000 MHz. Figures 4.1a, 4.2a, and 4.3a show the three uncontrolled RF environments that were considered for radiated emission testing. For each environment, the EMC analyzer and the BicoLOG 30100X EMC antenna were used to perform a scan of the background RF environment from 30 MHz to 1000 MHz using the four "Spectrum Sweep" state files from Table 3.1 to configure the EMC analyzer.

Figure 4.1a shows the first RF environment that was considered: an outdoor area on MSU's campus near Pavilion Drive. The nearby MSU Observatory (pictured in the background of Figure

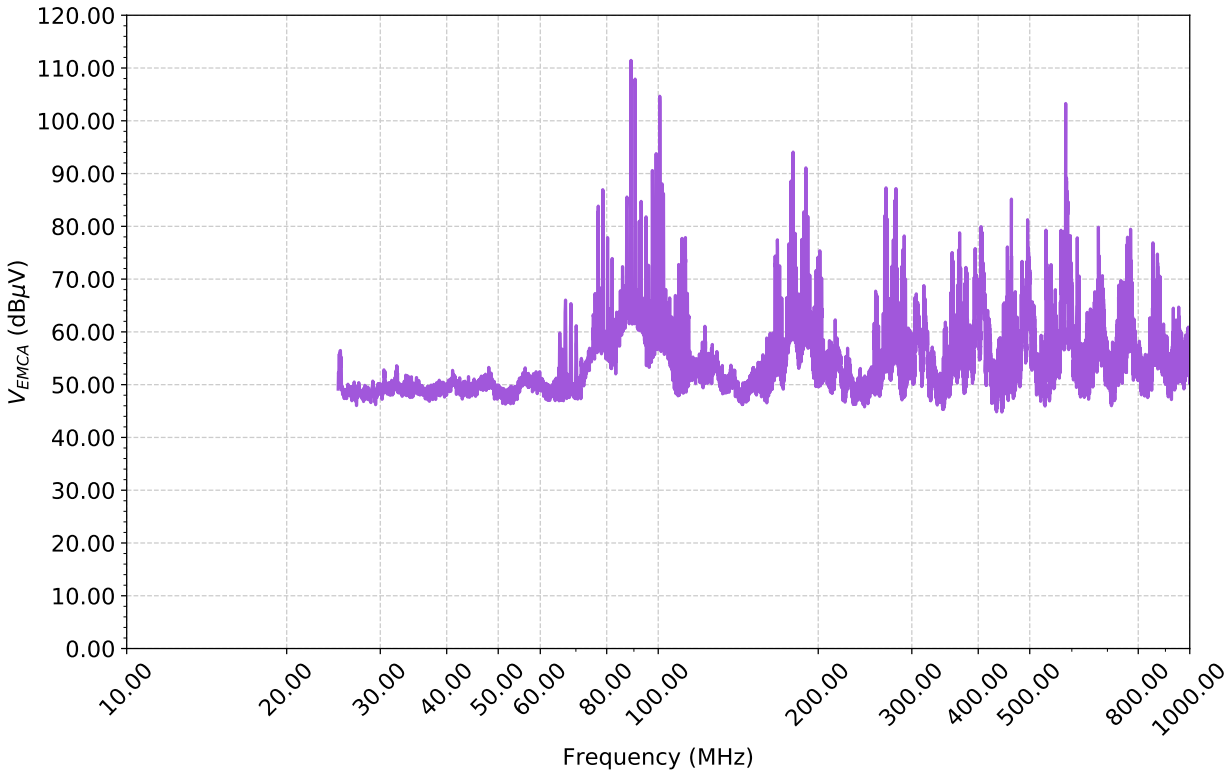
4.1a) was used to power the EMC analyzer. Figure 4.1b shows the background RF noise at this location. One can see that the noise floor in this environment is at 50 dBμV, which is significantly higher than the ideal noise floor of 20 dBμV from the characterization test setup (refer to Figure 3.11). Also notable are the FM broadcast signals, the strongest of which are visible between 90 and 100 MHz with amplitudes of over 100 dBμV. Figure 4.1b also shows many other background RF signals. According to the National Telecommunications and Information Administration Office of Spectrum Management [55], the strong signals observed just before 200 MHz are broadcast television channels 7 through 13, while the strong signals observed just before 300 MHz are most likely from a variety of nearby mobile telecommunications towers. Figure 4.1b also shows strong signals just before 600 MHz, the strongest of which has an amplitude level above 100 dBμV. These correspond to television broadcast channels 21 through 36 [55].

There are several reasons why the outdoor environment shown in Figure 4.1a was not chosen for radiated emissions testing. One reason was a simple lack of convenience, as weather and time constraints made collecting data at the location difficult. However, the main reason was the large amount of background RF noise that would have made identifying radiated emissions from an EUT very difficult. The issue of background RF noise is also a problem for creating the open area test sites (OATS) used for full compliance testing, as the number of ambient signals from things like digital broadcasts are always increasing [23].

The next RF environment that was considered is shown in Figure 4.2a. The tunnel shown in this figure connects the basements of Anthony Hall and the G.M. Trout Food Science and Human Nutrition Building on MSU's campus and is approximately 2.1 m wide and 2.5 m tall. This RF environment was considered for radiated emission testing because its location greatly reduces the amount of ambient RF signals, as shown in Figure 4.2b. One can see that the FM broadcast transmissions are still present, but their amplitudes have been reduced by more than 30 dB from what was measured at the outdoor location shown in Figure 4.1b. The measured noise floor in Figure 4.2b is very low, only slightly higher than the 20 dBμV noise floor present during the characterization measurements that were shown in Chapter 3.



(a)

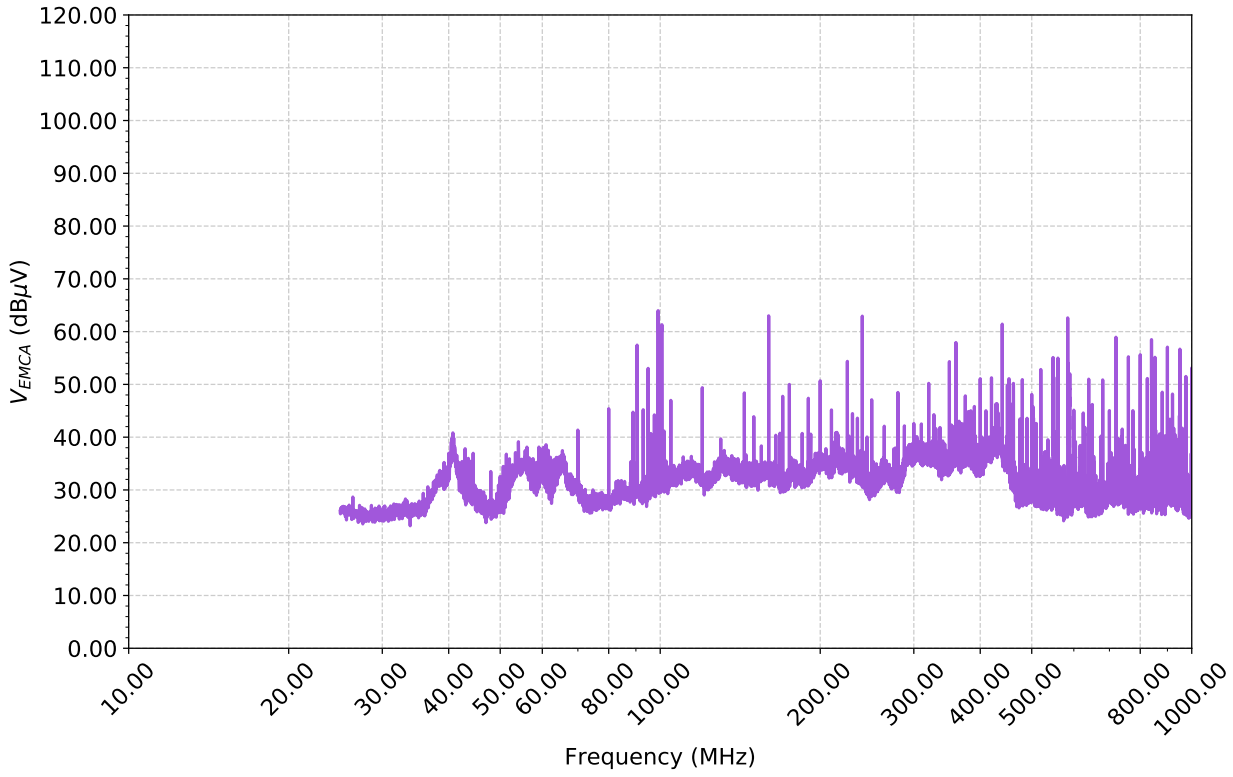


(b)

Figure 4.1: (a) The outdoor area that was considered for radiated emission testing and (b) the ambient RF signals measured at the outdoor area from 30 MHz to 1000 MHz (average of 3 trials).



(a)



(b)

Figure 4.2: (a) The underground tunnel that was considered for radiated emission testing and (b) the ambient RF signals measured in the underground tunnel from 30 MHz to 1000 MHz (average of 3 trials).

This RF environment was ultimately rejected for three main reasons. First, the area was not very accessible and made collecting measurements difficult because of the large amount of time it took to transport the test equipment required for the radiated emission tests. Second, the narrow dimensions of the tunnel meant that there was an increased chance for strong signal reflections from the EUT. These kind of reflections are difficult to predict and would have contributed to a large amount of inconsistency in radiated emissions measurements. The last reason is that this location is too unique. Most product designers will not have access to an underground tunnel for performing radiated emission measurements.

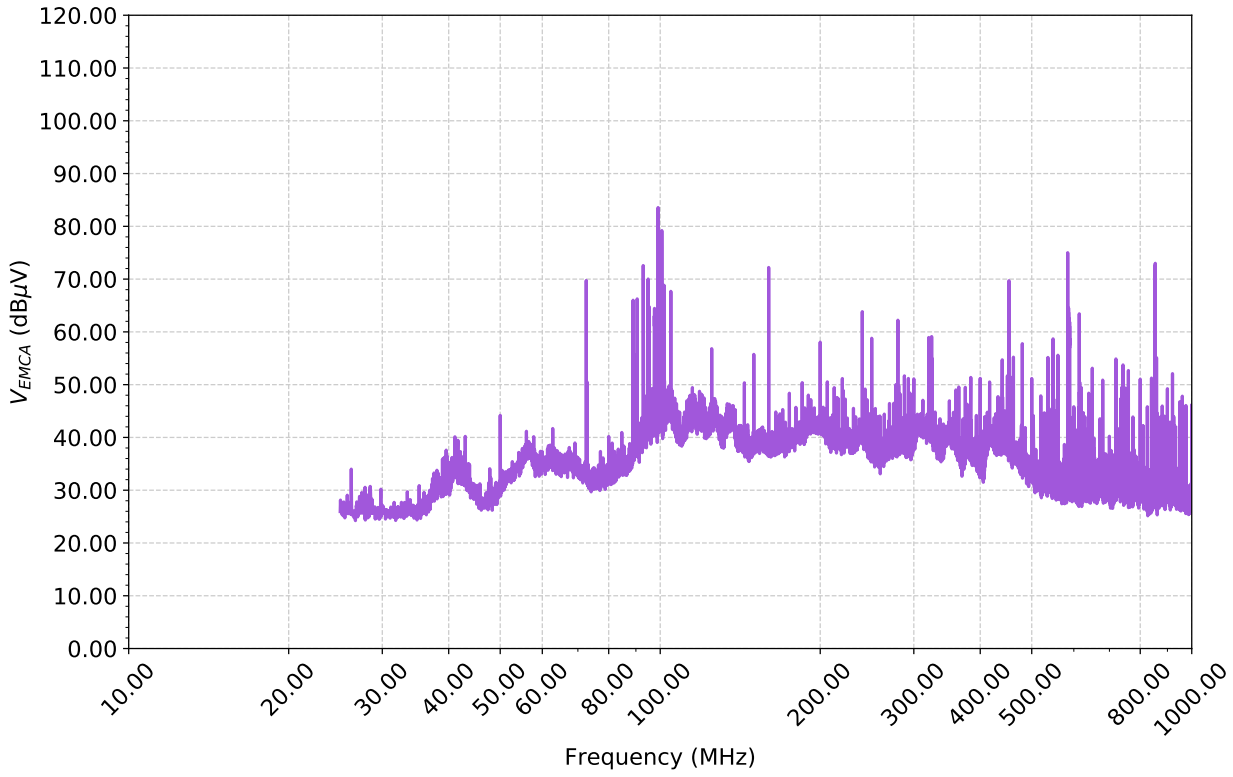
Figure 4.3a shows the hallway in the MSU Engineering Building that was chosen for performing the radiated emission measurements presented in this thesis. This hallway is located on the second floor of the MSU Engineering Building and has a width of approximately 3.7 m and a ceiling height of about 2.5 m. One can see from Figure 4.3b that although the noise floor is comparable to the underground tunnel, there are many more ambient RF signals present (especially in the frequency range from 80 MHz to 500 MHz). These additional signals greatly obscure the noise floor across most of the 30 MHz to 1000 MHz frequency range, save for the small region from 30 MHz to 40 MHz. However, the largest ambient RF signals are still reduced by more than 20 dB compared to the results from the outside area shown in Figure 4.1b.

The RF environment shown in Figure 4.3a was chosen for two reasons. The main reason is that the area is likely comparable to the kind of uncontrolled RF environment that would be readily available to an electronic product designer. The other main reason is that this location allowed for testing at 10 m while also being very close to where the test equipment was stored. This proximity to the test equipment storage location greatly reduced setup time, allowing for a greater number of radiated emission tests to be performed.

Once the RF environment shown in Figure 4.3a was selected, a procedure was developed for performing consistent radiated emission measurements at a distance of 1 m and 10 m. These distances were chosen so that the effects of the uncontrolled RF environment that were discussed in Section 2.2 could be investigated. Figure 4.4 shows the test setup used for the radiated emissions



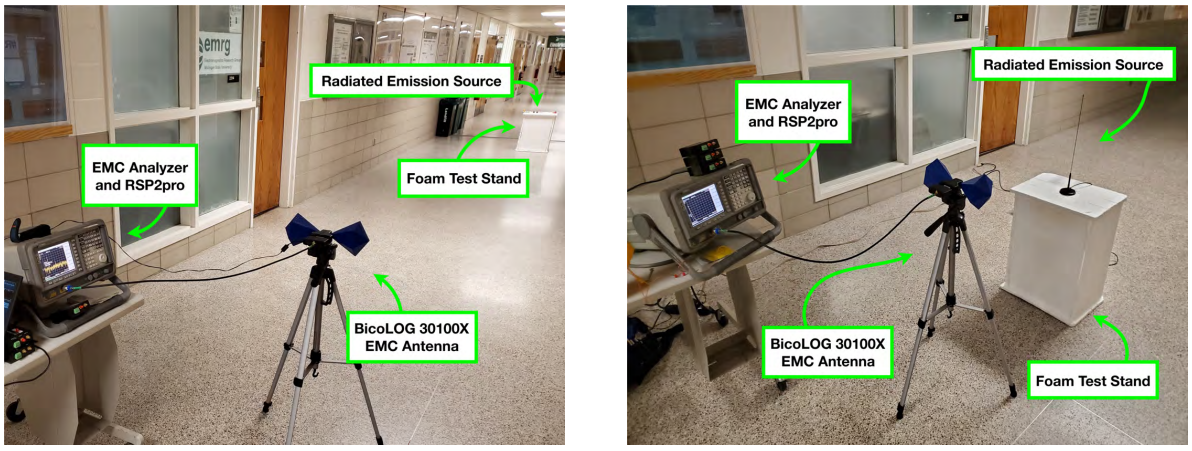
(a)



(b)

Figure 4.3: (a) The indoor hallway that was ultimately chosen for radiated emission testing and (b) the ambient RF signals measured in the hallway from 30 MHz to 1000 MHz (average of 3 trials).

measurement procedure for both distances. The test setup consisted of the EMC analyzer and the RSP2pro units, which were placed on a movable desk, and the BicoLOG 30110X EMC antenna, which was mounted on a camera tripod. For both of the distances tested, the radiated emissions source was placed on a foam test stand that was the same height as the EMC antenna. Since the goal of the radiated emission test procedure was to compare the performance of the RSP2pro and the EMC analyzer, the antenna was only used at a single fixed height. When performing an actual radiated emission test with an EUT, the antenna should be vertically scanned when possible so that the strongest radiated emissions can be found [2]. Figure 4.5 shows the RF signal chain that was used by both test setups shown in Figure 4.4.



(a) (b)

Figure 4.4: The radiated emissions test setup consisting of the EMC analyzer, BicoLOG 30100X EMC antenna, RSP2pro, and radiated emissions source at (a) 10 m and (b) 1 m.

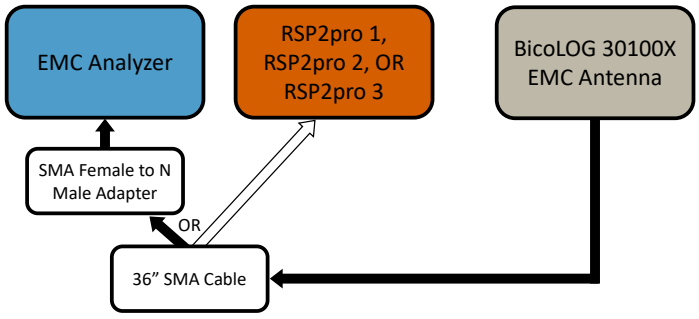


Figure 4.5: Diagram showing the RF signal chains used for the radiated emissions test setup.

4.1.1 Narrowband Source: Whip Antenna

Three types of signals that are very common with electrical systems are continuous wave (CW) narrowband signals, repetitive broadband signals, and single event broadband signals [56]. The first two types were tested for the results presented in this thesis. This section describes the radiated emissions source that was designed to create CW narrowband radiated emissions. Table 4.1 describes the frequency and amplitude of the eleven input data points that were used by the CW source shown in Figure 4.6. These specific frequencies were primarily selected due to the output limitations of the broadband source described in Section 4.1.2.

Frequency (MHz)	Amplitude (dBμV)
40	100
50	
80	
100	
120	
160	
240	
320	
360	
480	50
800	100

Table 4.1: Input data points used by the CW radiated emissions source shown in Figure 4.6.

The VHF/UHF "whip" antenna shown in Figure 4.6 was the radiated emissions source that was used to transmit the CW signals from the HP 8657A signal generator. According to the manufacturer [57], this omni-directional whip antenna is designed for transmitting RF signals in the frequency ranges of 136 to 174 MHz and 400 to 520 MHz. This antenna also has a magnetic base so that it can be attached to metallic surfaces like the exterior of a vehicle [57]. In order to understand how this antenna would perform when transmitting signals at other frequencies in the 30 MHz to 1000 MHz band, a network analyzer was used to measure its VSWR. The VSWR of the antenna was measured while it was attached to a metallic surface (Figure 4.7a) and once again when it was placed on the foam test stand (Figure 4.8a). The results of each measurement are shown in

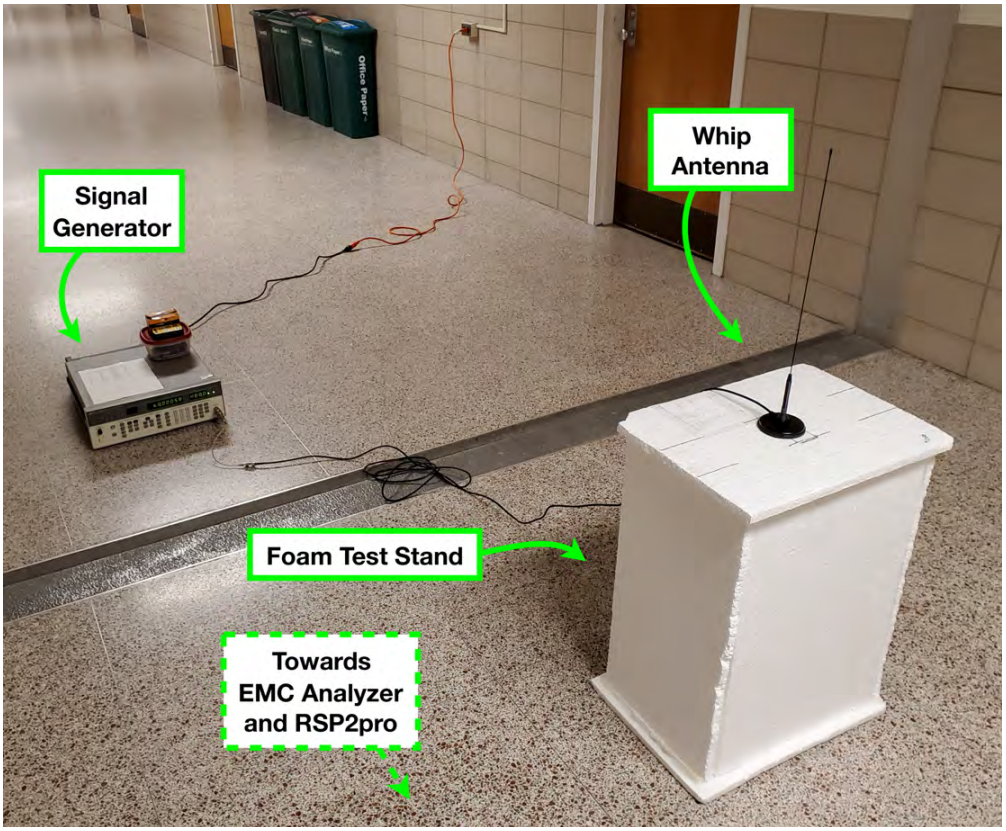
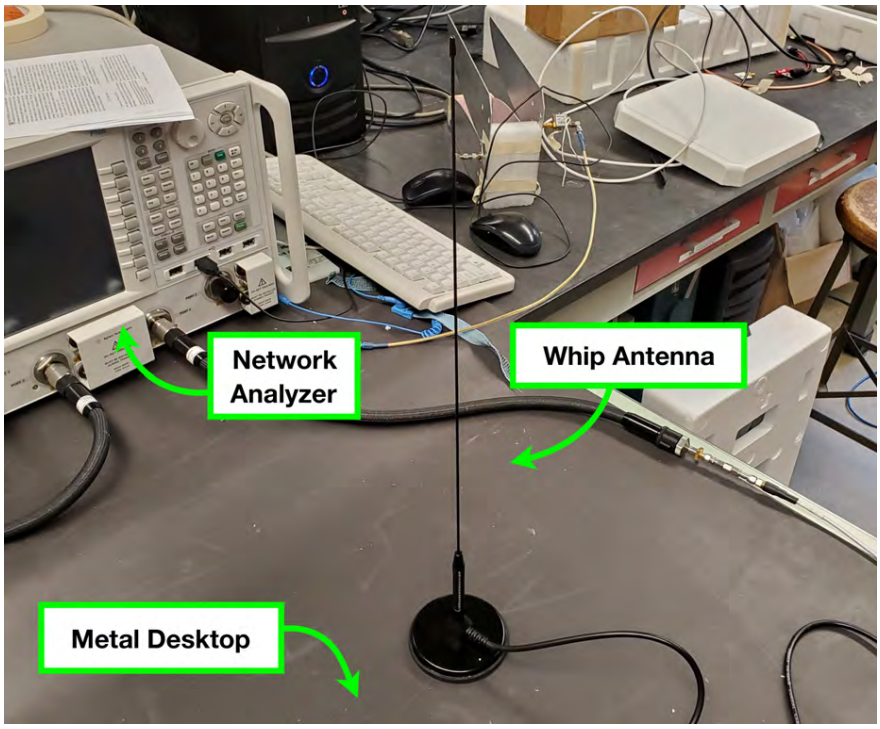


Figure 4.6: Detailed view of the whip antenna arrangement used to simulate CW narrowband radiated emissions from an EUT.

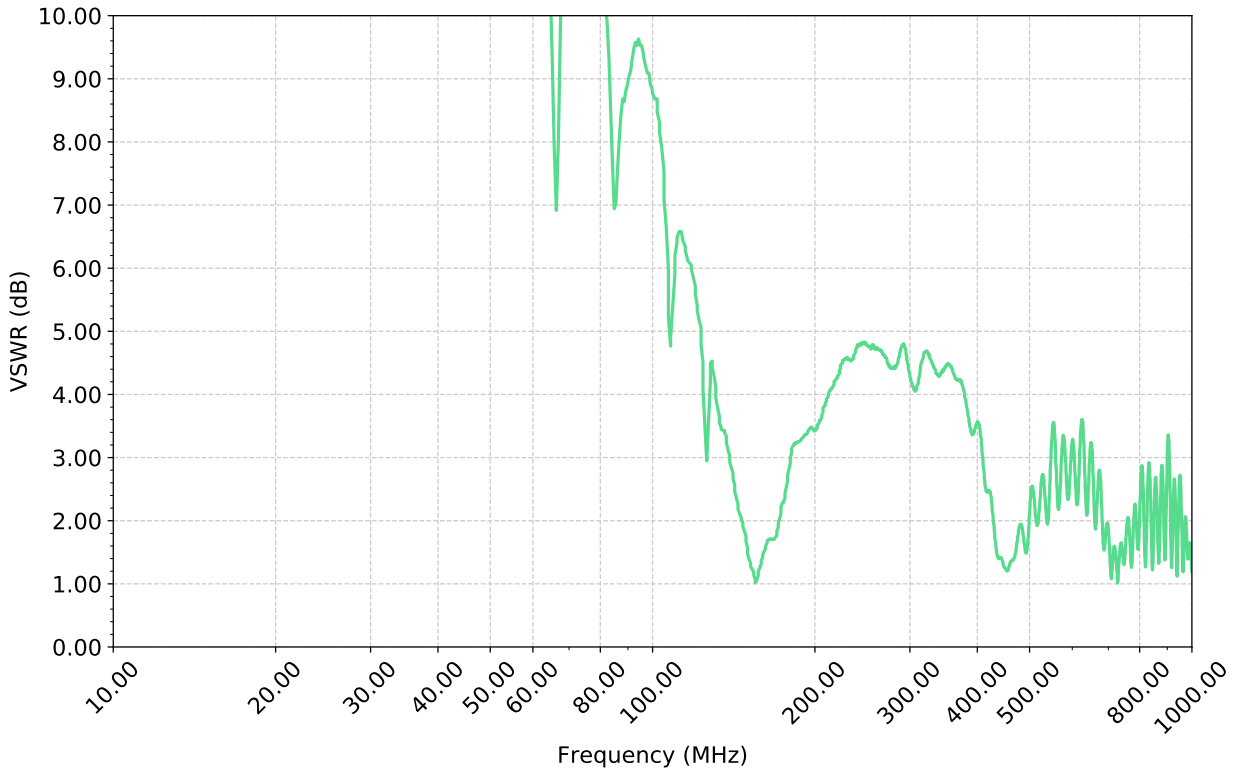
Figures 4.7b and 4.8b. From these figures, one can see that using the magnetic base greatly effects the performance of the whip antenna.

When attached to a metallic surface, the measured VSWR of the whip antenna approaches a value of one at two seperate frequencies: 154 MHz and 450 MHz. This is to be expected based upon the manufacturer specifications [57]. An antenna is often considered to have a good impedance match with its source when the VSWR is less than two [58], and one can see from Figure 4.7b that over most of the 30 MHz to 1000 MHz range the whip antenna has a poor impedance match. This is especially true for frequencies below 100 MHz, and this is the reason why only three frequencies below 100 MHz were tested (refer to Table 4.1).

One can see from Figure 4.8b that the measured VSWR was significantly different when the antenna was not magnetically attached to a metallic surface. In this configuration, there are several

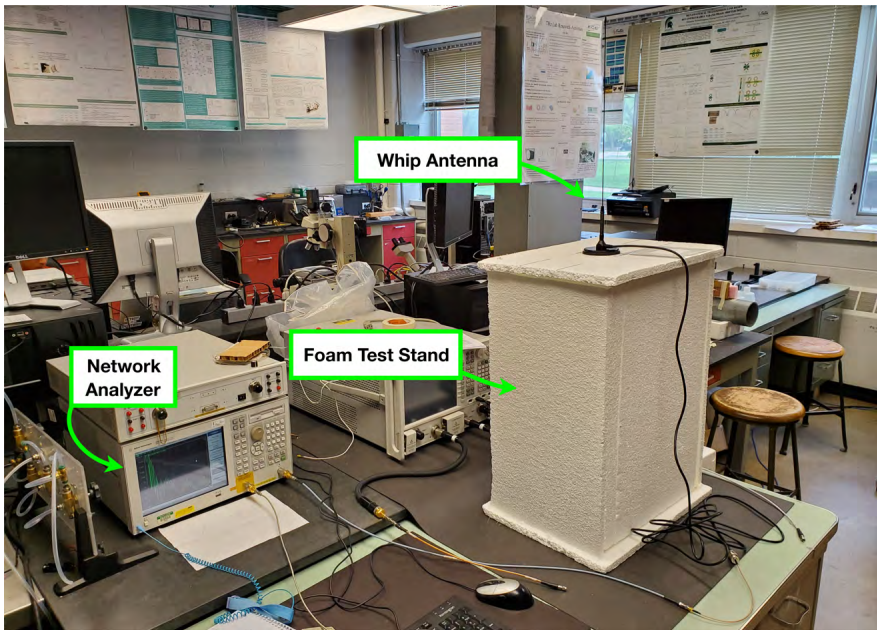


(a)

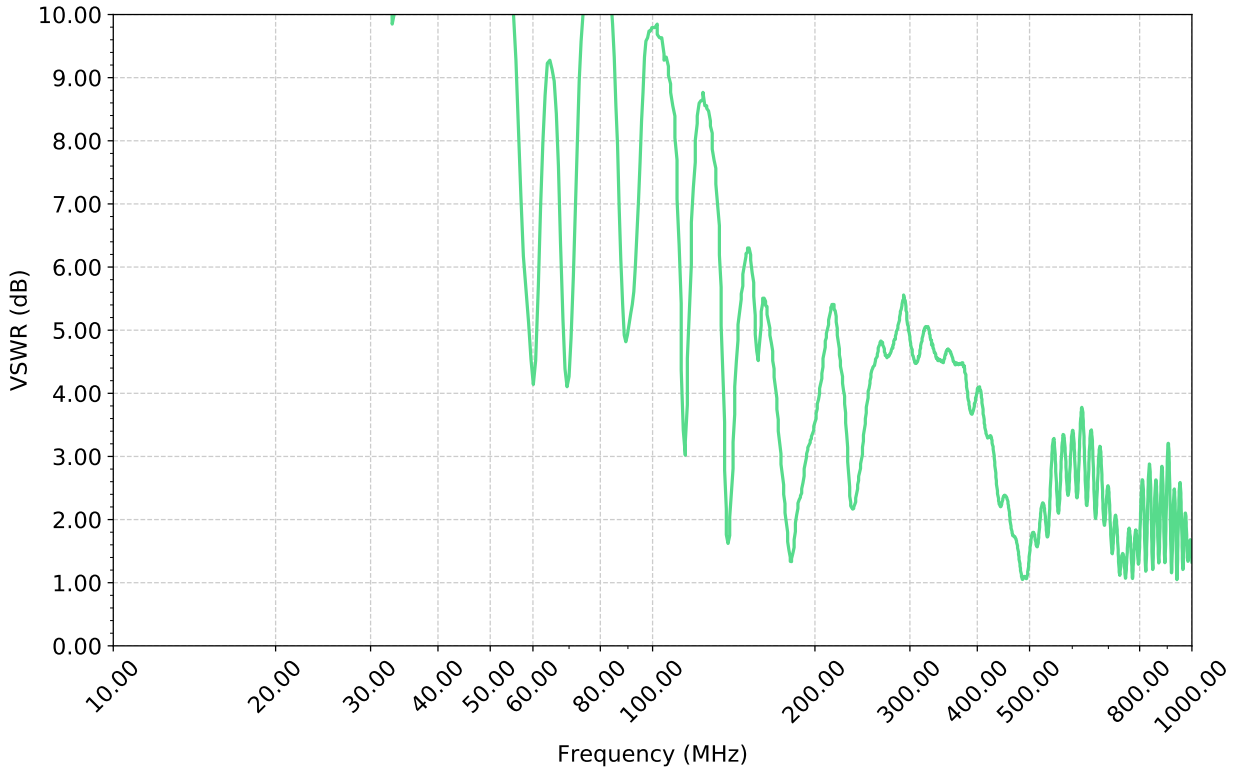


(b)

Figure 4.7: (a) The whip antenna used to create CW narrowband radiated emissions attached to a metal surface and (b) its measured VSWR from 30 MHz to 1000 MHz (average of 5 measurements).



(a)



(b)

Figure 4.8: (a) The whip antenna used to create CW narrowband radiated emissions on a non-metallic surface and (b) its measured VSWR from 30 MHz to 1000 MHz (average of 5 measurements).

other narrow frequency regions with low VSWR values and the two largest nulls shift slightly up in frequency, with new center frequency values of approximately 180 MHz and 480 MHz. During testing, it was found that the radiated emissions created with the antenna at 480 MHz were capable of overloading the EMC analyzer and the RSP2pro, even at a distance of 10 m. This is why the input data point for 480 MHz shown in Table 4.1 is the only input data point with an amplitude of 50 dBμV. Note that the measured VSWR can vary greatly depending on the orientation of the antenna (particularly with the antennas position relative to other objects in the near field) and so the values shown in both Figures 4.7b and 4.8b should only be considered typical values for the specific configurations shown in Figures 4.7a and 4.8a.

4.1.2 Broadband Source: Custom Equipment Under Test (EUT)

Repetitive broadband radiated emissions can easily be created by common electronic system components such as digital clocks [56]. These types of emissions are usually from the harmonics created by an electric signal at some fundamental frequency. These harmonics occur at integer multiples of the fundamental frequency and can have radiated emissions that are larger than those caused by the electric signal at the fundamental frequency. Digital clock signals cause particularly large radiated emissions because of their fast rise times [2]. While fast rise times are desirable from a digital signal processing perspective, Ott [2] explains that "repetitive, high-frequency signals with large currents and fast rise/fall times will have large spectra content." Clock signals typically have a 50% duty cycle, which means that the odd harmonics of the fundamental frequency will dominate the spectral content of the clock signal [4] and thus the odd harmonics will create the strongest radiated emissions.

The large spectral content of clock signals can be exploited to create a circuit that is capable of creating radiated emissions at a user-selectable frequency. A custom EUT that does this was created for use in this research. The design was based upon a similar circuit [7] that uses a 10 MHz square wave oscillator to demonstrate that common-mode currents are the dominant source of the radiated emissions above 30 MHz that are created by system cabling. As explained by Paul and

Bush [7], the source and load sides of the circuit are powered via battery to show that devices not connected to the commercial power grid can create large radiated emissions from common-mode currents. A simplified block diagram of the custom EUT based on the design in [7] is shown in Figure 4.9. Figure 4.10 shows a detailed view of the source side of the EUT. The load side of the custom EUT is simply a logic inverter with an input capacitance of 15 pF and its own isolated 5V power supply. The design files for the custom EUT are available in the Digital Appendix that is included with the digital version of this thesis (refer to Appendix B for more information).

The custom EUT is able to produce signals at multiple different frequencies using a high-frequency multiplier IC that uses a phased-lock loop (PLL) implementation. PLLs are mixed signal devices consisting of a phase detector, amplifier, and voltage-controlled oscillator [54]. According to Horowitz [54], PLLs are typically used in telecommunications devices to set the frequency of each operating channel. In the custom EUT shown in Figure 4.9, the PT7C4511 PLL IC is used with three different crystal oscillators to produce multiple output frequencies. The small tan rotary switches shown in Figure 4.10 are used to select the crystal oscillator used by the PLL (10, 20, or 30 MHz). The red and white toggle switches shown in Figure 4.10 are then used to control the digital logic inputs of the PLL IC according to the data sheet [59]. The PLL IC uses these inputs to determine the frequency multiplier applied to the input frequency to create the desired output frequency. The harmonic frequencies of several output frequencies are shown in Table 4.2 and were used to measure radiated emissions across the entire 30 MHz to 1000 MHz spectrum.

The third column in Table 4.2 shows the harmonic frequencies (and thus the radiated emission frequencies) that were measured. This table also shows how the EUT's ribbon cable (with a length of 41 cm) compares to the wavelength of each of the measured frequencies. Radiated emissions become stronger when the conductors in the EUT become comparable to a wavelength [4], so the harmonic frequencies above 120 MHz should produce the strongest radiated emissions.

Both the time-domain and frequency-domain characteristics of the custom EUT were characterized using the setups shown in Figure 4.11a and 4.11b. A 1 GHz oscilloscope with 500 MHz probes

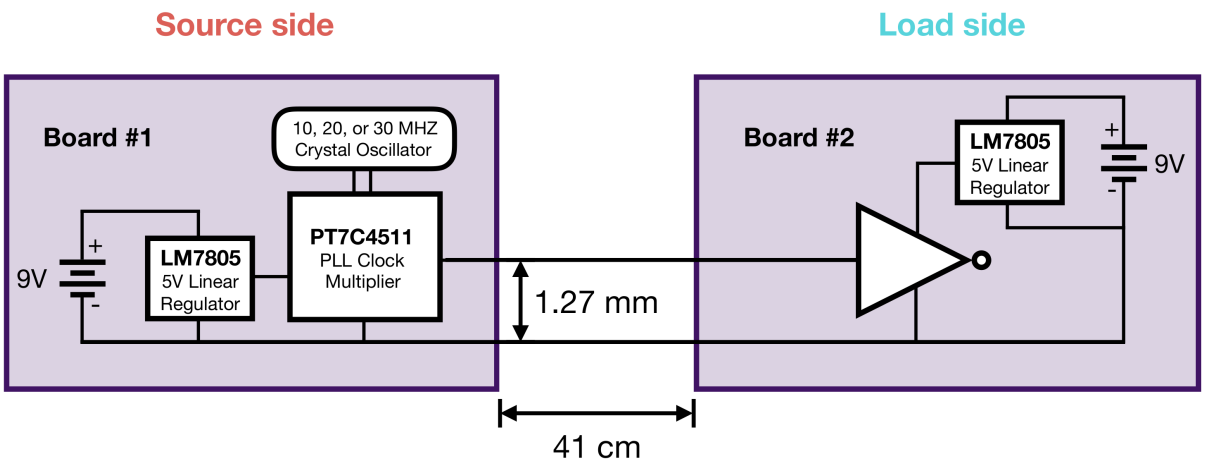


Figure 4.9: Simplified block diagram of the custom EUT constructed to create wideband radiated emissions at multiple frequencies. Refer to Appendix A for the custom EUT circuit diagrams.

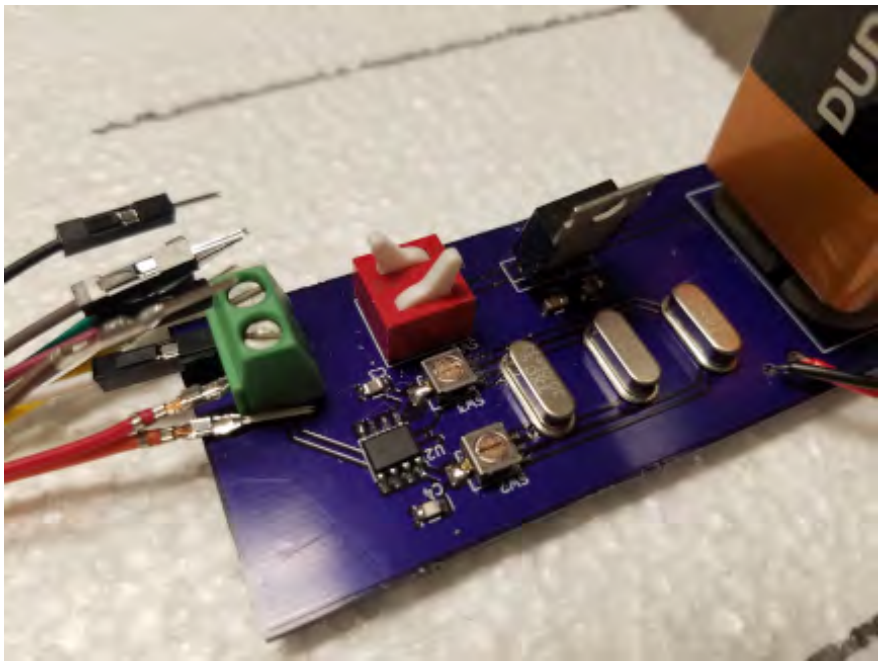


Figure 4.10: Detailed view of the source side shown in Figure 4.9.

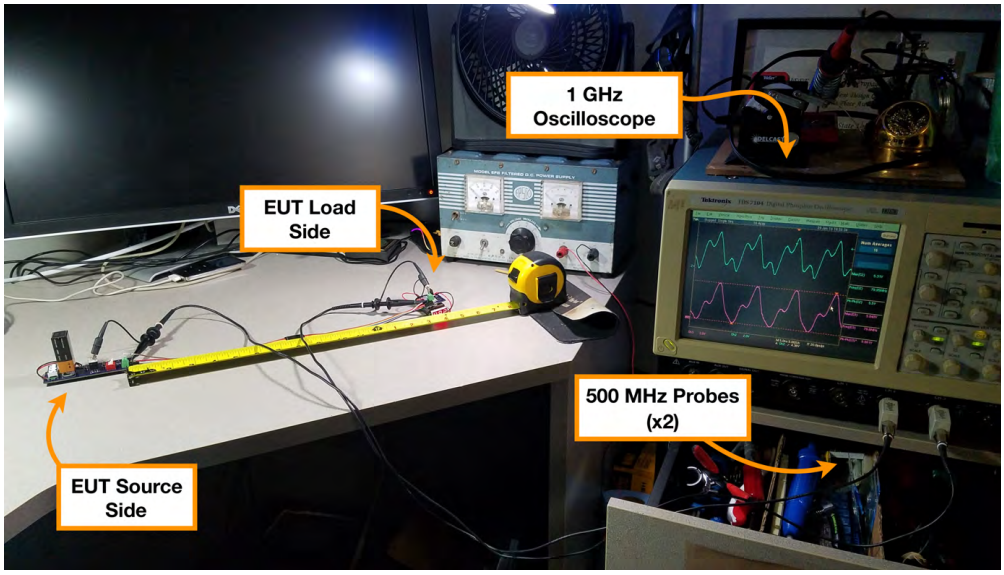
EUT PLL Frequency (MHz)	Harmonic Number	Harmonic Frequency (MHz)	λ (m)	EUT Cable Length in terms of λ (denominator rounded to nearest whole number)
40	1	40	7.49	λ/18
50	1	50	6.00	λ/15
40	2	80	3.75	λ/9
50	2	100	3.00	λ/7
40	3	120	2.50	λ/6
50	3	150	2.00	λ/5
160	1	160	1.87	λ/5
120	2	240	1.25	λ/3
120	3	360	0.83	λ/2
120	4	480	0.62	λ/2
160	5	800	0.37	λ/1

Table 4.2: Details for the frequencies that were created by the custom EUT to test radiated emissions across the CISPR-22 Class B frequency band.

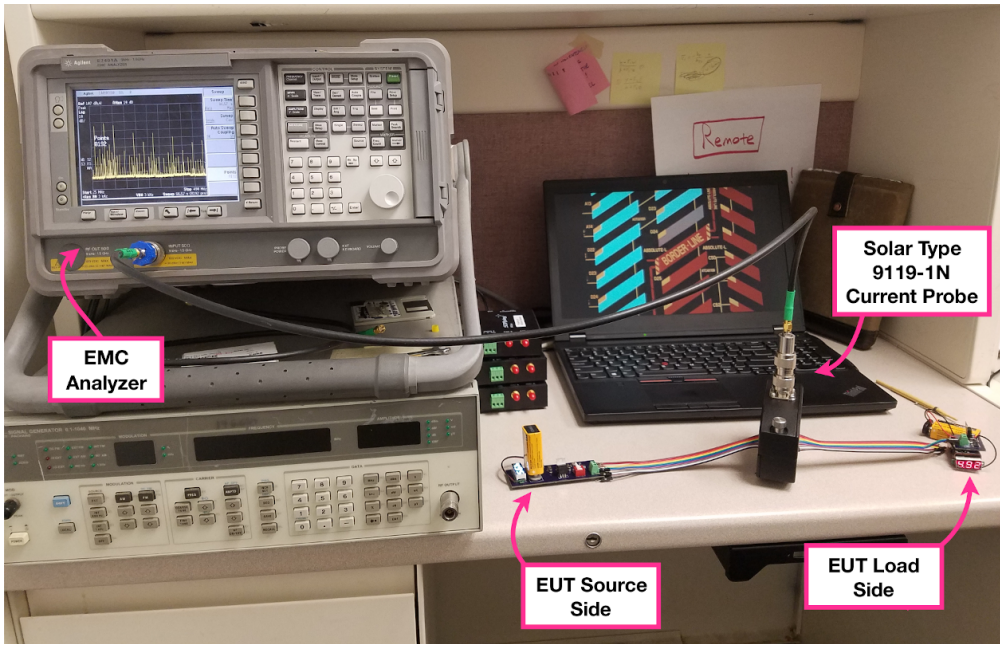
was used to measure the time-domain characteristics of the signal created by the PLL IC at each end of the ribbon cable connecting the source and load sides of the custom EUT. A current probe was then used to measure the common-mode current on the ribbon cable so that the magnitude of the harmonics of the PLL IC signal could be determined. The current probe was placed at the center of the ribbon cable following the method used by Paul and Bush [7], who found that the common-mode current on a ribbon cable at VHF and UHF frequencies remains mostly constant along the entire length of the cable. This procedure was repeated for all of the EUT PLL frequencies shown in the first column of Table 4.2.

Figures 4.12, 4.13, 4.14, and 4.15 show the results from the custom EUT characterization procedure. Each of these figures shows the time-domain characteristics at the top and the frequency-domain characteristics at the bottom. Also, the most significant harmonics have been marked in each of the frequency-domain plots. The current probe that was used has a very flat response from 10 MHz to 500 MHz: its transfer impedance decreases by less than 10 dB over this entire frequency range. This allows for a fair comparison between the magnitudes of the harmonics that were measured between 10 MHz and 500 MHz.

Figure 4.12 shows the characteristics of the custom EUT when set to a frequency of 40 MHz. One can see that the signal received by the load side of the EUT is significantly distorted when



(a)



(b)

Figure 4.11: (a) The test setup used to measure the time-domain characteristics of the custom EUT and (b) the test setup with the current probe that was used to measure the frequency-domain characteristics of the custom EUT.

compared to the signal that was measured on the source side of the EUT. Also notable is the nearly 3.5 ns increase in the rise time between the source and the load. This indicates that the signal is becoming very spread out as it travels along the ribbon cable. The frequency characteristics shown in Figure 4.12 indicate that the fundamental frequency of 40 MHz was not the strongest signal present on the ribbon cable, but that the second harmonic at 80 MHz was. The magnitudes of the marked harmonics are displayed to the right of the plot, where V_f indicates the magnitude of the harmonic frequency and V_{xf} indicates the magnitude of the xth harmonic. These magnitudes show that (with the exception of the second harmonic) the odd numbered harmonics were significantly stronger than the even harmonics. This is because the duty cycle of the signal coming from the source side of the EUT is nearly 50%. The first, second, and third harmonics shown in Figure 4.12 were measured during the broadband radiated emissions testing.

Figure 4.13 shows the characteristics of the custom EUT when set to a frequency of 50 MHz. One can see that the signal received by the load side of the EUT is significantly distorted when compared to the signal that was measured on the source side of the EUT. The rise time on the source side of the ribbon cable is also significantly shorter than the rise time observed at the load side. The frequency characteristics in Figure 4.13 show that the fundamental frequency of 50 MHz was the strongest signal present on the ribbon cable. Interestingly, the magnitudes of the marked harmonics that are displayed to the right of the plot show that the even harmonics are stronger at the frequencies above 450 MHz. The first, second, and third harmonics shown in Figure 4.13 were measured during the broadband radiated emissions testing.

Figure 4.14 shows the characteristics of the custom EUT when set to a frequency of 120 MHz. One can see that the signal received by the load side of the EUT is significantly reduced when compared to the signal that was measured on the source side of the EUT. However, the rise time of the signal has increased by less than 1 ns. This indicates that the amplitude is being reduced by some other means. Table 4.2 shows that at 120 MHz, the ribbon cable is equivalent to one-sixth of the signal wavelength. This means that the ribbon cable can quite effectively radiate 120 MHz signals, and so the load side attenuation is likely due to radiation. The frequency characteristics shown in

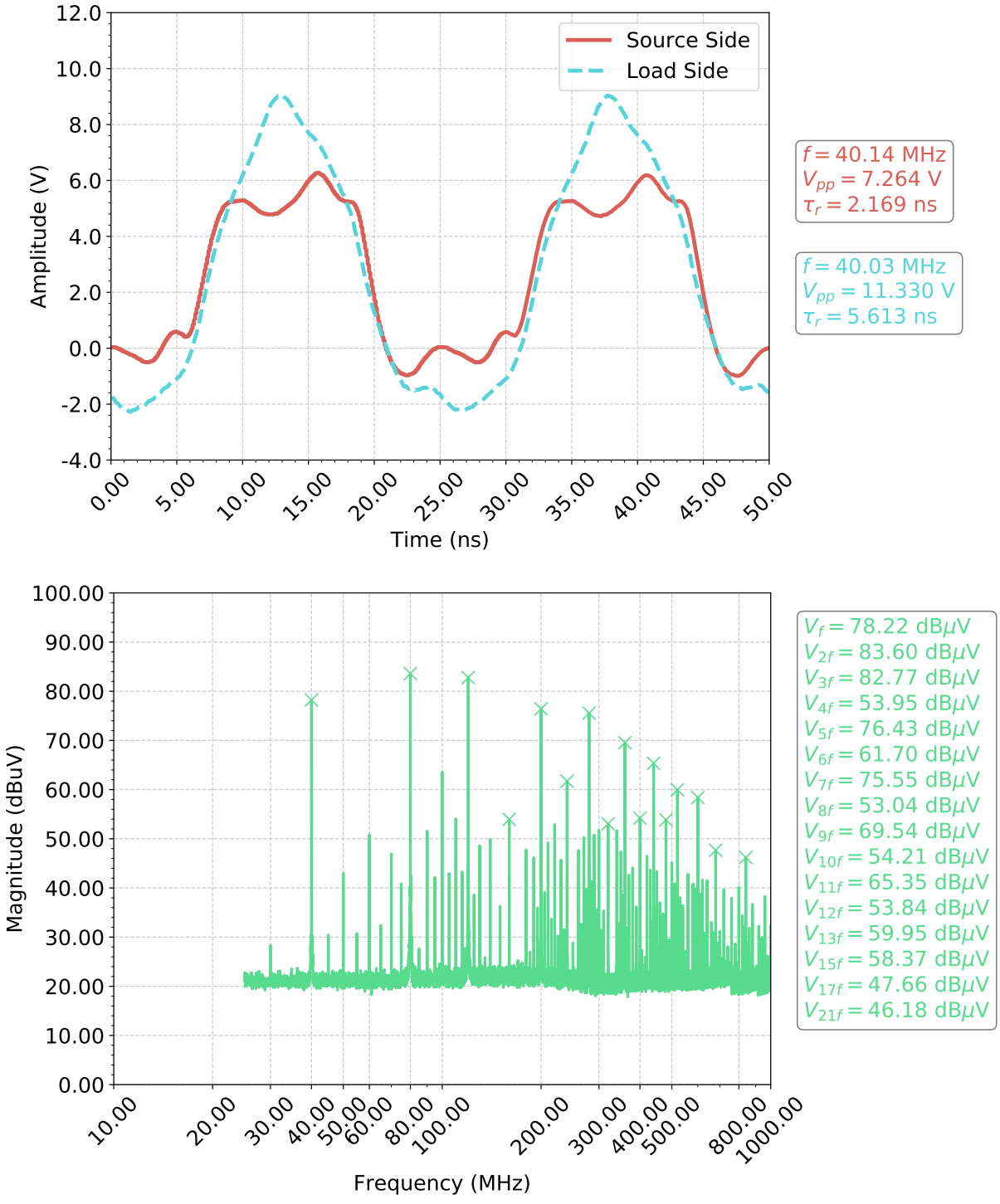


Figure 4.12: Time and frequency characteristics of the EUT when set to the base frequency $f = 40$ MHz. The significant harmonics of f have been marked with an "x" and their values are shown.

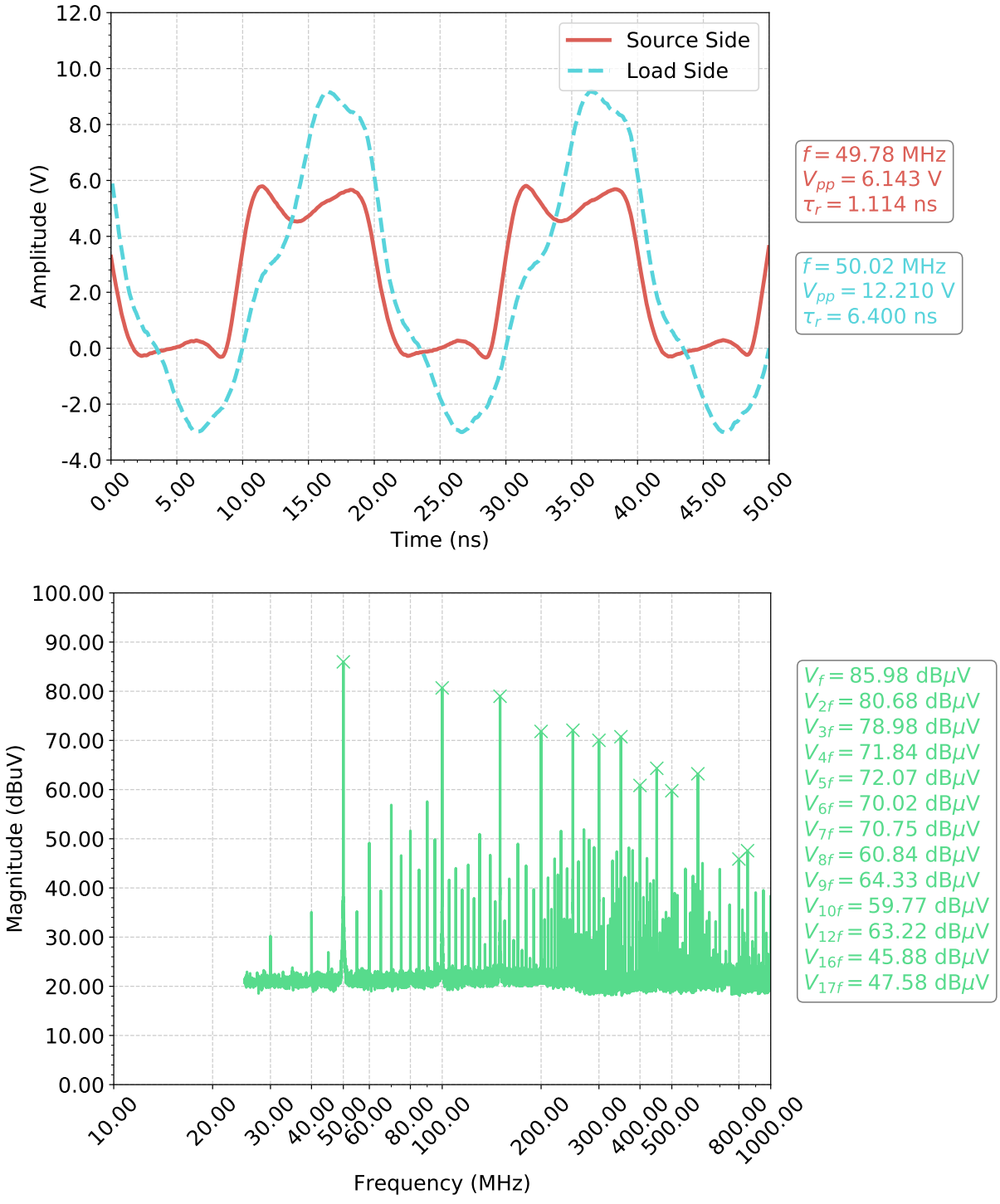


Figure 4.13: Time and frequency characteristics of the EUT when set to the base frequency $f = 50$ MHz. The significant harmonics of f have been marked with an "x" and their values are shown.

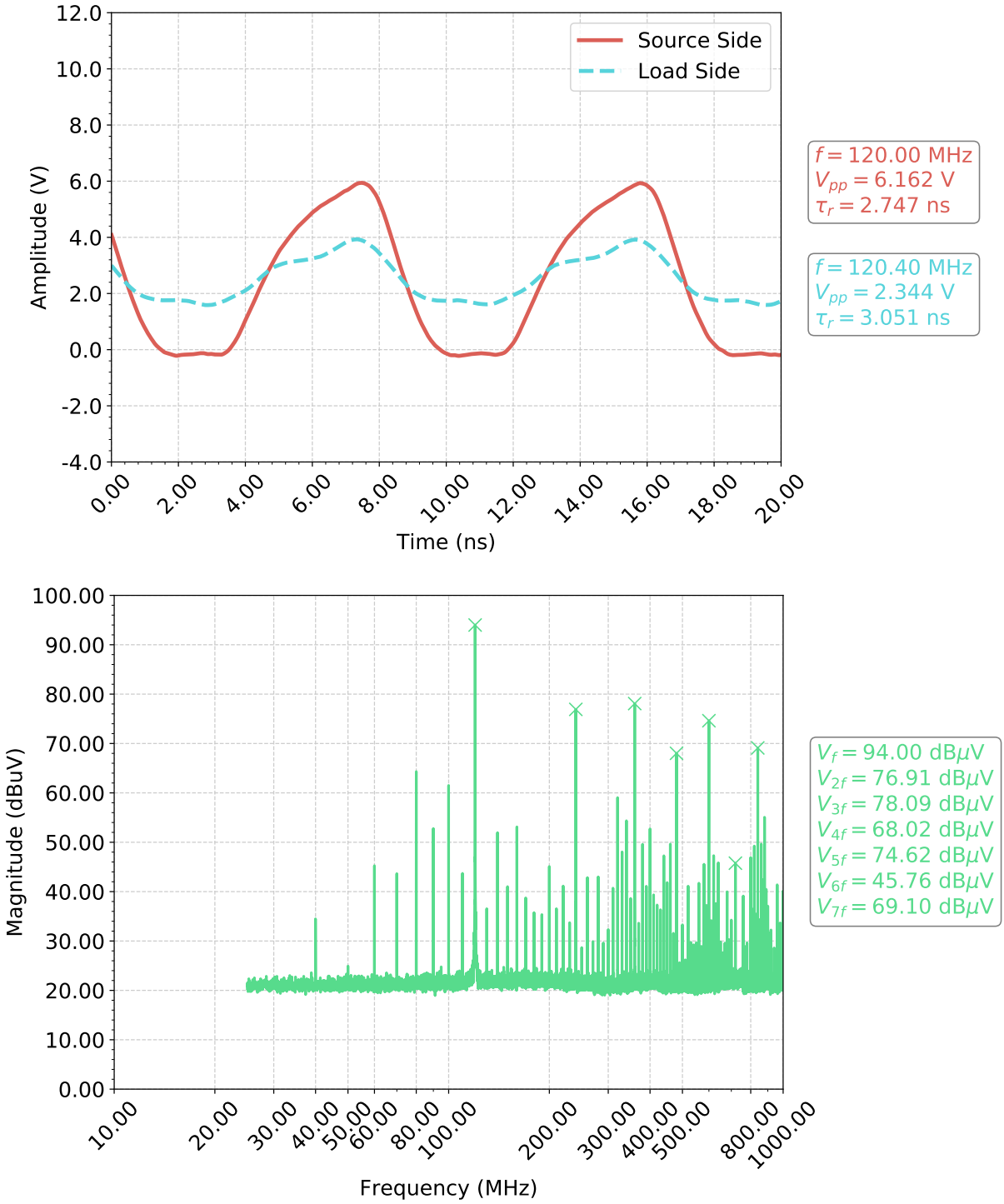


Figure 4.14: Time and frequency characteristics of the EUT when set to the base frequency $f = 120$ MHz. The significant harmonics of f have been marked with an "x" and their values are shown.

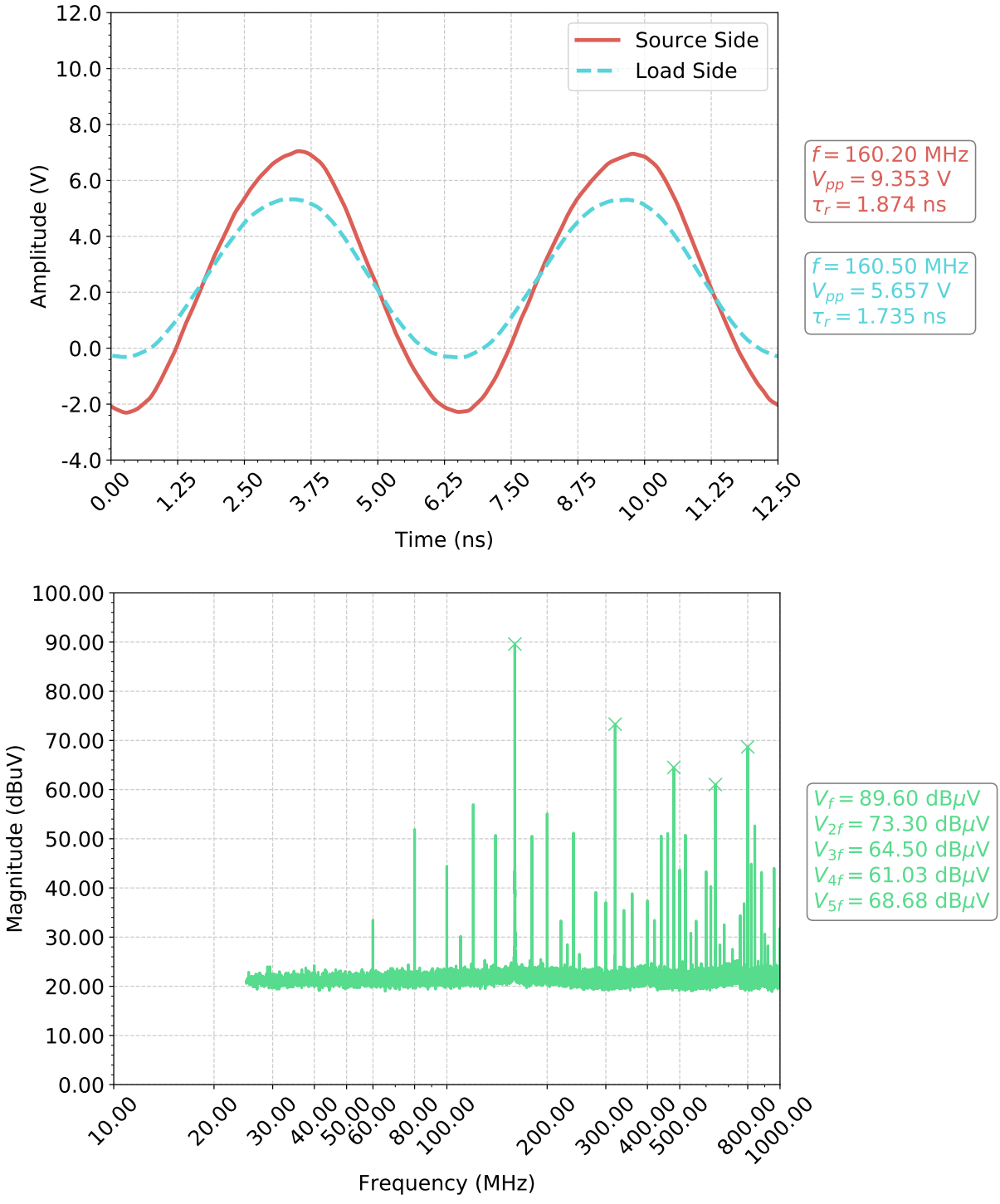


Figure 4.15: Time and frequency characteristics of the EUT when set to the base frequency $f = 160$ MHz. The significant harmonics of f have been marked with an "x" and their values are shown.

Figure 4.14 support this, as the magnitude of the 120 MHz signal is the strongest observed yet at over 90 dBμV. The magnitudes of the marked harmonics displayed to the right of the plot show that (with the exception of the second harmonic) the odd numbered harmonics are significantly stronger than the even harmonics, again due to the 50% duty cycle of the EUT signal. The second, third, and fourth harmonics shown in Figure 4.14 were measured during the broadband radiated emissions testing. Note that the 120 MHz signal measured for radiated emissions testing was the third harmonic of the 40 MHz signal shown in Figure 4.12, as the fundamental frequency of the 120 MHz signal shown in Figure 4.14 overloaded the EMC analyzer and the RSP2pro.

Figure 4.15 shows the characteristics of the custom EUT when set to a frequency of 160 MHz. Interestingly, at this frequency the signal created by the PT7C4511 PLL IC is almost perfectly sinusoidal. As was observed at 120 MHz, the signal received by the load side of the EUT is significantly reduced when compared to the signal that was measured on the source side of the EUT. Also, the rise time of the load side signal has increased by less than 0.1 ns. Table 4.2 shows that at 160 MHz, the ribbon cable is equivalent to one-fifth of the signal wavelength. This means that the ribbon cable can effectively radiate 160 MHz signals, and so the load side attenuation is likely due to the signal being radiated. The frequency characteristics shown in Figure 4.15 support this, as the magnitude of the 160 MHz signal is very strong (nearly 90 dBμV). The magnitudes of the marked harmonics displayed to the right of the plot show that (with the exception of the second harmonic) the odd numbered harmonics are still stronger than the even harmonics despite the sinusoidal shape of the 160 MHz signal. The first and fifth harmonics shown in Figure 4.15 were measured during the broadband radiated emissions testing.

Following the method described by Paul and Bush [7], the common-mode current measured by the current probe (I_p) can be used to predict the radiated emissions of the EUT at a specific frequency f with the equation

$$|E_{c_{max}}| = \left(6.28\times10^{-7}\right)\left(I_p L f / R\right) \tag{4.1}$$

where L is the length of the conductors, R represents the measurement distance in m, and $|E_{c_{max}}|$ represents the maximum electric field created by the common-mode currents along the conductors

in units of V/m. Note that here, I_p has units of microamps. To compare $|E_{c_{max}}|$ to the CISPR-22 Class B limits, it is converted to dBμV/m using Equation 3.5.

The voltages measured with the current probe (shown in the bottom of Figures 4.12, 4.13, 4.14, and 4.15) can be converted to the measured current I_p of Equation 4.1 using the transfer impedance of the current probe. The transfer impedance has units of dBΩ. Using V_p to represent the voltage measured by the current probe at a particular frequency, the measured current I_p can be calculated with the equation

$$I_p = V_p - Z_T \tag{4.2}$$

where Z_T is the current probe's transfer impedance at that frequency. Note that here I_p has units of dBμA. Also note that the measured current I_p is equivalent to twice the value of the common-mode current present on the ribbon cable [4].

Table 4.3 shows the predicted radiated emissions for the custom EUT at the frequencies that were tested (the frequencies tested are shown in column three). The PLL frequency used to create each radiated emission frequency is shown in column one for reference. One can see from the results in Table 4.3 that the radiated emissions created by the custom EUT are significantly higher than CISPR-22 Class B limits. It was discovered during initial testing that the radiated emissions created by the custom EUT were actually capable of overloading the EMC analyzer and the RSP2pro, so two ferrite cores were used on the ribbon cable to reduce the common-mode currents in the ribbon cable and dampen the radiated emissions. The ferrite cores were placed around the ribbon cable on the source side of the EUT as shown in Figure 4.16.

Figure 4.16 also shows the configuration of the custom EUT that was used for all of the broadband radiated emission tests. The custom EUT was placed on a foam test stand that was the same height as the EMC antenna. The center of the 41 cm cable was placed in the center of the foam test stand. The source and load boards each had their own voltage indicator to ensure that all tests were performed with both boards operating at 5 V.

EUT PLL Frequency (MHz)	Harmonic Number	Radiated Emission Frequency (MHz)	V_p (dBμV)	Z_T (dBΩ)	I_p (dBμA)	$\lvert E_{c_{max}} \rvert$ (10 m, dBμV/m)	CISPR-22 Class B Limit (10 m, dBμV/m)	$\lvert E_{c_{max}} \rvert$ (1 m, dBμV/m)	CISPR-22 Class B Limit (1 m, dBμV/m)
40	1	40	78.22	-10.9	89.1	89.4	30	109.4	50
50	1	50	85.98	-11.4	97.4	99.6	30	119.6	50
40	2	80	83.60	-12.1	95.7	102.0	30	122.0	50
50	2	100	80.68	-12.7	93.4	101.6	30	121.6	50
40	3	120	82.77	-12.9	95.7	105.5	30	125.5	50
50	3	150	78.98	-13.2	92.2	103.9	30	123.9	50
160	1	160	89.60	-13.4	103.0	115.3	30	135.3	50
120	2	240	76.91	-14.5	91.4	107.3	37	127.3	57
120	3	360	76.91	-14.3	91.2	107.1	37	127.1	57
120	4	480	68.02	-14.6	82.6	104.5	37	124.5	57
160	5	800	68.68	-14.3	83.0	109.3	37	129.3	57

Table 4.3: The conversion of the voltages measured by the current probe into the predicted maximum radiated emissions, $\lvert E_{c_{max}} \rvert$, using Equations 4.1 and 4.2 with distances of 10 m and 1 m.

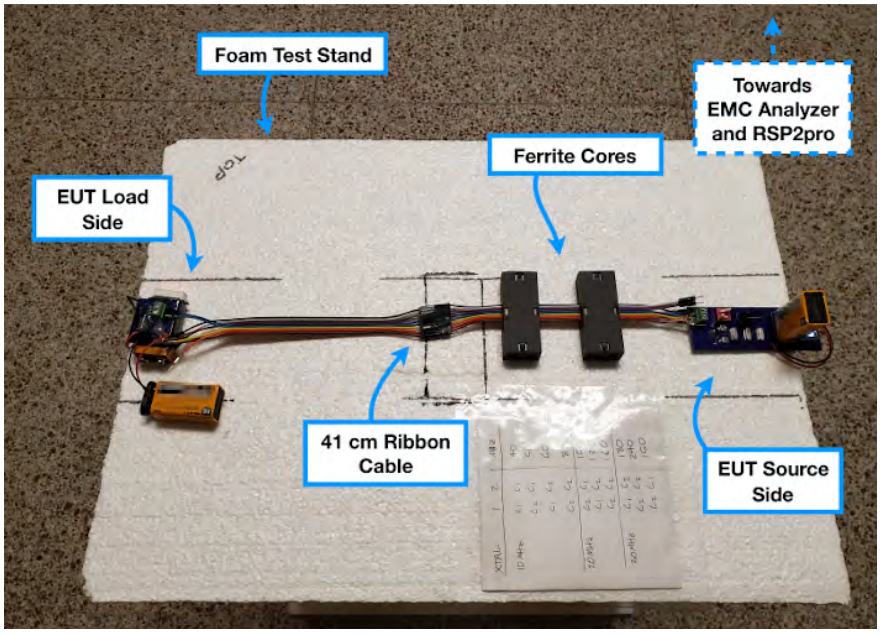


Figure 4.16: Detailed view of how the custom EUT was arranged during broadband radiated emission tests.

4.2 Narrowband Results

Figures 4.17 and 4.18 show the average measured amplitude difference between the EMC analyzer and each of the three RSP2pro units for every data point collected from using the input data points from Table 4.1 at test distances of 10 m and 1 m (respectively). Each of the plots shown in these figures corresponds to one of the RSP2pro units that was tested. Recall that the average offset value $\overline{V}_d$ is defined in Equation 3.13 as

$$\overline{V}_d = \overline{V}_R - \overline{V}_E$$

where $\overline{V}_R$ represents the average voltage measured by the RSP2pro at particular input data point and $\overline{V}_E$ represents the average voltage measured by the EMC analyzer at that same input data point. Each input data point was measured three times for the results presented in this section. Recall that the EMC analyzer amplitude has an absolute measurement accuracy of ±0.30 dB [43] for all of the input data points tested. By using the EMC analyzer for comparison, the margin of error of ±2.00 dB developed in Section 3.3 can be assessed. Each plot in Figures 4.17 and 4.18 also shows the measured SNR at each data point using the rightmost y-axis. The black dots in these plots represent the measured SNR.

From the results for the measurements performed at a distance of 10 m shown in Figure 4.17, one can see that all three RSP2pro units had an average offset greater than 2.00 dB. Some frequencies performed within a ±2.00 dB margin of the EMC analyzer, but with all three RSP2pro units the radiated emissions measured at 40, 160, and 360 MHz were outside this margin. Additionally, RSP2pro 1's measurement at 50 MHz was off by more than +10.00 dB from the EMC analyzer measurement. The SNR data in these plots show that at every frequency tested, the SNR experienced by all three RSP2pro units was comparable to the SNR observed in Section 3.3. The average SNR value shown in Figure 4.17 was about 10 dB higher for frequencies above 200 MHz for all three RSP2pro units.

Figure 4.18 shows the results for the measurements performed at a distance of 1 m with the CW radiated emissions source. One can see that compared to the 10 m results in Figure 4.17, all three

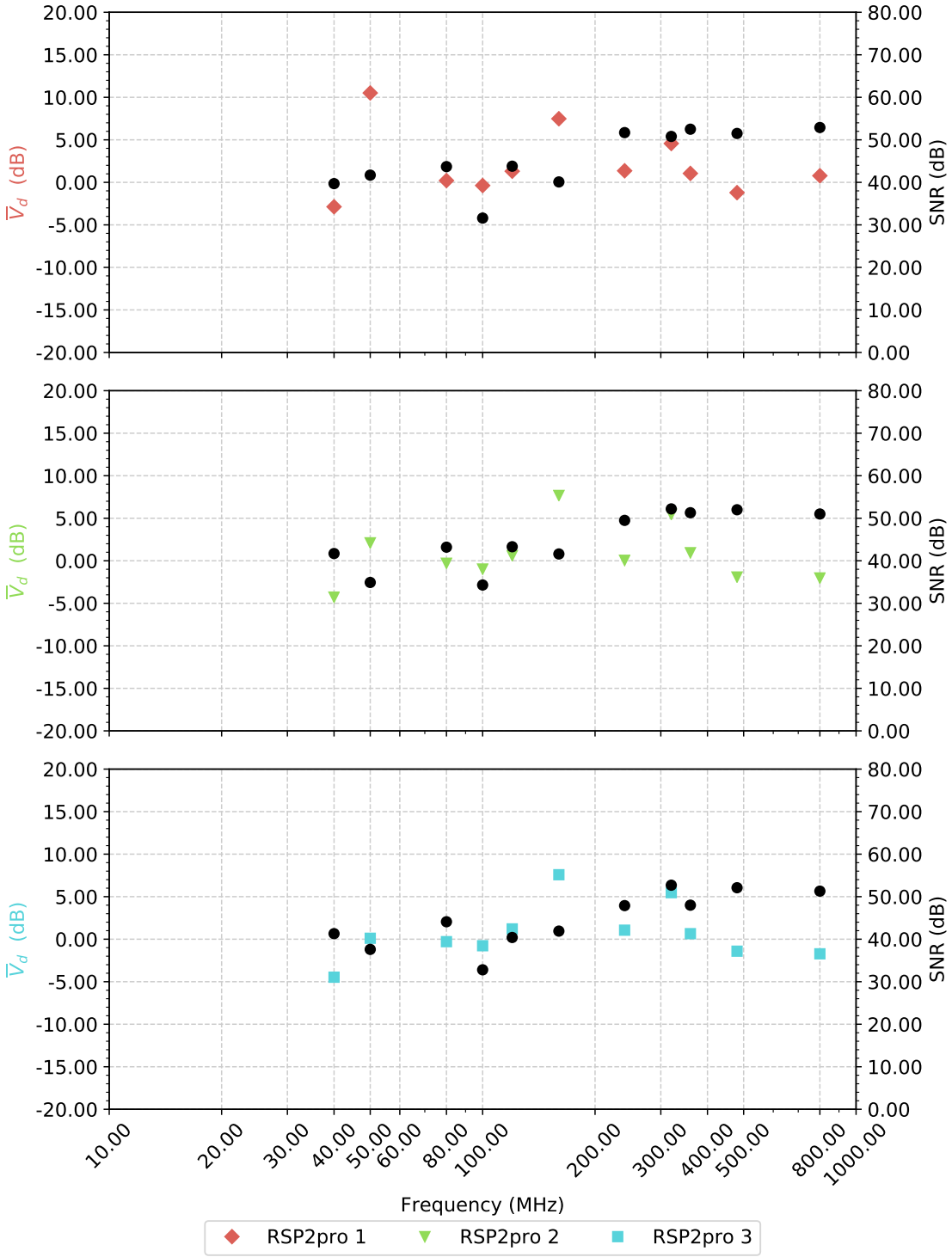


Figure 4.17: Average RSP2pro voltage measurement offset from Equation 3.13 for the input data points shown in Table 4.1 measured at a distance of 10 m. Each data point represents the average of three measurements.

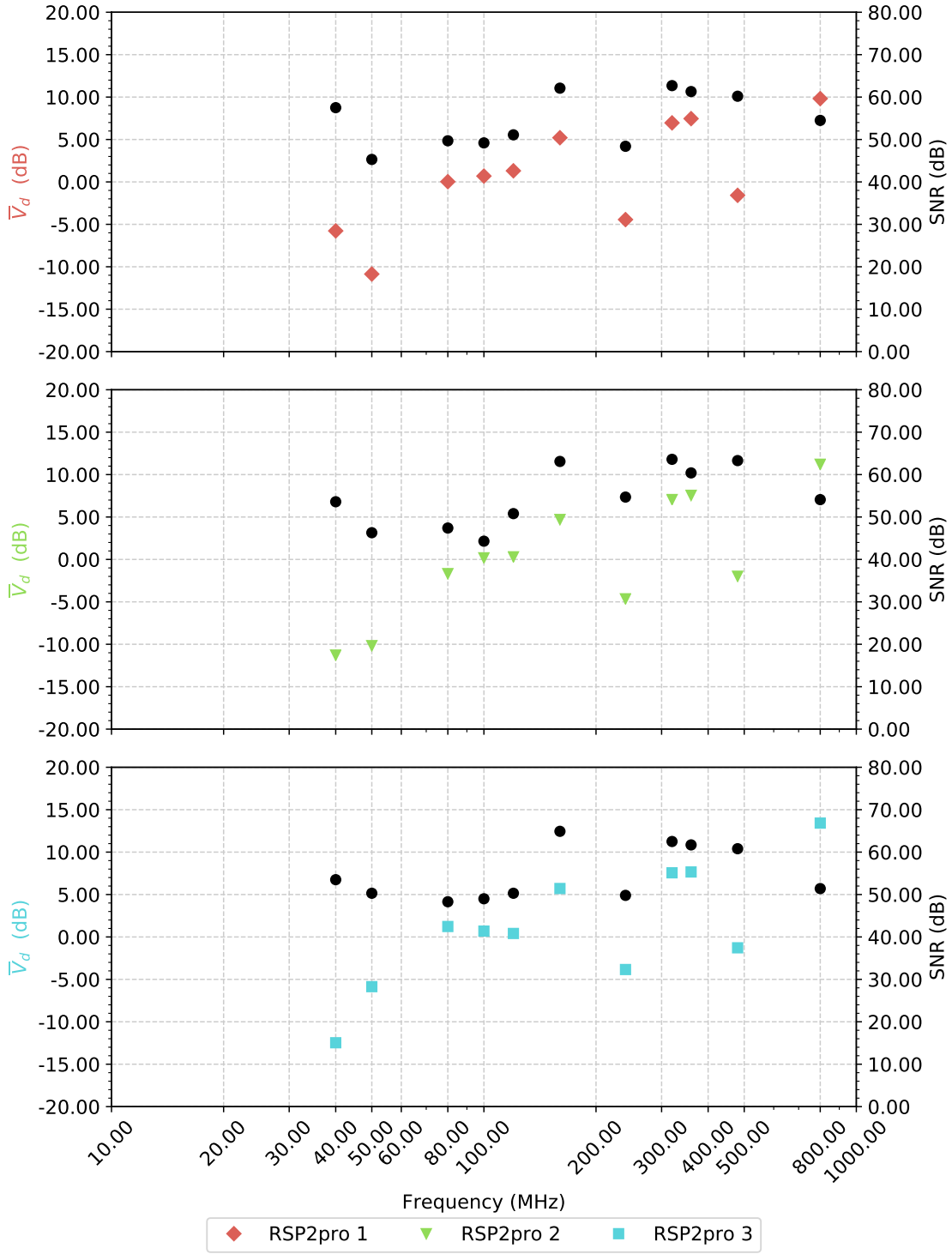


Figure 4.18: Average RSP2pro voltage measurement offset from Equation 3.13 for the input data points shown in Table 4.1 measured at a distance of 1 m. Each data point represents the average of three measurements.

RSP2pro units had a much larger average offset from the EMC analyzer at multiple frequencies, especially below 80 MHz and above 200 MHz. Interestingly, there is a region between 60 and 150 MHz where all three RSP2pro units had an average offset of approximately 2 dB. Also notable is that all three RSP2pro units measured a lower level of radiated emissions when the frequency was below 60 MHz while nearly 80% of the measured radiated emissions above 80 MHz where larger than what was measured by the EMC analyzer. The larger overall average offset at most frequencies and the short measurement distance of 1 m indicates that the RSP2pro ADCs were mostly likely being overloaded. The SNR data in Figure 4.18 shows that at every frequency tested, the SNR was about 10 dB higher than what was measured at 10 m.

4.3 Broadband Results

Figures 4.19 and 4.20 show the average measured amplitude difference between the EMC analyzer and each of the three RSP2pro units for each of the radiated emission frequencies shown in Table 4.3 at test distances of 10 m and 1 m (respectively). Each of the plots shown in these figures corresponds to one of the RSP2pro units that was tested. Recall that the average offset value $\overline{V}_d$ is defined in Equation 3.13 as

$$\overline{V}_d = \overline{V}_R - \overline{V}_E$$

where $\overline{V}_R$ represents the average voltage measured by the RSP2pro at particular input data point and $\overline{V}_E$ represents the average voltage measured by the EMC analyzer at that same input data point. Each input data point was measured three times for the results presented in this section. Recall that the EMC analyzer amplitude has an absolute measurement accuracy of ±0.30 dB [43] for all of the input data points tested. By using the EMC analyzer for comparison, the margin of error of ±2.00 dB developed in Section 3.3 can be assessed. Each plot in Figures 4.19 and 4.20 also shows the measured SNR at each data point using the rightmost y-axis. The black dots in these plots represent the measured SNR.

Figure 4.19 shows the results from measuring radiated emissions from the custom EUT at a distance of 10 m. One can see that the RSP2pro measurements above 100 MHz are significantly

lower than the EMC analyzer measurements for all three units. All three RSP2pro units also exhibit a very large offset at a frequency of 40 MHz, where they differ from the EMC analyzer measurement by 20 dB. While the RSP2pro units measure consistently with one another, they are only within the ±2.00 dB margin at the frequencies between 50 and 100 MHz. The measured SNR is comparable to what was seen from the narrowband CW radiated emissions at 10 m shown in Figure 4.17.

The broadband radiated emissions measurements at 1 m shown in Figure 4.20 are by far the results with the most deviation from the EMC analyzer measurements for all three RSP2pro units. Note that the plots in this figure have a maximum value of 0 dB, not 20 dB. Much like the results from 10 m shown in Figure 4.19, Figure 4.20 shows that the measurements were consistent across all three units. However, at every frequency (with the exception of 40 MHz) all three RSP2pro units measured more than 10 dB lower than the EMC analyzer. In the frequency range from 50 MHz to 200 MHz, most of the measurements from all three units were 20 dB less than what was measured by the EMC analyzer. The SNR data shown in this figure is very similar to the SNR that was observed when measuring the same broadband emissions at 10 m. However, it is very likely that the RSP2pro ADCs were overloaded when measuring broadband radiated emissions at 1 m. This is implied by the large deviation from the EMC analyzer voltage measurements at the frequencies that were tested.

The results in this section can be compared to the narrowband CW results from Section 4.2 to conclude that the RSP2pro units are more accurate when measuring CW radiated emissions. This could be because the RSP2pro is designed to be a radio receiver, and so the SDRuno settings have been tailored to measure the correct power level when measuring narrowband communication signals instead of broadband noise signals. To further investigate the validity of this conclusion, the predictions from Table 4.3 can be compared to the actual radiated emissions measured by the three RSP2pro units and the EMC analyzer. By adding a term for the insertion loss from the SMA cable to Equation 2.6, the measured voltages can be converted into field strengths and compared to the predictions:

$$E\left(dB\mu V/m\right) = V\left(dB\mu V\right) + AF\left(dB\ 1/m\right) - IL\left(dB\right) \tag{4.3}$$

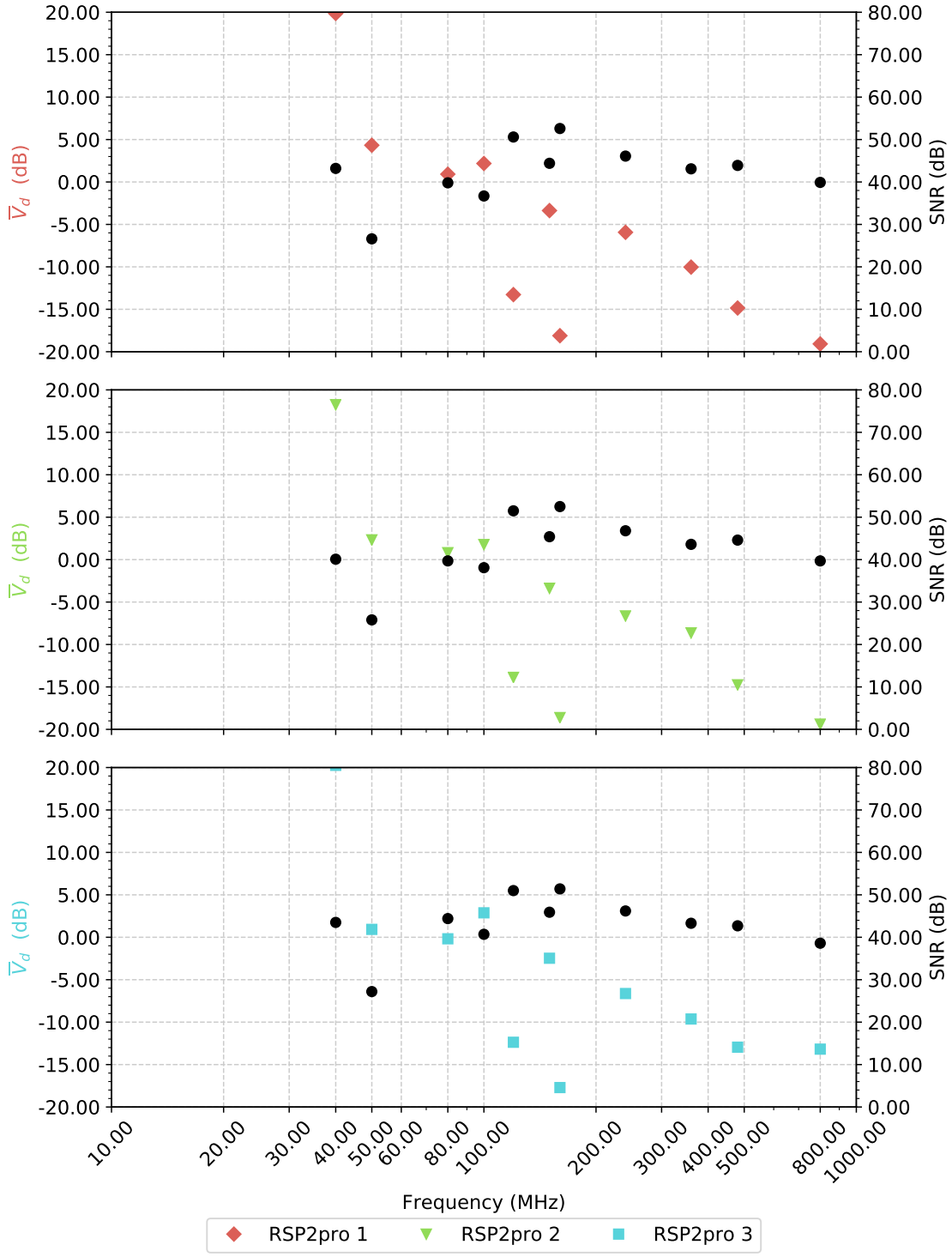


Figure 4.19: Average RSP2pro voltage measurement offset from Equation 3.13 for the custom EUT radiated emission frequencies shown in Table 4.3 measured at a distance of 10 m. Each data point represents the average of three measurements.

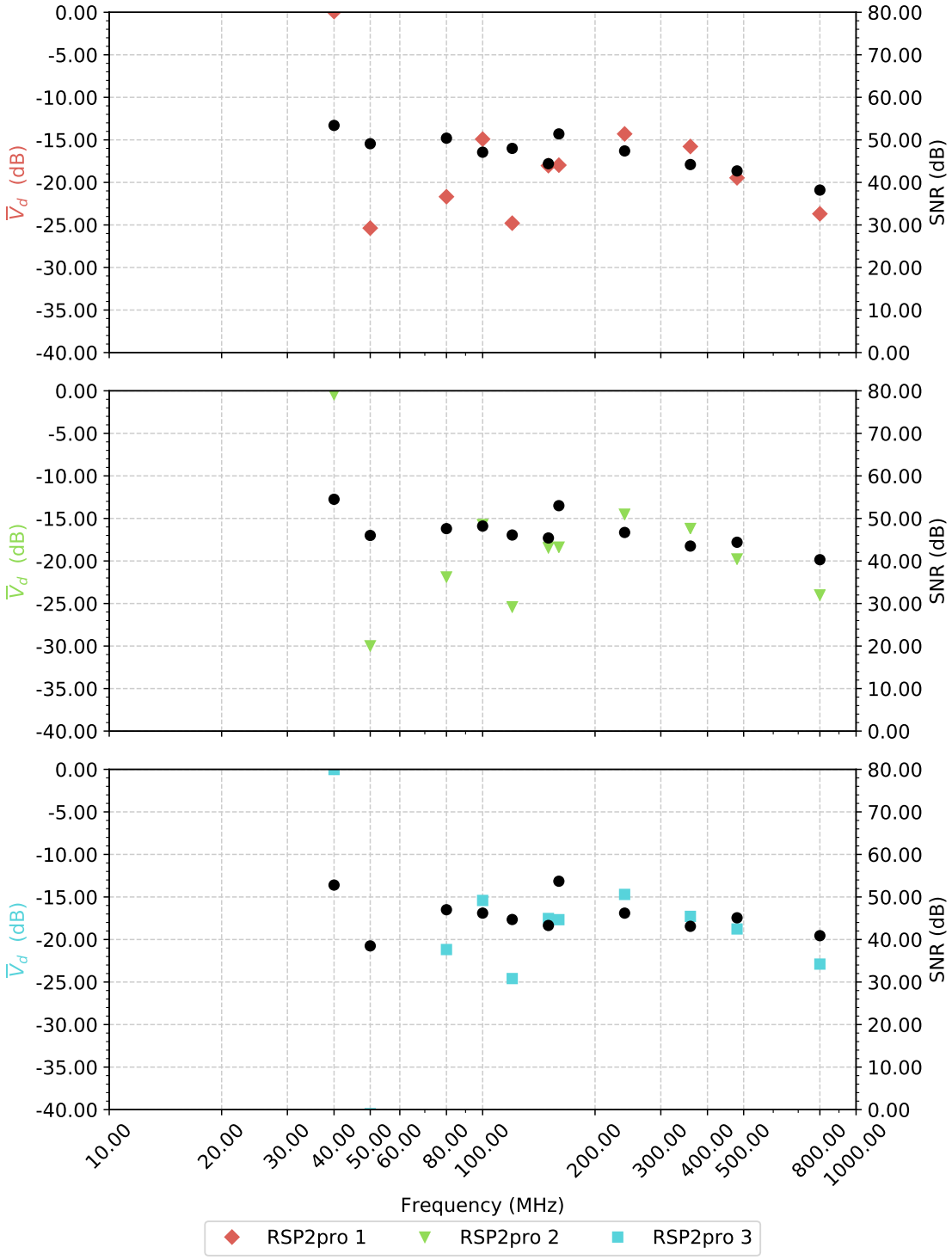


Figure 4.20: Average RSP2pro voltage measurement offset from Equation 3.13 for the custom EUT radiated emission frequencies shown in Table 4.3 measured at a distance of 1 m. Each data point represents the average of three measurements.

where IL represents the measured insertion loss of the SMA cable at a specific frequency in dB. Table 4.4 and 4.5 show the measured radiated emissions at 10 m and 1 m (respectively) that were converted from the measured voltages using Equation 4.3. The second to last column of each table shows the predicted radiated emissions from Table 4.3. Note that for brevity, only the measured voltage for the EMC analyzer (V_E) is shown in Table 4.4 and 4.5. The values of the variables in Equation 4.3 used to convert the measured voltage into an electric field are shown in each row. The values of AF correspond to the BicoLOG 30100X antenna factor.

Table 4.4 shows that the measured radiated emissions of the EMC analyzer (E_E) and all three RSP2pro units (E_{R1}, E_{R2}, E_{R3}) at a distance of 10 m. All of the measured radiated emissions are significantly lower than the predicted radiated emissions, $|E_{c_{max}}|$. This could be for any number of reasons, but two main causes are the uncontrolled RF environment and the ferrite chokes used to dampen the radiated emissions. Table 4.4 shows that, with the exception of 800 MHz, all three RSP2pro units were well within 3.00 dB of one another at all frequencies. However, the radiated emissions measured from all three units varied greatly from the values measured by the EMC analyzer.

Table 4.5 shows that the measured radiated emissions of the EMC analyzer (E_E) and all three RSP2pro units (E_{R1}, E_{R2}, E_{R3}) at a distance of 1 m. These results are very similar to the ones shown in Table 4.4. All of the measured radiated emissions are significantly lower than the predicted radiated emissions, $|E_{c_{max}}|$. One can also see from Table 4.5 that, with the exception of 50 MHz, all three RSP2pro units were well within ±3.00 dB of one another at all frequencies. However, the radiated emissions measured from all three units varied greatly from the values measured by the EMC analyzer.

Radiated Emission Frequency (MHz)	V_E (dBμV)	AF (dB 1/m)	IL (dB)	E_E (dBμV/m)	E_{R1} (dBμV/m)	E_{R2} (dBμV/m)	E_{R3} (dBμV/m)	$\lvert E_{c_{max}} \rvert$ (10 m, dBμV/m)	CISPR-22 Class B Limit (10 m, dBμV/m)
40	55.34	-4.3488	0.06	50.93	70.79	69.19	71.19	89.4	30
50	59.17	-6.9106	0.08	52.18	56.51	54.51	53.11	99.6	30
80	77.48	-10.5282	0.09	66.86	67.78	67.68	66.68	102.0	30
100	76.82	-12.1900	0.10	64.53	66.71	66.31	67.41	101.6	30
120	110.87	-13.6064	0.10	97.16	83.89	83.29	84.79	105.5	30
150	99.07	-15.8782	0.10	83.09	79.72	79.72	80.62	103.9	30
160	121.31	-16.8276	0.16	104.32	86.21	85.71	86.61	115.3	30
240	101.03	-16.8276	0.17	84.03	78.10	77.40	77.40	107.3	37
360	105.43	-19.2258	0.19	86.01	75.98	77.38	76.38	107.1	37
480	104.65	-19.4839	0.23	84.94	70.09	70.19	71.99	104.5	37
800	95.48	-17.2552	0.32	77.90	58.82	58.52	64.72	109.3	37

Table 4.4: Measured radiated emissions from the custom EUT at 10 m compared to the predicted radiated emissions.

Radiated Emission Frequency (MHz)	V_E (dBμV)	AF (dB 1/m)	IL (dB)	E_E (dBμV/m)	E_{R1} (dBμV/m)	E_{R2} (dBμV/m)	E_{R3} (dBμV/m)	$\lvert E_{c_{max}} \rvert$ (1 m, dBμV/m)	CISPR-22 Class B Limit (1 m, dBμV/m)
40	90.92	-4.3488	0.06	86.51	86.59	86.09	86.49	109.4	50
50	97.07	-6.9106	0.08	90.08	64.71	60.11	49.61	119.6	50
80	101.98	-10.5282	0.09	91.36	69.68	69.48	70.18	122.0	50
100	97.11	-12.1900	0.10	84.82	69.91	69.11	69.41	121.6	50
120	121.19	-13.6064	0.10	107.48	82.69	82.09	82.89	125.5	50
150	121.73	-15.8782	0.10	105.75	87.72	87.32	88.22	123.9	50
160	121.87	-16.8276	0.16	104.88	86.91	86.51	87.21	135.3	50
240	104.20	-16.8276	0.17	87.20	72.90	72.70	72.50	127.3	57
360	111.18	-19.2258	0.19	91.76	75.98	75.58	74.48	127.1	57
480	112.16	-19.4839	0.23	92.45	72.99	72.69	73.69	124.5	57
800	107.49	-17.2552	0.32	89.91	66.22	65.92	67.02	129.3	57

Table 4.5: Measured radiated emissions from the custom EUT at 1 m compared to the predicted radiated emissions.

CHAPTER 5

A COMPLETE PRECOMPLIANCE TEST SETUP

This chapter discusses the combination of the RSP2pro and two common television antennas, which together create a complete precompliance test setup. First, the specifics of how the antennas were selected and the modifications that were made to them are described in Section 5.1. That section also includes a description of the test procedure used for characterizing each antenna. The results of this characterization procedure are discussed in Section 5.2 and Section 5.3. Finally, Section 5.4 assesses the utility of the RSP2pro test setup compared to the utility of the traditional setup by measuring an EUT with known radiated emissions. This assessment includes a discussion of the relative cost, measurement accuracy, and overall convenience of using each test setup.

5.1 Antenna Selection, Modifications, and Procedure

As mentioned in Section 2.2, a low-cost precompliance test setup like the one shown in Figure 2.11 typically uses some type of VHF or UHF antenna in place of a traditional EMC antenna. While this has the major disadvantage of not allowing the use of an antenna factor, it does greatly reduce the cost of the test setup. According to André and Wyatt [25], common television antennas work well in the VHF and UHF frequency bands. Figure 5.1 shows two common television antennas that were purchased online for less than $20.00. The "rabbit ear" antenna shown in Figure 5.1a has adjustable elements and can be used for measuring signals in the VHF frequency range from 65 to 200 MHz [25]. Figure 5.1b shows the "bowtie" UHF antenna that is usable for signals from 300 MHz to 800 MHz [25]. In Sections 5.2 and 5.3, the usefulness of each antenna in the frequency ranges described by André and Wyatt [25] is assessed.

Before either of the antennas shown in Figure 5.1 was used, support hardware was created so that consistent measurements could be made with each antenna. This included a custom-designed antenna holder that allows each antenna to be placed in a horizontal or vertical orientation. Figure 5.2 shows the CAD models for the antenna holders. Two different versions were required, as each



(a) (b)

Figure 5.1: (a) The VHF "rabbit ear" television antenna and (b) the UHF "bowtie" television antenna.

antenna has a unique shape that must be cradled by the holder so that the antenna cannot move. The white holder in Figure 5.2 was designed for the rabbit ear antenna while the green holder was designed for the bowtie antenna. While these holders were specifically designed for the antennas purchased for this thesis, it should be noted that these holders will work with other antennas of the same style. This is because most television antennas are very similar in shape and size. The holder models shown in Figure 5.2 were 3D-printed using black PLA material (the final holders are shown being used in Figure 5.1). This was ultimately a poor choice, as PLA has a very low melting point. During several outdoor measurement tests, the holders actually deformed due to exposure to bright sunlight. Using boiling water, the holders were able to be reshaped. This problem can be avoided by using a more resilient material to construct the antenna holders. The design files for each antenna holder are available in the Digital Appendix that is included with the digital version of this thesis (refer to Appendix B for more information).

To allow the antennas to be placed in either the vertical or horizontal orientation, each of the antenna holders was design with an L-shaped mounting point that uses a recessed nut to attach a standard camera tripod adapter. Figure 5.3 shows how the camera tripod adapters attach to the

L-shaped mounting point. The recessed nuts can be seen in Figure 5.3a and the camera tripod adapters can be seen in Figure 5.3b.

The last hardware accessory that was designed for the antennas was a clip that prevents the angular adjustment of the rabbit ear antenna elements. The CAD model for this clip is shown in Figure 5.4. The clip forces the antenna elements of the rabbit ear antenna to remain separated by 180°. This makes the antenna more like a center-fed dipole. The clip also allows for more consistent measurements since the elements are able to be placed in the same position for each test.

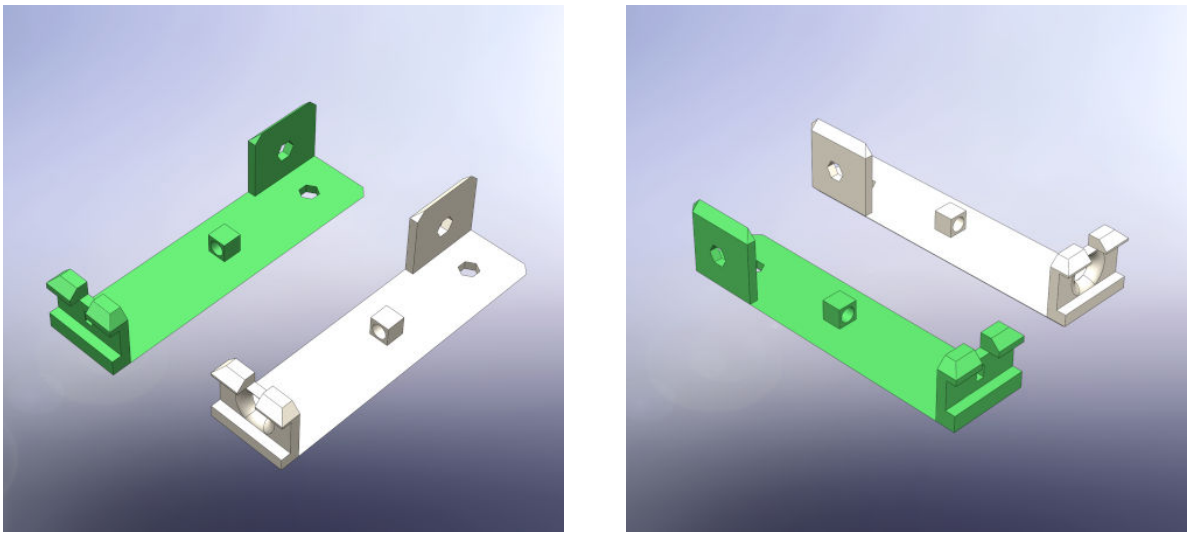


Figure 5.2: CAD models for the antenna holders that are shown in Figure 5.1.



(a) (b)

Figure 5.3: (a) The antenna holder's dual attachment points and (b) the antenna holder with both camera tripod adapters attached.

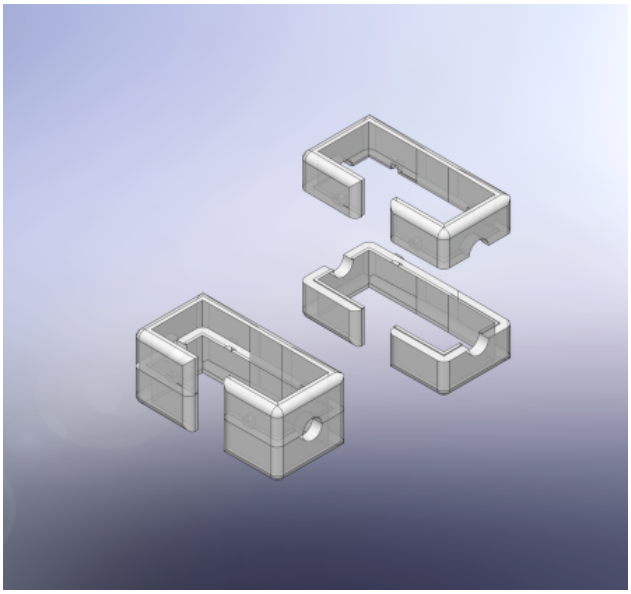


Figure 5.4: CAD model of the antenna clip that was used to fix the angular position of rabbit ear antenna elements to 180°.

The clip was 3D-printed using a stereolithography process. A detailed view of the final clip can be seen in Figures 5.6b and 5.6b. The resin material used by the stereolithography process worked much more reliably than the PLA that was used to create the antenna holders.

The television antennas each come with a long twinlead cable that is designed for being attached directly to a balun. A balun is a type of matching transformer that gets its name from a combination of the words "BALanced" and "UNbalanced." Baluns are most often used to interface balanced center-fed antennas to unbalanced coaxial cables [60]. Off-the-shelf televison baluns are also capable of improving the impedance match between television antennas and the 50 Ω impedance used by most spectrum analyzers and receivers [3]. The baluns considered for this thesis are shown in Figure 5.5a. These are common television baluns that use a 4:1 transformer to match the 300 Ω television antennas to the 75 Ω impedance of television coaxial cable.

Figure 5.5a shows that each balun uses a different construction to achieve the desired matching behavior. This means that it is important to characterize the antennas that are used with the specific balun that is available. The balun shown on the left side of Figure 5.5a was used for this thesis.

Figure 5.5b shows how the balun was originally wired. The two inputs of the 4:1 transformer element were connected to capacitors that were attached to the screw-terminals on the balun. This was where the blades on the television antennas' twin-lead cables would normally be attached. Figures 5.5c and 5.5d show how the original design was modified. The screw-terminals were removed and the capacitor leads were fed through two small holes that were drilled into the balun enclosure. This modification was inspired by Wyatt [3], who recommends shortening the television antenna twin-leads as much as possible to improve the impedance match of the antenna to the balun. Figure 5.5e shows the bowtie antenna with its leads completely removed and how the slot in the 3D-printed antenna holder allows the balun to be directly soldered to the antenna.

The rabbit ear antenna required further modifications to be used with the balun and 3D-printed antenna holder. Figure 5.6a shows how the body of the rabbit ear antenna was first reduced in size by removing its plastic mounting rod. This figure also shows the white twin-lead cable that was originally attached to the antenna. Figure 5.6b shows the connectors of each antenna element after removing the twin-lead cable. This figure also shows how the 3D-printed clip from Figure 5.4 was attached to the rabbit ear antenna body to lock the antenna elements in place. Lastly, Figure 5.6c shows the wires that were attached to the connectors in Figure 5.6b. These wires are much shorter than the original twin-lead cable and allow the rabbit ear antenna to be soldered directly to the modified balun (like the bowtie antenna shown in Figure 5.5e).

After the antennas were modified to better match the 50 Ω impedance of the EMC analyzer and the RSP2pro, each was characterized by first measuring the VSWR with a network analyzer. The VSWR was measured to determine the impedance mismatch losses of the television antennas and to determine if the usable frequency ranges for each antenna as described by André and Wyatt [25] are correct. The antennas were then used to measure the radiated emissions from the custom EUT described in Section 4.1.2 at a distance of 1 m using the setup shown in Figure 4.4b. These measurements were performed by sweeping across the entire CISPR-22 Class B frequency range with the EMC analyzer and the four "Spectrum Sweep" state files described in Table 3.1. Identical measurements were then performed using the BicoLOG 30100X EMC antenna so a direct



(a)



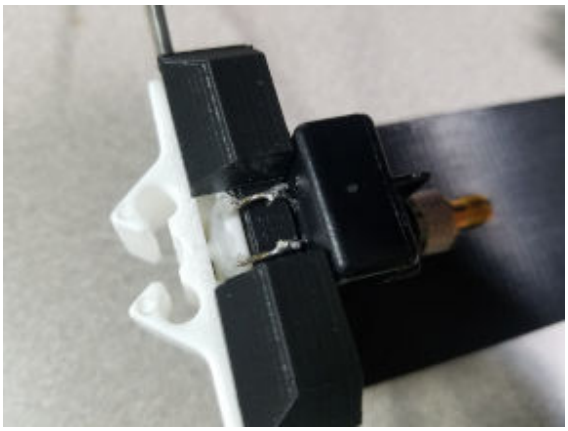
(b)



(c)

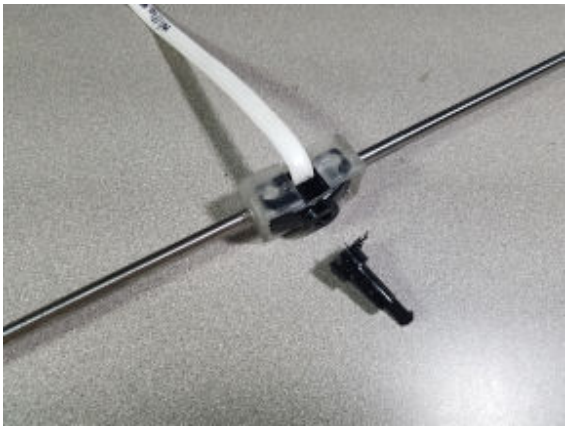


(d)

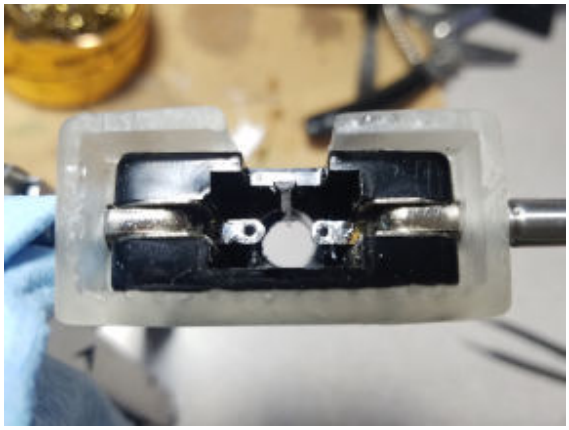


(e)

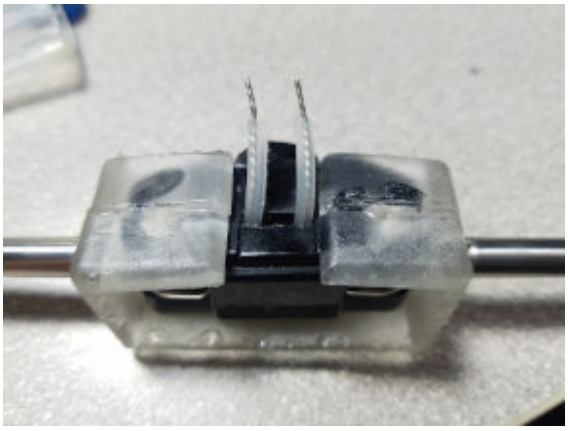
Figure 5.5: (a) Two 4:1 televison baluns (the one on the left was used), (b) internal components of the balun, (c) balun with the screw-terminals removed, (d) the final modified 4:1 balun, and (e) the connection between the modified bowtie antenna and the modified balun.



(a)



(b)



(c)

Figure 5.6: (a) Removal of the mounting post, (b) the exposed antenna element connectors, and (c) the final modified rabbit ear antenna.

comparison of its performance could be made with each television antenna. Figure 5.7 shows the RF signal chains used for these measurements.

Note that it is a misconception that the minimum VSWR of a transmission line connected to an antenna occurs at the frequency where the antenna is resonant [58]. A minimum VSWR at a specific frequency only indicates that there is minimal return loss between the antenna and the transmission line. For this reason, the direct comparison to the BicoLOG 30100X antenna is a much better indicator of the resonant frequencies of each television antenna. According to [58], a VSWR of as much as 6:1 is still acceptable for signal reception in the VHF and UHF frequency bands (although a VSWR of less than 2:1 is generally preferred).

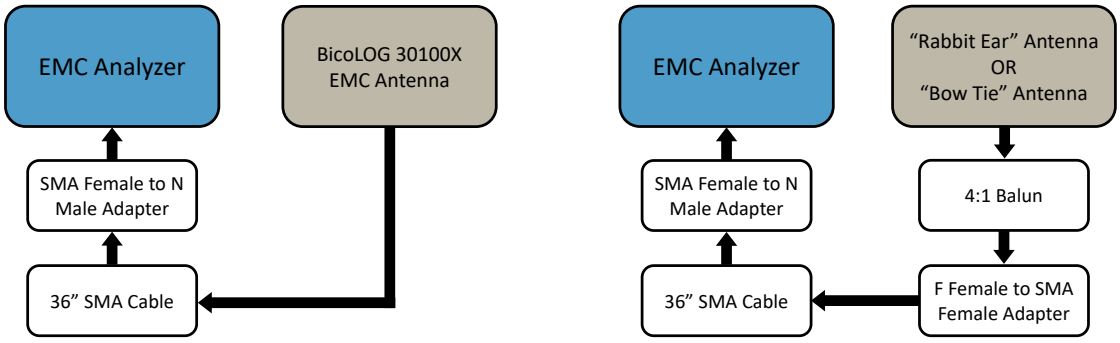


Figure 5.7: Diagram showing the RF signal chains used during the television antenna comparison test setup.

5.2 VHF "Rabbit Ear" Antenna Performance

Three different element lengths (EL) were chosen for the rabbit ear television antenna: 0.375 m, 0.667 m, and 0.991 m. These element lengths were carefully measured each time a measurement was performed to ensure consistency between measurements. Note that the longer element lengths made it impossible to use the rabbit ear antenna in the vertical orientation, and so the antenna was only characterized in the horizontal orientation. Each of these lengths are shown from shortest EL to longest EL in Figures 5.8a, 5.8b, and 5.8c.

Figure 5.8 shows the measured VSWR for each of the three element lengths that were tested. One can see that as EL increases, the VSWR improves at lower frequencies. This is best observed in the region from 60 MHz to 80 MHz, where the VSWR improves by more than four as EL increases. The element lengths shown in Figures 5.8b and 5.8c performed best in this region and the region from 100 MHz to 200 MHz. Interestingly, the shortest element length that was tested has a VSWR of less than three between approximately 250 and 350 MHz. However, all three element lengths showed a very oscillatory VSWR behavior above 200 MHz, which indicates that the usable region of 65 to 200 MHz for this antenna described by André and Wyatt [25] is accurate.

Using Equations 3.10 and 3.11, the measured VSWR of each element length shown in Figure 5.8 can be used to plot the measurement error due to impedance mismatch when the antenna is

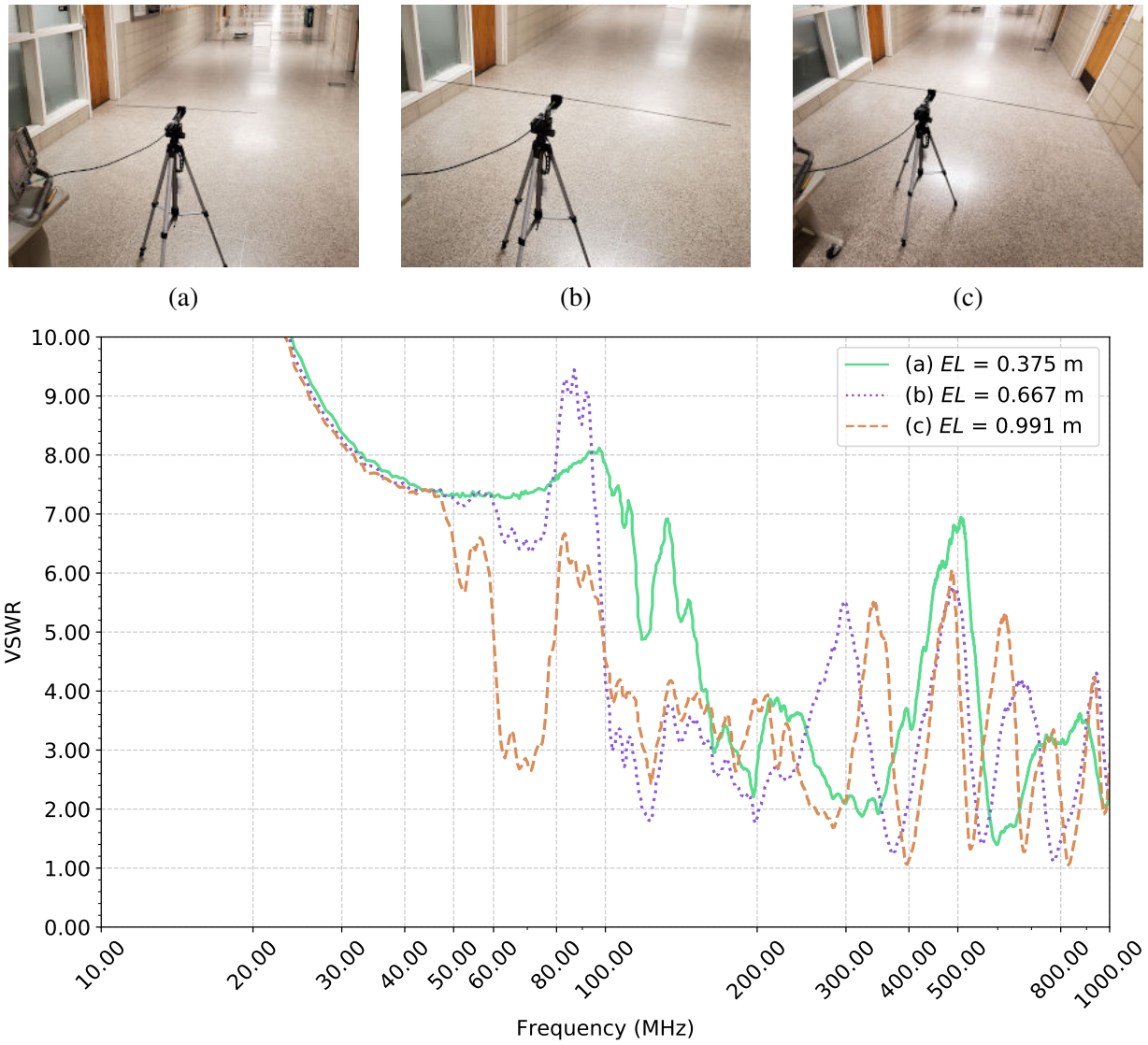


Figure 5.8: Measured VSWR of the VHF rabbit ear antenna with (a) $EL = 0.375$ m, (b) $EL = 0.667$ m, and (c) $EL = 0.991$ m. Each trace is the average of five trials.

used with the EMC analyzer (ME_E). Figure 5.9 shows the value of ME_E for all three rabbit ear element lengths tested. One can see that the mismatch error is less than 2.00 dB across the entire frequency range, indicating that the modified balun is doing a good job of matching the rabbit ear antenna to a 50 Ω system. This figure also shows that the element length can effect the mismatch error by up to ±0.5 dB at multiple frequencies.

Figures 5.10, 5.11, and 5.12 show the radiated emissions measured by the rabbit ear antenna at each of the three element lengths compared to the radiated emissions measured by the BicoLOG 30100X EMC antenna. In each of these figures, the BicoLOG 30100X results are shown at the top and the other three plots show the results from using the rabbit ear antenna (with the element length EL increasing from plot (b) to plot (d)). The source of the radiated emissions being measured was the custom EUT that was presented in Section 4.1.2.

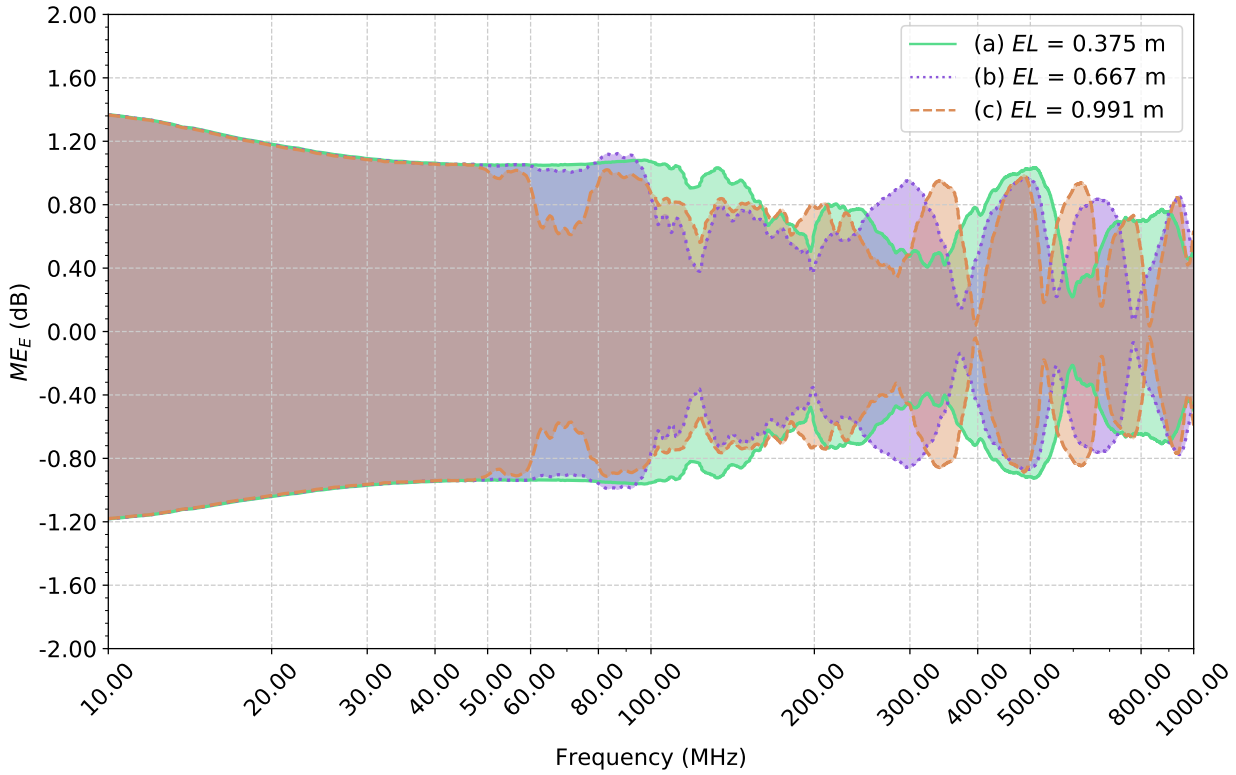


Figure 5.9: The amount of potential mismatch error with the EMC analyzer ME_E for all three of the rabbir ear antenna element lengths EL that were tested.

Figure 5.10 shows the results from measuring the custom EUT at 1 m with a base frequency setting of 40 MHz. One can see that the results from the BicoLOG 30100X shown in Figure 5.10a indicate that it is significantly more sensitive than the rabbit ear antenna. Figures 5.10b, 5.10c, and 5.10d show that the element length of the rabbit ear antenna does not appear to significantly affect which frequencies the antenna is sensitive to. The one exception is in Figure 5.10b, where the radiated emissions above 200 MHz are detected much better by the shortest antenna than by the other two element lengths.

Figure 5.11 shows the results from measuring the custom EUT at 1 m with a base frequency setting of 50 MHz. Figures 5.11b, 5.11c, and 5.11d show that the antenna with the longest element length is much more sensitive to the fundamental frequency of 50 MHz than the other length antennas. However, all three antennas were able to measure the strongest harmonic at 150 MHz with a margin of less than 10 dB.

Figure 5.12 shows the results from measuring the custom EUT at 1 m with a base frequency setting of 120 MHz. Figures 5.12b, 5.12c, and 5.12d show that the shortest element length was much more sensitive to frequencies above 400 MHz. However, this frequency range is not recommended because of the previously discussed VSWR behavior above 200 MHz. The plots in this figure also illustrate how the BicoLOG 30100X EMC antenna is significantly more sensitive than the rabbit ear antenna at all of the frequencies tested.

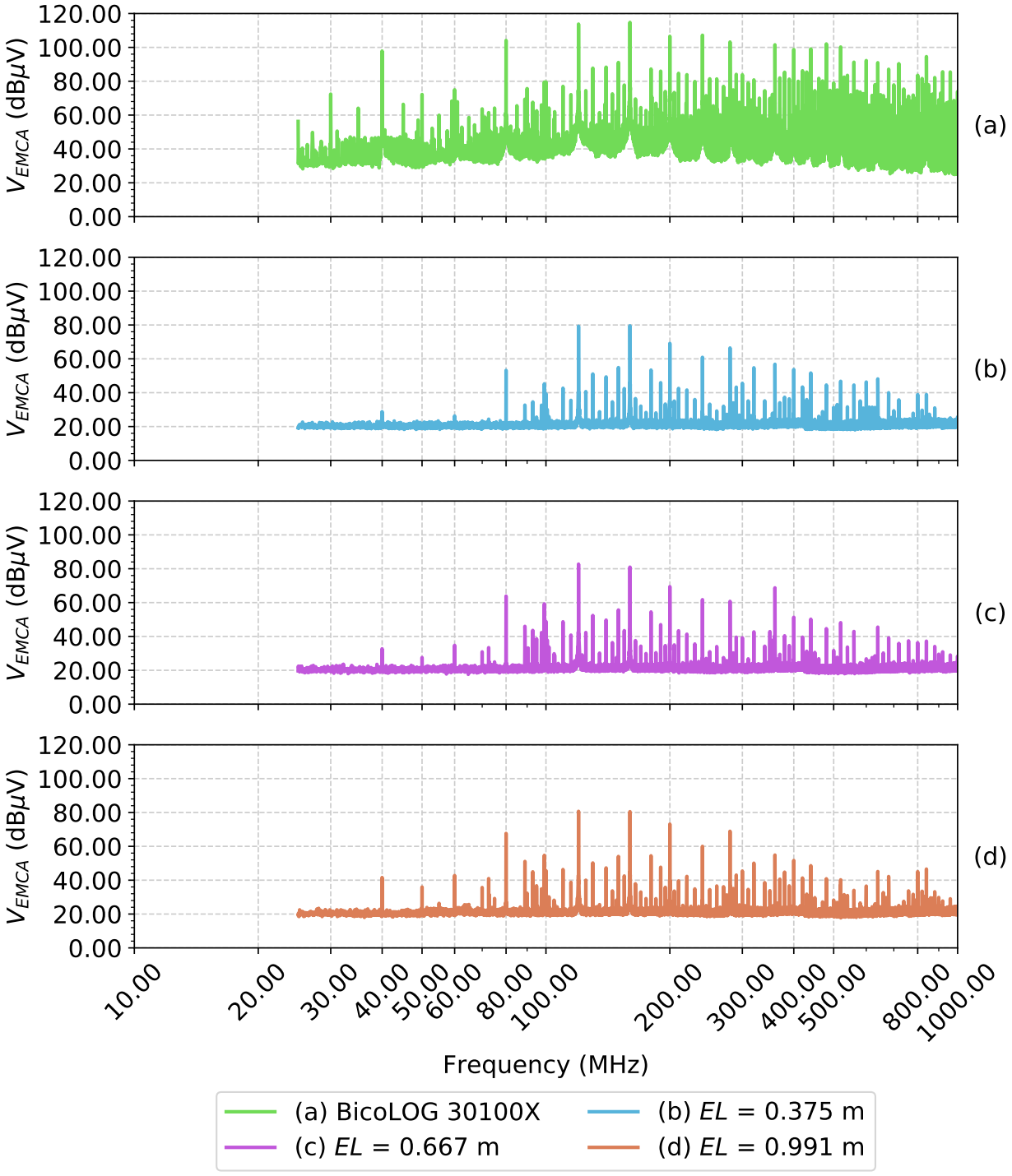


Figure 5.10: Radiated emissions from the custom EUT set to 40 MHz measured by using (a) the BicoLOG 30100X EMC antenna and by using the rabbit ear antenna with (b) EL = 0.375 m, (c) EL = 0.667 m, and (d) EL = 0.991 m.

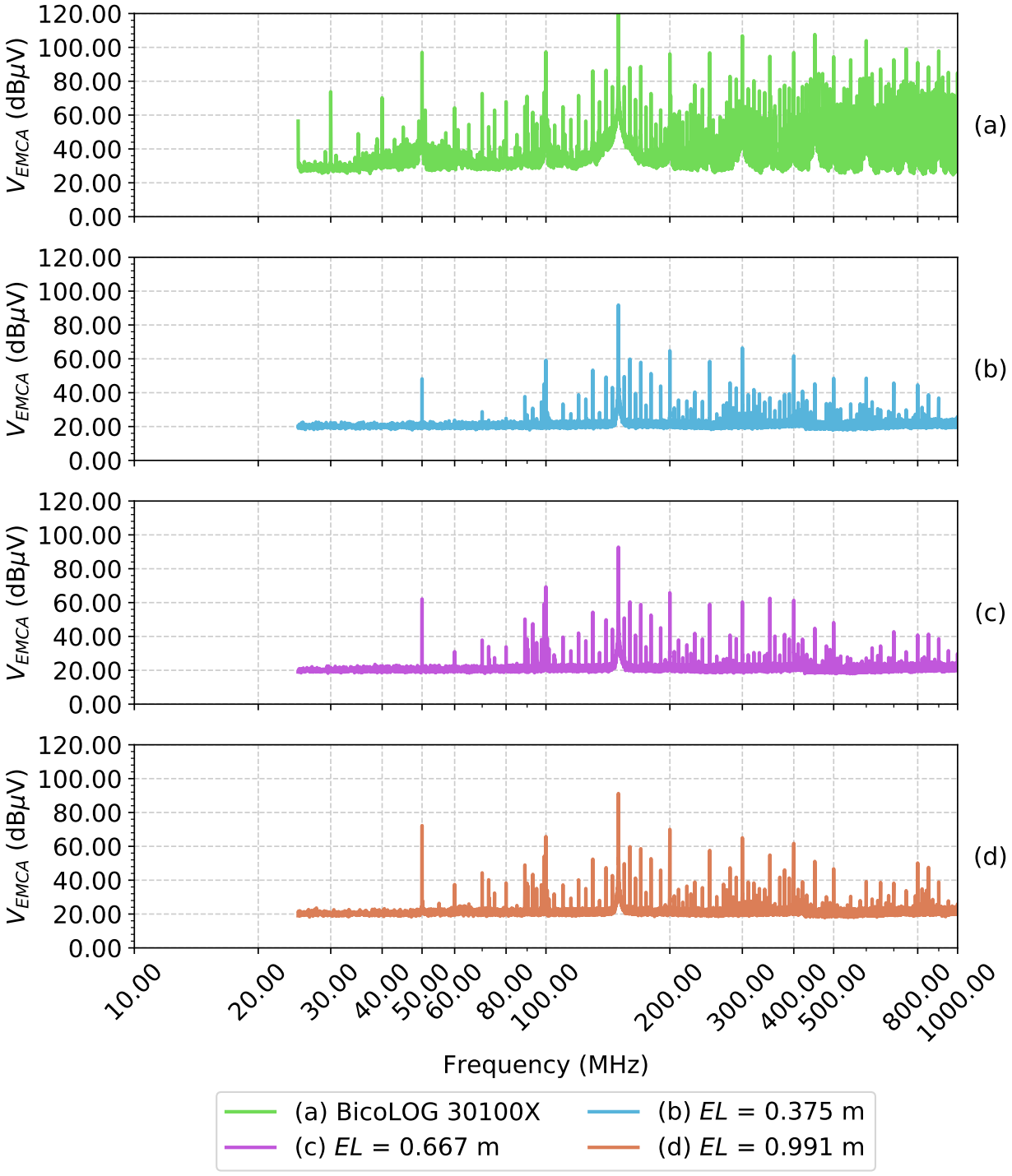


Figure 5.11: Radiated emissions from the custom EUT set to 50 MHz measured by using (a) the BicoLOG 30100X EMC antenna and by using the rabbit ear antenna with (b) EL = 0.375 m, (c) EL = 0.667 m, and (d) EL = 0.991 m.

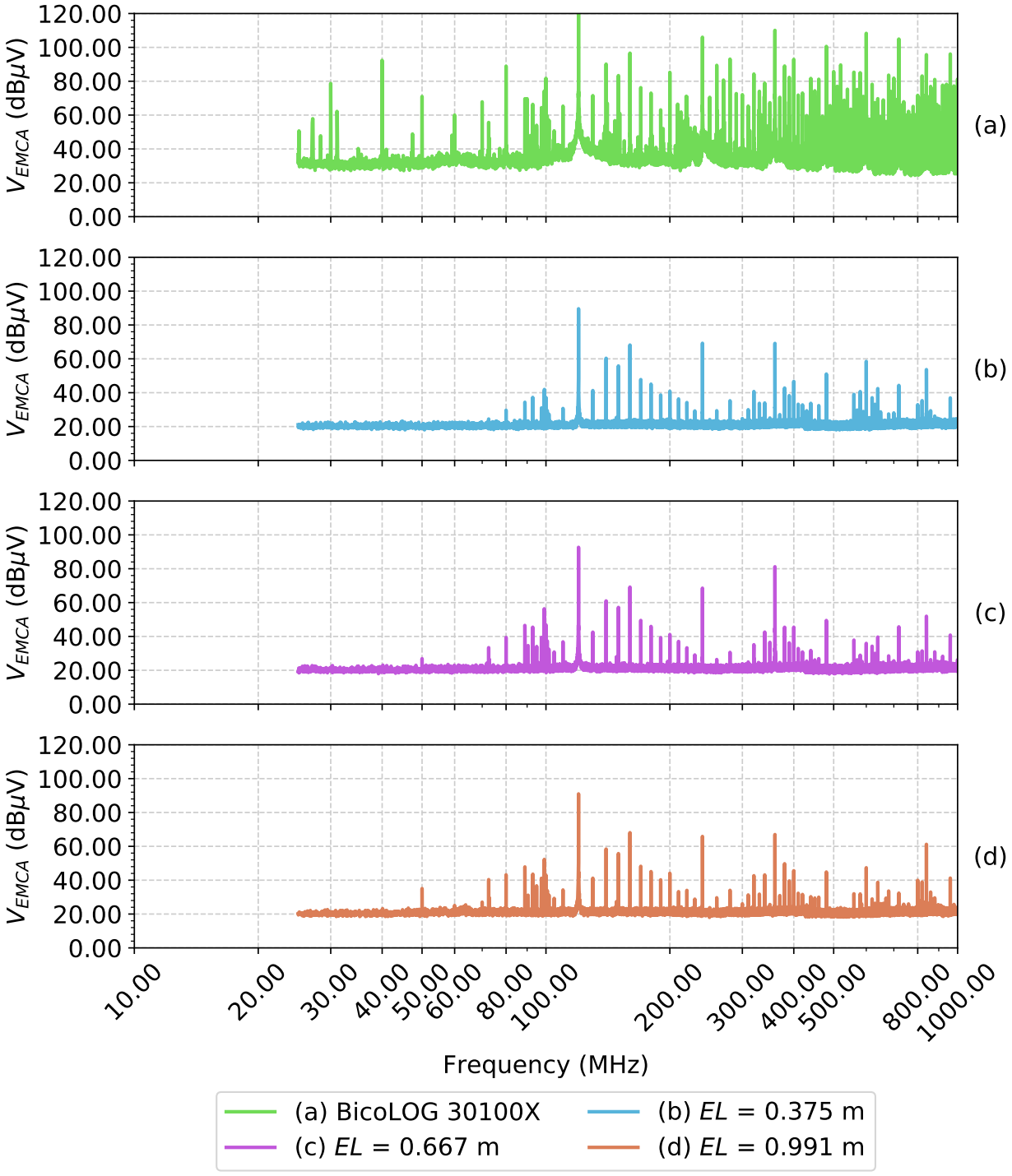


Figure 5.12: Radiated emissions from the custom EUT set to 120 MHz measured by using (a) the BicoLOG 30100X EMC antenna and by using the rabbit ear antenna with (b) EL = 0.375 m, (c) EL = 0.667 m, and (d) EL = 0.991 m.

5.3 UHF "Bow Tie" Antenna Performance

Figure 5.13 shows the measured VSWR for both the horizontal orientation (Figure 5.13a) and the vertical orientation (Figure 5.13b) of the bowtie antenna. The measured VSWR was less than six in the region from approximately 350 MHz to 1000 MHz. This result is very close to the usable range of 300 MHz to 800 MHz described by André and Wyatt [25].

Using Equations 3.10 and 3.11, the measured VSWR of each element length shown in Figure 5.13 can be used to find the amount of mismatch error from using each orientation of the bowtie antenna with the EMC analyzer (ME_E). Figure 5.14 shows the value of ME_E for both orientations of the bowtie antenna across the entire CISPR-22 Class B frequency band. As with the rabbit ear antenna, one can see from this figure that the mismatch error is less than 2.00 dB at all frequencies. This is due to the improved impedance match between the antenna and the SMA cable using the modified balun. Most notable is the sharp null that occurs just past 500 MHz. Here, the value of ME_E is nearly zero indicating an almost perfect impedance match with the EMC analyzer. Since the results from each orientation are so similar, only the horizontal orientation was used for radiated emission measurements.

Figures 5.15 and 5.16 show the radiated emissions measured by the bowtie antenna in each orientation compared to the radiated emissions measured by the BicoLOG 30100X EMC antenna. In each of these figures, the BicoLOG 30100X results are shown at the top and the other plots show the results from using the bowtie antenna. The source of the radiated emissions being measured was the custom EUT that was presented in Section 4.1.2.

Figure 5.15 shows the results from measuring the custom EUT at 1 m with a base frequency setting of 120 MHz. One can see that the bowtie antenna is much less sensitive than the BicoLOG 30100X EMC antenna. However, the bowtie antenna was able to measure the radiated emissions caused by the harmonics of the 120 MHz signal. On average, each measured harmonic was approximately 40 dB lower than what was measured by the BicoLOG 30100X EMC antenna. Also notable is how the bowtie antenna was able to measure the fundamental frequency at 120 MHz, well outside the suggested usable frequency range of 300 MHz to 1000 MHz. This is more indicative

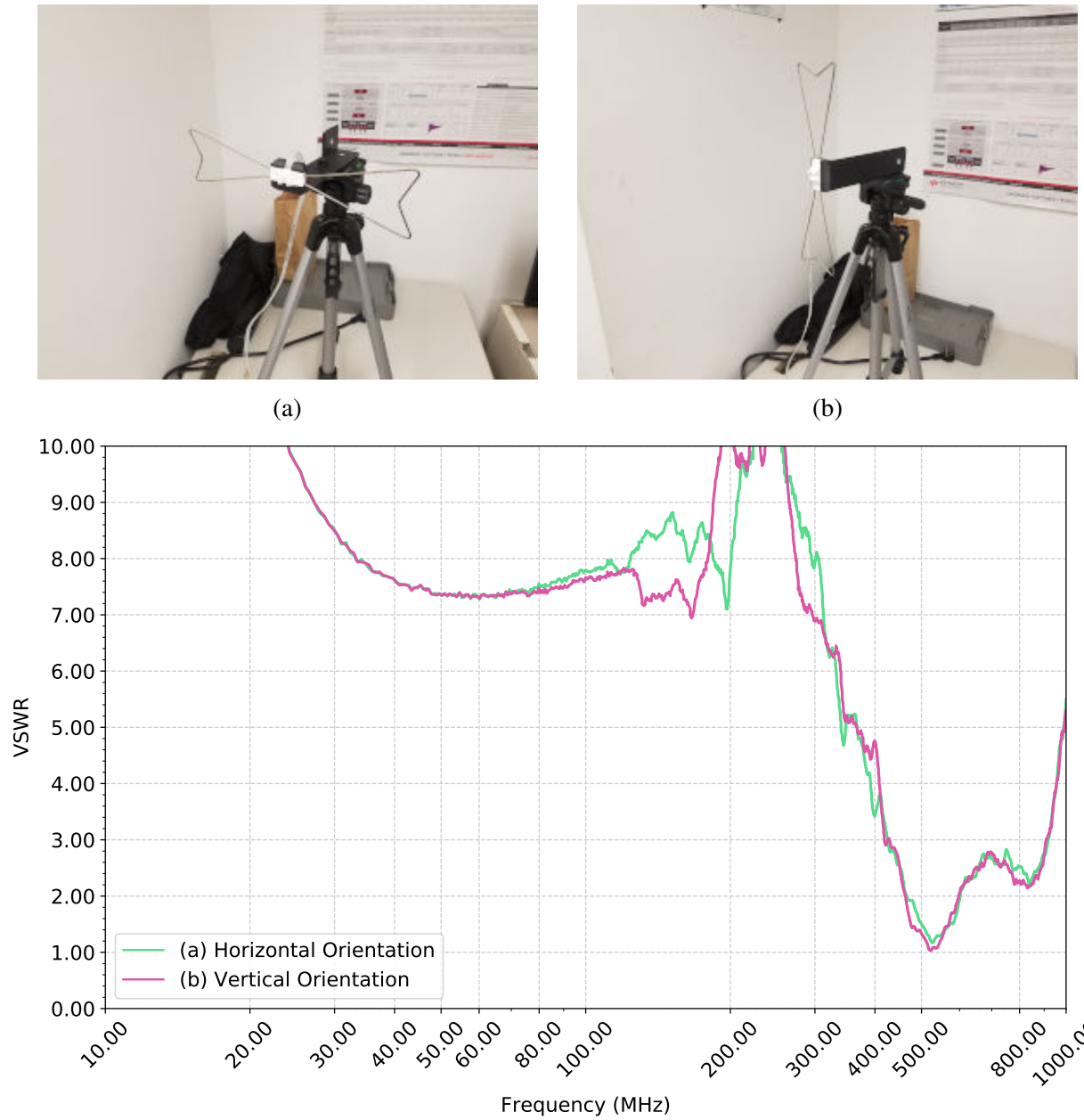


Figure 5.13: Measured VSWR of the UHF bowtie antenna in (a) the horizontal orientation and (b) the vertical orientation. Each trace is the average of five trials.

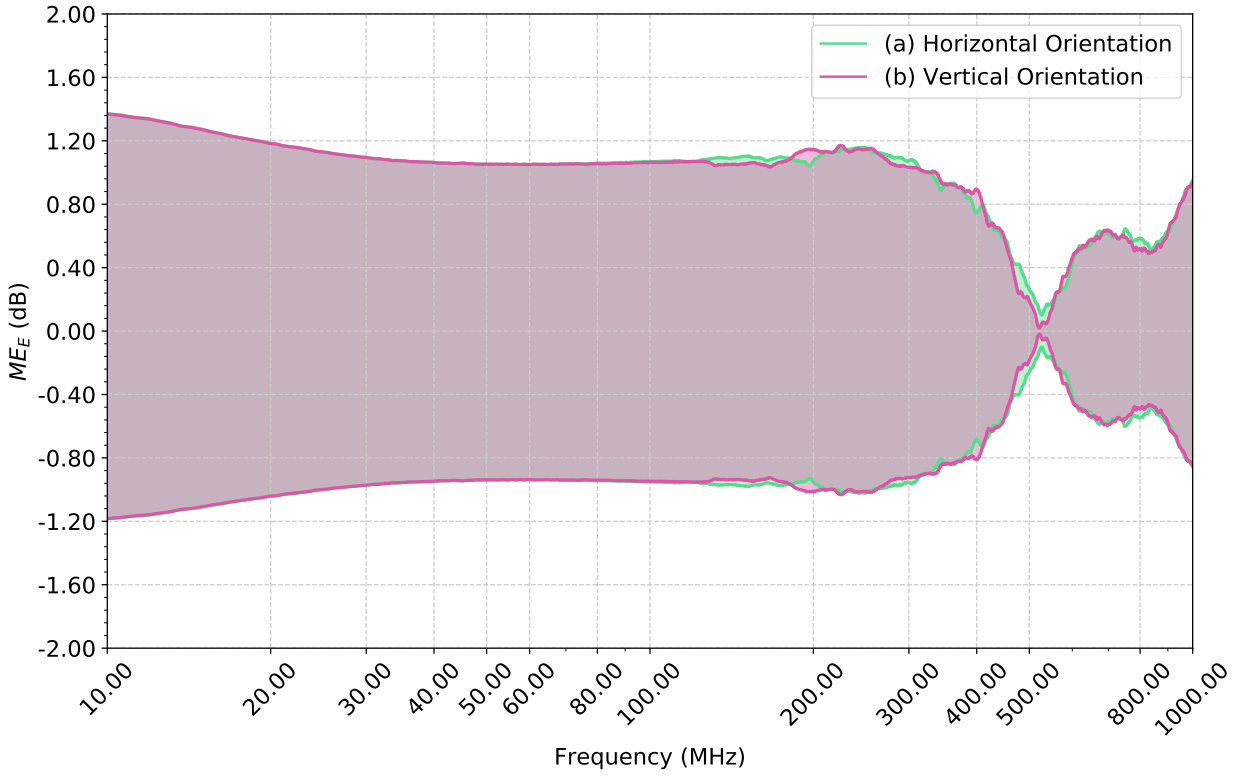


Figure 5.14: The amount of potential mismatch error with the EMC analyzer ME_E for the horizontal and vertical orientations of the bowtie antenna.

of the strong emissions created by the custom EUT and less indicative of the bowtie antenna being sensitive below 300 MHz.

Figure 5.16 shows the results from measuring the custom EUT at 1 m with a base frequency setting of 160 MHz. As was observed with the 120 MHz results, the bowtie antenna was able to measure the major harmonics created by the 160 MHz signal despite its very weak sensitivity relative to the BicoLOG 30100X EMC antenna. Both Figures 5.15 and 5.16 show that the bowtie is unable to receive the strong FM radio signals around 100 MHz that are received by the EMC antenna.

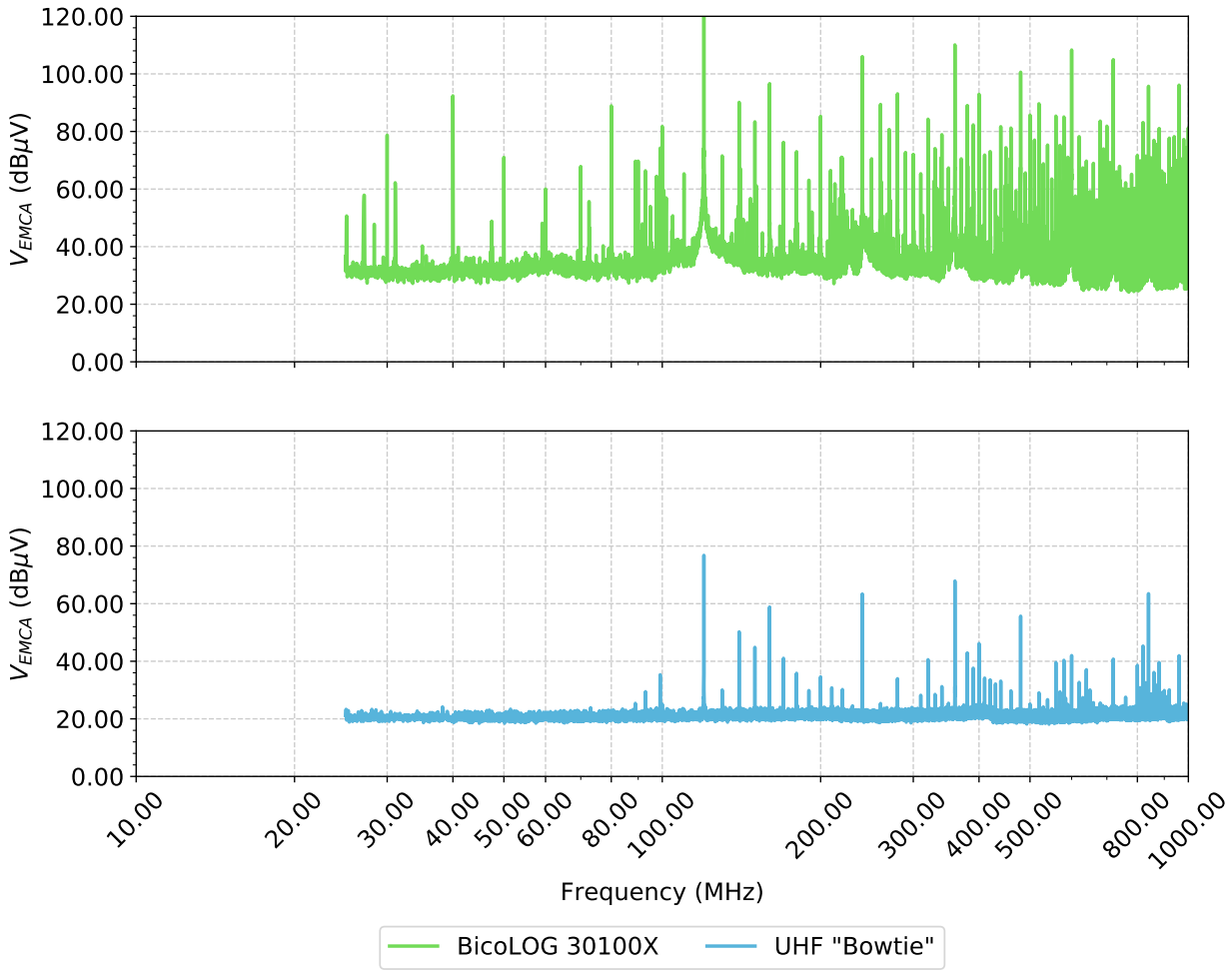


Figure 5.15: Radiated emissions from the custom EUT set to 120 MHz measured by using (a) the BicoLOG 30100X EMC antenna and (b) the bowtie antenna in the horizontal orientation.

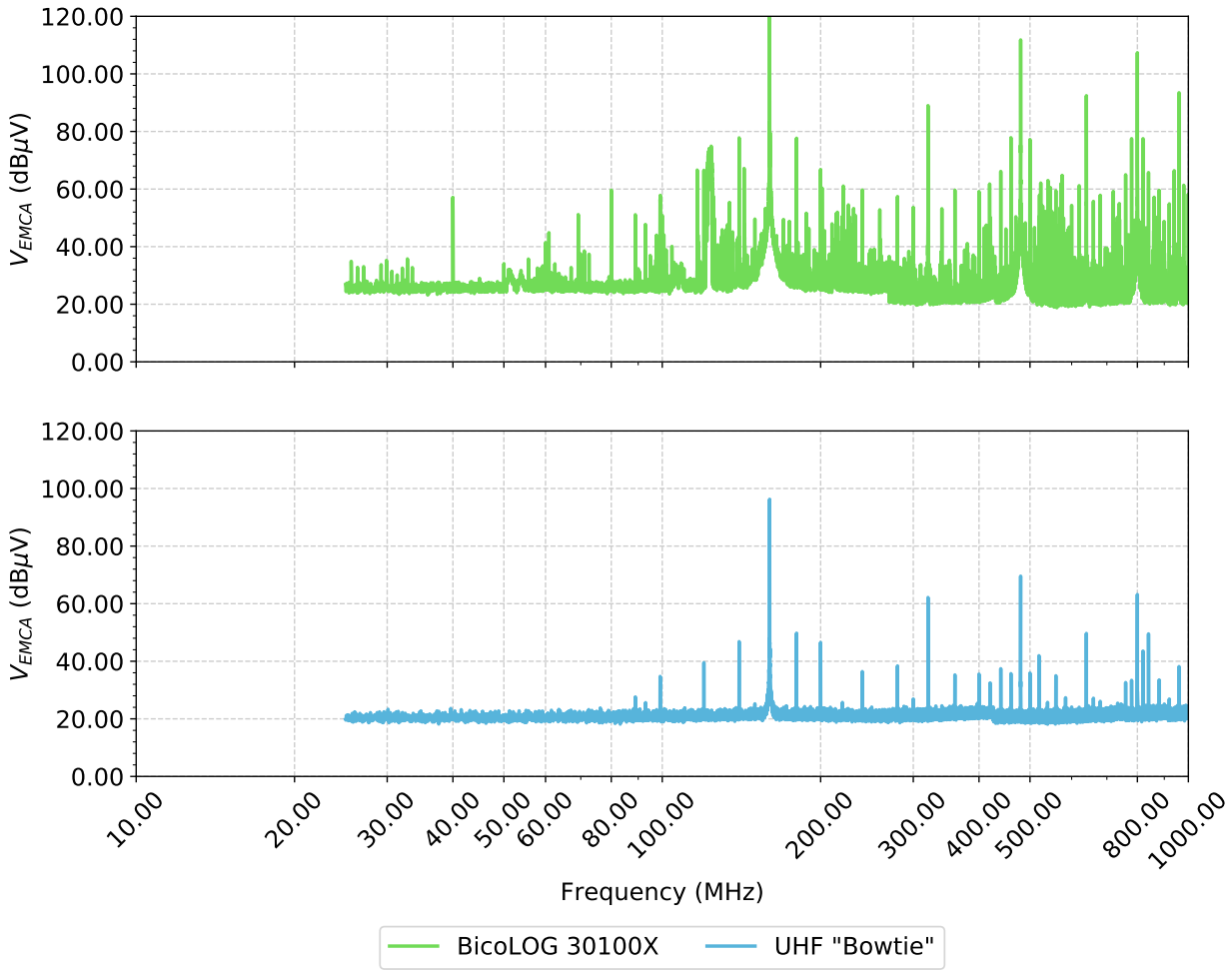


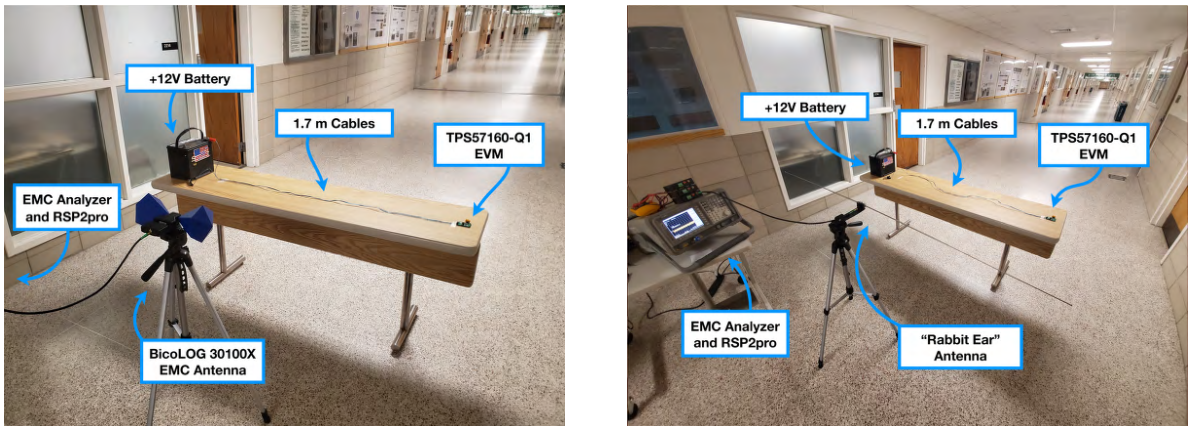
Figure 5.16: Radiated emissions from the custom EUT set to 160 MHz measured by using (a) the BicoLOG 30100X EMC antenna and (b) the bowtie antenna in the horizontal orientation.

5.4 Comparison to a More Traditional Test Setup

The last section of this thesis directly compares a more traditional precompliance test setup with the low-cost precompliance test setup. Figure 5.17a shows the more traditional test setup consisting of the EMC analyzer and the BicoLOG 30100X EMC antenna while Figure 5.17b shows the complete low-cost setup consisting of the RSP2pro and the rabbit ear antenna with an element length of 0.991 m. This element length was chosen based upon the radiated emissions from the EUT. Figure 5.19 shows the RF signal chains used by each test setup.

The EUT that was measured for the results presented in this section was the TPS57161-Q1 evaluation module from Texas Instruments. This module is based around a high-frequency buck converter that is designed to convert a 12 V_{DC} input to a 5 V_{DC} output [61]. The radiated emissions for this evaluation module were measured by Texas Instruments [61] in a controlled RF environment with a biconnical EMC antenna like the one shown in Figure 3.2a. The results from [61] can be used to directly access the effectiveness of both test setups shown in Figure 5.17. Radiated emissions at two frequencies, approximately 41 and 50 MHz, were measured for comparison to the results from [61].

Figures 5.20a and 5.20b show the RF spectrum as measured by the more traditional test setup near 41 MHz when the EUT is powered off and then turned on, respectively. The results here are



(a) (b)

Figure 5.17: (a) The traditional precompliance test setup and (b) the low-cost test setup.

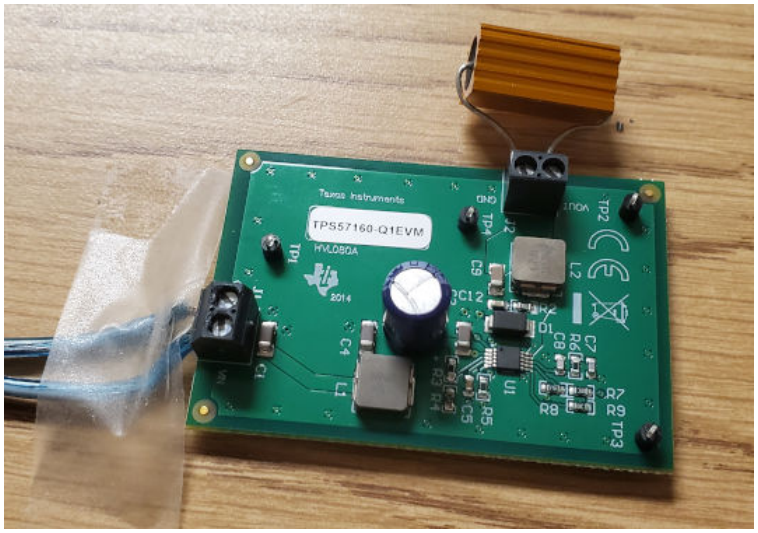
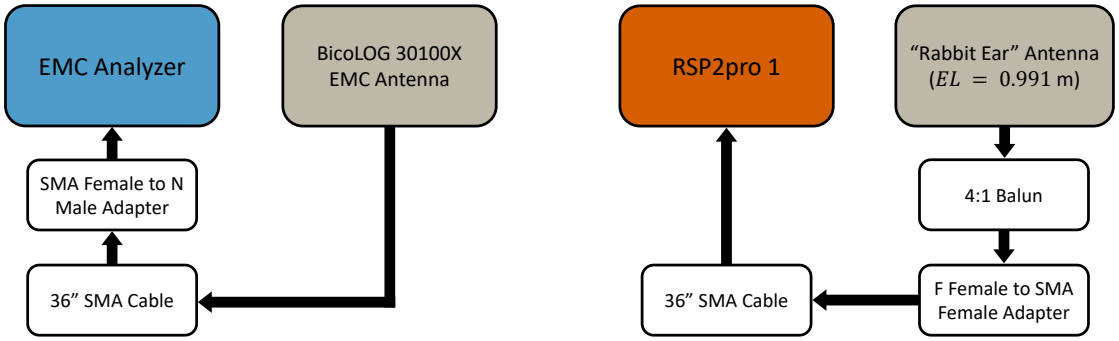


Figure 5.18: Detailed view of the EUT showing the gold 5 Ω resistive load.



(a) (b)

Figure 5.19: Diagram showing the RF signal chains of (a) the traditional test setup and (b) the low-cost test setup.

shown in dBm so that they can be directly compared to the same measurement in SDRuno (which can only display magnitude in terms of dBm). One can see that the radiated emissions from the EUT were approximately -55.00 dBm at a frequency near 49.950 MHz. Figures 5.21a and 5.21b shows the same radiated emissions as measured by RSP2pro 1. One can see from the background measurement in Figure 5.21a that the noise floor observed by the RSP2pro is almost 15 dB lower than the noise floor measured by the EMC analyzer in Figure 5.20a. This is likely due in part to the the low sensitivity of the rabbit ear antenna when compared to the BicoLOG 30100X EMC

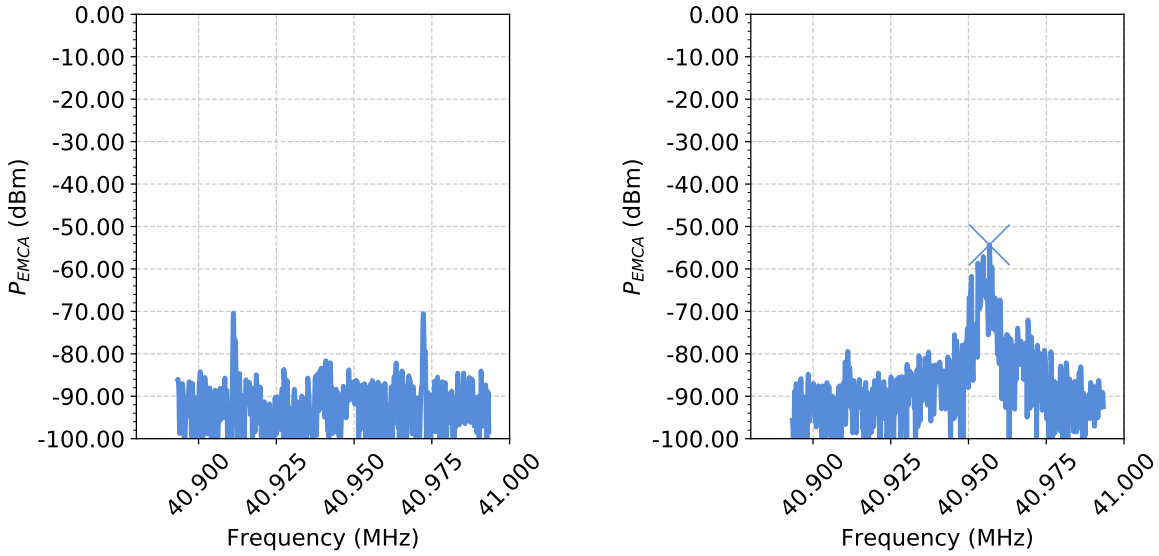
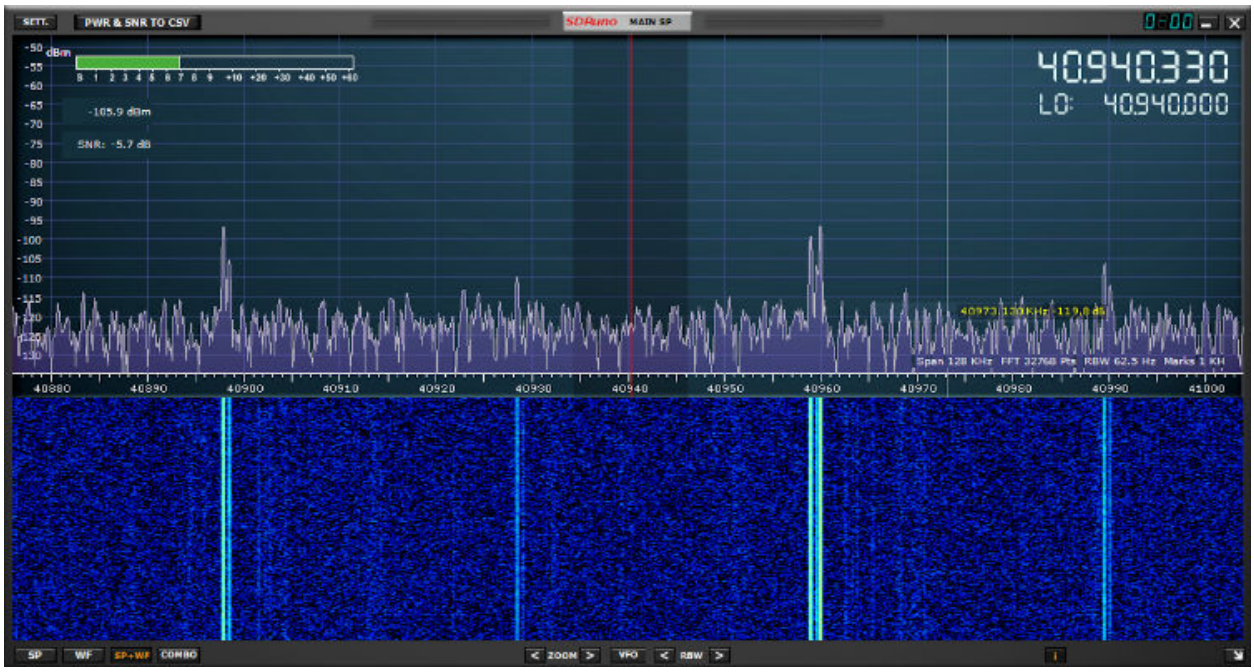


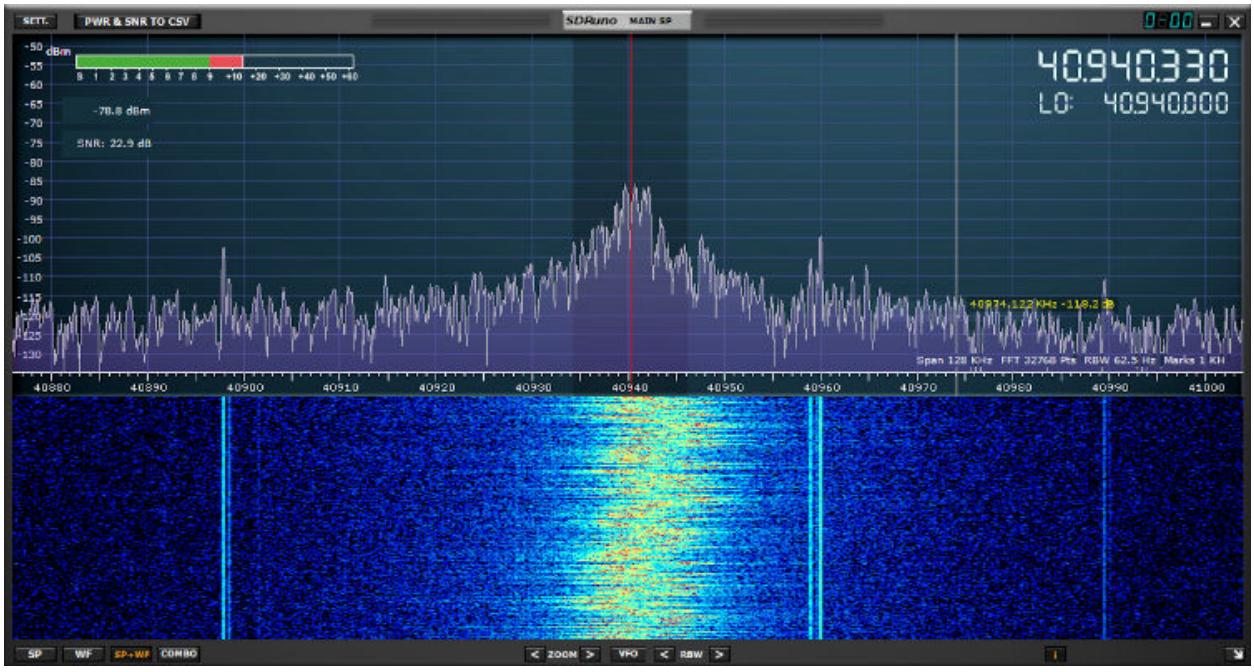
Figure 5.20: (a) The background measurement with the EUT off and (b) the radiated emissions from the EUT near 41 MHz.

antenna. Figure 5.21b shows the radiated emissions from the EUT. The level measured by SDRuno is much lower than what was measured by the traditional test setup, with a peak value of only -78.8 dBm. However, the waterfall plot in the lower half of this figure clearly shows how the emissions from the EUT are continuous in nature. The waterfall plot also better illustrates the bandwidth of the radiated emission.

Figures 5.22a and 5.22b show the RF spectrum as measured by the more traditional test setup near 50 MHz when the EUT is powered off and then turned on, respectively. The results here are shown in dBm so that they can be directly compared to the same measurement in SDRuno (which can only display magnitude in terms of dBm). Compared to the emissions near 41 MHz, one can see that the radiated emissions measured by the more traditional test setup are much weaker, having a magnitude closer to -70.00 dBm. Figures 5.23a and 5.23b shows the same radiated emissions as measured by RSP2pro 1. One can see from the background measurement in Figure 5.23a that the noise floor observed by the RSP2pro is more than 15 dB lower than the noise floor measured by the EMC analyzer in Figure 5.22a. However, Figure 5.23b shows that the radiated emissions from the EUT as measured by SDRuno are much lower than what was measured by the traditional test



(a)



(b)

Figure 5.21: (a) The background measurement with the EUT off and (b) the radiated emissions from the EUT near 41 MHz.

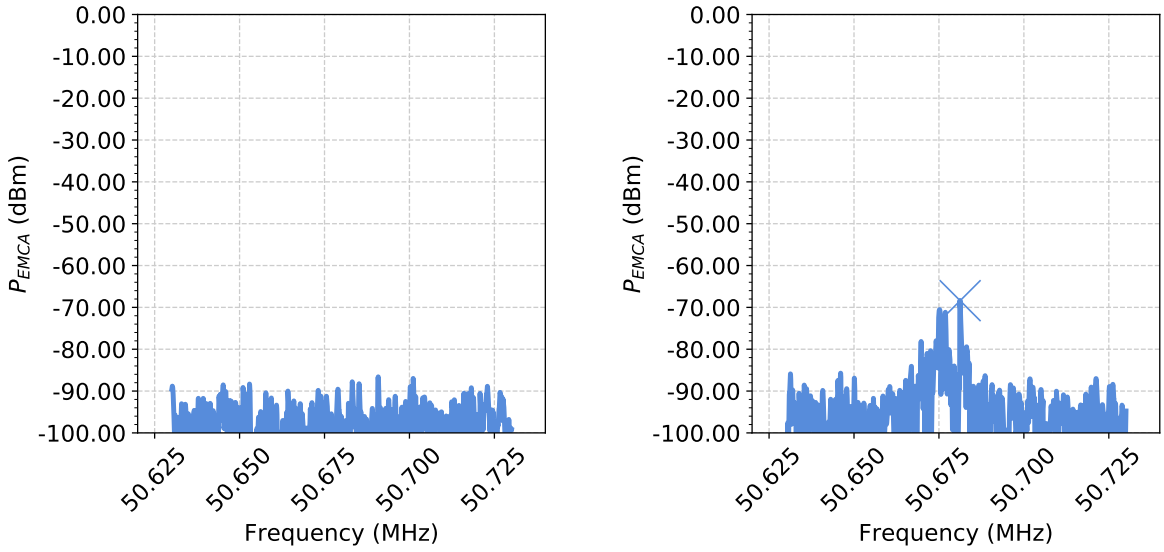


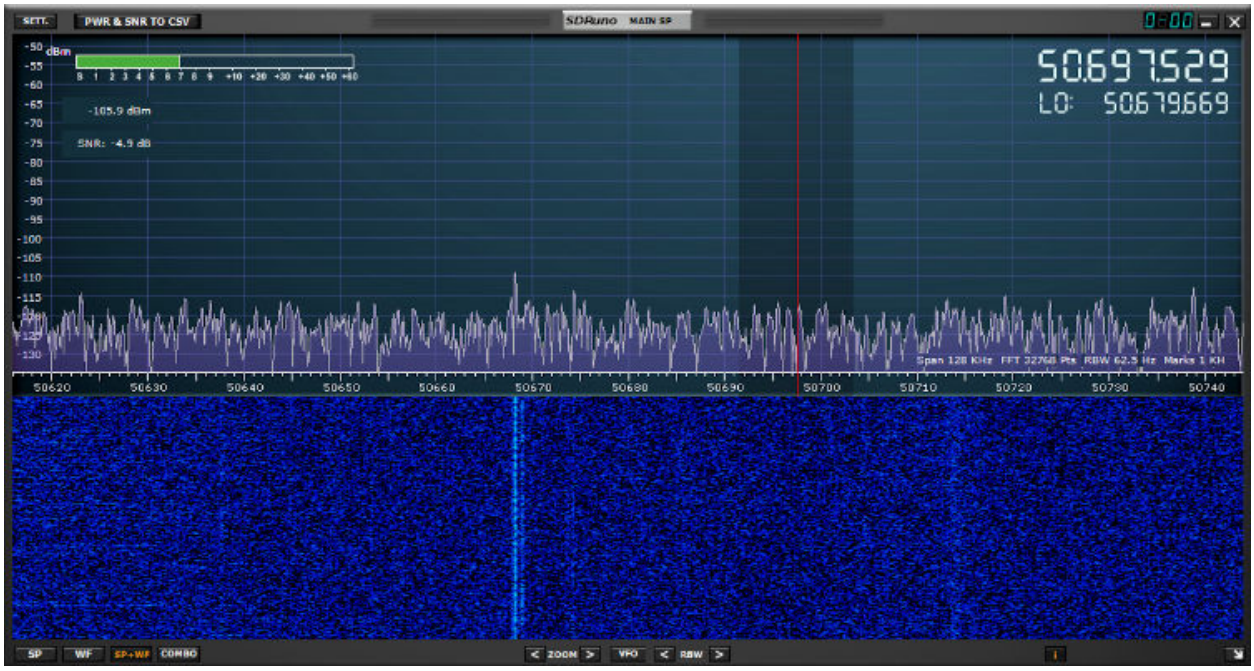
Figure 5.22: (a) The background measurement with the EUT off and (b) the radiated emissions from the EUT near 50 MHz.

setup, with a peak value of only -92.1 dBm.

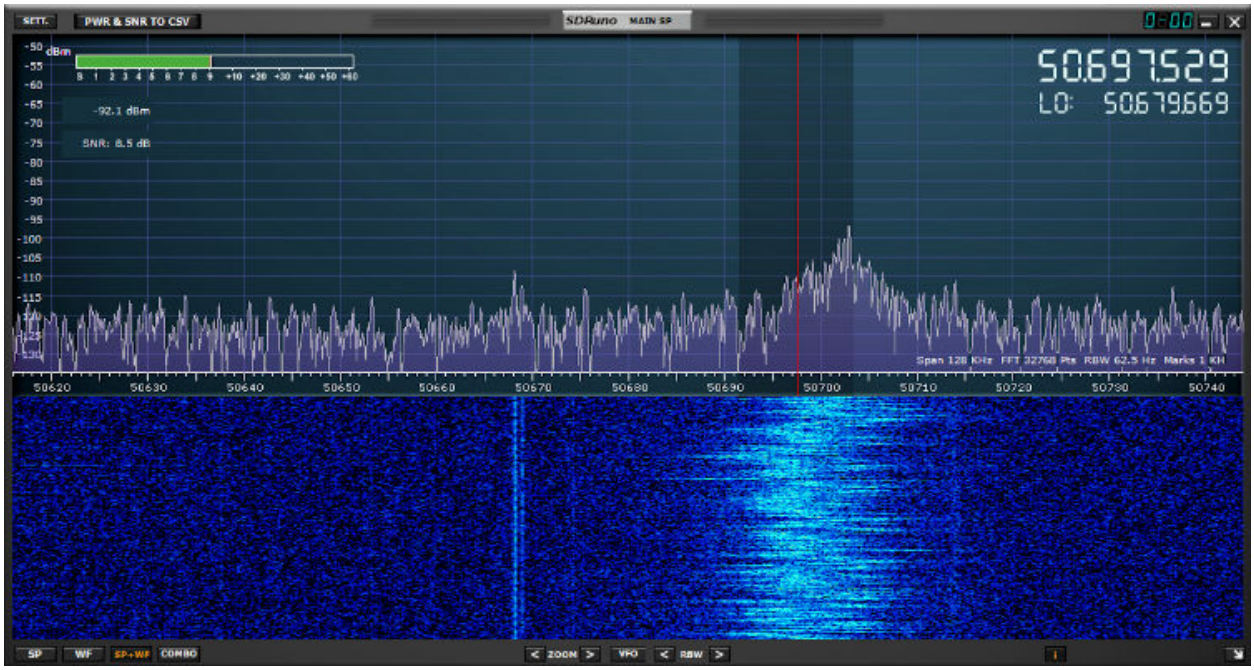
The results from the radiated emission measurements in this section have been summarized in Tables 5.1 and 5.2. These tables also compare the measured radiated emissions with the emissions measured by Texas Instruments from [61]. The radiated emissions measured by the traditional test setup were converted to field strength using the same process that was used in Section 4.3. Since the antenna factor of the rabbit ear antenna is unknown, the same could not be done for the measurements recorded with the low-cost setup.

Table 5.1 shows how the background RF signals at each data point compared. The measurement from [61] (shown in column three) was significantly lower than the level measured by the traditional test setup. This is due to the uncontrolled RF environment used in this thesis. For both data points, the background level measured by the low-cost setup was significantly lower than the other two test setups. This is because of the low sensitivity of the rabbit ear antenna. Again, note that the low-cost test setup measurement is not a field strength but the equivalent magnitude that results from converting SDRuno's dBm measurement to dBμV using Equation 3.9.

Table 5.2 compares the radiated emission measurements at each data point. As was observed



(a)



(b)

Figure 5.23: (a) The background measurement with the EUT off and (b) the radiated emissions from the EUT near 50 MHz.

Data Point	Approximate Frequency (MHz)	Reference Measurement (dBμV/m)	Traditional Setup Measurement (dBμV/m)	Low-cost Setup Measurement (dBμV)
Background RF @ 1	40.95	4.13	16.53	1.10
Background RF @ 2	50.68	2.17	4.01	1.10

Table 5.1: Comparison of the radiated emissions measured from the EUT shown in Figure 5.18 with several different test setups.

Data Point	Approximate Frequency (MHz)	Reference Measurement (dBμV/m)	Traditional Setup Measurement (dBμV/m)	Low-cost Setup Measurement (dBμV)
1	40.95	34.13	48.33	28.20
2	50.68	16.41	31.69	14.90

Table 5.2: Comparison of the radiated emissions measured from the EUT shown in Figure 5.18 with several different test setups.

in Table 5.1, the measurement from [61] (shown in column three) was significantly lower than the level measured by the traditional test setup. For both data points, the background level measured by the low-cost setup was significantly lower than the other two test setups. This is because of the low sensitivity of the rabbit ear antenna. Again, note that the low-cost test setup measurement is not a field strength but the equivalent magnitude that results from converting SDRuno's dBm measurement to dBμV using Equation 3.9.

The results from this section show that the low-cost test setup consisting of the common television antenna and the RSP2pro is capable of identifying the radiated emissions from a product when used properly. However, without knowing the antenna factor of the television antennas, the conversion from the RSP2pro measurement to the radiated emission field strength cannot be performed. This makes the low-cost test setup suitable for performing multiple bench-top radiated emission tests throughout the electronic product design process to check for problem frequencies. Once the problem frequencies have been identified, the low-cost test setup can be used to assess

the effectiveness of changes that are made to the product. The main benefits from the low-cost test setup come from its low cost and compact size relative to the traditional test setup.

The RSP2pro units used in this thesis were purchased for approximately $192.00 each. The television antennas, balun, 3D-printed mounting hardware, and camera tripod accessories all cost less than $50.00. By comparison, the Bicolog30100X EMC antenna that was used in this thesis cost just under $3000.00. While the specific EMC analyzer used in this thesis was provided by the MSU ECE department, an eBay search at the time of writing for "E7401A EMC analyzer" brings up several listings for used units that range in price from $5500.00 to $6999.99. Averaging these prices to get an estimated value of $6250.00, the total cost of the more traditional precompliance test setup pictured in Figure 5.17a is nearly $9250.00 while the low-cost setup shown in Figure 5.17b cost less than $250.00. Note that this price does not include the cost of the necessary host computer to use SDRuno. This author assumes that most electronic product designers have access to a computer, and the SDRUno software is free (even for commercial use).

Just comparing the E7401A EMC analyzer and the RSP2pro reveals several other advantages of the low-cost setup. The EMC analyzer has a rectangular footprint of 409 mm by 373 mm and a weight of 12.6 kg [43]. By comparison, the RSP2pro has a rectangular footprint of only 99 mm by 87 mm and a weight of 0.296 kg [42]. This makes the RSP2pro extremely portable and very convenient for performing quick bench-top measurements. The RSP2pro is also fully powered via USB and only consumes about 850 mW [42] while the EMC analyzer requires 120 V AC and can consume up to 300 W during operation [43].

The major disadvantages with the RSP2pro are mostly related to its reliance on SDRuno. Performing a full spectrum sweep with a spectrum analyzer is much more convenient than manually tuning to each frequency that must be tested. Additionally, only 10 MHz of the RF spectrum can be observed at a time within SDRuno. Viewing 10 MHz of the spectrum at a time with SDRuno requires the use of the zero-IF mode, which introduces spurious mixing and intermodulation products that can be mistaken for radiated emissions. The difference between each of these noise signals can be determined by changing the center frequency in SDRuno. The spurious mixing products will also

change in frequency, but intermodulation products (and more importantly, the radiated emissions being measured) will not [62]. To determine the difference between intermodulation products and the radiated emissions that one is trying to measure, a 10 dB attenuator should be added to the RF signal chain. Intermodulation products will be reduced by much more than 10 dB while the actual RF signals being measured (and the spurious mixing products) will be reduced by 10 dB [62]. These types of false signals are not an issue when using a high quality spectrum analyzer like the E7401A.

CHAPTER 6

CONCLUSIONS

This thesis examines the potential for using the RSP2pro software-defined radio (SDR) for the precompliance testing of radiated emissions. A margin of error for the signal level measured by the RSP2pro was successfully developed by directly comparing the performance of the RSP2pro to a traditional spectrum analyzer. This thesis also examines the use of low-cost television antennas for identifying radiated emissions across the CISPR-22 Class B frequency band from 30 MHz to 1000 MHz. By combining the RSP2pro with these antennas, a complete low-cost precompliance test setup was demonstrated. The low-cost setup introduced in this thesis can be used by electronic product designers from all backgrounds to quickly and easily test a product for radiated emissions throughout the design process.

This work presented in this thesis can be greatly expanded upon, particularly with regards to the characterization of the television antennas. Developing a method for either simulating or measuring the antenna factors of these antennas would allow the measurements taken with the low-cost test setup to be directly converted to field strength and compared to the radiated emission limits. Further characterization of the RSP2pro would allow for a better understanding of the relationship between the measured SNR in SDRuno and the accuracy of the measured signal power. The greatest avenue for more research would be finding a software replacement for SDRuno that has a user interface more similar to a spectrum analyzer. Such a software could implement the functionality of a tracking generator so that the RSP2pro's 10 MHz measurement bandwidth could be used to span much larger frequency ranges.

At the time of writing, SDRplay is working on a software interface for their SDRs that has many of the capabilities usually found on a spectrum analyzer. While the SDRs offered by SDRplay are currently the only software-centric SDRs with absolute power measurement capabilities, perhaps others will become available as designers realize the full potential of the SDR-based spectrum analyzer concept.

APPENDICES

APPENDIX A

EUT CIRCUIT DIAGRAMS AND BILL OF MATERIALS

Figures A.1 and A.2 show the circuit diagrams for each of the PCB boards that together create the custom EUT from Section 4.1.2. The schematics, bill of materials (BOM), and the gerber files for each PCB board can be found in the Digital Appendix (refer to Appendix B).

The custom EUT PCBs were designed using KiCad, an open-source electric design automation software available for multiple operating systems. Learn more about it at: https://www.kicad-pcb.org/.

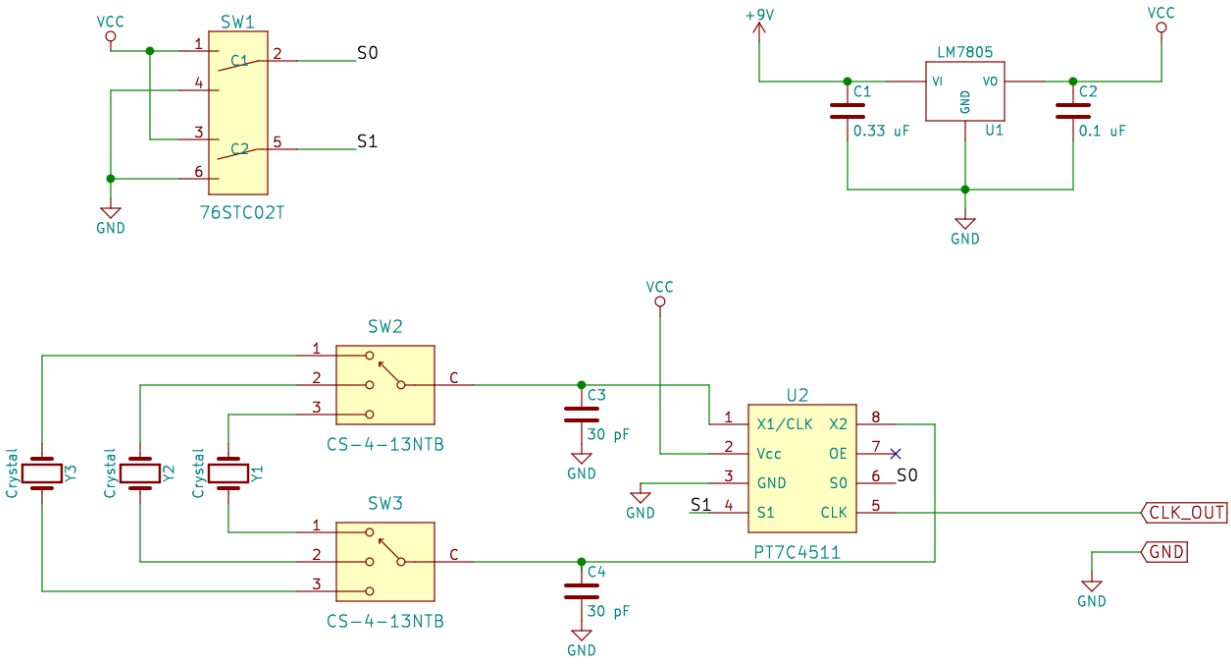


Figure A.1: Electrical circuit for the source side of the custom EUT from Section 4.1.2.

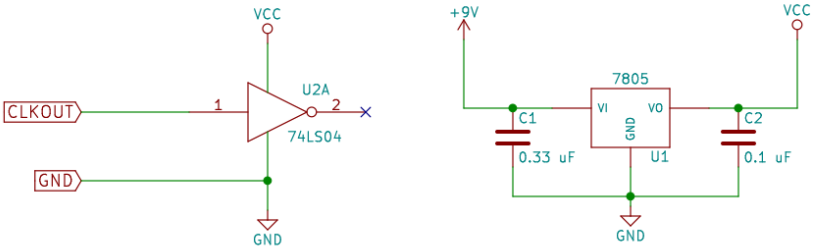


Figure A.2: Electrical circuit for the load side of the custom EUT from Section 4.1.2.

APPENDIX B

DIGITAL APPENDIX CONTENTS LIST

The digital version of this thesis includes a Digital Appendix in the form of a single compressed archive. The Digital Appendix includes the following subfolders and files:

- Section 3.2 Input Data Sets (from Table 3.2)
  - agilent_11955A.csv
  - agilent_11955A_minus10.csv
  - agilent_11955A_plus10.csv
  - agilent_11956A.csv
  - agilent_11956A_minus10.csv
  - agilent_11956A_plus10.csv
  - bicolog30100X.csv
  - bicolog30100X_minus10.csv
  - bicolog30100X_plus10.csv
- Section 4.1.2 Design Files
  - custom_eut_BOM.xlsx
  - custom_eut_load_gerbers.zip
  - custom_eut_load_schematic.pdf
  - custom_eut_source_gerbers.zip
  - custom_eut_source_schematic.pdf

- Section 5.1 Design Files
  - antenna_clip_body1.STL
  - antenna_clip_body2.STL
  - antenna_holder_bowtie.STL
  - antenna_holder_rabbit.STL

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